

gilmore-lie-groups: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00001	225	█ 1.000	$U(2) \rightarrow H_4$	$U(2) \rightarrow H_4$	13.4 The contraction $U(2) \rightarrow H_4$
gilmore-lie-groups_F00002	63	█ 0.978	$S_0(3)$	$SO(3)$	13. The group $SO(3)$ is three dimensional. Show that every matrix in $SO(3)$ can be
gilmore-lie-groups_F00003	63	█ 1.000	$S_0(4)$	$SO(4)$	degeneracy under the Lie group $SO(4)$. All bound states with the same principal
gilmore-lie-groups_F00004	224	█ 0.997	$S_0(4,1)$	$SO(4,1)$	This group can be contracted from either $SO(4,1)$ or $SO(3,2)$.
gilmore-lie-groups_F00005	215	█ 0.943	$S_0(4,2)$	$SO(4,2)$	Show that the conformal group on space-time is $SO(4,2)$.
gilmore-lie-groups_F00006	15	■ 0.764	$(+,-,\times,\div)$	$(+,-,\times,\div)$	four standard operations of arithmetic $(+,-,\times,\div)$ as well as extraction of the n th
gilmore-lie-groups_F00007	15	■ 0.764	n	n	four standard operations of arithmetic $(+,-,\times,\div)$ as well as extraction of the n th
gilmore-lie-groups_F00008	17	█ 1.000	G	G	if and only if its Galois group G contains a chain of subgroups $G = G_0 \supset G_1 \supset$
gilmore-lie-groups_F00009	17	█ 1.000	$G=G_{\{0\}} \supset G_{\{1\}} \supset$	$G = G_0 \supset G_1 \supset$	if and only if its Galois group G contains a chain of subgroups $G = G_0 \supset G_1 \supset$
gilmore-lie-groups_F00010	17	█ 1.000	$\cdots \supset G_{\{\omega\}}=I$	$\cdots \supset G_\omega = I$	$\cdots \supset G_\omega = I$ with the properties:
gilmore-lie-groups_F00011	17	█ 1.000	$G_{\{i+1\}}$	G_{i+1}	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00012	17	█ 1.000	$G_{\{i\}}$	G_i	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00013	17	█ 1.000	$G_{\{i\}} / G_{\{i+1\}}$	G_i/G_{i+1}	(ii) each factor group G_i/G_{i+1} is commutative.
gilmore-lie-groups_F00014	17	█ 0.998	$G=\left\{g_{\{1\}}, g_{\{2\}}, \ldots\right\}$	$G = \{g_1, g_2, \ldots\}$	A group G is defined as follows. It consists of a set of operations $G = \{g_1, g_2, \ldots\}$,
gilmore-lie-groups_F00015	18	█ 1.000	$g_{\{i\}} \in G, g_{\{j\}} \in G$	$g_i \in G, g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in G$.
gilmore-lie-groups_F00016	18	█ 1.000	$g_{\{i\}} \cdot g_{\{j\}} \in G$	$g_i \cdot g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in G$.
gilmore-lie-groups_F00017	18	█ 1.000	$g_{\{i\}} \in G, g_{\{j\}} \in G, g_{\{k\}} \in G$	$g_i \in G, g_j \in G, g_k \in G$	(ii) Associativity: for all $g_i \in G, g_j \in G, g_k \in G$,
gilmore-lie-groups_F00018	18	■ 0.835	I	I	(iii) Identity: there is a group operation, I (identity operator), with the property that
gilmore-lie-groups_F00019	18	█ 1.000	$g_{\{i\}}$	g_i	(iv) Inverse: every group operation g_i has an inverse (called g_i^{-1}):
gilmore-lie-groups_F00020	18	█ 1.000	$g_{\{i\}}^{-1}$	g_i^{-1}	(iv) Inverse: every group operation g_i has an inverse (called g_i^{-1}):
gilmore-lie-groups_F00021	18	█ 1.000	$z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}$	$z_1, z_2, \ldots, z_n$	is the group that permutes the roots $z_1, z_2, \ldots, z_n$ among themselves and leaves the
gilmore-lie-groups_F00022	18	■ 0.658	$P_{\{n\}}$	P_n	This group, called the permutation group P_n or the symmetric group S_n , has $n!$
gilmore-lie-groups_F00023	18	■ 0.658	$S_{\{n\}}$	S_n	This group, called the permutation group P_n or the symmetric group S_n , has $n!$

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gilmore-lie-groups_F00024	18	0.658	$n!$	$n!$	This group, called the permutation group P_n or the symmetric group S_n , has $n!$
gilmore-lie-groups_F00025	18	0.996	$n \times n$	$n \times n$	of $n \times n$ matrices. These matrices have one nonzero element, $+1$, in each row
gilmore-lie-groups_F00026	18	1.000	$6=3!3 \times 3$	$6 = 3!3 \times 3$	and each column. For example, the $6 = 3!3 \times 3$ matrices for the permutation
gilmore-lie-groups_F00027	18	1.000	$S_{\{3\}}$	S_3	representation of S_3 are
gilmore-lie-groups_F00028	19	0.998	$z_{\{1\}}$	z_1	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups_F00029	19	0.998	$z_{\{2\}}, z_{\{2\}}$	z_2, z_2	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups_F00030	19	0.999	$z_{\{3\}}$	z_3	z_3 , and z_3 is replaced by z_1
gilmore-lie-groups_F00031	19	0.768	$\times$	$\times$	stitutions in S_n) and $\times$ represents the multiplication operation among the matrices
gilmore-lie-groups_F00032	19	1.000	$G \rightarrow \Gamma(G)$	$G \rightarrow \Gamma(G)$	tion of a group, $G \rightarrow \Gamma(G)$, is that the product of matrices representing two group
gilmore-lie-groups_F00033	19	0.988	$\left(\Gamma(g_i) \times \Gamma(g_j)\right)$	$(\Gamma(g_i) \times \Gamma(g_j))$	operations $(\Gamma(g_i) \times \Gamma(g_j))$ is equal to the matrix representing the product of these
gilmore-lie-groups_F00034	19	0.999	$\left(\Gamma(g_i \cdot g_j)\right)$	$(\Gamma(g_i \cdot g_j))$	operations in the group $(\Gamma(g_i \cdot g_j))$ for all group operations $g_i, g_j \in G$.
gilmore-lie-groups_F00035	19	0.999	$g_i, g_j \in G$	$g_i, g_j \in G$	operations in the group $(\Gamma(g_i \cdot g_j))$ for all group operations $g_i, g_j \in G$.
gilmore-lie-groups_F00036	19	0.874	3×3	3×3	since knowledge of the 3×3 matrix uniquely identifies the original group operation
gilmore-lie-groups_F00037	19	1.000	H	H	A subgroup H of the group G is a subset of group operations in G that is closed
gilmore-lie-groups_F00038	19	1.000	$A_{\{3\}}$	A_3	particular subgroup is denoted A_3 (alternating group). It consists of those oper-
gilmore-lie-groups_F00039	19	1.000	$A_{\{3\}} \subset S_{\{3\}}$	$A_3 \subset S_3$	$A_3 \subset S_3$ and the three two-element subgroups $S_2(ij)$ of S_3 are illustrated in Fig. 1.1.
gilmore-lie-groups_F00040	19	1.000	$S_{\{2\}}(ij)$	$S_2(ij)$	$A_3 \subset S_3$ and the three two-element subgroups $S_2(ij)$ of S_3 are illustrated in Fig. 1.1.
gilmore-lie-groups_F00041	20	0.706	$(123) \cdot (123) = (321)$	$(123) \cdot (123) = (321)$	of operations in this subset do not lie in this subset: $(123) \cdot (123) = (321)$, $(123) \cdot$
gilmore-lie-groups_F00042	20	1.000	$G, H_{\{1\}} \subset G$	$G, H_1 \subset G$	Two subgroups of G , $H_1 \subset G$ and $H_2 \subset G$ are conjugate if there is a group
gilmore-lie-groups_F00043	20	1.000	$H_{\{2\}} \subset G$	$H_2 \subset G$	Two subgroups of G , $H_1 \subset G$ and $H_2 \subset G$ are conjugate if there is a group
gilmore-lie-groups_F00044	20	1.000	$g \in G$	$g \in G$	element $g \in G$ with the property
gilmore-lie-groups_F00045	20	1.000	$S_{\{2\}}(12)$	$S_2(12)$	Example The subgroups $S_2(12)$ and $S_2(13)$ are conjugate in S_3 since
gilmore-lie-groups_F00046	20	1.000	$S_{\{2\}}(13)$	$S_2(13)$	Example The subgroups $S_2(12)$ and $S_2(13)$ are conjugate in S_3 since
gilmore-lie-groups_F00047	20	1.000	$G=S_{\{3\}}$	$G = S_3$	operation in $G = S_3$ serves merely to permute the group operations in A_3 among
gilmore-lie-groups_F00048	20	1.000	$H \subset G$	$H \subset G$	A subgroup $H \subset G$ which is self-conjugate under all operations in G is called
gilmore-lie-groups_F00049	21	1.000	f	f	A mapping f from a group G with group operations $g_1, g_2, \dots$ and group multi-
gilmore-lie-groups_F00050	21	1.000	$g_{\{1\}}, g_{\{2\}}, \dots$	$g_1, g_2, \dots$	A mapping f from a group G with group operations $g_1, g_2, \dots$ and group multi-

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gilmore-lie-groups_F00051	21	■ 0.999	$h_{\{1\}}, h_{\{2\}}, \ldots$	$h_1, h_2, \dots$	plication $\cdot$ to a group H with group operations $h_1, h_2, \dots$ and group multiplication
gilmore-lie-groups_F00052	21	■ 1.000	$h_{\{i\}} \in H$	$h_i \in H$	ments in G may map to a single group element in H . Every element $h_i \in H$ has
gilmore-lie-groups_F00053	21	■ 1.000	$g_{\{j\}} \in G$	$g_j \in G$	the same number of inverse images $g_j \in G$. If each group element $h \in H$ has a
gilmore-lie-groups_F00054	21	■ 1.000	$h \in H$	$h \in H$	the same number of inverse images $g_j \in G$. If each group element $h \in H$ has a
gilmore-lie-groups_F00055	21	■ 0.933	$g \in G \left(h_{\{1\}} = f \left(g_{\{1\}} \right) \right)$	$g \in G (h_1 = f(g_1))$	unique inverse image $g \in G (h_1 = f(g_1))$ and $h_2 = f(g_2), h_1 = h_2 \Rightarrow g_1 = g_2$ the
gilmore-lie-groups_F00056	21	■ 0.933	$\left(h_{\{2\}} = f \left(g_{\{2\}} \right) \right), h_{\{1\}} = h_{\{2\}} \Rightarrow g_{\{1\}} = g_{\{2\}}$	$h_2 = f(g_2), h_1 = h_2 \Rightarrow g_1 = g_2$	unique inverse image $g \in G (h_1 = f(g_1))$ and $h_2 = f(g_2), h_1 = h_2 \Rightarrow g_1 = g_2$ the
gilmore-lie-groups_F00057	21	■ 1.000	$S_{\{2\}}$	S_2	Example The 3:1 mapping f of S_3 onto S_2 given by
gilmore-lie-groups_F00058	22	■ 0.987	h	h	product of an element h in the subgroup H with a group element in a “quotient,”
gilmore-lie-groups_F00059	22	■ 1.000	G / H	G/H	or <i>coset</i> (denoted G/H). A coset is a subset of G . If the <i>order</i> of G is $ G $ (S_3
gilmore-lie-groups_F00060	22	■ 1.000	$ G \left(S_{\{3\}} \right)$	$ G (S_3)$	or <i>coset</i> (denoted G/H). A coset is a subset of G . If the <i>order</i> of G is $ G $ (S_3
gilmore-lie-groups_F00061	22	■ 0.637	$3! = 6$	$3! = 6$	has $3! = 6$ group elements, so the order of S_3 is 6), then the order of G/H is
gilmore-lie-groups_F00062	22	■ 0.866	$ G / H = G / H $	$ G/H = G / H $	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00063	22	■ 0.866	$H = A_{\{3\}} = \{I$	$H = A_3 = \{I$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00064	22	■ 0.866	$\}$	$\}$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00065	22	■ 0.992	$H = S_{\{2\}}(23) = \{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$ we have
gilmore-lie-groups_F00066	22	■ 1.000	$ G / H $	$ G / H $	The choice of the $ G / H $ group elements in the quotient space is not unique. For
gilmore-lie-groups_F00067	22	■ 0.692	$G / H = S_{\{3\}} / A_{\{3\}} = \{I$	$G/H = S_3/A_3 = \{I$	the subgroup A_3 we could equally well have chosen $G/H = S_3/A_3 = \{I, (13)\}$
gilmore-lie-groups_F00068	22	■ 0.997	$\{I, (23)\}$	$\{I, (23)\}$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups_F00069	22	■ 0.997	$S_{\{2\}}(23)$	$S_2(23)$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups_F00070	22	■ 0.997	$G / H = S_{\{3\}} / S_{\{2\}}(23) =$	$G/H = S_3/S_2(23) =$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups_F00071	22	■ 0.556	$\{I, (123), (321)\}$	$\{I, (123), (321)\}$	$\{I, (123), (321)\}$.
gilmore-lie-groups_F00072	22	■ 1.000	$S_{\{3\}} / A_{\{3\}}$	S_3/A_3	G/H . Since A_3 is an invariant subgroup of S_3 , the coset S_3/A_3 is a group, and
gilmore-lie-groups_F00073	22	■ 0.630	$\operatorname{group} G / H: G = G / H \times H$	$\operatorname{group} G/H: G = G/H \times H$	$\operatorname{group} G/H: G = G/H \times H$.
gilmore-lie-groups_F00074	22	■ 1.000	k	k	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups_F00075	22	■ 1.000	$I_{\{k\}}$	I_k	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups_F00076	22	■ 1.000	$z_{\{i\}}$	z_i	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups_F00077	23	■ 0.992	$I_{\{k\}} (k=1, 2, \ldots, n)$	$I_k (k = 1, 2, \dots, n)$	The n functions $I_k (k = 1, 2, \dots, n)$ of the n roots $(z_1, z_2, \dots, z_n)$ are symmetric:

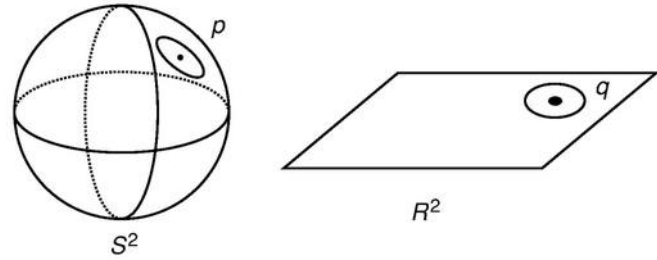
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gilmore-lie-groups_ F00078	23	■ 0.992	$\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$(z_1, z_2, \dots, z_n)$	The n functions I_k ($k = 1, 2, \dots, n$) of the n roots $\boxed{z_1, z_2, \dots, z_n}$ are symmetric:
gilmore-lie-groups_ F00079	23	■ 1.000	$f\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$f(z_1, z_2, \dots, z_n)$	any function $f(z_1, z_2, \dots, z_n)$ that is invariant under S_n can be written as a function
gilmore-lie-groups_ F00080	23	■ 1.000	$I_{\{1\}}, I_{\{2\}}, \ldots, I_{\{n\}}$	$I_1, I_2, \dots, I_n$	of the invariants $\boxed{I_1, I_2, \dots, I_n}$. The invariants are easily expressed in terms of
gilmore-lie-groups_ F00081	23	■ 1.000	$G_{\{\omega-1\}} \supset G_{\{\omega\}}=I$	$G_{\omega-1} \supset G_{\omega} = I$	group-subgroup pair in this chain is isolated: $\boxed{G_{\omega-1} \supset G_{\omega} = I}$. The character
gilmore-lie-groups_ F00082	23	■ 1.000	$G_{\{\omega-1\}} / G_{\{\omega\}}=G_{\{\omega-1\}}$	$G_{\omega-1}/G_{\omega} = G_{\omega-1}$	table for the commutative group $G_{\omega-1}/G_{\omega} = G_{\omega-1}$ is constructed. This lists
gilmore-lie-groups_ F00083	23	■ 1.000	$\left G_{\{\omega-1\}}\right / \left G_{\{\omega\}}\right $	$ G_{\omega-1} / G_{\omega} $	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups_ F00084	23	■ 1.000	$G_{\{\omega-1\}}$	$G_{\omega-1}$	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups_ F00085	23	■ 1.000	$G_{\{\omega\}}=I$	$G_{\omega} = I$	(i) symmetric under $\boxed{G_{\omega} = I}$,
gilmore-lie-groups_ F00086	23	■ 1.000	$G_{\{\omega-2\}} \supset G_{\{\omega-1\}}$	$G_{\omega-2} \supset G_{\omega-1}$	Next, the next pair of groups $G_{\omega-2} \supset G_{\omega-1}$ is isolated. Starting from the set of
gilmore-lie-groups_ F00087	23	■ 1.000	$G_{\{\omega-2\}}$	$G_{\omega-2}$	(ii) not all symmetric under $G_{\omega-2}$.
gilmore-lie-groups_ F00088	23	■ 1.000	$G=G_{\{0\}} \supset$	$G = G_0 \supset$	This bootstrap procedure continues until the last group-subgroup pair $G = G_0 \supset$
gilmore-lie-groups_ F00089	23	■ 1.000	$G_{\{1\}}$	G_1	G_1 is treated. At this stage the last set of functions can be solved by radicals.
gilmore-lie-groups_ F00090	24	■ 1.000	$\Gamma^{\{1\}}, \Gamma^{\{2\}}$	Γ^1, Γ^2	ducible representations Γ^1, Γ^2 are
gilmore-lie-groups_ F00091	24	■ 1.000	$r_{\{1\}}-r_{\{2\}}$	$r_1 - r_2$	That is, the function $\boxed{r_1 - r_2}$ is mapped into itself by the identity, and into its negative
gilmore-lie-groups_ F00092	24	■ 0.899	$\left(r_{\{1\}}-r_{\{2\}}\right)$	$(r_1 - r_2)$	As a result, $\boxed{r_1 - r_2}$ is not symmetric under the action of the group S_2 . It transforms
gilmore-lie-groups_ F00093	24	■ 1.000	$\Gamma^{\{2\}}$	Γ^2	under the irreducible representation Γ^2 , not the identity representation Γ^1 .
gilmore-lie-groups_ F00094	24	■ 1.000	$\Gamma^{\{1\}}$	Γ^1	under the irreducible representation Γ^2 , not the identity representation Γ^1 .
gilmore-lie-groups_ F00095	24	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)^{\{2\}}$	$(r_1 - r_2)^2$	Since the square $(r_1 - r_2)^2$ is symmetric (transforms under the identity repre-
gilmore-lie-groups_ F00096	24	■ 0.999	$I_{\{1\}}, I_{\{2\}}$	I_1, I_2	sentation of S_2), it can be expressed in terms of the two invariants I_1, I_2 as follows
gilmore-lie-groups_ F00097	25	■ 1.000	D	D	where D is the discriminant of the quadratic equation. Since $(r_1 - r_2) = \pm\sqrt{D}$,
gilmore-lie-groups_ F00098	25	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)=\pm \sqrt{D}$	$(r_1 - r_2) = \pm\sqrt{D}$	where D is the discriminant of the quadratic equation. Since $\boxed{(r_1 - r_2) = \pm\sqrt{D}}$,
gilmore-lie-groups_ F00099	25	■ 1.000	x	x	to move the origin to the mean value of the roots by defining a new variable, x , in
gilmore-lie-groups_ F00100	25	■ 1.000	z	z	terms of z (see Eq. (1.15)) by a Tschirnhaus transformation
gilmore-lie-groups_ F00101	26	■ 1.000	$S_{\{3\}} / A_{\{3\}}=S_{\{2\}}$	$S_3/A_3 = S_2$	subgroups) is satisfied. Since $S_3/A_3 = S_2$ is commutative and $A_3/I = A_3$ is com-
gilmore-lie-groups_ F00102	26	■ 1.000	$A_{\{3\}} / I=A_{\{3\}}$	$A_3/I = A_3$	subgroups) is satisfied. Since $S_3/A_3 = S_2$ is commutative and $A_3/I = A_3$ is com-
gilmore-lie-groups_ F00103	26	■ 1.000	$A_{\{3\}} \supset I$	$A_3 \supset I$	We begin the solution with the last group-subgroup pair in this chain: $A_3 \supset I$.
gilmore-lie-groups_ F00105	35	■ 1.000	ω	ω	Next, we take cube roots of these quantities. These are unique up to a factor of ω

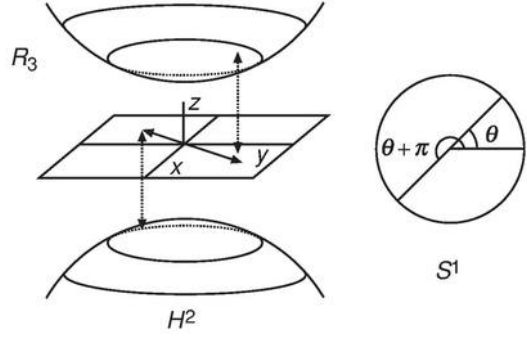
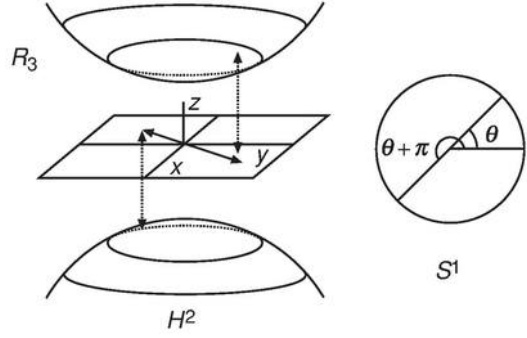
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gilmore-lie-groups_ F00110	26	■ 1.000	$(123)^{-1}$	$(123)^{-1}$	For example, the action of $(123)^{-1}$ on v_2 is
gilmore-lie-groups_ F00111	26	■ 1.000	$v_{\{2\}}$	v_2	For example, the action of $(123)^{-1}$ on v_2 is
gilmore-lie-groups_ F00112	27	■ 1.000	$v_{\{1\}}$	v_1	Since v_1 is symmetric under both A_3 and S_3 , it can be expressed in terms of the
gilmore-lie-groups_ F00113	27	■ 1.000	$v_{\{3\}}$	v_3	The remaining functions, v_2 and v_3 , are symmetric under I but not under A_3 .
gilmore-lie-groups_ F00114	27	■ 1.000	$S_{\{3\}} \supset A_{\{3\}}$	$S_3 \supset A_3$	We now proceed to the next group-subgroup pair: $S_3 \supset A_3$. To construct func-
gilmore-lie-groups_ F00115	27	■ 1.000	$S_{\{2\}}=S_{\{3\}} / A_{\{3\}}$	$S_2 = S_3/A_3$	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the
gilmore-lie-groups_ F00116	27	■ 1.000	$v_{\{2\}}^{\{3\}}$	v_2^3	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the
gilmore-lie-groups_ F00117	27	■ 1.000	$v_{\{3\}}^{\{3\}}$	v_3^3	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the
gilmore-lie-groups_ F00118	27	■ 1.000	$J_{\{1\}}, J_{\{2\}}$	J_1, J_2	Since J_1, J_2 are symmetric under S_3 , they can be expressed in terms of the invariants
gilmore-lie-groups_ F00119	27	■ 0.995	$I_{\{1\}}, I_{\{2\}}, I_{\{3\}}$	I_1, I_2, I_3	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write
gilmore-lie-groups_ F00120	27	■ 0.995	$J_{\{1\}}$	J_1	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write
gilmore-lie-groups_ F00121	27	■ 0.995	$J_{\{2\}}$	J_2	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write
gilmore-lie-groups_ F00122	27	■ 1.000	$I_{\{1\}}=s_{\{1\}}+s_{\{2\}}+$	$I_1 = s_1 + s_2 +$	These relations can be computed, but they simplify considerably if $I_1 = s_1 + s_2 +$
gilmore-lie-groups_ F00123	27	■ 1.000	$s_{\{3\}}=0$	$s_3 = 0$	$s_3 = 0$. This can be accomplished by shifting the origin using a Tschirnhaus trans-
gilmore-lie-groups_ F00124	28	■ 1.000	$J_{\{1\}}=v_{\{2\}}^{\{3\}}+v_{\{3\}}^{\{3\}}$	$J_1 = v_2^3 + v_3^3$	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'
gilmore-lie-groups_ F00125	28	■ 1.000	$J_{\{2\}}=v_{\{2\}}^{\{3\}} v_{\{3\}}^{\{3\}}$	$J_2 = v_2^3 v_3^3$	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'
gilmore-lie-groups_ F00126	28	■ 1.000	$I_{\{2\}}^{\{\prime\}}, I_{\{3\}}^{\{\prime\}}$	I_2', I_3'	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'
gilmore-lie-groups_ F00127	28	■ 1.000	$v_{\{2\}}^{\{3\}}, v_{\{3\}}^{\{3\}}$	v_2^3, v_3^3	The resolvent quadratic equation whose solution provides v_2^3, v_3^3 is
gilmore-lie-groups_ F00128	28	■ 1.000	$s_{\{1\}}, s_{\{2\}}, s_{\{3\}}$	s_1, s_2, s_3	Finally, the roots s_1, s_2, s_3 are linearly related to v_1, v_2, v_3 by
gilmore-lie-groups_ F00129	28	■ 1.000	$v_{\{1\}}, v_{\{2\}}, v_{\{3\}}$	v_1, v_2, v_3	Finally, the roots s_1, s_2, s_3 are linearly related to v_1, v_2, v_3 by
gilmore-lie-groups_ F00130	29	■ 1.000	$z=z^{\{\prime\}}+\frac{1}{4} I_{\{1\}}$	$z = z' + \frac{1}{4} I_1$	of coordinates through the Tschirnhaus transformation $z = z' + \frac{1}{4} I_1$
gilmore-lie-groups_ F00131	29	■ 1.000	$S_{\{4\}}$	S_4	The Galois group is S_4 . This has the subgroup chain shown in Fig. 1.5. The
gilmore-lie-groups_ F00132	29	■ 1.000	$A_{\{4\}}$	A_4	alternating group A_4 consists of the twelve group operations that have determinant
gilmore-lie-groups_ F00133	29	■ 0.957	$V_{\{4\}}$	V_4	<i>group, Klein four-group</i>) V_4 is $\{I, (12)(34), (13)(24), (14)(23)\}$. The chain
gilmore-lie-groups_ F00134	29	■ 0.957	$\{I, (12)(34), (13)(24), (14)(23)\}$	$\{I, (12)(34), (13)(24), (14)(23)\}$	<i>group, Klein four-group</i>) V_4 is $\{I, (12)(34), (13)(24), (14)(23)\}$. The chain
gilmore-lie-groups_ F00135	30	■ 1.000	$S_{\{4\}} / A_{\{4\}}=S_{\{2\}}$	$S_4/A_4 = S_2$	(i) A_4 is invariant in S_4 and $S_4/A_4 = S_2$,

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gilmore-lie-groups_ F00136	30	■ 0.987	$A_{\{4\}} / V_{\{4\}} = C_{\{3\}} = \{I$	$A_4/V_4 = C_3 = \{I$	(ii) V_4 is invariant in A_4 and $A_4/V_4 = C_3 = \{I, (234), (432)\}$,
gilmore-lie-groups_ F00137	30	■ 0.987	$V_{\{4\}} / I = V_{\{4\}} = \{I$	$V_4/I = V_4 = \{I$	(iii) I is invariant in V_4 and $V_4/I = V_4 = \{I, (12)(34), (13)(24), (14)(23)\}$.
gilmore-lie-groups_ F00143	30	■ 0.971	$w_{\{2\}}, w_{\{3\}}, w_{\{4\}}$	w_2, w_3, w_4	These basis vectors are symmetric under I but the basis vectors w_2, w_3, w_4 are not
gilmore-lie-groups_ F00144	30	■ 1.000	$A_{\{4\}} \supset V_{\{4\}}$	$A_4 \supset V_4$	We now advance to the next group-subgroup pair: $A_4 \supset V_4$. It is a simple matter
gilmore-lie-groups_ F00145	30	■ 1.000	$A_{\{4\}} / V_{\{4\}}$	A_4/V_4	(ii) permuted among themselves by A_4 and the group A_4/V_4 .
gilmore-lie-groups_ F00146	30	■ 0.999	$w_{\{1\}} = I_{\{1\}}$	$w_1 = I_1$	These functions are $w_1 = I_1$ and w_2^2, w_3^2, w_4^2 . In the coordinate system in which the
gilmore-lie-groups_ F00147	30	■ 0.999	$w_{\{2\}}^2, w_{\{3\}}^2, w_{\{4\}}^2$	w_2^2, w_3^2, w_4^2	These functions are $w_1 = I_1$ and w_2^2, w_3^2, w_4^2 . In the coordinate system in which the
gilmore-lie-groups_ F00148	30	■ 1.000	$w_{\{j\}}^2 (j=2,3,4)$	$w_j^2 (j=2,3,4)$	It is clear that the three $w_j^2 (j=2,3,4)$ are permuted among themselves by the
gilmore-lie-groups_ F00149	30	■ 1.000	$C_{\{3\}} = A_{\{4\}} / V_{\{4\}}$	$C_3 = A_4/V_4$	factor group $C_3 = A_4/V_4$, which is a subgroup of the Galois group of a resolvent
gilmore-lie-groups_ F00150	31	■ 1.000	$J_{\{k\}}$	J_k	Since the three J_k are invariant under C_3 , they can be expressed in terms of the
gilmore-lie-groups_ F00151	31	■ 1.000	$C_{\{3\}}$	C_3	Since the three J_k are invariant under C_3 , they can be expressed in terms of the
gilmore-lie-groups_ F00152	31	■ 1.000	$S_{\{4\}} \supset A_{\{4\}}$	$S_4 \supset A_4$	chain: $S_4 \supset A_4$, with $S_4/A_4 = S_2$. The cubic is solved by introducing the resolvent
gilmore-lie-groups_ F00153	31	■ 1.000	$y_{\{2\}}, y_{\{3\}}, y_{\{4\}}$	y_2, y_3, y_4	If the three solutions of the resolvent cubic equation are called y_2, y_3, y_4 , then
gilmore-lie-groups_ F00154	31	■ 1.000	$w_{\{2\}} w_{\{3\}} w_{\{4\}} = 8 I_{\{3\}}^{\prime}$	$w_2 w_3 w_4 = 8 I_3'$	A simple computation shows that $w_2 w_3 w_4 = 8 I_3'$. The signs $\pm \sqrt{y_j}$ are chosen so
gilmore-lie-groups_ F00155	31	■ 1.000	$\pm \sqrt{y_{\{j\}}}$	$\pm \sqrt{y_j}$	A simple computation shows that $w_2 w_3 w_4 = 8 I_3'$. The signs $\pm \sqrt{y_j}$ are chosen so
gilmore-lie-groups_ F00156	31	■ 1.000	$8 I_{\{3\}}^{\prime}$	$8 I_3'$	that their product is $8 I_3'$. The simple linear relation between the roots t_i and the
gilmore-lie-groups_ F00157	31	■ 1.000	$t_{\{i\}}$	t_i	that their product is $8 I_3'$. The simple linear relation between the roots t_i and the
gilmore-lie-groups_ F00158	31	■ 1.000	$I_{\{1\}}$	I_1	invariants I_1 and functions $w_j(I')$ is easily inverted:
gilmore-lie-groups_ F00159	31	■ 1.000	$w_{\{j\}} \left(I^{\prime} \right)$	$w_j(I')$	invariants I_1 and functions $w_j(I')$ is easily inverted:
gilmore-lie-groups_ F00160	31	■ 1.000	$w_{\{j\}}$	w_j	where the w_j are square roots of the solutions of the resolvent cubic equation whose
gilmore-lie-groups_ F00161	31	■ 1.000	$S_{\{5\}}$	S_5	is sufficient to study the structure of its Galois group S_5 . The alternating subgroup
gilmore-lie-groups_ F00162	32	■ 1.000	$A_{\{5\}}$	A_5	A_5 of order 60 is an invariant subgroup. S_5 has no invariant subgroups except A_5
gilmore-lie-groups_ F00163	32	■ 1.000	$S_{\{5\}} / A_{\{5\}} = S_{\{2\}}$	$S_5/A_5 = S_2$	Although $S_5/A_5 = S_2$ is commutative, $A_5/I = A_5$ is not. Therefore the quintic
gilmore-lie-groups_ F00164	32	■ 1.000	$A_{\{5\}} / I = A_{\{5\}}$	$A_5/I = A_5$	Although $S_5/A_5 = S_2$ is commutative, $A_5/I = A_5$ is not. Therefore the quintic
gilmore-lie-groups_ F00165	33	■ 1.000	$t_{\{1\}}^{\prime} = -3, t_{\{2\}}^{\prime} = -2, t_{\{3\}}^{\prime} =$	$t_1' = -3, t_2' = -2, t_3' =$	Next, we introduce linear combinations of the four roots $t_1' = -3, t_2' = -2, t_3' =$
gilmore-lie-groups_ F00166	33	■ 0.961	$1, t_{\{4\}}^{\prime} = 4$	$1, t_4' = 4$	$1, t_4' = 4$

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gilmore-lie-groups_F00167	33	■ 1.000	$y=y^{\{\prime\prime\prime\}}+\frac{1}{3} J_{\{1\}}=y^{\{\prime\prime\prime\}}+\frac{1}{3}(4+16+100)=y^{\{\prime\prime\prime\}}+40$	$y = y' + \frac{1}{3}J_1 = y' + \frac{1}{3}(4 + 16 + 100) = y' + 40$	$y = y' + \frac{1}{3}J_1 = y' + \frac{1}{3}(4 + 16 + 100) = y' + 40$. The roots are now
gilmore-lie-groups_F00168	34	■ 0.990	$v_{\{2\}}^{\{3\}}+v_{\{3\}}^{\{3\}}, v_{\{2\}}^{\{3\}} v_{\{3\}}^{\{3\}}$	$v_2^3 + v_3^3, v_2^3 v_3^3$	We can express $v_2^3 + v_3^3, v_2^3 v_3^3$ in terms of J_2', J_3' :
gilmore-lie-groups_F00169	34	■ 0.990	$J_{\{2\}}^{\{\prime\prime\prime\}}, J_{\{3\}}^{\{\prime\prime\prime\}}$	J_2', J_3'	We can express $v_2^3 + v_3^3, v_2^3 v_3^3$ in terms of J_2', J_3' :
gilmore-lie-groups_F00170	34	■ 1.000	$x=x^{\{\prime\prime\prime\}}+\frac{1}{2} K_{\{1\}}$	$x = x' + \frac{1}{2}K_1$	A Tschirnhaus transformation $x = x' + \frac{1}{2}K_1$ produces the auxiliary quadratic
gilmore-lie-groups_F00171	35	■ 1.000	$y_{\{1\}}, y_{\{2\}}, y_{\{3\}}$	y_1, y_2, y_3	The values y_1, y_2, y_3 of the resolvent cubic are complex linear combinations of
gilmore-lie-groups_F00172	35	■ 0.982	$v_{\{2\}}, v_{\{3\}}$	v_2, v_3	v_2, v_3
gilmore-lie-groups_F00173	35	■ 1.000	$w_{\{2\}} w_{\{3\}} w_{\{4\}}=8 I_{\{3\}}^{\{\prime\prime\prime\}}=80$	$w_2 w_3 w_4 = 8I_3' = 80$	Since $w_2 w_3 w_4 = 8I_3' = 80$, an even number of these signs must be negative. The
gilmore-lie-groups_F00174	36	■ 1.000	$S_{\{4\}} / A_{\{4\}}, A_{\{4\}} / V_{\{4\}}, V_{\{4\}}$	$S_4/A_4, A_4/V_4, V_4$	1. Compute $S_4/A_4, A_4/V_4, V_4$ and show that they are commutative.
gilmore-lie-groups_F00175	36	■ 0.986	$V_{\{8\}}$	V_8	2. Construct the group V_8 with the property $S_4 \supset V_8 \supset V_4$ (see Fig. 1.5). (Hint: include
gilmore-lie-groups_F00176	36	■ 0.986	$S_{\{4\}} \supset V_{\{8\}} \supset V_{\{4\}}$	$S_4 \supset V_8 \supset V_4$	2. Construct the group V_8 with the property $S_4 \supset V_8 \supset V_4$ (see Fig. 1.5). (Hint: include
gilmore-lie-groups_F00177	36	■ 1.000	$z^{\{3\}}-7 z+6=0((z-1)(z-2)(z+3)=0)$	$z^3 - 7z + 6 = 0((z - 1)(z - 2)(z + 3) = 0)$	3. For the cubic equation $z^3 - 7z + 6 = 0((z - 1)(z - 2)(z + 3) = 0)$ show
gilmore-lie-groups_F00178	36	■ 1.000	$\left(x-v_{\{2\}}^{\{3\}}\right)\left(x-v_{\{3\}}^{\{3\}}\right)=x^{\{2\}}-162 x+$	$(x - v_2^3)(x - v_3^3) = x^2 - 162x +$	Show that the resolvent equation for v_2^3, v_3^3 is $(x - v_2^3)(x - v_3^3) = x^2 - 162x +$
gilmore-lie-groups_F00179	36	■ 0.996	$9261=0$	$9261 = 0$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00180	36	■ 0.996	$v_{\{2\}}^{\{3\}}, v_{\{3\}}^{\{3\}}=81 \pm i 30 \sqrt{3}$	$v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00181	36	■ 0.996	$v_{\{2\}}, v_{\{3\}}=\frac{1}{2}(3 \pm$	$v_2, v_3 = \frac{1}{2}(3 \pm$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00182	36	■ 1.000	$i 5 \sqrt{3}$	$i5\sqrt{3}$	$i5\sqrt{3}$). Invert Eq. (1.43) to determine the three roots of the original equation:
gilmore-lie-groups_F00183	36	■ 0.543	$x+y, x-y, x y$	$x + y, x - y, xy$	to construct intervals of lengths $x + y, x - y, xy$ and x/y using ruler and compass.
gilmore-lie-groups_F00184	36	■ 0.543	x / y	x/y	to construct intervals of lengths $x + y, x - y, xy$ and x/y using ruler and compass.
gilmore-lie-groups_F00185	36	■ 1.000	$\sqrt{x}$	$\sqrt{x}$	It is also possible to construct intervals of length $\sqrt{x}$ by these means. The set of all
gilmore-lie-groups_F00186	36	■ 1.000	$x+i y=(x, y)$	$x + iy = (x, y)$	forms a subset of the numbers $x + iy = (x, y)$ in the complex plane. If a number is
gilmore-lie-groups_F00187	36	■ 1.000	K	K	number satisfies an algebraic equation of degree K with integer coefficients, where
gilmore-lie-groups_F00188	36	■ 1.000	$K=2^{\{n\}}$	$K = 2^n$	$K = 2^n$ must be some power of two.
gilmore-lie-groups_F00189	36	■ 1.000	π	π	a. Square a circle? For the circle of radius 1 the area is π . Squaring a circle means
gilmore-lie-groups_F00190	36	■ 1.000	$x^{\{2\}}-\pi=0$	$x^2 - \pi = 0$	finding an interval of length x , where $x^2 - \pi = 0$. This is of degree 2 but π is not
gilmore-lie-groups_F00191	36	■ 1.000	$1^{\{3\}}=1$	$1^3 = 1$	b. Double the cube? A cube with edge length 1 has volume $1^3 = 1$. A cube with
gilmore-lie-groups_F00192	36	■ 1.000	$x^{\{3\}}-2=0$	$x^3 - 2 = 0$	twice the volume has edge length x , where x satisfies $x^3 - 2 = 0$. Although the

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gilmore-lie-groups_F00193	36	■ 1.000	$3 \neq 2^n$	$3 \neq 2^n$	coefficients are integers this equation is of degree $3 \neq 2^n$ for any integer n . Argue
gilmore-lie-groups_F00194	36	■ 1.000	$3 \ \theta$	3θ	c. Trisect an angle? If 3θ is some angle, the trigonometric functions of 3θ and
gilmore-lie-groups_F00195	36	■ 1.000	$\frac{1}{3}(3 \ \theta)=\theta$	$\frac{1}{3}(3\theta) = \theta$	$\frac{1}{3}(3\theta) = \theta$ are related by
gilmore-lie-groups_F00196	37	■ 1.000	$\cos (3 \ \theta)$	$\cos(3\theta)$	Whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00197	37	■ 1.000	$\cos (\theta)$	$\cos(\theta)$	Whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00198	37	■ 1.000	$\left(x^2+a \ x+b\right)(x+c)=0$	$(x^2 + ax + b)(x + c) = 0$	cubic factors into the form $\boxed{x^2 + ax + b)(x + c) = 0}$, where a, b, c are rational.
gilmore-lie-groups_F00199	37	■ 1.000	$a, \ b, \ c$	a, b, c	cubic factors into the form $(x^2 + ax + b)(x + c) = 0$, where a, b, c are rational.
gilmore-lie-groups_F00200	37	■ 1.000	$\cos (3 \ \theta)=0, \ c=0$	$\cos(3\theta) = 0, c = 0$	For example, if $\boxed{\cos(3\theta) = 0, c = 0}$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00201	37	■ 1.000	$a=0$	$a = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00202	37	■ 1.000	$b=-3 \ / \ 4$	$b = -3/4$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00203	37	■ 1.000	$\cos (\theta)=0$	$\cos(\theta) = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00204	37	■ 0.986	$\pm \sqrt{3} \ / \ 2$	$\pm\sqrt{3}/2$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2$ (+), $3\pi/2$ (0), or $5\pi/2$ (−).
gilmore-lie-groups_F00205	37	■ 0.986	$3 \ \theta=\pi \ / \ 2(+), \ 3 \ \pi \ / \ 2(0)$	$3\theta = \pi/2(+), 3\pi/2(0)$	or $\pm\sqrt{3}/2$ for $\boxed{3\theta = \pi/2(+), 3\pi/2(0)}$, or $5\pi/2$ (−).
gilmore-lie-groups_F00206	37	■ 0.986	$5 \ \pi \ / \ 2(-)$	$5\pi/2(-)$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2$ (+), $3\pi/2$ (0), or $5\pi/2$ (−).
gilmore-lie-groups_F00207	38	■ 1.000	$g_{\{i\}}, \ g_{\{j\}}, \ g_{\{k\}}, \ \ldots$	$g_i, g_j, g_k, \dots$	Definition A set $\boxed{g_i, g_j, g_k, \dots}$ (called group elements or group operations)
gilmore-lie-groups_F00208	38	■ 1.000	$g_{\{i\}} \ \circ g_{\{j\}} \ \in \ G$	$g_i \circ g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \circ g_j \in G$.
gilmore-lie-groups_F00209	38	■ 1.000	e	e	(iii) Identity: there is an operator e (the identity operation) with the property that for
gilmore-lie-groups_F00210	38	■ 1.000	$g_{\{i\}} \ \in \ G$	$g_i \in G$	every group operation $\boxed{g_i \in G}$
gilmore-lie-groups_F00211	39	■ 1.000	$2 \ \times \ 2$	2×2	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups_F00212	39	■ 1.000	$S \ L(2 \ ; \ \mathbb{R})$	$SL(2; \mathbb{R})$	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups_F00213	39	■ 1.000	$\alpha, \ \beta, \ \gamma, \ \delta$	$\alpha, \beta, \gamma, \delta$	where $\alpha, \beta, \gamma, \delta$ are real numbers. This set forms a group under matrix multiplica-
gilmore-lie-groups_F00214	39	■ 0.797	A	A	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00215	39	■ 0.797	B	B	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00216	39	■ 0.797	$A \ \circ B=C$	$A \circ B = C$	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00217	39	■ 0.797	$\circ$	$\circ$	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00218	39	■ 0.922	C	C	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups_F00219	39	■ 0.922	$\operatorname{det}(A)=+1$	$\det(A) = +1$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,

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gilmore-lie-groups-F00220	39	■ 0.922	$\operatorname{det}(B)=+1$	$\det(B) = +1$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups-F00221	39	■ 1.000	$\operatorname{det}(C)=\operatorname{det}(A)\operatorname{det}(B)=+1$	$\det(C) = \det(A) \det(B) = +1$	then $\det(C) = \det(A) \det(B) = +1$.
gilmore-lie-groups-F00222	39	■ 0.998	$(A \circ B) \circ C$	$(A \circ B) \circ C$	(ii) Associativity: $(A \circ B) \circ C$ and $A \circ (B \circ C)$ are given explicitly by
gilmore-lie-groups-F00223	39	■ 0.998	$A \circ (B \circ C)$	$A \circ (B \circ C)$	(ii) Associativity: $(A \circ B) \circ C$ and $A \circ (B \circ C)$ are given explicitly by
gilmore-lie-groups-F00224	39	■ 1.000	$g_i \rightarrow g(x)$	$g_i \rightarrow g(x)$	in the group with a point in some topological space: $g_i \rightarrow g(x)$. In other words, the
gilmore-lie-groups-F00225	39	■ 1.000	i	i	index i depends on one or more continuous real variables.
gilmore-lie-groups-F00226	39	■ 1.000	$S^2 \subset R^3: x^2+y^2+z^2=1$	$S^2 \subset R^3: x^2 + y^2 + z^2 = 1$	example, every point on the surface of a unit sphere $S^2 \subset R^3: x^2 + y^2 + z^2 = 1$,
gilmore-lie-groups-F00227	39	■ 1.000	R^2	R^2	has a neighborhood that looks, over small distances, like a piece of the plane R^2
gilmore-lie-groups-F00228	39	■ 1.000	S^2	S^2	(see Fig. 2.1). Locally, the two spaces S^2 and R^2 are topologically equivalent but
gilmore-lie-groups-F00229	39	■ 1.000	M^n	M^n	Definition An n -dimensional differentiable manifold M^n consists of the follow-
gilmore-lie-groups-F00230	39	■ 1.000	T	T	(i) A topological space T . This includes a collection of open sets U_α (a topology) that
gilmore-lie-groups-F00231	39	■ 1.000	U_{α}	U_α	(i) A topological space T . This includes a collection of open sets U_α (a topology) that
gilmore-lie-groups-F00232	39	■ 0.896	$T: \cup_{\alpha} U_{\alpha}=T$	$T: \cup_\alpha U_\alpha = T$	cover $T: \cup_\alpha U_\alpha \equiv T$.
gilmore-lie-groups-F00233	40	—	p	p	
gilmore-lie-groups-F00234	40	■ 1.000	ϕ_{α}	ϕ_α	(ii) A collection of charts ϕ_α , with $\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$. Each ϕ_α is a homeomorphism of
gilmore-lie-groups-F00235	40	■ 1.000	$\phi_{\alpha}(U_{\alpha})=V_{\alpha} \subset R^n$	$\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$	(ii) A collection of charts ϕ_α , with $\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$. Each ϕ_α is a homeomorphism of
gilmore-lie-groups-F00236	40	■ 1.000	V_{α}	V_α	U_α to V_α .
gilmore-lie-groups-F00237	40	■ 1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1}: \phi_{\beta}(U_{\alpha} \cap U_{\beta}) \rightarrow \phi_{\alpha}(U_{\alpha} \cap U_{\beta})$	$\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta)$	(iii) Smoothness conditions. The homeomorphisms $\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta)$
gilmore-lie-groups-F00238	40	■ 0.988	U_{β}	U_β	U_β) of open sets in R^n to open sets in R^n are 1:1, invertible, and differentiable.
gilmore-lie-groups-F00239	40	■ 0.988	R^n	R^n	U_β) of open sets in R^n to open sets in R^n are 1:1, invertible, and differentiable.
gilmore-lie-groups-F00240	40	■ 1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1}$	$\phi_\alpha \circ \phi_\beta^{-1}$	functions” $\phi_\alpha \circ \phi_\beta^{-1}$ map $R^n \rightarrow R^n$, all the definitions of elementary multivariable
gilmore-lie-groups-F00241	40	■ 1.000	$R^n \rightarrow R^n$	$R^n \rightarrow R^n$	functions” $\phi_\alpha \circ \phi_\beta^{-1}$ map $R^n \rightarrow R^n$, all the definitions of elementary multivariable
gilmore-lie-groups-F00242	40	■ 0.697	C^k	C^k	replaced by other adjectives (C^k , smooth, analytic, . . .) in the definition above.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00243	40	■ 1.000	$x_1 \rightarrow 0$	$x_1 \rightarrow 0$	lem, as is clear when one asks: “what happens as $x_1 \rightarrow 0$?” We will consider this
gilmore-lie-groups_F00244	40	■ 1.000	$\times S^1$	$\times S^1$	fold R^2 (plane) $\times S^1$ (circle) (see Fig. 2.2). This is not at all obvious, but will become
gilmore-lie-groups_F00245	41	—	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$	
gilmore-lie-groups_F00246	41	—	$H^2 \times S^1$	$H^2 \times S^1$	
gilmore-lie-groups_F00247	41	■ 0.698	$\left. U_{\alpha} \right)$	$U_{\alpha})$	$U_{\alpha})$ has a finite subcover: $\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$.
gilmore-lie-groups_F00248	41	■ 0.698	$\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	$\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	$U_{\alpha})$ has a finite subcover: $\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$.
gilmore-lie-groups_F00249	41	■ 1.000	$\left(\left x - x' \right ^2 = \left x_1 - x_1' \right ^2 + \cdots + \right.$	$\left(\left x - x' \right ^2 = \left x_1 - x_1' \right ^2 + \cdots + \right.$	In spaces R^n with a Euclidean notion of distance ($ x - x' ^2 = x_1 - x_1' ^2 + \cdots +$
gilmore-lie-groups_F00250	41	■ 1.000	$\left. \left x_n - x_n' \right ^2 \right)$	$ x_n - x_n' ^2$	$ x_n - x_n' ^2$), this definition is equivalent to an older definition of compact spaces:
gilmore-lie-groups_F00251	42	■ 1.000	$\left(g(x), x \in M^n \right)$	$(g(x), x \in M^n)$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups_F00252	42	■ 1.000	$g(x) \circ g(y) =$	$g(x) \circ g(y) =$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups_F00253	42	■ 1.000	$g(z)$	$g(z)$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_F00254	42	■ 1.000	$z \in M^n$	$z \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_F00255	42	■ 1.000	$x \in M^n$	$x \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_F00256	42	■ 1.000	$y \in M^n$	$y \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_F00257	42	■ 1.000	$z = \phi(x, y)$	$z = \phi(x, y)$	through a function $z = \phi(x, y)$.
gilmore-lie-groups_F00258	42	■ 1.000	$g(x) \circ g(y) = g(z)$	$g(x) \circ g(y) = g(z)$	defined by $g(x) \circ g(y) = g(z)$, is differentiable.
gilmore-lie-groups_F00259	42	■ 1.000	$y = \psi(x)$	$y = \psi(x)$	(ii) Smoothness of the group inversion map The group inversion map $y = \psi(x)$, defined
gilmore-lie-groups_F00260	42	■ 1.000	$g(x)^{-1} = g(y)$	$g(x)^{-1} = g(y)$	by $g(x)^{-1} = g(y)$, is differentiable.

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gilmore-lie-groups_F00261	42	■ 1.000	$g(\phi(x, y))$	$g(\phi(x, y))$	$g(\phi(x, y))$
gilmore-lie-groups_F00262	42	■ 1.000	ϕ	ϕ	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_F00263	42	■ 1.000	$x_{\{1\}}$	x_1	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_F00264	42	■ 1.000	$y_{\{1\}}$	y_1	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_F00265	42	■ 1.000	$x_{\{2\}}-x_{\{3\}}$	$x_2 - x_3$	bounded away from the x_2-x_3 plane $x_1 = 0$ and the y_2-y_3 plane $y_1 = 0$. In the
gilmore-lie-groups_F00266	42	■ 1.000	$x_{\{1\}}=0$	$x_1 = 0$	bounded away from the x_2-x_3 plane $x_1 = 0$ and the y_2-y_3 plane $y_1 = 0$. In the
gilmore-lie-groups_F00267	42	■ 1.000	$y_{\{2\}}-y_{\{3\}}$	$y_2 - y_3$	bounded away from the x_2-x_3 plane $x_1 = 0$ and the y_2-y_3 plane $y_1 = 0$. In the
gilmore-lie-groups_F00268	42	■ 1.000	$y_{\{1\}}=0$	$y_1 = 0$	bounded away from the x_2-x_3 plane $x_1 = 0$ and the y_2-y_3 plane $y_1 = 0$. In the
gilmore-lie-groups_F00269	43	■ 0.983	$\left(y_{\{1\}}, y_{\{2\}}, y_{\{3\}}\right)$	(y_1, y_2, y_3)	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_F00270	43	■ 0.983	$\left[g\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)\right]^{-1}$	$[g(x_1, x_2, x_3)]^{-1}$	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_F00271	43	■ 0.998	$\left(z_{\{1\}}, z_{\{2\}}, z_{\{3\}}\right)=(1,0,0)$	$(z_1, z_2, z_3) = (1, 0, 0)$	$(z_1, z_2, z_3) = (1, 0, 0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_F00272	43	■ 0.998	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)$	(x_1, x_2, x_3)	$(z_1, z_2, z_3) = (1, 0, 0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_F00273	43	■ 1.000	$[g(x)]^{-1}=g(y)=g(\psi(x))$	$[g(x)]^{-1} = g(y) = g(\psi(x))$	The inverse mapping $[g(x)]^{-1} = g(y) = g(\psi(x))$ is
gilmore-lie-groups_F00274	43	■ 0.964	(x, y, z , θ)	(x, y, z , θ)	The rotation matrix is parameterized by a point on a circle. Two points (x, y, z , θ)
gilmore-lie-groups_F00275	43	■ 0.882	$(-x, -y, - z , \theta+\pi)$	$(-x, -y, - z , \theta + \pi)$	and $(-x, -y, - z , \theta + \pi)$ map to the same matrix in $SL(2; \mathbb{R})$. The manifold that
gilmore-lie-groups_F00276	43	■ 1.000	$H^{2+} \times S^1$	$H^{2+} \times S^1$	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_F00277	43	■ 1.000	H^{2+}	H^{2+}	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_F00278	43	■ 1.000	$\phi(x, y)$	$\phi(x, y)$	operation that obeys the group axioms. The combinatorial operation $\phi(x, y)$ de-
gilmore-lie-groups_F00279	43	■ 1.000	$g(x) \circ g(y)=g(z)=g(\phi(x, y))$	$g(x) \circ g(y) = g(z) = g(\phi(x, y))$	fined by $g(x) \circ g(y) = g(z) = g(\phi(x, y))$ is differentiable in both sets of variables.
gilmore-lie-groups_F00280	44	■ 1.000	$[g(x)]^{-1}=$	$[g(x)]^{-1} =$	In addition, the mapping $y = \psi(x)$ of a group operation to its inverse $[g(x)]^{-1} =$
gilmore-lie-groups_F00281	44	■ 1.000	$g(y)=g(\psi(x))$	$g(y) = g(\psi(x))$	$g(y) = g(\psi(x))$ is also differentiable.
gilmore-lie-groups_F00282	44	■ 0.863	$\operatorname{SL}(2 ; \mathbb{R})$	$SL(2; \mathbb{R})$	1. Construct the analytic mapping $\phi(x, y)$ for the parameterization of $SL(2; \mathbb{R})$ illus-
gilmore-lie-groups_F00283	44	■ 1.000	M	M	3. Convince yourself that every matrix M in the group $SL(n; \mathbb{R})$ can be written as the
gilmore-lie-groups_F00284	44	■ 1.000	$S L(n ; \mathbb{R})$	$SL(n; \mathbb{R})$	3. Convince yourself that every matrix M in the group $SL(n; \mathbb{R})$ can be written as the
gilmore-lie-groups_F00285	44	■ 1.000	S	S	product of an $n \times n$ real symmetric unimodular matrix S and an orthogonal matrix
gilmore-lie-groups_F00286	44	■ 0.755	O	O	O in $SO(n)$: $M = SO$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_F00287	44	■ 0.755	$S O(n): M=S O$	$SO(n) : M = SO$	O in $SO(n)$: $M = SO$. Devise an algorithm for constructing these matrices. Show

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gilmore-lie-groups_ F00288	44	■ 1.000	$S=\left(M\ M^t\right)^{\wedge 1 / 2}$	$S=\left(M M^t\right)^{1 / 2}$	$S=\left(M M^t\right)^{1 / 2}$ and $O=S^{-1} M$. How do you compute the square root of a matrix?
gilmore-lie-groups_ F00289	44	■ 1.000	$O=S^{\wedge-1}\} M$	$O=S^{-1} M$	$S=\left(M M^t\right)^{1 / 2}$ and $O=S^{-1} M$. How do you compute the square root of a matrix?
gilmore-lie-groups_ F00290	44	■ 1.000	$(x, y, z) \rightarrow\left(x^{\wedge\{prime\}}, y^{\wedge\{prime\}}, z^{\wedge\{prime\}}\right)$	$(x, y, z) \rightarrow\left(x', y', z'\right)$	4. Construct the most general linear transformation $(x, y, z) \rightarrow\left(x', y', z'\right)$ that leaves
gilmore-lie-groups_ F00291	44	■ 1.000	$M_{\{1\}}$	M_1	where the real symmetric matrices M_1 and M_2 satisfy
gilmore-lie-groups_ F00292	44	■ 1.000	$M_{\{2\}}$	M_2	where the real symmetric matrices M_1 and M_2 satisfy
gilmore-lie-groups_ F00293	44	■ 0.995	$[S\ 0(1,1)]$	$[SO(1,1)]$	5. Construct the group of linear transformations $[SO(1,1)]$ that leaves invariant the
gilmore-lie-groups_ F00294	44	■ 0.972	$(c\ t)^{\wedge 2}-x^{\wedge 2}$	$(c t)^2-x^2$	quantity $(c t)^2-x^2$. Compare this with the group of linear transformations $[SO(2)]$
gilmore-lie-groups_ F00295	44	■ 0.972	$S\ 0(2)$	$SO(2)$	quantity $(c t)^2-x^2$. Compare this with the group of linear transformations $[SO(2)]$
gilmore-lie-groups_ F00296	44	■ 0.999	$x^{\wedge 2}+y^{\wedge 2}$	x^2+y^2	that leaves invariant the radius of the circle x^2+y^2 . (This comparison involves map-
gilmore-lie-groups_ F00297	45	■ 1.000	$(c\ t)^{\wedge 2}-x^{\wedge 2}-y^{\wedge 2}-z^{\wedge 2}$	$(c t)^2-x^2-y^2-z^2$	$(c t)^2-x^2-y^2-z^2$. This is the Lorentz group $O(3,1)$. Four disconnected mani-
gilmore-lie-groups_ F00298	45	■ 1.000	$0(3,1)$	$O(3,1)$	$(c t)^2-x^2-y^2-z^2$. This is the Lorentz group $O(3,1)$. Four disconnected mani-
gilmore-lie-groups_ F00299	45	■ 0.535	$S\ L(2 ; \mathbb{C})$	$SL(2 ; \mathbb{C})$	7. The group of 2×2 complex matrices with determinant $+1$ is named $SL(2 ; \mathbb{C})$. Ma-
gilmore-lie-groups_ F00300	45	■ 0.788	$\left[\begin{array}{ll}\alpha & \beta \\ \gamma & \delta\end{array}\right]$	$\left[\begin{array}{ll}\alpha & \beta \\ \gamma & \delta\end{array}\right]$	trices in this group have the structure $\left[\begin{array}{ll}\alpha & \beta \\ \gamma & \delta\end{array}\right]$, where $\alpha, \beta, \gamma, \delta$ are complex numbers
gilmore-lie-groups_ F00301	45	■ 0.987	$\alpha \delta-\beta \gamma=1$	$\alpha \delta-\beta \gamma=1$	and $\alpha \delta-\beta \gamma=1$. Define the matrix X by
gilmore-lie-groups_ F00302	45	■ 0.987	x	X	and $\alpha \delta-\beta \gamma=1$. Define the matrix X by
gilmore-lie-groups_ F00303	45	■ 1.000	$\mathbf{x}$	$\mathbf{x}$	where $\mathbf{x}$ is the three vector $\mathbf{x}=(x, y, z)$ and $\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$ are the
gilmore-lie-groups_ F00304	45	■ 1.000	$\mathbf{x}=(x, y, z)$	$\mathbf{x}=(x, y, z)$	where $\mathbf{x}$ is the three vector $\mathbf{x}=(x, y, z)$ and $\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$ are the
gilmore-lie-groups_ F00305	45	■ 1.000	$\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$	$\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$	where $\mathbf{x}$ is the three vector $\mathbf{x}=(x, y, z)$ and $\sigma=\left(\sigma_1, \sigma_2, \sigma_3\right)=\left(\sigma_x, \sigma_y, \sigma_z\right)$ are the
gilmore-lie-groups_ F00306	45	■ 1.000	$X^{\dagger} \equiv\left(X^t\right)^{*}=X$	$X^{\dagger} \equiv\left(X^t\right)^{*}=X$	a. Show that X is hermitian: $X^{\dagger} \equiv\left(X^t\right)^{*}=X$.
gilmore-lie-groups_ F00307	45	■ 1.000	$g \in S L(2 ; \mathbb{C})$	$g \in S L(2 ; \mathbb{C})$	c. If $g \in S L(2 ; \mathbb{C})$, show that $g^{\dagger} X g=X'=H\left(x', y', z', c t'\right)$.
gilmore-lie-groups_ F00308	45	■ 1.000	$g^{\dagger} X g=X'=H\left(x', y', z', c t'\right)$	$g^{\dagger} X g=X'=H\left(x', y', z', c t'\right)$	c. If $g \in S L(2 ; \mathbb{C})$, show that $g^{\dagger} X g=X'=H\left(x', y', z', c t'\right)$.
gilmore-lie-groups_ F00309	45	■ 0.964	$\left(x', y', z', c t'\right)$	$\left(x', y', z', c t'\right)$	d. How are the new space-time coordinates $\left(x', y', z', c t'\right)$ related to the original
gilmore-lie-groups_ F00310	45	■ 0.918	$(x, y, z, c t)$	$(x, y, z, c t)$	coordinates $(x, y, z, c t)$? (They are linearly related by coefficients that are bilinear
gilmore-lie-groups_ F00311	45	■ 1.000	g	g	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^{\dagger}$.)
gilmore-lie-groups_ F00312	45	■ 1.000	$\alpha^*, \beta^*, \gamma^*, \delta^*$	$\alpha^*, \beta^*, \gamma^*, \delta^*$	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^{\dagger}$.)
gilmore-lie-groups_ F00313	45	■ 1.000	$g^{\dagger}$	$g^{\dagger}$	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^{\dagger}$.)
gilmore-lie-groups_ F00314	45	■ 0.990	$t^{\wedge\{prime\}}=t$	$t'=t$	e. Find the subgroup of $SL(2 ; \mathbb{C})$ that leaves $t'=t$. (It is $SU(2) \subset S L(2 ; \mathbb{C})$).

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gilmore-lie-groups_ F08315	45	■ 0.990	$S U(2) \subset S L(2 ; \mathbb{C})$	$SU(2) \subset SL(2; \mathbb{C})$	e. Find the subgroup of $SL(2; \mathbb{C})$ that leaves $t' = t$. (It is $SU(2) \subset SL(2; \mathbb{C})$).
gilmore-lie-groups_ F08316	45	■ 0.952	$g=k \ h$	$g = kh$	f. For any $g \in SL(2; \mathbb{C})$ write $g = kh$, where $h \in SU(2)$, $h^\dagger = h^{-1}$, h has the
gilmore-lie-groups_ F08317	45	■ 0.952	$h \in S U(2), h^{\dagger}=h^{-1}, h$	$h \in SU(2), h^\dagger = h^{-1}, h$	f. For any $g \in SL(2; \mathbb{C})$ write $g = kh$, where $h \in SU(2)$, $h^\dagger = h^{-1}$, h has the
gilmore-lie-groups_ F08318	45	■ 0.461	$h=\operatorname{EXP}\left(\frac{i}{2} \sigma \cdot \theta\right)$	$h = \operatorname{EXP}\left(\frac{i}{2} \sigma \cdot \theta\right)$	form $h = \operatorname{EXP}(\frac{i}{2} \sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2)$, $k^\dagger = k^{+1}$, k has the form
gilmore-lie-groups_ F08319	45	■ 0.461	$k \in \operatorname{SL}(2 ; \mathbb{C}) / \operatorname{SU}(2), k^{\dagger}=k^{+1}, k$	$k \in \operatorname{SL}(2; \mathbb{C})/SU(2), k^\dagger = k^{+1}, k$	form $h = \operatorname{EXP}(\frac{i}{2} \sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2), k^\dagger = k^{+1}, k$ has the form
gilmore-lie-groups_ F08320	45	■ 1.000	$k=\operatorname{EXP}\left(\frac{1}{2} \sigma \cdot \mathbf{b}\right)$	$k = \operatorname{EXP}\left(\frac{1}{2} \sigma \cdot \mathbf{b}\right)$	$k = \operatorname{EXP}(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08321	45	■ 1.000	$\mathbf{b}$	$\mathbf{b}$	$k = \operatorname{EXP}(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08322	45	■ 1.000	θ	θ	$k = \operatorname{EXP}(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08323	45	■ 0.899	$k^{\dagger} H(x, y, z, c t) k=H\left(x^{\prime}, y^{\prime}, z^{\prime}, c t^{\prime}\right)$	$k^\dagger H(x, y, z, ct) k = H(x', y', z', ct')$	and $\mathbf{b}$ are real. Construct $k^\dagger H(x, y, z, ct) k = H(x', y', z', ct')$. If this is too diffi-
gilmore-lie-groups_ F08324	45	■ 0.893	$\mathbf{b}=(0,0, b)$	$\mathbf{b} = (0, 0, b)$	cult, choose $\mathbf{b}$ along the z -axis, $\mathbf{b} = (0, 0, b)$.
gilmore-lie-groups_ F08325	45	■ 0.458	$k\left(b^{\prime}\right)$	$k(b')$	h. Applying $k(b')$ after applying $k(b)$ results in (a) $k(b' + b)$, (b) two successive
gilmore-lie-groups_ F08326	45	■ 0.458	$k(b)$	$k(b)$	h. Applying $k(b')$ after applying $k(b)$ results in (a) $k(b' + b)$, (b) two successive
gilmore-lie-groups_ F08327	45	■ 0.458	(a) $k\left(b^{\prime}+b\right)$	$(a) k(b' + b)$	h. Applying $k(b')$ after applying $k(b)$ results in (a) $k(b' + b)$, (b) two successive
gilmore-lie-groups_ F08328	45	■ 0.981	$\mathbf{b}^{\prime}$	$\mathbf{b}'$	i. If $\mathbf{b}$ and $\mathbf{b}'$ are not colinear, $k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$. Compute $\mathbf{b}'', \theta$. The angle θ is
gilmore-lie-groups_ F08329	45	■ 0.981	$k\left(\mathbf{b}^{\prime}\right) k(\mathbf{b})=k\left(\mathbf{b}^{\prime \prime}\right) h(\theta)$	$k(\mathbf{b}') k(\mathbf{b}) = k(\mathbf{b}'') h(\theta)$	i. If $\mathbf{b}$ and $\mathbf{b}'$ are not colinear, $k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$. Compute $\mathbf{b}'', \theta$. The angle θ is
gilmore-lie-groups_ F08330	45	■ 0.981	$\mathbf{b}^{\prime \prime}, \theta$	$\mathbf{b}'', \theta$	i. If $\mathbf{b}$ and $\mathbf{b}'$ are not colinear, $k(\mathbf{b}')k(\mathbf{b}) = k(\mathbf{b}'')h(\theta)$. Compute $\mathbf{b}'', \theta$. The angle θ is
gilmore-lie-groups_ F08331	45	■ 1.000	$\theta \rightarrow \theta^{\prime}=\theta+k+f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$, where k is a real number, $0 \leq k < 2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_ F08332	45	■ 1.000	$0 \leq k < 2 \pi$	$0 \leq k < 2\pi$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$, where k is a real number, $0 \leq k < 2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_ F08333	45	■ 1.000	$f(\theta)$	$f(\theta)$	$\theta \rightarrow \theta' = \theta + k + f(\theta)$, where k is a real number, $0 \leq k < 2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_ F08334	45	■ 0.991	$f(\theta+2 \pi)=f(\theta)$	$f(\theta + 2\pi) = f(\theta)$	$f(\theta + 2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08335	45	■ 0.991	1: 1	1 : 1	$f(\theta + 2\pi) = f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08336	45	■ 1.000	$f(\theta): d f(\theta) / d \theta>-1$	$f(\theta): df(\theta)/d\theta > -1$	on $f(\theta): df(\theta)/d\theta > -1$ everywhere. Does this set of transformations form a group?
gilmore-lie-groups_ F08337	46	■ 0.769	(a, b, c, d)	(a, b, c, d)	9. Rational fractional transformations (a, b, c, d) map points on the real line (real pro-
gilmore-lie-groups_ F08338	46	■ 1.000	$R P^1$	RP^1	jective line RP^1) to the real line as follows:
gilmore-lie-groups_ F08339	46	■ 1.000	$(\lambda a, \lambda b, \lambda c, \lambda d)=\lambda(a, b, c, d)(\lambda \neq 0)$	$(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d)(\lambda \neq 0)$	The transformations (a, b, c, d) and $(\lambda a, \lambda b, \lambda c, \lambda d) = \lambda(a, b, c, d)(\lambda \neq 0)$ generate
gilmore-lie-groups_ F08340	46	■ 1.000	A, B, C, D	A, B, C, D	values of A, B, C, D .
gilmore-lie-groups_ F08341	46	■ 0.668	$(\lambda, 0, 0, \lambda)$	$(\lambda, 0, 0, \lambda)$	b. Show that the transformations $(\lambda, 0, 0, \lambda)$ map x to itself.

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gilmore-lie-groups_F00342	46	■ 0.999	$x^{\{\prime\}}$ $\rightarrow$ x	$x' \rightarrow x$	c. Construct the inverse transformation $x' \rightarrow x$, and show that it is $\lambda(d, -b, -c, a)$
gilmore-lie-groups_F00343	46	■ 0.999	$\lambda(d, -b, -c, a)$	$\lambda(d, -b, -c, a)$	c. Construct the inverse transformation $x' \rightarrow x$, and show that it is $\lambda(d, -b, -c, a)$
gilmore-lie-groups_F00344	46	■ 1.000	$\lambda \neq 0$	$\lambda \neq 0$	provided $\lambda \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups_F00345	46	■ 1.000	$D = a d - b c \neq 0$	$D = ad - bc \neq 0$	provided $\lambda \neq 0$. Such transformations exist if $D = ad - bc \neq 0$.
gilmore-lie-groups_F00346	46	■ 1.000	$x^{\{\prime\}} = (a, b, c, d) \quad x = \lambda(a, b, c, d) x$	$x' = (a, b, c, d)x = \lambda(a, b, c, d)x$	d. Show that the transformation degeneracy $x' = (a, b, c, d)x = \lambda(a, b, c, d)x$ can
gilmore-lie-groups_F00347	46	■ 1.000	a, b, c, d	a, b, c, d	be lifted by requiring that the four parameters a, b, c, d describing these transfor-
gilmore-lie-groups_F00348	46	■ 1.000	$D = a d - b c = 1$	$D = ad - bc = 1$	mations satisfy the constraint $D = ad - bc = 1$.
gilmore-lie-groups_F00349	46	■ 0.980	(y, z)	(y, z)	e. It is useful to introduce homogeneous coordinates (y, z) and define the real projec-
gilmore-lie-groups_F00350	46	■ 1.000	$x = y / z$	$x = y/z$	tive coordinate x as the ratio of these homogeneous coordinates: $x = y/z$. If the
gilmore-lie-groups_F00351	46	■ 0.477	x_1, x_2, x_3	x_1, x_2, x_3	distinct points x_1, x_2, x_3 on the real line to the three standard positions $(0, 1, \infty)$,
gilmore-lie-groups_F00352	46	■ 0.477	∞	∞	distinct points x_1, x_2, x_3 on the real line to the three standard positions $(0, 1, \infty)$,
gilmore-lie-groups_F00353	46	■ 0.996	$D = 1$	$D = 1$	What matrix in $SL(2; \mathbb{R})$ describes this mapping? (Careful of the condition $D = 1$.)
gilmore-lie-groups_F00354	46	■ 1.000	$\left(x_1^{\{\prime\}}, x_2^{\{\prime\}}, x_3^{\{\prime\}}\right)$	(x'_1, x'_2, x'_3)	points (x_1, x_2, x_3) to any other triple of distinct points (x'_1, x'_2, x'_3) .
gilmore-lie-groups_F00355	46	■ 1.000	R^n	RP^n	10. The real projective space RP^n is the space of all straight lines through the origin
gilmore-lie-groups_F00356	46	■ 0.987	R^{n+1}	R^{n+1}	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_F00357	46	■ 0.987	$SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	$SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_F00358	46	■ 0.987	$x^{\{\prime\}} \in R^{n+1}$	$x' \in R^{n+1}$	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_F00359	46	■ 1.000	$x^{\{\prime\}} \neq 0 \rightarrow x \neq 0$	$x' \neq 0 \leftrightarrow x \neq 0$	with $x' \neq 0 \leftrightarrow x \neq 0$ and $x' = 0 \leftrightarrow x = 0$. A straight line through the origin contains
gilmore-lie-groups_F00360	46	■ 1.000	$x^{\{\prime\}} = 0 \rightarrow x = 0$	$x' = 0 \leftrightarrow x = 0$	with $x' \neq 0 \leftrightarrow x \neq 0$ and $x' = 0 \leftrightarrow x = 0$. A straight line through the origin contains
gilmore-lie-groups_F00361	46	■ 1.000	$x \neq 0$	$x \neq 0$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_F00362	46	■ 1.000	$y \neq 0$	$y \neq 0$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_F00363	46	■ 1.000	$y = \lambda x$	$y = \lambda x$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_F00364	46	■ 1.000	y	y	factor can always be chosen so that y is in the unit sphere in R^{n+1} : $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups_F00365	46	■ 1.000	$R^{n+1}: y \in S^n \subset R^{n+1}$	$R^{n+1}: y \in S^n \subset R^{n+1}$	factor can always be chosen so that y is in the unit sphere in R^{n+1} : $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups_F00366	46	■ 1.000	λ	λ	In fact, two values of λ can be chosen: $\lambda = \pm 1 / (\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_F00367	46	■ 1.000	$\lambda = \pm 1 / (\sum_{i=1}^{n+1} x_i^2)^{1/2}$	$\lambda = \pm 1 / \left(\sum_{i=1}^{n+1} x_i^2 \right)^{1/2}$	In fact, two values of λ can be chosen: $\lambda = \pm 1 / (\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_F00368	46	■ 1.000	R^3	R^3	In fact, two values of λ can be chosen: $\lambda = \pm 1 / (\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight

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gilmore-lie-groups-F00369	46	■ 0.983	(x, y, z)	(x, y, z)	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups-F00370	46	■ 0.983	$(X, Y)=$	$(X, Y) =$	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups-F00371	46	■ 1.000	$(x / z, y / z)$	$(x/z, y/z)$	$(x/z, y/z)$ if $z \neq 0$. Straight lines through the origin of R^3 are mapped to straight lines
gilmore-lie-groups-F00372	46	■ 1.000	$z \neq 0$	$z \neq 0$	$(x/z, y/z)$ if $z \neq 0$. Straight lines through the origin of R^3 are mapped to straight lines
gilmore-lie-groups-F00373	46	■ 1.000	$x \rightarrow x^{\prime}=M x, M \in S L(3 ; \mathbb{R})$	$x \rightarrow x^{\prime}=M x, M \in S L(3 ; \mathbb{R})$	in R^3 by $x \rightarrow x^{\prime}=M x, M \in S L(3 ; \mathbb{R})$. Show that the homogeneous coordinates
gilmore-lie-groups-F00374	47	■ 1.000	$x^{\prime}$	$x^{\prime}$	representing the two lines containing x and $x^{\prime}$ are related by the linear fractional
gilmore-lie-groups-F00375	47	■ 1.000	$R P^n \rightarrow R P^n$	$R P^n \rightarrow R P^n$	Generalize for linear fractional transformations $R P^n \rightarrow R P^n$.
gilmore-lie-groups-F00376	47	■ 0.989	$S L(2 ; \mathbb{R}) / S O(2) \simeq \left[\begin{array}{cc} z+x & y \\ y & z-x \end{array}\right]$	$S L(2 ; \mathbb{R}) / S O(2) \simeq \left[\begin{array}{cc} z+x & y \\ y & z-x \end{array}\right]$	11. The hyperbolic two-space $S L(2 ; \mathbb{R}) / S O(2) \simeq \left[\begin{array}{cc} z+x & y \\ y & z-x \end{array}\right]$ consists of the algebraic sub-
gilmore-lie-groups-F00377	47	■ 1.000	$x=r \cos \phi, y=r \sin \phi$	$x=r \cos \phi, y=r \sin \phi$	b. Introduce polar coordinates $x=r \cos \phi, y=r \sin \phi$, and show
gilmore-lie-groups-F00378	47	■ 0.633	$S O(3) / S O(2)$	$S O(3) / S O(2)$	d. Repeat this calculation for $S O(3) / S O(2)$. This space is a sphere $S^2 \subset R^3$: the
gilmore-lie-groups-F00379	47	■ 0.633	$S^2 \subset R^3$	$S^2 \subset R^3$	d. Repeat this calculation for $S O(3) / S O(2)$. This space is a sphere $S^2 \subset R^3$: the
gilmore-lie-groups-F00380	47	■ 1.000	$z^2+\left(x^2+y^2\right)=1$	$z^2+\left(x^2+y^2\right)=1$	algebraic manifold in R^3 that satisfies $z^2+\left(x^2+y^2\right)=1$ and inherits the metric
gilmore-lie-groups-F00381	47	■ 1.000	$d s^2=d z^2+\left(d x^2+d y^2\right)$	$d s^2=d z^2+\left(d x^2+d y^2\right)$	$d s^2=d z^2+\left(d x^2+d y^2\right)$ from this Euclidean space. Show that the metric and
gilmore-lie-groups-F00382	47	■ 1.000	H^2	H^2	measure on S^2 are obtained from the results above for H^2 by the substitutions
gilmore-lie-groups-F00383	47	■ 1.000	$1+r^2 \rightarrow 1-r^2$	$1+r^2 \rightarrow 1-r^2$	$1+r^2 \rightarrow 1-r^2$. Show that the disk $0 \leq r \leq 1, 0 \leq \phi \leq 2 \pi$ maps onto the up-
gilmore-lie-groups-F00384	47	■ 1.000	$0 \leq r \leq 1, 0 \leq \phi \leq 2 \pi$	$0 \leq r \leq 1, 0 \leq \phi \leq 2 \pi$	$1+r^2 \rightarrow 1-r^2$. Show that the disk $0 \leq r \leq 1, 0 \leq \phi \leq 2 \pi$ maps onto the up-
gilmore-lie-groups-F00385	47	■ 1.000	$r=0$	$r=0$	per hemisphere of the sphere, with $r=0$ mapping to the north pole and $r=1$
gilmore-lie-groups-F00386	47	■ 1.000	$r=1$	$r=1$	per hemisphere of the sphere, with $r=0$ mapping to the north pole and $r=1$
gilmore-lie-groups-F00387	47	■ 1.000	$\phi=0$	$\phi=0$	equator along the longitude $\phi=0$ is $s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$ and the volume of
gilmore-lie-groups-F00388	47	■ 1.000	$s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$	$s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$	equator along the longitude $\phi=0$ is $s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$ and the volume of
gilmore-lie-groups-F00389	47	■ 1.000	$V=\int_{r=0}^1 \int_{\phi=0}^{2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$	$V=\int_{r=0}^1 \int_{\phi=0}^{2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$	the hemisphere surface is $V=\int_{r=0}^1 \int_{\phi=0}^{2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$
gilmore-lie-groups-F00390	47	■ 1.000	2π	2π	2π .
gilmore-lie-groups-F00391	48	■ 0.969	$G L(n ; \mathbb{F})$	$G L(n ; \mathbb{F})$	typically encountered. These include the general linear groups $G L(n ; \mathbb{F})$
gilmore-lie-groups-F00392	48	■ 1.000	$\mathbb{F}$	$\mathbb{F}$	of nonsingular $n \times n$ matrices over the fields $\mathbb{F}$ of real numbers, complex
gilmore-lie-groups-F00393	48	■ 1.000	$H_1 \subset G$	$H_1 \subset G$	Theorem If $H_1 \subset G$ and $H_2 \subset G$ are subgroups of G then their intersection
gilmore-lie-groups-F00394	48	■ 1.000	$H_{12}=H_1 \cap H_2$	$H_{12}=H_1 \cap H_2$	$H_{12}=H_1 \cap H_2$ is a subgroup of G .

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gilmore-lie-groups-F00395	48	■ 1.000	$H_{\{1\}} \cap H_{\{2\}}$	$H_1 \cap H_2$	Proof Verify that the four group axioms are satisfied for all operations in $H_1 \cap H_2$.
gilmore-lie-groups-F00396	48	■ 0.950	$H_{\{1\}}$	H_1	Example If H_1 is the two-dimensional subgroup of $SL(2; \mathbb{R})$ described in (3.1)
gilmore-lie-groups-F00397	48	■ 0.817	$H_{\{2\}}$	H_2	above and H_2 is the one-dimensional subgroup of 2×2 orthogonal matrices
gilmore-lie-groups-F00398	49	■ 1.000	$\pm I_{\{2\}}$	$\pm I_2$	discrete group operations $\pm I_2$.
gilmore-lie-groups-F00399	49	■ 0.950	$(\mathbb{F} = \mathbb{R})$	$(\mathbb{F} = \mathbb{R})$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups-F00400	49	■ 0.950	$\mathbb{F} = \mathbb{C}$	$\mathbb{F} = \mathbb{C}$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups-F00401	49	■ 0.950	$\mathbb{F} = \mathbb{Q}$	$\mathbb{F} = \mathbb{Q}$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups-F00402	49	■ 1.000	$1, \mathcal{I}, J, K$	$1, \mathcal{I}, J, K$	The four basis vectors $1, \mathcal{I}, J, K$ for this map are related to the four Pauli spin
gilmore-lie-groups-F00403	49	■ 0.900	$[\operatorname{det}(\mathrm{M}) = +1]$	$[\det(\mathrm{M}) = +1]$	$[\det(\mathrm{M}) = +1]$ are imposed (12).
gilmore-lie-groups-F00404	49	■ 0.999	$GL(1; \mathbb{Q})$	$GL(1; \mathbb{Q})$	the real, complex, or quaternion fields. The group $GL(1; \mathbb{Q})$ consists of 1×1
gilmore-lie-groups-F00405	49	■ 0.999	1×1	1×1	the real, complex, or quaternion fields. The group $GL(1; \mathbb{Q})$ consists of 1×1
gilmore-lie-groups-F00406	50	■ 1.000	A_i^j	A_i^j	The determinant of an $n \times n$ matrix A with matrix elements A_i^j is defined by
gilmore-lie-groups-F00407	50	■ 1.000	$\epsilon^{i_1 i_2 \dots i_n}$	$\epsilon^{i_1 i_2 \dots i_n}$	Here $\epsilon^{i_1 i_2 \dots i_n}$ and its covariant version are the Levi–Civita symbols: +1 for an even
gilmore-lie-groups-F00408	50	■ 0.719	$1, 2, \dots, n; -1$	$1, 2, \dots, n; -1$	permutation of the integers $1, 2, \dots, n; -1$ for an odd permutation; and 0 if two
gilmore-lie-groups-F00409	50	■ 1.000	i_*	i_*	or more values of the indices i_* are equal. With this definition there is no difficulty
gilmore-lie-groups-F00410	50	■ 1.000	$UT(p, q)$	$UT(p, q)$	2. $UT(p, q)$. Upper triangular groups. The $n \times n$ ($n = p + q$) matrix is parti-
gilmore-lie-groups-F00411	50	■ 1.000	$n \times n (n = p + q)$	$n \times n (n = p + q)$	2. $UT(p, q)$. Upper triangular groups. The $n \times n$ ($n = p + q$) matrix is parti-
gilmore-lie-groups-F00412	50	■ 1.000	$UT(1, 1)$	$UT(1, 1)$	Example The action of transformations in $UT(1, 1)$ on the plane R^2 is as
gilmore-lie-groups-F00413	50	■ 0.991	$y = 0$	$y = 0$	The x -axis $y = 0$ remains invariant. It is an invariant subspace ($y = 0 \rightarrow y' = 0$),
gilmore-lie-groups-F00414	50	■ 0.991	$y = 0 \rightarrow y' = 0$	$y = 0 \rightarrow y' = 0$	The x -axis $y = 0$ remains invariant. It is an invariant subspace ($y = 0 \rightarrow y' = 0$),
gilmore-lie-groups-F00415	50	■ 1.000	$x = 0$	$x = 0$	mapped into itself by all group operations in $UT(1, 1)$. The y -axis $x = 0$ is not
gilmore-lie-groups-F00416	50	■ 1.000	$V_p \oplus V_q$	$V_p \oplus V_q$	invariant. More generally, if $UT(p, q)$ acts on the direct sum vector space $V_p \oplus V_q$,
gilmore-lie-groups-F00417	50	■ 1.000	V_q	V_q	the subspace V_q is invariant while V_p is not. For lower triangular matrices reverse
gilmore-lie-groups-F00418	50	■ 1.000	V_p	V_p	the subspace V_q is invariant while V_p is not. For lower triangular matrices reverse
gilmore-lie-groups-F00419	50	■ 1.000	q	q	p and q .
gilmore-lie-groups-F00420	51	■ 1.000	$HT(p, q)$	$HT(p, q)$	3. $HT(p, q)$. This is a subgroup of $UT(p, q)$ obtained by imposing the additional
gilmore-lie-groups-F00421	51	■ 1.000	$HT(1, 1) (m_{22} = 1)$	$HT(1, 1) (m_{22} = 1)$	Example Affine transformations in $HT(1, 1) (m_{22} = 1)$ act on the x -axis by $x \rightarrow$

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gilmore-lie-groups_F00422	51	■ 1.000	$x \rightarrow$	$x \rightarrow$	Example Affine transformations in $HT(1, 1)$ ($m_{22} = 1$) act on the x -axis by $x \rightarrow$
gilmore-lie-groups_F00423	51	■ 0.997	$x^{\{\prime\}}=a\ x+b$	$x' = ax + b$	$x' = ax + b$:
gilmore-lie-groups_F00424	51	■ 0.999	$U\ T(p,\ q,\ r)$	$UT(p, q, r)$	4. $UT(p, q, r)$. This matrix group consists of upper triangular matrices that are the
gilmore-lie-groups_F00425	51	■ 1.000	$U\ T(p,\ q+r)\ \cap\ U\ T(p+q,\ r)$	$UT(p, q + r) \cap UT(p + q, r)$	intersection of the matrix groups $UT(p, q + r) \cap UT(p + q, r)$.
gilmore-lie-groups_F00426	52	■ 0.460	$S\ U(1,1)$	$SU(1, 1)$	where the 2×2 matrix $SU(1, 1)$ is defined below in (3.30). Matrix groups with the
gilmore-lie-groups_F00427	52	■ 1.000	$\operatorname{Sol}(n)=U\ T(1,1,1,\ \ldots,\ 1)$	$Sol(n) = UT(1, 1, 1, \dots, 1)$	5. $Sol(n) = UT(1, 1, 1, \dots, 1)$. Solvable groups are strictly upper triangular.
gilmore-lie-groups_F00428	52	■ 0.971	$U\ T(1,1,1)$	$UT(1, 1, 1)$	Example We consider the subgroup of 3×3 matrices in $UT(1, 1, 1)$ of the
gilmore-lie-groups_F00429	52	■ 1.000	$\hat{n}=a^{\dagger}a$	$\hat{n} = a^{\dagger}a$	of the photon number operator ($\hat{n} = a^{\dagger}a$), the creation ($a^{\dagger}$) and annihilation (a)
gilmore-lie-groups_F00430	52	■ 1.000	$a^{\dagger}$	$a^{\dagger}$	of the photon number operator ($\hat{n} = a^{\dagger}a$), the creation ($a^{\dagger}$) and annihilation (a)
gilmore-lie-groups_F00431	52	■ 1.000	a	a	of the photon number operator ($\hat{n} = a^{\dagger}a$), the creation ($a^{\dagger}$) and annihilation (a)
gilmore-lie-groups_F00432	52	■ 0.994	$I=a\ a^{\dagger}-a^{\dagger}a=\left[a,\ a^{\dagger}\right]$	$I = aa^{\dagger} - a^{\dagger}a = [a, a^{\dagger}]$	operators, and their commutator ($I = aa^{\dagger} - a^{\dagger}a = [a, a^{\dagger}]$). We will use this iden-
gilmore-lie-groups_F00433	52	■ 1.000	$\operatorname{Nil}(n)$	$Nil(n)$	6. $Nil(n)$. Nilpotent groups are subgroups of $Sol(n)$ whose diagonal matrix
gilmore-lie-groups_F00434	52	■ 1.000	$\operatorname{Sol}(n)$	$Sol(n)$	6. $Nil(n)$. Nilpotent groups are subgroups of $Sol(n)$ whose diagonal matrix
gilmore-lie-groups_F00435	52	■ 0.998	$\operatorname{Nil}(3)$	$Nil(3)$	Example Matrices in $Nil(3)$ of the form
gilmore-lie-groups_F00436	52	■ 0.976	$a^{\dagger},\ a,\ I$	$a^{\dagger}, a, I$	are closely related to the photon creation and annihilation operators ($a^{\dagger}, a, I$) and
gilmore-lie-groups_F00437	52	■ 0.716	$(p$	$(p$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_F00438	52	■ 0.716	$q)$	$q)$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_F00439	52	■ 0.716	$[p,\ q]=\hbar\ /\ i$	$[p, q] = \hbar/i$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_F00440	52	■ 1.000	$\langle p\ \ q\rangle=\frac{1}{\sqrt{2}}\ e^{\{2\ \pi\ i\ p\ q\ /\ h\}}$	$\langle p q \rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / h}$	The set of change of basis transformations $\langle p q \rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / h}$ encountered in
gilmore-lie-groups_F00441	52	■ 0.995	$A(p,\ q)$	$A(p, q)$	7. $A(p, q)$. This group consists of matrices that are the sum of an identity matrix
gilmore-lie-groups_F00442	52	■ 0.998	$(p,\ q)$	(p, q)	and the upper right-hand off-diagonal block of a (p, q) blocked matrix. Its matrix
gilmore-lie-groups_F00443	53	■ 1.000	$A\ B=B\ A$	$AB = BA$	This group is abelian or commutative: $AB = BA$ for all elements (matrices) in this
gilmore-lie-groups_F00444	53	■ 1.000	$A(1,1)$	$A(1, 1)$	Example We consider the translation subgroup $A(1, 1)$ of the affine group of
gilmore-lie-groups_F00445	53	■ 0.798	$x\ \rightarrow\ x^{\{\prime\}}=x+a$	$x \rightarrow x' = x + a$	transformations of the x -axis (3.9): $x \rightarrow x' = x + a$. Successive transformations
gilmore-lie-groups_F00446	53	■ 1.000	$M^{\dagger}\ G\ M=G$	$M^{\dagger}GM = G$	all satisfy the bilinear or quadratic constraint condition $M^{\dagger}GM = G$. If G is sym-
gilmore-lie-groups_F00447	53	■ 0.679	$G=I_{\{n\}}$	$G = I_n$	metric positive-definite we can set $G = I_n$ (8). If G is nonsingular and symmetric
gilmore-lie-groups_F00448	53	■ 0.573	$G=I_{\{p,\ q\}}$	$G = I_{p, q}$	but indefinite we can set $G = I_{p, q}$ (9). If G is nonsingular and antisymmetric, we

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gilmore-lie-groups_ F00449	53	■ 1.000	$I_{\{n\}}$	I_n	which we can take as I_n
gilmore-lie-groups_ F00450	54	■ 0.936	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	terize the matrix groups $GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.
gilmore-lie-groups_ F00451	54	■ 1.000	C^2	C^2	two-dimensional linear vector space C^2 (of spin states, for example), and quaternion
gilmore-lie-groups_ F00452	54	■ 1.000	$\mathbb{Q}$	$\mathbb{Q}$	over $\mathbb{Q}$
gilmore-lie-groups_ F00453	54	■ 0.808	$SU(1; \mathbb{Q})$	$SU(1; \mathbb{Q})$	The group $SU(1; \mathbb{Q})$ is the subgroup of $GL(1; \mathbb{Q})$ (3.5) subject to the condition
gilmore-lie-groups_ F00454	54	■ 1.000	R^4	R^4	in R^4
gilmore-lie-groups_ F00455	54	■ 0.992	$G = I_{\{p, q\}}, p+q=n$	$G = I_{p,q}, p+q=n$	$G = I_{p,q}, p+q=n$. This is a diagonal matrix with p elements $+1$ and q ele-
gilmore-lie-groups_ F00456	54	■ 1.000	$p \neq 0, q \neq 0$	$p \neq 0, q \neq 0$	The manifolds that parameterize these groups are noncompact when $p \neq 0, q \neq 0$.
gilmore-lie-groups_ F00457	54	■ 1.000	$x^2 + y^2 + z^2 - (ct)^2$	$x^2 + y^2 + z^2 - (ct)^2$	Example The Lorentz group preserves the invariant $x^2 + y^2 + z^2 - (ct)^2$ and is
gilmore-lie-groups_ F00458	55	■ 0.999	$N \times N$	$N \times N$	10. Antisymmetric metric-preserving groups The metric G is an $N \times N$ non-
gilmore-lie-groups_ F00459	55	■ 1.000	$\operatorname{det}(G) = \operatorname{det}(G^t) = \operatorname{det}(-G) = (-1)^N \operatorname{det}(G), N$	$\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$	Since $\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$ must be even: $N = 2n$. The
gilmore-lie-groups_ F00460	55	■ 1.000	$N = 2n$	$N = 2n$	Since $\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$ must be even: $N = 2n$. The
gilmore-lie-groups_ F00461	55	■ 1.000	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	This consists of n copies of the matrix $i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ along the diagonal. Symplectic
gilmore-lie-groups_ F00462	55	■ 0.990	$\operatorname{Sp}(2n; \mathbb{R})$	$Sp(2n; \mathbb{R})$	transformations in $Sp(2n; \mathbb{R})$ leave invariant the form of the classical hamiltonian
gilmore-lie-groups_ F00463	55	■ 0.935	$\operatorname{Sp}(2; \mathbb{R}) \subset GL(2; \mathbb{R})$	$Sp(2; \mathbb{R}) \subset GL(2; \mathbb{R})$	Example The symplectic group $Sp(2; \mathbb{R}) \subset GL(2; \mathbb{R})$ satisfies the constraint
gilmore-lie-groups_ F00464	55	■ 0.510	$ad - bc = +1$	$ad - bc = +1$	The constraint is $ad - bc = +1$. Thus, $Sp(2; \mathbb{R}) = SL(2; \mathbb{R})$.
gilmore-lie-groups_ F00465	55	■ 0.510	$\operatorname{Sp}(2; \mathbb{R}) = SL(2; \mathbb{R})$	$Sp(2; \mathbb{R}) = SL(2; \mathbb{R})$	The constraint is $ad - bc = +1$. Thus, $Sp(2; \mathbb{R}) = SL(2; \mathbb{R})$.
gilmore-lie-groups_ F00466	55	■ 1.000	$MGM^t = G$	$MGM^t = G$	Partitioning M into blocks and imposing the condition $MGM^t = G$, we find
gilmore-lie-groups_ F00467	56	■ 0.999	$E(3)$	$E(3)$	Euclidean group $E(3)$ whose action on the coordinates (x, y, z) of a point in
gilmore-lie-groups_ F00468	56	■ 0.607	$\mathbf{t}$	$\mathbf{t}$	That is, the coordinates are rotated by the matrix A and translated by the vector $\mathbf{t}$. By
gilmore-lie-groups_ F00469	56	■ 0.792	$A \in SO(3, 1), AI_{3,1}A^t = I_{3,1}$	$A \in SO(3, 1), AI_{3,1}A^t = I_{3,1}$	($A \in SO(3, 1), AI_{3,1}A^t = I_{3,1}$ (3.17)) and space-time displacements is isomorphic
gilmore-lie-groups_ F00470	56	■ 0.608	$\mathbf{v}$	$\mathbf{v}$	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00471	56	■ 0.608	$(\mathbf{t})$	$(\mathbf{t})$	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00472	56	■ 0.608	(t_4)	(t_4)	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00473	57	■ 0.991	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2; \mathbb{R}) \subset GL(2; \mathbb{R})$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2; \mathbb{R}) \subset GL(2; \mathbb{R})$	trices $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2; \mathbb{R}) \subset GL(2; \mathbb{R})$ satisfies the constraint $ad - bc = +1$, which
gilmore-lie-groups_ F00474	57	■ 0.998	$U(1, 1)$	$U(1, 1)$	taking the intersection of $U(1, 1)$ with $SL(2; \mathbb{C})$:

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gilmore-lie-groups_ F00475	57	■ 1.000	$a^{*} a - b^{*} b = +1$	$a^{*} a - b^{*} b = +1$	where $\boxed{a^{*} a - b^{*} b = +1}$.
gilmore-lie-groups_ F00476	57	■ 1.000	$U(n)$	$U(n)$	The unitary group $U(n)$ consists of $n \times n$ complex matrices that obey the constraint
gilmore-lie-groups_ F00477	57	■ 1.000	$U^{\dagger} U = I_n$	$U^{\dagger} U = I_n$	$U^{\dagger} U = I_n$. For some purposes it is useful to represent this group as a group of real
gilmore-lie-groups_ F00478	57	■ 1.000	$2n \times 2n$	$2n \times 2n$	is a real $2n \times 2n$ matrix M . This matrix inherits the constraint that comes with the
gilmore-lie-groups_ F00479	57	■ 1.000	$M^t M = I_{2n}$	$M^t M = I_{2n}$	unitary group, $U^{\dagger} U = I_n$. This constraint now appears in the form $M^t M = I_{2n}$.
gilmore-lie-groups_ F00480	57	■ 1.000	$\{ \}^{\dagger}$	$\dagger$	We have been able to replace $\dagger$ by t since the matrices are real, and must replace
gilmore-lie-groups_ F00481	57	■ 1.000	$\{ \}^t$	t	We have been able to replace $\dagger$ by t since the matrices are real, and must replace
gilmore-lie-groups_ F00482	57	■ 1.000	I_{2n}	I_{2n}	I_n by I_{2n} since the matrices are $2n \times 2n$. In other words, the matrices M obey the
gilmore-lie-groups_ F00483	57	■ 0.712	$SO(2n)$	$SO(2n)$	an <i>orthogonal</i> representation of the <i>unitary</i> group. It is a subgroup of $SO(2n)$. This
gilmore-lie-groups_ F00484	57	■ 1.000	$OU(2n)$	$OU(2n)$	matrix group is called $OU(2n)$. Symbolically,
gilmore-lie-groups_ F00485	58	■ 0.999	$U(n; \mathbb{Q}) = Sp(n)$	$U(n; \mathbb{Q}) = Sp(n)$	For the metric-preserving quaternion group $\boxed{U(n; \mathbb{Q}) = Sp(n)}$ whose matrices obey
gilmore-lie-groups_ F00486	58	■ 1.000	$M^{\dagger} M = I_{2n}$	$M^{\dagger} M = I_{2n}$	constraint in the form $M^{\dagger} M = I_{2n}$. In other words, the matrices M obey the condi-
gilmore-lie-groups_ F00487	58	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	tion that determines unitary groups (over $\mathbb{C}$). This group of $2n \times 2n$ matrices forms
gilmore-lie-groups_ F00488	58	■ 0.945	$SU(2n)$	$SU(2n)$	a <i>unitary</i> representation of the <i>symplectic</i> group. It is a subgroup of $SU(2n)$. This
gilmore-lie-groups_ F00489	58	■ 1.000	$USp(2n)$	$USp(2n)$	matrix group is called $USp(2n)$. Symbolically,
gilmore-lie-groups_ F00490	58	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	because they lack the geometric structure contributed by the continuous fields $\mathbb{R}$,
gilmore-lie-groups_ F00491	58	■ 1.000	$GL(n; \mathbb{Z})$	$GL(n; \mathbb{Z})$	(i) $GL(n; \mathbb{Z})$: if $m \in GL(n; \mathbb{Z})$, $\det(m) = \pm 1$.
gilmore-lie-groups_ F00492	58	■ 1.000	$m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$	$m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$	(i) $GL(n; \mathbb{Z})$: if $\boxed{m \in GL(n; \mathbb{Z}), \det(m) = \pm 1}$.
gilmore-lie-groups_ F00493	58	■ 1.000	$SL(n; \mathbb{Z})$	$SL(n; \mathbb{Z})$	(ii) $SL(n; \mathbb{Z})$: if $m \in SL(n; \mathbb{Z})$, $\det(m) = +1$.
gilmore-lie-groups_ F00494	58	■ 1.000	$m \in SL(n; \mathbb{Z}), \det(m) = +1$	$m \in SL(n; \mathbb{Z}), \det(m) = +1$	(ii) $SL(n; \mathbb{Z})$: if $\boxed{m \in SL(n; \mathbb{Z}), \det(m) = +1}$.
gilmore-lie-groups_ F00495	58	■ 1.000	$PSL(n; \mathbb{Z}), n$	$PSL(n; \mathbb{Z}), n$	(iii) $PSL(n; \mathbb{Z}), n$ even: $PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$.
gilmore-lie-groups_ F00496	58	■ 1.000	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \left\{ I_n, -I_n \right\}$	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$	(iii) $PSL(n; \mathbb{Z}), n$ even: $\boxed{PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}}$
gilmore-lie-groups_ F00497	58	■ 0.999	$n=2$	$n = 2$	For $n = 2$ these groups of matrices have the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with a, b, c, d all integers.
gilmore-lie-groups_ F00498	58	■ 0.999	$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	For $n = 2$ these groups of matrices have the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with a, b, c, d all integers.
gilmore-lie-groups_ F00499	58	■ 1.000	$\det(m) = n$	$\det(m) = n$	If $\det(m) = n$, with n an integer, then $\det(m^{-1}) = 1/n$. Since the determinant of
gilmore-lie-groups_ F00500	58	■ 1.000	$\det(m^{-1}) = 1/n$	$\det(m^{-1}) = 1/n$	If $\det(m) = n$, with n an integer, then $\det(m^{-1}) = 1/n$. Since the determinant of
gilmore-lie-groups_ F00501	58	■ 1.000	$\det(m) =$	$\det(m) =$	any matrix composed of integers must be an integer, the condition is that $\det(m) =$

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gilmore-lie-groups_ F00502	58	■ 0.714	± 1	± 1	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_ F00503	58	■ 0.714	$GL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_ F00504	58	■ 0.714	$SL(2; \mathbb{Z}) \subset$	$SL(2; \mathbb{Z}) \subset$	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_ F00505	58	■ 1.000	$PSL(2; \mathbb{Z})$	$PSL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$. The modular group $PSL(2; \mathbb{Z})$ is obtained by identifying each pair of
gilmore-lie-groups_ F00506	58	■ 0.906	$SL(2; \mathbb{Z})$	$SL(2; \mathbb{Z})$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_ F00507	58	■ 0.906	$\left[\begin{array}{ll} -a & -b \\ -c & -d \end{array} \right] \simeq \left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_ F00508	59	■ 1.000	$F(n)$	$F(n)$	where $F(n)$ is the n th Fibonacci number, defined recursively by
gilmore-lie-groups_ F00509	59	■ 1.000	$n=1(F(0)=0, F(1)=1)$	$n = 1(F(0) = 0, F(1) = 1)$	and by comparison of initial conditions for $n = 1$ ($F(0) = 0, F(1) = 1$). Many
gilmore-lie-groups_ F00510	59	■ 0.998	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	in the groups $GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$, etc.
gilmore-lie-groups_ F00511	59	■ 0.998	$GL(n; \mathbb{R})$	$GL(n; \mathbb{R})$	with $GL(n; \mathbb{R})$.
gilmore-lie-groups_ F00512	59	■ 1.000	$M^t I_n M = I_n$	$M^t I_n M = I_n$	Imposing quadratic constraints, for example $M^t I_n M = I_n$, generates a sub-
gilmore-lie-groups_ F00513	59	■ 1.000	$O(n; \mathbb{Z})$	$O(n; \mathbb{Z})$	group, $O(n; \mathbb{Z})$, consists of $n \times n$ matrices in which all but one matrix element
gilmore-lie-groups_ F00514	59	■ 1.000	$\operatorname{det}(m) = +1$	$\det(m) = +1$	Finally, the multilinear condition $\det(m) = +1$ defines the unimodular subgroup
gilmore-lie-groups_ F00515	60	■ 1.000	A_n	A_n	ple, the alternating group A_n consists of unimodular matrices in P_n :
gilmore-lie-groups_ F00516	60	■ 0.932	$O(2; \mathbb{Z})$	$O(2; \mathbb{Z})$	Example The group $O(2; \mathbb{Z})$ consists of the $8 = 2^2 \times 2!$ matrices
gilmore-lie-groups_ F00517	60	■ 0.932	$8=2^2 \times 2!$	$8 = 2^2 \times 2!$	Example The group $O(2; \mathbb{Z})$ consists of the $8 = 2^2 \times 2!$ matrices
gilmore-lie-groups_ F00518	60	■ 0.974	$O(3; \mathbb{Z})$	$O(3; \mathbb{Z})$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups_ F00519	60	■ 0.974	$2^3 \times 3! = 48$	$2^3 \times 3! = 48$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups_ F00520	60	■ 0.974	$6=3!$	$6 = 3!$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups_ F00521	60	■ 1.000	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	Its alternating subgroup $A_3 \subset S_3 \subset O(3; \mathbb{Z})$ consists of the three matrices with
gilmore-lie-groups_ F00522	60	■ 0.997	G_2, F_4, E_6, E_7, E_8	G_2, F_4, E_6, E_7, E_8	G_2, F_4, E_6, E_7, E_8 and their real forms (encountered in Chapters 10 and 11), and
gilmore-lie-groups_ F00523	60	■ 1.000	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	integers, $GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$, and $PSL(n; \mathbb{Z})$. Matrix groups over finite fields are
gilmore-lie-groups_ F00524	60	■ 1.000	$PSL(n; \mathbb{Z})$	$PSL(n; \mathbb{Z})$	integers, $GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$, and $PSL(n; \mathbb{Z})$. Matrix groups over finite fields are
gilmore-lie-groups_ F00525	61	■ 1.000	$\mathcal{IJ} = -\mathcal{K}$	$\mathcal{IJ} = -\mathcal{K}$	where $\mathcal{IJ} = -\mathcal{K}$:
gilmore-lie-groups_ F00526	61	■ 0.997	$\sigma_x, \sigma_y, \sigma_z$	$\sigma_x, \sigma_y, \sigma_z$	Here $\sigma_x, \sigma_y, \sigma_z$ are the Pauli spin matrices, and i is the usual square root of -1 . Show
gilmore-lie-groups_ F00527	61	■ 0.924	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{IJ} + \mathcal{JI} = 0$	that any pair of the unit quaternions anticommute: i.e., $\mathcal{IJ} = \mathcal{JI} + \mathcal{I} = 0$.

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gilmore-lie-groups_F00528	61	■ 1.000	$\mathcal{I}, J, K$	$\mathcal{I}, J, K$	2. Show that the unit quaternions $\mathcal{I}, J, K$ generate a group of order 8 under multi-
gilmore-lie-groups_F00529	61	■ 0.464	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	3. Show that $SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$.
gilmore-lie-groups_F00530	61	■ 1.000	$p+q=n$	$p + q = n$	in Example (9), for suitable positive integers p and q , with $p + q = n$.
gilmore-lie-groups_F00531	61	■ 0.996	$SL_i(n; \mathbb{C})$	$SL_i(n; \mathbb{C})$	6. Show that it is possible to define subgroups $SL_i(n; \mathbb{C})$ of $GL(n; \mathbb{C})$ by the con-
gilmore-lie-groups_F00532	61	■ 0.996	$GL(n; \mathbb{C})$	$GL(n; \mathbb{C})$	6. Show that it is possible to define subgroups $SL_i(n; \mathbb{C})$ of $GL(n; \mathbb{C})$ by the con-
gilmore-lie-groups_F00533	61	■ 1.000	ϕ, λ, r	ϕ, λ, r	where ϕ, λ, r are real and $r \neq 0$. Show that the dimensions of these three subgroups
gilmore-lie-groups_F00534	61	■ 1.000	$r \neq 0$	$r \neq 0$	where ϕ, λ, r are real and $r \neq 0$. Show that the dimensions of these three subgroups
gilmore-lie-groups_F00535	61	■ 1.000	$2n^2 - 1$	$2n^2 - 1$	are $2n^2 - 1$ and that $SL_3(n; \mathbb{C})$ is disconnected. It consists of two topologically iden-
gilmore-lie-groups_F00536	61	■ 1.000	$SL_3(n; \mathbb{C})$	$SL_3(n; \mathbb{C})$	are $2n^2 - 1$ and that $SL_3(n; \mathbb{C})$ is disconnected. It consists of two topologically iden-
gilmore-lie-groups_F00537	61	■ 1.000	$SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$	$SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$	Show that $SL(n; \mathbb{C}) = SL_1(n; \mathbb{C}) \cap SL_2(n; \mathbb{C})$. Do these results extend under the field
gilmore-lie-groups_F00538	61	■ 1.000	$\mathbb{C} \rightarrow \mathbb{R}$	$\mathbb{C} \rightarrow \mathbb{R}$	restriction $\mathbb{C} \rightarrow \mathbb{R}$? and the field extension $\mathbb{C} \rightarrow \mathbb{Q}$?
gilmore-lie-groups_F00539	61	■ 1.000	$\mathbb{C} \rightarrow \mathbb{Q}$	$\mathbb{C} \rightarrow \mathbb{Q}$	restriction $\mathbb{C} \rightarrow \mathbb{R}$? and the field extension $\mathbb{C} \rightarrow \mathbb{Q}$?
gilmore-lie-groups_F00540	61	■ 0.936	$\left[\begin{array}{cc} -1 & a \\ 0 & 1 \end{array} \right], a \in \mathbb{R}$	$\left[\begin{array}{cc} -1 & a \\ 0 & 1 \end{array} \right], a \in \mathbb{R}$	7. A subgroup of $UT(1, 1)$ includes matrices of the form $\begin{bmatrix} -1 & a \\ 0 & 1 \end{bmatrix}$, $a \in \mathbb{R}$. Show that the
gilmore-lie-groups_F00541	61	■ 1.000	$\mathbb{R}^1$	$\mathbb{R}^1$	underlying group manifold consists of two copies of the real line $\mathbb{R}^1$. If matrices of
gilmore-lie-groups_F00542	61	■ 1.000	$\left[\begin{array}{cc} 1 & a \\ 0 & -1 \end{array} \right]$	$\left[\begin{array}{cc} 1 & a \\ 0 & -1 \end{array} \right]$	the form $\begin{bmatrix} 1 & a \\ 0 & -1 \end{bmatrix}$ are also included, then the parameterizing manifold consists of how
gilmore-lie-groups_F00547	84	■ 1.000	$n(n-1) / 2$	$n(n-1)/2$	Show that a useful set of basis vectors (matrices) consists of the $n(n-1)/2$ matrices
gilmore-lie-groups_F00549	62	■ 1.000	$\mathbf{F} = d\mathbf{p}/dt$	$\mathbf{F} = d\mathbf{p}/dt$	9. Newton's equations of motion are $\mathbf{F} = d\mathbf{p}/dt$. In the Lorentz gauge Maxwell's equa-
gilmore-lie-groups_F00551	62	■ 0.814	$SU(2)$	$SU(2)$	10. Show that the group of 2×2 matrices $SU(2)$ is parameterized by two complex num-
gilmore-lie-groups_F00552	62	■ 1.000	$c_1 = a_1 + ib_1$	$c_1 = a_1 + ib_1$	bers $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$, so that
gilmore-lie-groups_F00553	62	■ 1.000	$c_2 = a_2 + ib_2$	$c_2 = a_2 + ib_2$	bers $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$, so that
gilmore-lie-groups_F00554	62	■ 1.000	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	subject to the condition $a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$. Convince yourself (a) that topolog-
gilmore-lie-groups_F00555	62	■ 0.991	$S^3 \subset \mathbb{R}^4$	$S^3 \subset \mathbb{R}^4$	$S^3 \subset \mathbb{R}^4$; and (b) algebraically it is equivalent to $SU(1; \mathbb{Q})$ (cf. (3.16)).
gilmore-lie-groups_F00556	62	■ 1.000	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	subject to the condition $a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$. Identify the parameterizing manifold.
gilmore-lie-groups_F00557	62	■ 1.000	$\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array} \right]$	$\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array} \right]$	in the form $\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row
gilmore-lie-groups_F00558	62	■ 1.000	$m_{11}^2 + x^2 = 1$	$m_{11}^2 + x^2 = 1$	in the form $\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row
gilmore-lie-groups_F00559	62	■ 1.000	$m_{11} = \pm\sqrt{1-x^2}$	$m_{11} = \pm\sqrt{1-x^2}$	in the form $\begin{bmatrix} m_{11} & x \\ m_{21} & m_{22} \end{bmatrix}$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row

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gilmore-lie-groups_F00560	63	■ 1.000	$m_{\{21\}} m_{\{11\}} + m_{\{22\}} x = 0$	$m_{21}m_{11} + m_{22}x = 0$	is orthogonal to the first, so that $\overline{m_{21}m_{11} + m_{22}x} = 0$. As a result, we find
gilmore-lie-groups_F00561	63	■ 0.994	$m_{\{i\ j\}} \geq j$	$m_{ij} \geq j$	$m_{ij} \geq j$ in terms of the three parameters (x, y, z) .
gilmore-lie-groups_F00562	63	■ 0.495	$Z_{\{2\}}$	Z_2	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .
gilmore-lie-groups_F00563	63	■ 0.495	$Z_{\{1\}}$	Z_1	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .
gilmore-lie-groups_F00564	63	■ 0.495	(x, y)	(x, y)	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .
gilmore-lie-groups_F00565	63	■ 0.854	$M \in GL(n; \mathbb{Z})$	$M \in GL(n; \mathbb{Z})$	15. If $\overline{M \in GL(n; \mathbb{Z})}$, show that $\det(M)$ must be ± 1 .
gilmore-lie-groups_F00566	63	■ 0.854	$\operatorname{det}(M)$	$\det(M)$	15. If $M \in GL(n; \mathbb{Z})$, show that $\det(M)$ must be ± 1 .
gilmore-lie-groups_F00567	63	■ 1.000	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	16. Show that the orders of $\overline{O(n; \mathbb{Z}) \supset S_n \supset A_n}$ are $2^n \times n!$, $n!$, $\frac{1}{2}n!$.
gilmore-lie-groups_F00568	63	■ 1.000	$2^n \times n!, n!, \frac{1}{2}n!$	$2^n \times n!, n!, \frac{1}{2}n!$	16. Show that the orders of $O(n; \mathbb{Z}) \supset S_n \supset A_n$ are $2^n \times n!$, $n!$, $\frac{1}{2}n!$.
gilmore-lie-groups_F00569	63	■ 1.000	$\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$	$\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$	17. Estimate the Fibonacci number $F(n)$ from the eigenvalues $\overline{\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})}$ of the
gilmore-lie-groups_F00570	63	■ 0.999	$F(0)=0, F(1)=1$	$F(0) = 0, F(1) = 1$	(other than $F(0) = 0, F(1) = 1$) are introduced?
gilmore-lie-groups_F00571	63	■ 1.000	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array} \right]$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array} \right]$	$GL(2; \mathbb{Z})$ (for example, $\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array} \right]$) and other initial conditions.
gilmore-lie-groups_F00572	63	■ 1.000	$ n\ l\ m\rangle$	$ nlm\rangle$	19. The energy levels $ nlm\rangle$ of the nonrelativistic hydrogen atom exhibit an n^2 -fold
gilmore-lie-groups_F00573	63	■ 1.000	n^2	n^2	19. The energy levels $ nlm\rangle$ of the nonrelativistic hydrogen atom exhibit an n^2 -fold
gilmore-lie-groups_F00574	63	■ 1.000	$E(n\ l\ m) = -E_0 / n^2 \left(E_0 = 13.6 \text{ eV} \right)$	$E(nlm) = -E_0/n^2$ ($E_0 = 13.6\text{eV}$)	quantum number n have the same energy $E(nlm) = -E_0/n^2$ ($E_0 = 13.6\text{ eV}$). If the
gilmore-lie-groups_F00575	63	■ 1.000	$SO(4) \downarrow SO(3)$	$SO(4) \downarrow SO(3)$	symmetry reduction $SO(4) \downarrow SO(3)$. The representations of $SO(4)$ that enter into
gilmore-lie-groups_F00576	63	■ 0.993	$n(n=1,2,3, \ldots)$	$n(n = 1, 2, 3, \dots)$	quantum number $\overline{n(n = 1, 2, 3, \dots)}$. The $SO(4)$ representation with quantum num-
gilmore-lie-groups_F00577	63	■ 0.980	$l, l=0,1,2 \ldots, n-1$	$l, l = 0, 1, 2 \dots, n - 1$	number $\overline{l, l = 0, 1, 2 \dots, n - 1}$, with $\sum_{l=0}^{l=n-1} (2l + 1) = n^2$. The $SO(3)$ multiplet
gilmore-lie-groups_F00578	63	■ 0.980	$\sum_{l=0}^{l=n-1} (2l+1) = n^2$	$\sum_{l=0}^{l=n-1} (2l + 1) = n^2$	number $l, l = 0, 1, 2 \dots, n - 1$, with $\sum_{l=0}^{l=n-1} (2l + 1) = n^2$. The $SO(3)$ multiplet
gilmore-lie-groups_F00579	63	■ 1.000	l	l	with quantum number l is $2l + 1$ -fold degenerate. An empirical hamiltonian with
gilmore-lie-groups_F00580	63	■ 1.000	$2l + 1$	$2l + 1$	with quantum number l is $2l + 1$ -fold degenerate. An empirical hamiltonian with
gilmore-lie-groups_F00581	63	■ 1.000	$+Ze$	$+Ze$	introduced into a Coulomb potential established by a central charge $+Ze$ can be
gilmore-lie-groups_F00582	64	■ 1.000	$\delta=0.28$	$\delta = 0.28$	trum with $\overline{\delta = 0.28}$ provides the filling ordering that accounts for Mendeleev's
gilmore-lie-groups_F00583	64	■ 0.608	$(n, l) \rightarrow 1s$	$(n, l) \rightarrow 1s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00584	64	■ 0.608	$2s$	$2s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00585	64	■ 0.608	$2p$	$2p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00586	64	■ 0.608	$3s$	$3s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$

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gilmore-lie-groups_F00587	64	0.608	3 p	$3p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00588	64	0.608	4 s	$4s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00589	64	0.608	3 d	$3d$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00590	64	0.608	4 p	$4p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00591	64	0.957	5 s, 4 d, 5 p ; 6 s, 4 f, 5 d, 6 p ; 7 s, 5 f, 6 d, 7 p ; 8 s, 6 f, 7 d, 8 p ; \ldots	$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$	<u>$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$</u> .
gilmore-lie-groups_F00592	64	1.000	$G_{\{2\}}$	G_2	21. G_1 and G_2 are two metrics on a real $2n$ -dimensional linear vector space that are
gilmore-lie-groups_F00593	64	1.000	2 n	$2n$	21. G_1 and G_2 are two metrics on a real $2n$ -dimensional linear vector space that are
gilmore-lie-groups_F00594	64	1.000	$M^{\{t\}} G_{\{i\}} M = G_{\{i\}}$	$M^t G_i M = G_i$	Show that the $2n \times 2n$ matrices M that satisfy the bilinear constraints $M^t G_i M = G_i$
gilmore-lie-groups_F00595	64	1.000	$y^{\{j\}} = x^{\{i\}} M_{\{i\}}^{\{j\}}$	$y^j = x^i M_i^j$	by a linear transformation: $y^j = x^i M_i^j$. Show that the derivatives are related by the
gilmore-lie-groups_F00596	64	0.990	$f(q, p)$	$f(q, p)$	23. The Poisson brackets between two functions $f(q, p)$ and $g(q, p)$ on a classical phase
gilmore-lie-groups_F00597	64	0.990	$g(q, p)$	$g(q, p)$	23. The Poisson brackets between two functions $f(q, p)$ and $g(q, p)$ on a classical phase
gilmore-lie-groups_F00598	65	0.666	(Q, P)	(Q, P)	b. Introduce a new coordinate system (Q, P) , related to the original by a linear
gilmore-lie-groups_F00599	65	1.000	$A^{\{t\}} C$	$A^t C$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00600	65	1.000	$B^{\{t\}} D$	$B^t D$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00601	65	1.000	$A^{\{t\}} D - B^{\{t\}} C = I_{\{n\}}$	$A^t D - B^t C = I_n$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and <u>$A^t D - B^t C = I_n$</u> .
gilmore-lie-groups_F00602	65	0.999	$\left[q_{\{j\}}, q_{\{k\}} \right] = \left[p_{\{j\}}, p_{\{k\}} \right] = 0$	$[q_j, q_k] = [p_j, p_k] = 0$	mechanical commutator bracket: <u>$[q_j, q_k] = [p_j, p_k] = 0$</u> and $[q_j, p_k] = i\hbar\delta_{jk}$.
gilmore-lie-groups_F00603	65	0.999	$\left[q_{\{j\}}, p_{\{k\}} \right] = i \hbar \delta_{\{j\} \{k\}}$	$[q_j, p_k] = i\hbar\delta_{jk}$	mechanical commutator bracket: $[q_j, q_k] = [p_j, p_k] = 0$ and $[q_j, p_k] = i\hbar\delta_{jk}$.
gilmore-lie-groups_F00604	65	1.000	$e^{+i k x}$	e^{+ikx}	with a similar expression for the wavefunction on the right. The exponential e^{+ikx}
gilmore-lie-groups_F00605	65	0.644	m	m	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00606	65	0.644	(+)	(+)	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00607	65	0.644	$\hbar k$	$\hbar k$	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00608	65	1.000	$E = (\hbar k)^2 / 2 m$	$E = (\hbar k)^2 / 2m$	energy $E = (\hbar k)^2 / 2m$. The complex number A_L is the probability amplitude for
gilmore-lie-groups_F00609	65	1.000	$A_{\{L\}}$	A_L	energy $E = (\hbar k)^2 / 2m$. The complex number A_L is the probability amplitude for
gilmore-lie-groups_F00610	65	1.000	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	momentum in the left-hand region is $\langle \hat{p} \rangle = (A_L ^2 - B_L ^2) \hbar k$, where the operator
gilmore-lie-groups_F00611	65	1.000	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$.

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gilmore-lie-groups_ F00612	65	—	$V_{\{L\}}=V_{\{R\}}$	$V_L = V_R$	
gilmore-lie-groups_ F00613	65	—	$\psi_{\{\sigma\}}(x)=A_{\{\sigma\}} e^{\{+i\ k\ x\}}+B_{\{\sigma\}} e^{\{-i\ k\ x\}}$	$\psi_{\sigma}(x) = A_{\sigma}e^{+ikx} + B_{\sigma}e^{-ikx}$	
gilmore-lie-groups_ F00614	65	—	$\sigma=L, R$	$\sigma = L, R$	
gilmore-lie-groups_ F00615	66	■ 1.000	$A_{\{L\}}, A_{\{R\}}$	A_L, A_R	b. Since the Schrödinger equation is second order the four amplitudes $A_L, A_R,$
gilmore-lie-groups_ F00616	66	■ 1.000	$B_{\{L\}}, B_{\{R\}}$	B_L, B_R	B_L, B_R are not independent. Only two are independent. Two linear relations exist
gilmore-lie-groups_ F00617	66	■ 1.000	E	E	of energy E . Show that $T(E) \in U(1, 1)$.
gilmore-lie-groups_ F00618	66	■ 1.000	$T(E) \in U(1,1)$	$T(E) \in U(1, 1)$	of energy E . Show that $T(E) \in U(1, 1)$.
gilmore-lie-groups_ F00619	66	■ 1.000	$T \in SU(1,1)$	$T \in SU(1, 1)$	c. Show that $T \in SU(1, 1)$ by appropriate choice of phase.
gilmore-lie-groups_ F00620	66	■ 1.000	$S \in U(2)$	$S \in U(2)$	a. Invoke conservation of momentum arguments to conclude $S \in U(2)$.
gilmore-lie-groups_ F00621	66	■ 1.000	$S(E)$	$S(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_ F00622	66	■ 1.000	$T(E)$	$T(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_ F00623	66	■ 1.000	$t_{\{11\}}(E)$	$t_{11}(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_ F00624	66	■ 1.000	$r_{\{j\}} / \left[\left(E - E_{\{j\}} \right) + i \left(\Gamma_{\{j\}} / 2 \right) \right]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$ describe resonances at energy E_j with characteristic de-
gilmore-lie-groups_ F00625	66	■ 1.000	$E_{\{j\}}$	E_j	$r_j / [(E - E_j) + i(\Gamma_j/2)]$ describe resonances at energy E_j with characteristic de-
gilmore-lie-groups_ F00626	66	■ 1.000	$\Gamma_{\{j\}} / \hbar$	$\Gamma_j / \hbar$	cay time $\Gamma_j / \hbar$.
gilmore-lie-groups_ F00627	66	■ 1.000	$V_{\{1\}}$	V_1	26. Two interaction regions V_1 and V_2 on the line are characterized by transfer matrices
gilmore-lie-groups_ F00628	66	■ 1.000	$V_{\{2\}}$	V_2	26. Two interaction regions V_1 and V_2 on the line are characterized by transfer matrices
gilmore-lie-groups_ F00629	66	■ 1.000	$T_{\{1\}}$	T_1	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region

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gilmore-lie-groups_F00630	66	■ 1.000	$T_{\{2\}}$	T_2	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00631	66	■ 1.000	$S_{\{1\}}$	S_1	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00632	67	■ 1.000	$S_{\{\text{Tot}\}}$	S_{Tot}	c. Show that S_{Tot} is unitary.
gilmore-lie-groups_F00633	67	■ 1.000	$1/\left(1-s_{12}s_{43}\right)$	$1/(1-s_{12}s_{43})$	by expanding the fraction $1/(1-s_{12}s_{43})$ as a geometric sum and interpreting each
gilmore-lie-groups_F00634	67	■ 1.000	$V_{\{1\}}^{\prime}$	V_1'	27. If the potentials V_1 and V_2 are modified to V_1' and V_2' their transfer matrices and
gilmore-lie-groups_F00635	67	■ 1.000	$V_{\{2\}}^{\prime}$	V_2'	27. If the potentials V_1 and V_2 are modified to V_1' and V_2' their transfer matrices and
gilmore-lie-groups_F00636	67	■ 1.000	$T_{\{i\}}(E) \rightarrow T_{\{i\}}^{\prime}(E)$	$T_i(E) \rightarrow T_i'(E)$	their scattering matrices will also be modified $T_i(E) \rightarrow T_i'(E)$ and $S_i(E) \rightarrow S_i'(E)$,
gilmore-lie-groups_F00637	67	■ 1.000	$S_{\{i\}}(E) \rightarrow S_{\{i\}}^{\prime}(E)$	$S_i(E) \rightarrow S_i'(E)$	their scattering matrices will also be modified $T_i(E) \rightarrow T_i'(E)$ and $S_i(E) \rightarrow S_i'(E)$,
gilmore-lie-groups_F00638	67	■ 1.000	$i=1,2$	$i=1,2$	$i=1,2$. It is possible that for some energy E , $S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$. Find the set of
gilmore-lie-groups_F00639	67	■ 1.000	$E, S_{\{\text{Tot}\}}^{\prime}(E)=S_{\{\text{Tot}\}}(E)$	$E, S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$	$i=1,2$. It is possible that for some energy E , $S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$. Find the set of
gilmore-lie-groups_F00640	67	■ 1.000	$S_{\{1\}}^{\prime}(E)$	$S_1'(E)$	all modified scattering matrices $S_1'(E)$ and $S_2'(E)$ with the property that the modi-
gilmore-lie-groups_F00641	67	■ 1.000	$S_{\{2\}}^{\prime}(E)$	$S_2'(E)$	all modified scattering matrices $S_1'(E)$ and $S_2'(E)$ with the property that the modi-
gilmore-lie-groups_F00642	67	■ 0.998	$S_{\{\text{Tot}\}}(E)$	$S_{\text{Tot}}(E)$	fied pair maps into the original total S -matrix $S_{\text{Tot}}(E)$. In fancy terms, find the fiber
gilmore-lie-groups_F00643	67	■ 0.999	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	in <u>$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$</u> . (Hint: if this seems daunting, note that to satisfy
gilmore-lie-groups_F00644	67	■ 0.997	$T_{\{1\}}(E) T_{\{2\}}(E)=T_{\{\text{Tot}\}}(E)=T_{\{1\}}^{\prime}(E) T_{\{2\}}^{\prime}(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$ we can take $T_1'(E) = T_1(E)R$ and $T_2'(E) =$
gilmore-lie-groups_F00645	67	■ 0.997	$T_{\{1\}}^{\prime}(E)=T_{\{1\}}(E) R$	$T_1'(E) = T_1(E)R$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$ we can take $T_1'(E) = T_1(E)R$ and $T_2'(E) =$
gilmore-lie-groups_F00646	67	■ 0.997	$T_{\{2\}}^{\prime}(E)=$	$T_2'(E) =$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1'(E)T_2'(E)$ we can take $T_1'(E) = T_1(E)R$ and $T_2'(E) =$
gilmore-lie-groups_F00647	67	■ 1.000	$R^{-1} T_{\{2\}}(E)$	$R^{-1}T_2(E)$	$R^{-1}T_2(E)$ for any $R \in U(1,1)$. The fiber in $U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$
gilmore-lie-groups_F00648	67	■ 1.000	$R \in U(1,1)$	$R \in U(1,1)$	$R^{-1}T_2(E)$ for any $R \in U(1,1)$. The fiber in $U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$
gilmore-lie-groups_F00649	67	■ 1.000	$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$	$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$	$R^{-1}T_2(E)$ for any $R \in U(1,1)$. The fiber in <u>$U(2,2) \supset U(1,1) \otimes U(1,1) \downarrow U(1,1)$</u>
gilmore-lie-groups_F00650	67	■ 0.932	$T_{\{\text{Tot}\}}(E)$	$T_{\text{Tot}}(E)$	over $T_{\text{Tot}}(E)$ consists of the matrices $(T_1(E)R, R^{-1}T_2(E))$. Now map this into the
gilmore-lie-groups_F00651	67	■ 0.932	$\left(T_{\{1\}}(E) R, R^{-1} T_{\{2\}}(E)\right)$	$(T_1(E)R, R^{-1}T_2(E))$	over $T_{\text{Tot}}(E)$ consists of the matrices $(T_1(E)R, R^{-1}T_2(E))$. Now map this into the
gilmore-lie-groups_F00652	67	■ 0.850	$\left(S_{\{1\}}^{\prime}(E), S_{\{2\}}^{\prime}(E)\right)$	$(S_1'(E), S_2'(E))$	fiber <u>$(S_1'(E), S_2'(E))$</u> over $S_{\text{Tot}}(E)$.
gilmore-lie-groups_F00653	68	■ 0.987	$n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{k\}}$	$n_1, n_2, \dots, n_k$	with <u>$n_1, n_2, \dots, n_k$</u> leads can be connected together by soldering some of the leads
gilmore-lie-groups_F00654	68	■ 0.718	$k S$	kS	direct sum of k S matrices of sizes $n_j \times n_j$ ($j=1,2,\dots,k$). Through appropriate
gilmore-lie-groups_F00655	68	■ 0.718	$n_{\{j\}} \times n_{\{j\}}(j=1,2, \ldots, k)$	$n_j \times n_j(j=1,2,\dots,k)$	direct sum of k S matrices of sizes <u>$n_j \times n_j(j=1,2,\dots,k)$</u> . Through appropriate
gilmore-lie-groups_F00656	68	■ 1.000	Γ	Γ	The matrix Γ that relates internal outputs to internal inputs describes the topology, or

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gilmore-lie-groups_F00657	68	■ 0.999	$\text{\texttt{\left[o_e\right]=S_{\text {Network }}\left[i_e\right]}}$	$[o_e] = S_{\text{Network}} [i_e]$	a. Show that the S matrix that describes the network, defined by $[o_e] = S_{\text{Network}} [i_e]$,
gilmore-lie-groups_F00658	68	■ 1.000	$\text{\texttt{S_{\text {Newtork }}}}$	S_{Newtork}	b. Show that S_{Newtork} is unitary: $S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$.
gilmore-lie-groups_F00659	68	■ 1.000	$\text{\texttt{S_{\text {Newtork }}}^{\dagger}=S_{\text {Newtork }}}, S_{\text {\text {Newtork }}} \subset U(d)$	$S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$	b. Show that S_{Newtork} is unitary: $S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$.
gilmore-lie-groups_F00660	68	■ 1.000	$\text{\texttt{U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k \otimes U\left(n_j\right) \otimes U\left(n_j\right)}}$	$U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k \otimes U(n_j)$	$U(\sum_{j=1}^k n_j) \supset \Pi_{j=1}^k \otimes U(n_j)$ and a projection to the total network scattering matrix
gilmore-lie-groups_F00661	68	■ 1.000	$\text{\texttt{U(d)}}$	$U(d)$	in $U(d)$, where d is the number of the network's external leads. The connectivity is
gilmore-lie-groups_F00662	68	■ 1.000	$\text{\texttt{d}}$	d	in $U(d)$, where d is the number of the network's external leads. The connectivity is
gilmore-lie-groups_F00663	68	■ 0.917	$\text{\texttt{\Pi_{j=1}^k \otimes U\left(n_j\right)}}$	$\Pi_{j=1}^k \otimes U(n_j)$	determined by the permutation matrix Γ . Determine the fiber in $\Pi_{j=1}^k \otimes U(n_j)$ over
gilmore-lie-groups_F00664	68	■ 1.000	$\text{\texttt{U}}$	U	30. All the matrices in this problem are square $n \times n$, with: H hermitian; U unitary; A
gilmore-lie-groups_F00665	70	■ 1.000	$\text{\texttt{b}}$	b	into the neighborhood of group element b by multiplying a , and every element in
gilmore-lie-groups_F00666	70	■ 1.000	$\text{\texttt{b a^{-1}}}$	ba^{-1}	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This
gilmore-lie-groups_F00667	70	■ 1.000	$\text{\texttt{a^{-1} b}}$	$a^{-1}b$	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This
gilmore-lie-groups_F00668	71	■ 0.953	$\text{\texttt{(a, b, c)}}$	(a, b, c)	done by allowing the parameters (a, b, c) to become infinitesimals and expanding
gilmore-lie-groups_F00669	71	■ 1.000	$\text{\texttt{\phi(x, y) \rightarrow \phi(x, \theta+\delta y)}}$	$\phi(x, y) \rightarrow \phi(x, 0 + \delta y)$	arguments, say $\phi(x, y) \rightarrow \phi(x, 0 + \delta y)$. This generates a left-invariant vector field.
gilmore-lie-groups_F00670	72	■ 1.000	$\text{\texttt{\epsilon}}$	ϵ	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_F00671	72	■ 1.000	$\text{\texttt{I+\epsilon X}}$	$I + \epsilon X$	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_F00672	72	■ 0.744	$\text{\texttt{X_{a}}}}$	X_a	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00673	72	■ 0.744	$\text{\texttt{X_{b}}}}$	X_b	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00674	72	■ 0.744	$\text{\texttt{X_{c}}}}$	X_c	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00675	72	■ 1.000	$\text{\texttt{X^{n}}}}$	X^n	With brute force, each of the matrices X^n is computed explicitly, a pattern is
gilmore-lie-groups_F00676	72	■ 1.000	$\text{\texttt{X^{0}=I_{2}}, X^{1}=X}}$	$X^0 = I_2, X^1 = X$	recognized, and the sum is carried out. The first few powers are $X^0 = I_2, X^1 = X$
gilmore-lie-groups_F00677	72	■ 1.000	$\text{\texttt{X^{2}=\theta^2 I_{2}}}}$	$X^2 = \theta^2 I_2$	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00678	72	■ 1.000	$\text{\texttt{X^{2}}}}$	X^2	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00679	72	■ 1.000	$\text{\texttt{X^{3}=X^{2} X^{1}}}}$	$X^3 = X^2 X^1$	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00680	72	■ 0.991	$\text{\texttt{X\left(=\theta^2 X\right), X^{4}}}}$	$X (= \theta^2 X), X^4$	must be proportional to $X (= \theta^2 X)$, X^4 is proportional to the identity, and so on.
gilmore-lie-groups_F00681	72	■ 0.995	$\text{\texttt{X^{0}=I_{2}}}}$	$X^0 = I_2$	equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00682	72	■ 0.995	$\text{\texttt{X^{1}=X}}$	$X^1 = X$	equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.

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gilmore-lie-groups_ F06683	72	■ 1.000	$X^{\{3\}}$	X^3	Similarly, X^3 can be replaced by a linear combination of X^2 and X , and then X^2
gilmore-lie-groups_ F06684	72	■ 0.759	$I_{\{2\}}$	I_2	replaced by I_2 and X . By induction, any function of the 2×2 matrix X can be
gilmore-lie-groups_ F06685	73	■ 1.000	$f_{\{0\}}, f_{\{1\}}$	f_0, f_1	Furthermore, the functions f_0, f_1 are not arbitrary functions of the three parameters
gilmore-lie-groups_ F06686	73	■ 1.000	$\theta^2 = a^2 + bc$	$\theta^2 = a^2 + bc$	X is $\theta^2 = a^2 + bc$. As a result, we know from general and simple considerations
gilmore-lie-groups_ F06687	73	■ 1.000	$f_{\{0\}}(\theta) = 1 + \theta^2/2! + \theta^4/4! + \theta^6/6! + \dots = \cosh \theta$	$f_0(\theta^2) = 1 + \theta^2/2! + \theta^4/4! + \theta^6/6! + \dots = \cosh \theta$	The two functions are $f_0(\theta^2) = 1 + \theta^2/2! + \theta^4/4! + \theta^6/6! + \dots = \cosh \theta$ and
gilmore-lie-groups_ F06688	73	■ 1.000	$f_{\{1\}}(\theta) = 1 + \theta^2/3! + \theta^4/5! + \theta^6/7! + \dots = \sinh(\theta)/\theta$	$f_1(\theta^2) = 1 + \theta^2/3! + \theta^4/5! + \theta^6/7! + \dots = \sinh(\theta)/\theta$	$f_1(\theta^2) = 1 + \theta^2/3! + \theta^4/5! + \theta^6/7! + \dots = \sinh(\theta)/\theta$. These arguments are ap-
gilmore-lie-groups_ F06689	73	■ 1.000	Y	Y	If X and Y are elements in the Lie algebra, then $g_1 = I + \epsilon X$ is an element in
gilmore-lie-groups_ F06690	73	■ 1.000	$g_{\{1\}} = I + \epsilon X$	$g_1 = I + \epsilon X$	If X and Y are elements in the Lie algebra, then $g_1 = I + \epsilon X$ is an element in
gilmore-lie-groups_ F06691	73	■ 1.000	$I + \epsilon \alpha X$	$I + \epsilon \alpha X$	the Lie group near the identity for ϵ sufficiently small. In fact, so also is $I + \epsilon \alpha X$
gilmore-lie-groups_ F06692	74	■ 1.000	α	α	for any real number α . We can form the product
gilmore-lie-groups_ F06693	74	■ 1.000	ϵ, δ	ϵ, δ	If X and Y are in the Lie algebra, then for any ϵ, δ sufficiently small, $g_1(\epsilon) =$
gilmore-lie-groups_ F06694	74	■ 1.000	$g_{\{1\}}(\epsilon) =$	$g_1(\epsilon) =$	If X and Y are in the Lie algebra, then for any ϵ, δ sufficiently small, $g_1(\epsilon) =$
gilmore-lie-groups_ F06695	74	■ 1.000	$\operatorname{EXP}(\epsilon X)$	$\operatorname{EXP}(\epsilon X)$	$\operatorname{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups_ F06696	74	■ 1.000	$g_{\{1\}}(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$	$g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$	$\operatorname{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups_ F06697	74	■ 1.000	$g_{\{2\}}(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$. Expanding the commutator to lowest order nonvanishing
gilmore-lie-groups_ F06698	74	■ 0.996	$g_{\{1\}}(\epsilon) = \operatorname{EXP}(\epsilon X)$	$g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$	Therefore, the commutator of two group elements, $g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$ and $g_2(\delta) =$
gilmore-lie-groups_ F06699	74	■ 0.996	$g_{\{2\}}(\delta) =$	$g_2(\delta) =$	Therefore, the commutator of two group elements, $g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$ and $g_2(\delta) =$
gilmore-lie-groups_ F06700	74	■ 1.000	$\operatorname{EXP}(\delta Y)$	$\operatorname{EXP}(\delta Y)$	$\operatorname{EXP}(\delta Y)$, which is in the group G , requires the commutator of the operators X and
gilmore-lie-groups_ F06701	74	■ 1.000	$Y, [X, Y] = (XY - YX)$	$Y, [X, Y] = (XY - YX)$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups_ F06702	74	■ 1.000	$\mathfrak{g}$	$\mathfrak{g}$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups_ F06703	74	■ 1.000	$g_{\{1\}} H g_{\{1\}}^{-1} \subset H$	$g_1 H g_1^{-1} \subset H$	If H is an invariant subgroup of G , then $g_1 H g_1^{-1} \subset H$. This means that if X is in
gilmore-lie-groups_ F06704	74	■ 1.000	X, Y, Z	X, Y, Z	If X, Y, Z are in the Lie algebra, then the Jacobi identity is satisfied
gilmore-lie-groups_ F06705	74	■ 0.996	$([X, Y] = XY - YX)$	$([X, Y] = XY - YX)$	$([X, Y] = XY - YX)$ and showing that the 12 terms so obtained cancel pairwise.
gilmore-lie-groups_ F06706	74	■ 1.000	XY	XY	for which composition (e.g., XY is well defined, as are all other pairwise products)
gilmore-lie-groups_ F06707	75	■ 1.000	$X \in \mathfrak{g}$	$X \in \mathfrak{g}$	(i) It is a linear vector space under vector addition and scalar multiplication. If $X \in \mathfrak{g}$ and
gilmore-lie-groups_ F06708	75	■ 1.000	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$ then every linear combination of X and Y is in $\mathfrak{g}$:
gilmore-lie-groups_ F06709	75	■ 1.000	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	(iii) The Jacobi identity is satisfied. If $X \in \mathfrak{g}, Y \in \mathfrak{g}$, and $Z \in \mathfrak{g}$, then

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gilmore-lie-groups-F00710	75	■ 1.000	$Z \in \mathfrak{g}$	$Z \in \mathfrak{g}$	(iii) The Jacobi identity is satisfied. If $X \in \mathfrak{g}$, $Y \in \mathfrak{g}$, and $Z \in \mathfrak{g}$, then
gilmore-lie-groups-F00711	75	■ 0.996	$X_{\{1\}}, X_{\{2\}}, \dots, X_{\{n\}}$	$X_1, X_2, \dots, X_n$	$X_1, X_2, \dots, X_n$, then we can ask several additional questions such as: Is there a
gilmore-lie-groups-F00712	75	■ 1.000	$\left(X_{\{i\}}, X_{\{j\}}\right)$	(X_i, X_j)	(X_i, X_j) ? We return to these questions shortly.
gilmore-lie-groups-F00713	76	■ 1.000	$C_{\{i\} \{j\}}^{\{k\}}$	C_{ij}^k	The coefficients C_{ij}^k in this expansion are called structure constants . The structure
gilmore-lie-groups-F00714	76	■ 0.865	$X_{\{a\}}, X_{\{b\}}, X_{\{c\}}$	X_a, X_b, X_c	X_a, X_b, X_c (4.4) in the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ for the Lie group $SL(2; \mathbb{R})$ are, from
gilmore-lie-groups-F00715	76	■ 0.865	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	X_a, X_b, X_c (4.4) in the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ for the Lie group $SL(2; \mathbb{R})$ are, from
gilmore-lie-groups-F00716	76	■ 1.000	Z	Z	change of basis affects the commutator of an arbitrary element Z in the algebra
gilmore-lie-groups-F00717	76	■ 1.000	$R(Z)$	$R(Z)$	The association of a matrix $R(Z)$ with each element of a Lie algebra is called
gilmore-lie-groups-F00718	77	■ 1.000	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	Example The regular representation of the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ is easily con-
gilmore-lie-groups-F00719	77	■ 0.844	a, b	a, b	The rows and columns of this 3×3 matrix are labeled by the indices a, b and c ,
gilmore-lie-groups-F00720	77	■ 0.844	c	c	The rows and columns of this 3×3 matrix are labeled by the indices a, b and c ,
gilmore-lie-groups-F00721	78	■ 0.899	$X_{\{a\}}, X_{\{b\}}$	X_a, X_b	the three generators X_a, X_b , and X_c of $\mathfrak{sl}(2; \mathbb{R})$ to either the nonsemisimple or the
gilmore-lie-groups-F00722	78	■ 0.997	$E(2)$	$E(2)$	$E(2)$ with structure
gilmore-lie-groups-F00723	78	■ 1.000	$P_{\{x\}}, P_{\{y\}}$	P_x, P_y	where the rows and columns are labeled successively by the basis vectors P_x, P_y , and
gilmore-lie-groups-F00724	78	■ 1.000	$L_{\{z\}}$	L_z	L_z . This regular representation has the block diagonal structure of a nonsemisimple
gilmore-lie-groups-F00725	78	■ 0.985	A, B	A, B	very useful inner product when A, B are $p \times q$ matrices is the Hilbert–Schmidt
gilmore-lie-groups-F00726	78	■ 0.985	$p \times q$	$p \times q$	very useful inner product when A, B are $p \times q$ matrices is the Hilbert–Schmidt
gilmore-lie-groups-F00727	79	■ 0.998	$X_{\{i\}}$	X_i	generators that commute with all operators in the Lie algebra (X_i , where $C_{i\star}^* = 0$)
gilmore-lie-groups-F00728	79	■ 0.998	$C_{\{i\} \star}^{\{*\}} = 0$	$C_{i\star}^* = 0$	generators that commute with all operators in the Lie algebra (X_i , where $C_{i\star}^* = 0$)
gilmore-lie-groups-F00729	79	■ 1.000	$C_{\{*\} \star}^{\{i\}} = 0$	$C_{**}^i = 0$	generators (X_i , where $C_{**}^i = 0$).
gilmore-lie-groups-F00730	79	■ 1.000	$V_{\{0\}}$	V_0	The subspace V_0 is a subalgebra of $\mathfrak{g}$. It is the largest nilpotent invariant subal-
gilmore-lie-groups-F00731	79	■ 1.000	$V_{\{-\}}$	V_-	The subspace V_- is also a subalgebra of $\mathfrak{g}$. It consists of compact (a topological
gilmore-lie-groups-F00732	79	■ 1.000	$V_{\{+\}}$	V_+	Finally, the subspace V_+ is not a subalgebra of $\mathfrak{g}$. It consists of noncompact
gilmore-lie-groups-F00733	80	■ 1.000	$X_{\{\pm\}} = X_{\{b\}} \pm X_{\{c\}}$	$X_{\pm} = X_b \pm X_c$	and $X_{\pm} = X_b \pm X_c$. In this basis the metric matrix is
gilmore-lie-groups-F00734	80	■ 1.000	$X_{\{-\}}$	X_-	In this representation it is clear that the operator X_- spans a one-dimensional
gilmore-lie-groups-F00735	80	■ 1.000	$X_{\{a\}}, X_{\{+\}}$	X_a, X_+	compact subalgebra in $\mathfrak{sl}(2; \mathbb{R})$ and the generators X_a, X_+ are noncompact.
gilmore-lie-groups-F00736	81	■ 0.644	$\alpha^{\{1\}}, \alpha^{\{2\}}, \dots, \alpha^{\{n\}}$	$\alpha^1, \alpha^2, \dots, \alpha^n$	ric and measure . We assume the coordinates of the identity are $(\alpha^1, \alpha^2, \dots, \alpha^n)$

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gilmore-lie-groups_ F00737	81	■ 0.840	$\alpha^1+d \alpha^1, \alpha^2+d \alpha^2, \ldots, \alpha^n+d \alpha^n$	$\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n +$	and the coordinates of a point near the identity are $(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n +$
gilmore-lie-groups_ F00738	81	■ 0.866	$d \alpha^n$	$d\alpha^n$	$d\alpha^n$). If $(x^1, x^2, \dots, x^n)$ represents some other group operation, then the point
gilmore-lie-groups_ F00739	81	■ 0.866	$\left(x^1, x^2, \ldots, x^n\right)$	$(x^1, x^2, \dots, x^n)$	$d\alpha^n$). If $(x^1, x^2, \dots, x^n)$ represents some other group operation, then the point
gilmore-lie-groups_ F00740	81	■ 0.831	$\alpha^1+d \alpha^1, \alpha^2+d \alpha^2, \ldots, \alpha^n+d \alpha^n$	$\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n$	$(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$ is mapped to the point $(x^1 + dx^1, x^2 +$
gilmore-lie-groups_ F00741	81	■ 0.831	$x^1+d x^1, x^2+$	$x^1 + dx^1, x^2 +$	$(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$ is mapped to the point $(x^1 + dx^1, x^2 +$
gilmore-lie-groups_ F00742	81	■ 1.000	$d x^2, \ldots, x^n+d x^n$	$dx^2, \dots, x^n + dx^n$	$(dx^2, \dots, x^n + dx^n)$ under left (right) multiplication by the group operation associ-
gilmore-lie-groups_ F00743	81	■ 0.997	$d x$	dx	ated with $(x^1, x^2, \dots, x^n)$. The displacements dx and $d\alpha$ are related by a position-
gilmore-lie-groups_ F00744	81	■ 0.997	$d \alpha$	$d\alpha$	ated with $(x^1, x^2, \dots, x^n)$. The displacements dx and $d\alpha$ are related by a position-
gilmore-lie-groups_ F00745	81	■ 1.000	$d s$	ds	Suppose now that the distance ds between the identity and a point with coordi-
gilmore-lie-groups_ F00746	81	■ 1.000	$\alpha^i+d \alpha^i$	$\alpha^i + d\alpha^i$	nates $\alpha^i + d\alpha^i$ infinitesimally close to the identity is given by
gilmore-lie-groups_ F00747	81	■ 0.671	$x, g_{rs}(x)$	$x, g_{rs}(x)$	Killing inner product. Can we define a metric at $x, g_{rs}(x)$, with the property that
gilmore-lie-groups_ F00748	81	■ 1.000	$x, g(x)$	$x, g(x)$	In order to enforce the invariance condition, the metric at $x, g(x)$, must be related
gilmore-lie-groups_ F00749	82	■ 0.976	$\left(d \alpha^1, d \alpha^2, d \alpha^3\right)$	$(d\alpha^1, d\alpha^2, d\alpha^3)$	The linear relation between the infinitesimals $(d\alpha^1, d\alpha^2, d\alpha^3)$ in the neighborhood
gilmore-lie-groups_ F00750	82	■ 0.999	$(d x, d y, d z)$	(dx, dy, dz)	of the identity and the infinitesimals (dx, dy, dz) in the neighborhood of the group
gilmore-lie-groups_ F00751	83	■ 1.000	$[X, Y]=(X Y-Y X)$	$[X, Y] = (XY - YX)$	tional combinatorial operation, the commutator $[X, Y] = (XY - YX)$, and obey the
gilmore-lie-groups_ F00752	83	■ 1.000	$g_1=(I+\epsilon X), g_1^{-1}=(I+\epsilon X)^{-1}=I-$	$g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I -$	1. Carry out the commutator calculation for $g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I -$
gilmore-lie-groups_ F00753	83	■ 1.000	$\epsilon X+\epsilon^2 X^2-\cdots$	$\epsilon X + \epsilon^2 X^2 - \dots$	$(\epsilon X + \epsilon^2 X^2 - \dots)$, with similar expressions for g_2 , to obtain the same result as in
gilmore-lie-groups_ F00754	83	■ 1.000	g_2	g_2	$\epsilon X + \epsilon^2 X^2 - \dots$, with similar expressions for g_2 , to obtain the same result as in
gilmore-lie-groups_ F00755	83	■ 1.000	$X^{\prime}$	X'	2. The inner product of two vectors X and X' in a linear vector space can be computed
gilmore-lie-groups_ F00756	83	■ 1.000	$(X, X)=2\left(a^2+b c\right)$	$(X, X) = 2(a^2 + bc)$	c. Use the result from Eq. (4.43) that $(X, X) = 2(a^2 + bc)$ to show
gilmore-lie-groups_ F00757	83	■ 1.000	$Y=e^X$	$Y = e^X$	a Lie algebra: $Y = e^X$. Show that “for Y sufficiently close to the identity” the matrix
gilmore-lie-groups_ F00758	83	■ 0.570	λ_i	λ_i	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00759	83	■ 0.570	$0<\lambda_i<+2$	$0 < \lambda_i < +2$	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00760	83	■ 0.570	$Y \in \operatorname{SL}(2 ; \mathbb{R})$	$Y \in \text{SL}(2; \mathbb{R})$	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00761	83	■ 0.570	$\operatorname{tr} Y<-2$	$\text{tr } Y < -2$	ues λ_i of Y all satisfy $0 < \lambda_i < +2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_ F00762	84	■ 1.000	$X \in \mathfrak{sl}(2 ; \mathbb{R})$	$X \in \mathfrak{sl}(2; \mathbb{R})$	expansion does not converge. That is, there is no 2×2 matrix $X \in \mathfrak{sl}(2; \mathbb{R})$ with the
gilmore-lie-groups_ F00763	84	■ 0.637	$\operatorname{tr} e^X<-2$	$\text{tr } e^X < -2$	property $\text{tr } e^X < -2$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00764	84	■ 0.952	$\mathbf{L} =$	$\mathbf{L} =$	4. The Lie algebra of $SO(3)$ is spanned by three 3×3 antisymmetric matrices $\mathbf{L} =$
gilmore-lie-groups_F00765	84	■ 1.000	$\left(L_1, L_2, L_3\right) = \left(X_{23}, X_{31}, X_{12}\right)$	$(L_1, L_2, L_3) = (X_{23}, X_{31}, X_{12})$	$(L_1, L_2, L_3) = (X_{23}, X_{31}, X_{12})$, with
gilmore-lie-groups_F00766	84	■ 1.000	$\mathbf{X} = \theta \cdot \mathbf{L}$	$\mathbf{X} = \theta \cdot \mathbf{L}$	where θ is the single invariant that can be constructed from the matrix $\mathbf{X} = \theta \cdot \mathbf{L}$:
gilmore-lie-groups_F00767	84	■ 1.000	$\theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2$	$\theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2$	$\theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2$. Show
gilmore-lie-groups_F00768	84	■ 1.000	$SO(n)$	$SO(n)$	5. The Lie algebra for the matrix group $SO(n)$ consists of antisymmetric $n \times n$ matrices.
gilmore-lie-groups_F00769	84	■ 1.000	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$ with matrix elements $(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$.
gilmore-lie-groups_F00770	84	■ 1.000	$\left(X_{ij}\right)_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$	$(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$	$X_{ij} = -X_{ji} (1 \leq i \neq j \leq n)$ with matrix elements $(X_{ij})_{\alpha\beta} = \delta_{i\alpha}\delta_{j\beta} - \delta_{i\beta}\delta_{j\alpha}$.
gilmore-lie-groups_F00771	84	■ 1.000	$\mathcal{X}_{ij} = x^i \partial_j - x^j \partial_i$	$\mathcal{X}_{ij} = x^i \partial_j - x^j \partial_i$	b. Show that the operators $\mathcal{X}_{ij} = x^i \partial_j - x^j \partial_i$ satisfy isomorphic commutation rela-
gilmore-lie-groups_F00772	84	■ 1.000	$\mathcal{B}_{ij} =$	$\mathcal{B}_{ij} =$	c. Show that bilinear products of boson creation and annihilation operators $\mathcal{B}_{ij} =$
gilmore-lie-groups_F00773	84	■ 1.000	$b_i^\dagger b_j - b_j^\dagger b_i (1 \leq i \neq j \leq n)$	$b_i^\dagger b_j - b_j^\dagger b_i (1 \leq i \neq j \leq n)$	$b_i^\dagger b_j - b_j^\dagger b_i (1 \leq i \neq j \leq n)$ satisfy isomorphic commutation relations.
gilmore-lie-groups_F00774	84	■ 0.963	$\mathcal{F}_{ij} =$	$\mathcal{F}_{ij} =$	d. Show that bilinear products of fermion creation and annihilation operators $\mathcal{F}_{ij} =$
gilmore-lie-groups_F00775	84	■ 1.000	$f_i^\dagger f_j - f_j^\dagger f_i (1 \leq i \neq j \leq n)$	$f_i^\dagger f_j - f_j^\dagger f_i (1 \leq i \neq j \leq n)$	$f_i^\dagger f_j - f_j^\dagger f_i (1 \leq i \neq j \leq n)$ satisfy isomorphic commutation relations.
gilmore-lie-groups_F00776	84	■ 1.000	D, Y, Z	D, Y, Z	6. The Jacobi identity for operators D, Y, Z (replace X by D in Eq. (4.17)) can be
gilmore-lie-groups_F00777	84	■ 0.998	$\mathfrak{so}(4)$	$\mathfrak{so}(4)$	7. For the matrix Lie algebra $\mathfrak{so}(4)$ the defining matrix representation consists of 4×4
gilmore-lie-groups_F00778	84	■ 0.998	4×4	4×4	7. For the matrix Lie algebra $\mathfrak{so}(4)$ the defining matrix representation consists of 4×4
gilmore-lie-groups_F00779	85	■ 1.000	$a_{ij} X_{ij}$	$a_{ij} X_{ij}$	element $a_{ij} X_{ij}$ in the Lie algebra:
gilmore-lie-groups_F00780	85	■ 1.000	$SO(p, q)$	$SO(p, q)$	stant? How does this result extend to $SO(n)$? to $SO(p, q)$? Is there a simple relation
gilmore-lie-groups_F00781	85	■ 1.000	$\operatorname{EXP}(V_-)$	$\operatorname{EXP}(V_-)$	$\operatorname{EXP}(V_-)$ is multiply periodic. Apply this construction to the noncompact groups
gilmore-lie-groups_F00782	85	■ 0.546	$SO(3, 1)$	$SO(3, 1)$	$SO(3, 1)$ and $SO(2, 2)$. Show $\operatorname{EXP}(V_-)$ is a two-sphere S^2 for $SO(3, 1)$ and a two-
gilmore-lie-groups_F00783	85	■ 0.546	$SO(2, 2)$	$SO(2, 2)$	$SO(3, 1)$ and $SO(2, 2)$. Show $\operatorname{EXP}(V_-)$ is a two-sphere S^2 for $SO(3, 1)$ and a two-
gilmore-lie-groups_F00784	85	■ 0.982	T^2	T^2	torus T^2 for $SO(2, 2)$.
gilmore-lie-groups_F00785	85	■ 0.752	$e^{\theta \cdot \mathbf{L}} \in SO(3)$	$e^{\theta \cdot \mathbf{L}} \in SO(3)$	operator $e^{\theta \cdot \mathbf{L}} \in SO(3)$. From this construct the invariant density $\rho(\theta)$ and the metric
gilmore-lie-groups_F00786	85	■ 0.752	$\rho(\theta)$	$\rho(\theta)$	operator $e^{\theta \cdot \mathbf{L}} \in SO(3)$. From this construct the invariant density $\rho(\theta)$ and the metric
gilmore-lie-groups_F00787	85	■ 1.000	$g_{\mu\nu}(\theta)$	$g_{\mu\nu}(\theta)$	tensor $g_{\mu\nu}(\theta)$. Give a reason for the strange behavior (singularities) that these invariant
gilmore-lie-groups_F00788	85	■ 1.000	$dV(x) = \ g(x)\ ^{1/2} d^n x$	$dV(x) = \ g(x)\ ^{1/2} d^n x$	standard result of Riemannian geometry that $dV(x) = \ g(x)\ ^{1/2} d^n x$. Discuss the
gilmore-lie-groups_F00789	85	■ 0.623	$(\mathbf{x}, \mathbf{x})$	$(\mathbf{x}, \mathbf{x})$	12. An inner product $(\mathbf{x}, \mathbf{x})$ is imposed on a real n -dimensional linear vector space.
gilmore-lie-groups_F00790	85	■ 1.000	$g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$	$g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$	It is represented by a real symmetric nonsingular $n \times n$ matrix $g_{rs} = (\mathbf{e}_r, \mathbf{e}_s)$, where

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gilmore-lie-groups_F00791	85	■ 1.000	$\mathbf{x}=\mathbf{e}_i x^i$	$\mathbf{x} = \mathbf{e}_i x^i$	$\mathbf{x} = \mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.
gilmore-lie-groups_F00792	85	■ 1.000	$g^{\{r\ s\}}$	g^{rs}	$\mathbf{x} = \mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.
gilmore-lie-groups_F00793	85	■ 1.000	$\mathbf{y}=G\ \mathbf{x},(\mathbf{y},\mathbf{y})=(\mathbf{x},\mathbf{x})$	$\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$	a. Lie group G preserves inner products. If $\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$. Show $G^t g G = g$.
gilmore-lie-groups_F00794	85	■ 1.000	$G^{\{t\}}\ g\ G=g$	$G^t g G = g$	a. Lie group G preserves inner products. If $\mathbf{y} = G\mathbf{x}, (\mathbf{y}, \mathbf{y}) = (\mathbf{x}, \mathbf{x})$. Show $G^t g G = g$.
gilmore-lie-groups_F00795	85	■ 1.000	$H^{\{t\}}\ g+g\ H=0$	$H^t g + g H = 0$	b. Show the Lie algebra H of G satisfies $H^t g + g H = 0$.
gilmore-lie-groups_F00796	85	■ 1.000	$X_{\{r\ s\}}=g_{\{r\ t\}}\ x^{\{t\}}\ \partial_{\{s\}}-g_{\{s\ t\}}\ x^{\{t\}}\ \partial_{\{r\}}$	$X_{rs} = g_{rt} x^t \partial_s - g_{st} x^t \partial_r$	c. Show that the infinitesimal generators of G are $X_{rs} = g_{rt} x^t \partial_s - g_{st} x^t \partial_r$.
gilmore-lie-groups_F00797	86	■ 1.000	$X_{\{r\ s\}}$	X_{rs}	d. Show that the operators X_{rs} satisfy commutation relations
gilmore-lie-groups_F00798	86	■ 0.887	$M=S\ O$	$M = SO$	13. Every real unimodular 2×2 matrix M can be written in the form $M = SO$, where S
gilmore-lie-groups_F00808	98	■ 0.986	$A^{\{t\}}=-A$	$A^t = -A$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00809	86	■ 0.999	$M\ M^{\{t\}}=S^{\{2\}}=e^{\{2\}\ \Sigma}$	$MM^t = S^2 = e^{2\Sigma}$	a. Show that $MM^t = S^2 = e^{2\Sigma}$.
gilmore-lie-groups_F00810	86	■ 1.000	$O=S^{\{-1\}}\ M=e^{\{-\Sigma\}}\ M$	$O = S^{-1} M = e^{-\Sigma} M$	b. Show $O = S^{-1} M = e^{-\Sigma} M$
gilmore-lie-groups_F00811	86	■ 1.000	Σ	Σ	c. Write S as a power series expansion in Σ .
gilmore-lie-groups_F00812	86	■ 1.000	$S-I_{\{2\}}$	$S - I_2$	d. Write Σ as a power series expansion in $S - I_2$.
gilmore-lie-groups_F00813	86	■ 1.000	$M=H\ U$	$M = HU$	14. Extend the result of the previous problem to complex $n \times n$ matrices $M = HU$, with
gilmore-lie-groups_F00814	86	■ 1.000	$H^{\{\dag\}}=H^{\{+1\}}$	$H^\dagger = H^{+1}$	M arbitrary but nonsingular, $H^\dagger = H^{+1}$ hermitian and $U^\dagger = U^{-1}$ unitary.
gilmore-lie-groups_F00815	86	■ 1.000	$U^{\{\dag\}}=U^{\{-1\}}$	$U^\dagger = U^{-1}$	M arbitrary but nonsingular, $H^\dagger = H^{+1}$ hermitian and $U^\dagger = U^{-1}$ unitary.
gilmore-lie-groups_F00816	86	■ 1.000	$\hbar k_{\{L\}}$	$\hbar k_L$	or right with momentum $\hbar k_L$ or $-\hbar k_R$, has the form (Gilmore, 2004)
gilmore-lie-groups_F00817	86	■ 1.000	$-\hbar k_{\{R\}}$	$-\hbar k_R$	or right with momentum $\hbar k_L$ or $-\hbar k_R$, has the form (Gilmore, 2004)
gilmore-lie-groups_F00818	86	■ 1.000	$m_{\{i\ j\}}$	m_{ij}	The real quantities m_{ij} are the four matrix elements of a group operation in $SL(2; \mathbb{R})$.
gilmore-lie-groups_F00819	86	■ 1.000	$\hbar k_{\{L\}}=\hbar k_{\{R\}}$	$\hbar k_L = \hbar k_R$	They are energy dependent. By appropriate choice of $\hbar k_L = \hbar k_R$ and the matrix
gilmore-lie-groups_F00820	86	■ 0.679	$\operatorname{SU}(1,1)$	$SU(1,1)$	Show that these three matrices span the Lie algebra of the group $SU(1, 1)$.
gilmore-lie-groups_F00821	87	■ 1.000	$\kappa_{\{R\}}$	κ_R	The parameters κ_R and κ_L describe the decay length of the exponentially decaying
gilmore-lie-groups_F00822	87	■ 1.000	$\kappa_{\{L\}}$	κ_L	The parameters κ_R and κ_L describe the decay length of the exponentially decaying
gilmore-lie-groups_F00823	87	■ 1.000	$\kappa_{\{L\}}=\kappa_{\{R\}}$	$\kappa_L = \kappa_R$	energy dependent. By appropriate choice of $\kappa_L = \kappa_R$ and the matrix elements m_{ij} ,
gilmore-lie-groups_F00824	87	■ 1.000	$k_{\{*\}}=\sqrt{2\ m\left(E-V_{\{*\}}\right)}\ /\ \hbar^{\{2\}}$	$k_* = \sqrt{2m(E - V_*)}/\hbar^2$	(Hint: $k_* = \sqrt{2m(E - V_*)}/\hbar^2$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)}/\hbar^2$ for $E < V_*$,
gilmore-lie-groups_F00825	87	■ 1.000	$E>V_{\{*\}}$	$E > V_*$	(Hint: $k_* = \sqrt{2m(E - V_*)}/\hbar^2$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)}/\hbar^2$ for $E < V_*$,
gilmore-lie-groups_F00826	87	■ 1.000	$\kappa_{\{*\}}=\sqrt{2\ m\left(V_{\{*\}}-E\right)}\ /\ \hbar^{\{2\}}$	$\kappa_* = \sqrt{2m(V_* - E)}/\hbar^2$	(Hint: $k_* = \sqrt{2m(E - V_*)}/\hbar^2$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)}/\hbar^2$ for $E < V_*$,

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gilmore-lie-groups_ F08827	87	■ 1.000	$E < V_*$	$E < V_*$	(Hint: $k_* = \sqrt{2m(E - V_*)/\hbar^2}$ for $E > V_*$ and $\kappa_* = \sqrt{2m(V_* - E)/\hbar^2}$ for $E < V_*$,
gilmore-lie-groups_ F08828	87	■ 0.978	$*=L, R$	$* = L, R$	<u>$* \equiv L, R$</u> .)
gilmore-lie-groups_ F08829	88	■ 0.998	$\operatorname{det}(M) \neq 0$	$\det(M) \neq 0$	groups $GL(n; \mathbb{F})$ have no defining condition (the only condition is $\det(M) \neq 0$),
gilmore-lie-groups_ F08830	88	■ 1.000	$I + \epsilon A$	$I + \epsilon A$	the constraint to matrices infinitesimally close to the identity: $I + \epsilon A$. The matrices
gilmore-lie-groups_ F08831	88	■ 1.000	$\mathfrak{gl}(n; \mathbb{F})$	$\mathfrak{gl}(n; \mathbb{F})$	1. $\mathfrak{gl}(n; \mathbb{F})$. This algebra consists of arbitrary $n \times n$ matrices over the field $\mathbb{F}$.
gilmore-lie-groups_ F08832	89	■ 1.000	$\mathfrak{ut}(p, q)$	$\mathfrak{ut}(p, q)$	2. $\mathfrak{ut}(p, q)$. Upper triangular algebras. The matrix algebra has the same structure
gilmore-lie-groups_ F08833	89	■ 0.998	$\mathfrak{ht}(p, q)$	$\mathfrak{ht}(p, q)$	3. $\mathfrak{ht}(p, q)$. The algebra for this class of groups is defined by the condition
gilmore-lie-groups_ F08834	90	■ 1.000	$(a, b) = (1, 0)$	$(a, b) = (1, 0)$	The identity is at <u>$(a, b) = (1, 0)$</u> . Its algebra is spanned by the two operators
gilmore-lie-groups_ F08835	90	■ 0.999	$\left[X_a, X_b\right] = X_b$	$[X_a, X_b] = X_b$	The commutation relations are given by $[X_a, X_b] = X_b$.
gilmore-lie-groups_ F08836	90	■ 1.000	$\mathfrak{ut}(p, q, r)$	$\mathfrak{ut}(p, q, r)$	4. $\mathfrak{ut}(p, q, r)$. This matrix algebra is identical in structure to the parent group.
gilmore-lie-groups_ F08837	90	■ 0.999	$\mathfrak{ut}(1, 2, 1)$	$\mathfrak{ut}(1, 2, 1)$	Example A very useful six-parameter subalgebra of $\mathfrak{ut}(1, 2, 1)$ is given by
gilmore-lie-groups_ F08838	90	■ 1.000	j	j	algebra are summarized in the table below. The operator in the i th row and j th
gilmore-lie-groups_ F08839	90	■ 1.000	$\left[X_i, X_j\right]$	$[X_i, X_j]$	column is $[X_i, X_j]$.
gilmore-lie-groups_ F08845	91	■ 1.000	X_{δ}	X_{δ}	matrix X_{δ} representing the operator $I = [a, a^{\dagger}]$ is not diagonal. It need not be, as
gilmore-lie-groups_ F08854	91	■ 1.000	$a^{\dagger} a^{\dagger}$	$a^{\dagger} a^{\dagger}$	and annihilation operators $a^{\dagger} a^{\dagger}$ and aa , single-photon creation and annihilation
gilmore-lie-groups_ F08855	91	■ 1.000	$a a$	aa	and annihilation operators $a^{\dagger} a^{\dagger}$ and aa , single-photon creation and annihilation
gilmore-lie-groups_ F08863	90	■ 1.000	$\hat{n} = a^{\dagger} a + \frac{1}{2}$	$\hat{n} = a^{\dagger} a + \frac{1}{2}$	and the algebra of the photon energy operator <u>$\hat{n} = a^{\dagger} a + \frac{1}{2}$</u> , two-photon creation
gilmore-lie-groups_ F08864	91	■ 0.996	$I = [a, a^{\dagger}]$	$I = [a, a^{\dagger}]$	operators $a^{\dagger}$ and a , and the identity operator $I = [a, a^{\dagger}]$. We observe that the 4×4
gilmore-lie-groups_ F08865	91	■ 0.887	$\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$	$\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$	5. $\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$. This matrix algebra is also identical in structure to
gilmore-lie-groups_ F08866	91	■ 1.000	$\mathfrak{ut}(1, 1, 1)$	$\mathfrak{ut}(1, 1, 1)$	its parent group. A very useful four-parameter subalgebra of $\mathfrak{ut}(1, 1, 1)$ is given by
gilmore-lie-groups_ F08867	91	■ 0.936	$\mathfrak{nil}(n)$	$\mathfrak{nil}(n)$	6. $\mathfrak{nil}(n)$. Nilpotent matrices have an upper triangular structure, with +1 along the
gilmore-lie-groups_ F08868	91	■ 1.000	$\eta = 0$	$\eta = 0$	This is easily seen by setting $\eta = 0$ in (5.9). As a result, a number of difficult
gilmore-lie-groups_ F08869	91	■ 1.000	$\mathfrak{a}(p, q)$	$\mathfrak{a}(p, q)$	7. $\mathfrak{a}(p, q)$. The matrix algebra for the commutative group of Example (7) in Chap-
gilmore-lie-groups_ F08870	92	■ 0.998	$O(3)$	$O(3)$	antihermitian matrices. The Lie algebras for the groups $O(3)$ and $U(2)$ are
gilmore-lie-groups_ F08871	92	■ 0.998	$U(2)$	$U(2)$	antihermitian matrices. The Lie algebras for the groups $O(3)$ and $U(2)$ are
gilmore-lie-groups_ F08872	92	■ 0.728	σ_{μ}	σ_{μ}	The four 2×2 matrices σ_{μ} are called Pauli spin matrices .
gilmore-lie-groups_ F08873	92	■ 1.000	$\mathfrak{so}(2, 1)$	$\mathfrak{so}(2, 1)$	example, the algebra $\mathfrak{so}(2, 1)$ for the Lorentz group in the plane is

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gilmore-lie-groups_ F08874	93	■ 1.000	$\mathfrak{so}(3,1)$	$\mathfrak{so}(3,1)$	By an identical argument the matrix Lie algebra for the Lorentz group $\mathfrak{so}(3, 1)$ is
gilmore-lie-groups_ F08875	93	■ 1.000	$M=-G^{-1} M^{\dagger} G$	$M = -G^{-1} M^{\dagger} G$	Since G is nonsingular, $M = -G^{-1} M^{\dagger} G$ and
gilmore-lie-groups_ F08876	93	■ 1.000	$(p+q) \times (p+q)$	$(p+q) \times (p+q)$	In the case that the $(p+q) \times (p+q)$ matrix G has singular block diagonal struc-
gilmore-lie-groups_ F08877	93	■ 0.983	$\left[\begin{smallmatrix} g & 0 \\ 0 & 0 \end{smallmatrix}\right]$	$\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$	ture $\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$ with $\det(g) \neq 0$, this constraint reduces to
gilmore-lie-groups_ F08878	93	■ 0.983	$\operatorname{det}(g) \neq 0$	$\det(g) \neq 0$	ture $\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$ with $\det(g) \neq 0$, this constraint reduces to
gilmore-lie-groups_ F08879	93	■ 1.000	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1, 1, 1, 0)$ the Lie algebra is
gilmore-lie-groups_ F08880	94	■ 1.000	θ_i	θ_i	Here the parameters θ_i describe rotations about the i th coordinate axis and the
gilmore-lie-groups_ F08881	94	■ 1.000	s_4	s_4	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4} t$. If s_4 is set to zero
gilmore-lie-groups_ F08882	94	■ 1.000	$t^{\prime} = e^{s_4} t$	$t' = e^{s_4} t$	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4} t$. If s_4 is set to zero
gilmore-lie-groups_ F08883	94	■ 1.000	$E(3)=I \ S \ O(3)$	$E(3) = ISO(3)$	Euclidean group $E(3) \equiv ISO(3)$ (inhomogeneous rotation group in R^3).
gilmore-lie-groups_ F08884	94	■ 0.981	$I \ S \ O(3,1) (3.26)$	$ISO(3,1)(3.26)$	The matrix Lie algebra for the Poincaré group $ISO(3, 1)$ (3.26) is obtained by
gilmore-lie-groups_ F08885	94	■ 0.837	5×5	5×5	similar arguments using a singular 5×5 metric $G = \operatorname{diag}(1, 1, 1, -1, 0)$. The Lie
gilmore-lie-groups_ F08886	94	■ 0.837	$G=\operatorname{diag}(1,1,1,-1,0)$	$G = \operatorname{diag}(1, 1, 1, -1, 0)$	similar arguments using a singular 5×5 metric $G = \operatorname{diag}(1, 1, 1, -1, 0)$. The Lie
gilmore-lie-groups_ F08887	94	■ 0.988	$I \ S \ O(3,1)$	$ISO(3,1)$	“contraction” (cf., Chapter 13) from the Lie algebra of $ISO(3, 1)$:
gilmore-lie-groups_ F08888	94	■ 1.000	$\mathfrak{su}(n)$	$\mathfrak{su}(n)$	Lie algebras. The important algebra $\mathfrak{su}(n)$ is obtained from the intersection of $\mathfrak{u}(n)$
gilmore-lie-groups_ F08889	94	■ 1.000	$\mathfrak{u}(n)$	$\mathfrak{u}(n)$	Lie algebras. The important algebra $\mathfrak{su}(n)$ is obtained from the intersection of $\mathfrak{u}(n)$
gilmore-lie-groups_ F08890	94	■ 1.000	$\mathfrak{sl}(n; \mathbb{C})$	$\mathfrak{sl}(n; \mathbb{C})$	and $\mathfrak{sl}(n; \mathbb{C})$ (cf. (5.14)). For example
gilmore-lie-groups_ F08891	95	■ 1.000	$M_{ij}(i < j) (M \in \mathfrak{u}(n))$	$M_{ij}(i < j) (M \in \mathfrak{u}(n))$	each of the $n(n-1)/2$ complex matrix elements M_{ij} ($i < j$) ($M \in \mathfrak{u}(n)$) above the
gilmore-lie-groups_ F08892	95	■ 0.588	M_{ii}	M_{ii}	M_{ii} by a real 2×2 matrix representing an imaginary complex number ($a = 0$,
gilmore-lie-groups_ F08893	95	■ 0.761	M_{ij}	M_{ij}	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_ F08894	95	■ 0.761	$(i > j)$	$(i > j)$	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_ F08895	95	■ 0.999	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	$2n \times 2n$ matrix with the property $\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$. The dimension of $\mathfrak{ou}(2n)$
gilmore-lie-groups_ F08896	95	■ 0.999	$\mathfrak{o} \ \mathfrak{u}(2n)$	$\mathfrak{ou}(2n)$	$2n \times 2n$ matrix with the property $\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$. The dimension of $\mathfrak{ou}(2n)$
gilmore-lie-groups_ F08897	95	■ 0.640	$\mathcal{U}(n): 2 \times n(n-1) / 2+1 \times n=n^2$	$\square(n): 2 \times n(n-1)/2 + 1 \times n = n^2$	is the dimension of $\mathfrak{u}(n): 2 \times n(n-1)/2 + 1 \times n = n^2$.
gilmore-lie-groups_ F08898	95	■ 0.914	$S \ p(n)$	$Sp(n)$	The Lie algebra of $Sp(n)$ consists of $n \times n$ antihermitian matrices M over $\mathbb{Q}$:
gilmore-lie-groups_ F08899	95	■ 0.791	$\operatorname{Sp}(n)$	$\operatorname{Sp}(n)$	tained from the Lie algebra of $Sp(n)$ by replacing each of the $n(n-1)/2$ quaternion
gilmore-lie-groups_ F09900	95	■ 0.712	$M_{ij}(i < j) (M \in \mathfrak{sp}(n))$	$M_{ij}(i < j) (M \in \mathfrak{sp}(n))$	matrix elements M_{ij} ($i < j$) ($M \in \mathfrak{sp}(n)$) above the diagonal of M by a 2×2 com-

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gilmore-lie-groups_F00901	95	■ 0.998	$q_{\{0\}}=0, \ q_{\{i\}}$	$q_0 = 0, q_i$	matrix representing an imaginary quaternion ($q_0 = 0, q_i$ arbitrary in Eq. (3.4)). The
gilmore-lie-groups_F00902	95	■ 0.615	$i>j$	$i > j$	matrix elements M_{ij} below the diagonal ($i > j$) are obtained from the antihermitic-
gilmore-lie-groups_F00903	95	■ 1.000	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$ $\subset \mathfrak{u}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups_F00904	95	■ 1.000	$\mathfrak{u}(2n)$	$\mathfrak{usp}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups_F00905	95	■ 1.000	$\mathfrak{sp}(n)$	$\mathfrak{sp}(n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups_F00906	95	■ 1.000	$4 \times n(n-1) / 2 + n \times 3 = 2n(2n+1) / 2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$.
gilmore-lie-groups_F00907	96	■ 0.999	$(X, X)=0 \Rightarrow X=$	$(X, X) = 0 \Rightarrow X =$	is usually very useful. This inner product is positive–definite: $(X, X) = 0 \Rightarrow X =$
gilmore-lie-groups_F00908	96	■ 0.949	$X, X^{\prime}$	X, X'	0. For example, for the algebra $\mathfrak{so}(2, 1)$ (Eq. (5.18)), if X, X' are two 3×3 matrices
gilmore-lie-groups_F00909	96	■ 0.811	(X, Y)	(X, Y)	Lie algebra. The Cartan–Killing inner product (X, Y) is specifically defined as the
gilmore-lie-groups_F00910	97	■ 1.000	$\exp(\theta X)$	$\exp(\theta X)$	In the compact case, the group element $\exp(\theta X)$ periodically returns to the identity
gilmore-lie-groups_F00911	97	■ 1.000	$-\pi \leq \theta \leq +\pi$	$-\pi \leq \theta \leq +\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00912	97	■ 1.000	$-\pi$	$-\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00913	97	■ 1.000	$+\pi$	$+\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00914	97	■ 1.000	$-\infty < \theta < +\infty$	$-\infty < \theta < +\infty$	the noncompact case the group is parameterized by the entire line $-\infty < \theta < +\infty$.
gilmore-lie-groups_F00915	97	■ 1.000	S^1	S^1	The underlying manifolds for the two groups are the circle S^1 and the line R^1 .
gilmore-lie-groups_F00916	97	■ 1.000	R^k	R^k	the entire subspace in the Lie algebra, R^k , where k is the number of noncompact
gilmore-lie-groups_F00917	97	■ 1.000	$G=I_{\{n\}}, \ I_{\{p, \ q\}}$	$G = I_n, I_{p,q}$	the metric-preserving groups ($G = I_n, I_{p,q}$, nonsingular antisymmetric, general
gilmore-lie-groups_F00918	97	■ 0.989	$(A, A)=$	$(A, A) =$	Show that in this matrix Lie algebra an inner product can be defined by $(A, A) =$
gilmore-lie-groups_F00919	97	■ 1.000	$\operatorname{tr}(A)^2=a^2+c^2$	$\operatorname{tr}(A)^2 = a^2 + c^2$	$\operatorname{tr}(A)^2 = a^2 + c^2$.
gilmore-lie-groups_F00920	97	■ 1.000	$\mathfrak{ut}(1,1)$	$\mathfrak{ut}(1, 1)$	2. Show that the regular representation of the matrix Lie algebra $\mathfrak{ut}(1, 1)$ given in Prob-
gilmore-lie-groups_F00921	98	■ 1.000	$(A, A)=\operatorname{tr} R(A)^2=(a-c)^2$	$(A, A) = \operatorname{tr} R(A)^2 = (a - c)^2$	inner product in the regular representation is $(A, A) = \operatorname{tr} R(A)^2 = (a - c)^2$. The inner
gilmore-lie-groups_F00922	98	■ 0.998	$X_{\{a\}}+X_{\{c\}}$	$X_a + X_c$	product in the regular representation suggests that the linear combination $X_a + X_c$
gilmore-lie-groups_F00923	98	■ 1.000	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$ $\subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	3. Write down the algebra inclusions $\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$ explicitly in terms
gilmore-lie-groups_F00924	98	■ 1.000	$X_{\{\eta\}}, X_{\{r\}}, X_{\{l\}}, X_{\{\delta\}}$	$X_\eta, X_r, X_l, X_\delta$	the matrices $X_\eta, X_r, X_l, X_\delta$ defined in (5.9). Show that the two Lie algebras are
gilmore-lie-groups_F00925	98	■ 1.000	$\hat{E}=\left(a^{\dagger} a+\frac{1}{2}\right) \hbar \omega \rightarrow a^{\dagger} a+\frac{1}{2}$	$\hat{E} = \left(a^\dagger a + \frac{1}{2}\right) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$	is $\hat{E} = (a^\dagger a + \frac{1}{2}) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$ for $\hbar \omega = 1$.
gilmore-lie-groups_F00926	98	■ 1.000	$\hbar \omega=1$	$\hbar \omega = 1$	is $\hat{E} = (a^\dagger a + \frac{1}{2}) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$ for $\hbar \omega = 1$.
gilmore-lie-groups_F00927	98	■ 1.000	$X, X^{\prime} \in \mathfrak{h}$	$X, X' \in \mathfrak{h}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.

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gilmore-lie-groups_F00928	98	■ 1.000	$Y, Y^{\prime} \in \mathfrak{p}$	$Y, Y' \in \mathfrak{p}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00929	98	■ 1.000	$\left[X, X^{\prime}\right] \in \mathfrak{h},[X, Y] \in \mathfrak{p},\left[Y, Y^{\prime}\right] \in \mathfrak{h}$	$[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00930	98	■ 1.000	$\mathfrak{h}$	$\mathfrak{h}$	Conclude that the subspace $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$. Is $\mathfrak{p}$ a subalgebra (under what
gilmore-lie-groups_F00931	98	■ 1.000	$\mathfrak{p}$	$\mathfrak{p}$	Conclude that the subspace $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$. Is $\mathfrak{p}$ a subalgebra (under what
gilmore-lie-groups_F00932	98	■ 0.995	$X=\left[\begin{array}{cc} A & 0 \\ 0 & D \end{array}\right] \in \mathfrak{h}$	$X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$	If $X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$, then
gilmore-lie-groups_F00933	98	■ 0.995	$Y=\left[\begin{array}{cc} 0 & B \\ C & 0 \end{array}\right] \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$	If $X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$, then
gilmore-lie-groups_F00934	98	■ 1.000	$(X, Y)=0$	$(X, Y) = 0$	Show that X and Y are orthogonal under this inner product: $(X, Y) = 0$.
gilmore-lie-groups_F00935	98	■ 0.970	$\mathfrak{so}(p, q)$	$\mathfrak{so}(p, q)$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00936	98	■ 0.970	$\left[\begin{array}{cc} A & B \\ B^t & C \end{array}\right]$	$\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00937	98	■ 0.970	$p \times p$	$p \times p$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00938	98	■ 0.970	$q \times q$	$q \times q$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00939	98	■ 0.986	$C^t=-C$	$C^t = -C$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00940	98	■ 0.986	$X=\left[\begin{array}{cc} A & 0 \\ 0 & C \end{array}\right] \in \mathfrak{h}$	$X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00941	98	■ 0.986	$Y=\left[\begin{array}{cc} 0 & B \\ B^t & 0 \end{array}\right] \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00942	98	■ 0.988	$(X, X) \leq 0, \quad \text{quad}(X, X)=0 \Rightarrow X=0$	$(X, X) \leq 0, \quad (X, X) = 0 \Rightarrow X = 0$	• $(X, X) \leq 0, \quad (X, X) = 0 \Rightarrow X = 0$
gilmore-lie-groups_F00943	98	■ 1.000	$(Y, Y) \geq 0, \quad \text{quad}(Y, Y)=0 \Rightarrow Y=0$	$(Y, Y) \geq 0, \quad (Y, Y) = 0 \Rightarrow Y = 0$	• $(Y, Y) \geq 0, \quad (Y, Y) = 0 \Rightarrow Y = 0$
gilmore-lie-groups_F00944	99	■ 0.818	$\mathfrak{su}(p, q)$	$\mathfrak{su}(p, q)$	8. The Lie algebra $\mathfrak{su}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^\dagger & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00945	99	■ 0.818	$\left[\begin{array}{c} A \\ B^\dagger \\ C \end{array}\right]$	$\begin{bmatrix} A \\ B^\dagger \\ C \end{bmatrix}$	8. The Lie algebra $\mathfrak{su}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^\dagger & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00946	99	■ 0.999	$A^\dagger=-A, \quad C^\dagger=-C$	$A^\dagger = -A, C^\dagger = -C$	A and C satisfy $A^\dagger = -A, C^\dagger = -C$, and $\text{tr}(A + C) = 0$. If $X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and
gilmore-lie-groups_F00947	99	■ 0.999	$\text{tr}(A+C)=0$	$\text{tr}(A + C) = 0$	A and C satisfy $A^\dagger = -A, C^\dagger = -C$, and $\text{tr}(A + C) = 0$. If $X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and
gilmore-lie-groups_F00948	99	■ 1.000	$Y=\left[\begin{array}{cc} 0 & B \\ B^\dagger & 0 \end{array}\right] \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^\dagger & 0 \end{bmatrix} \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^\dagger & 0 \end{bmatrix} \in \mathfrak{p}$, then show once again that
gilmore-lie-groups_F00949	99	■ 1.000	$(X, X)=0 \Rightarrow X=0$	$(X, X) = 0 \Rightarrow X = 0$	Show that $(X, X) = 0 \Rightarrow X = 0$ and $(Y, Y) = 0 \Rightarrow Y = 0$.
gilmore-lie-groups_F00950	99	■ 1.000	$(Y, Y)=0 \Rightarrow Y=0$	$(Y, Y) = 0 \Rightarrow Y = 0$	Show that $(X, X) = 0 \Rightarrow X = 0$ and $(Y, Y) = 0 \Rightarrow Y = 0$.
gilmore-lie-groups_F00951	99	■ 1.000	$\mathfrak{sl}(n ; \mathbb{R})$	$\mathfrak{sl}(n; \mathbb{R})$	9. The Lie algebra for $\mathfrak{sl}(n; \mathbb{R})$ has a decomposition in terms of real antisymmetric and
gilmore-lie-groups_F00952	99	■ 1.000	$B^t=B$	$B^t = B$	traceless symmetric matrices $A^t = -A$ and $B^t = B$ with $\text{tr } B = 0$:

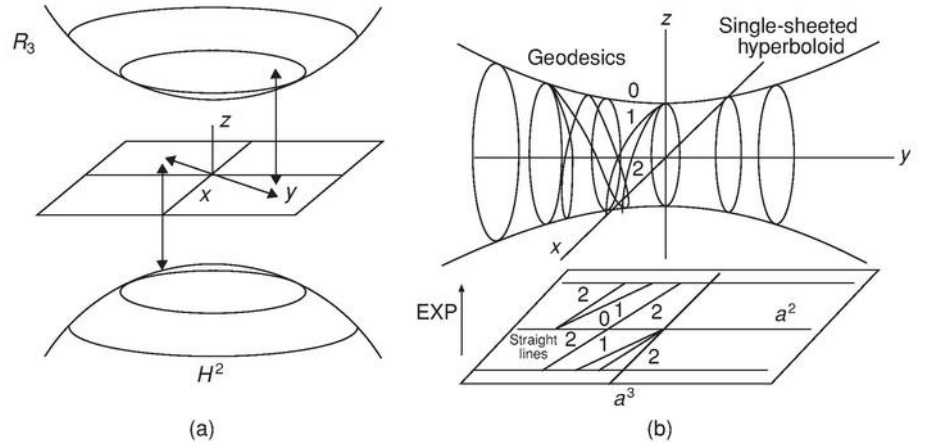
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gilmore-lie-groups_F00953	99	■ 1.000	$\operatorname{tr} B=0$	$\operatorname{tr} B = 0$	traceless symmetric matrices $A^t = -A$ and $B^t = B$ with $\operatorname{tr} B = 0$:
gilmore-lie-groups_F00954	99	■ 1.000	$(A, A)=0 \Rightarrow A=0$	$(A, A) = 0 \Rightarrow A = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(B, B) = 0 \Rightarrow B = 0$.
gilmore-lie-groups_F00955	99	■ 1.000	$(B, B)=0 \Rightarrow B=0$	$(B, B) = 0 \Rightarrow B = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(B, B) = 0 \Rightarrow B = 0$.
gilmore-lie-groups_F00956	99	■ 1.000	$A^{\dagger}=-A$	$A^{\dagger} = -A$	matrices $A^{\dagger} = -A$ and traceless hermitian matrices $H^{\dagger} = H$:
gilmore-lie-groups_F00957	99	■ 1.000	$H^{\dagger}=H$	$H^{\dagger} = H$	matrices $A^{\dagger} = -A$ and traceless hermitian matrices $H^{\dagger} = H$:
gilmore-lie-groups_F00958	99	■ 1.000	$(H, H)=0 \Rightarrow H=0$	$(H, H) = 0 \Rightarrow H = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(H, H) = 0 \Rightarrow H = 0$.
gilmore-lie-groups_F00959	99	■ 1.000	$\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$	$\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$	11. Assume that $\mathfrak{g}$ is a Lie algebra with a Cartan decomposition $\mathfrak{g} = \mathfrak{h} + \mathfrak{p}$, with commu-
gilmore-lie-groups_F00960	100	■ 1.000	$\mathfrak{g}'=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}'$	$\mathfrak{g}' = \mathfrak{h} + i\mathfrak{p} = \mathfrak{h} + \mathfrak{p}'$	defined by $\mathfrak{g}' = \mathfrak{h} + i\mathfrak{p} = \mathfrak{h} + \mathfrak{p}$ then
gilmore-lie-groups_F00961	100	■ 1.000	$\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$	$\mathfrak{g} = \mathfrak{sl}(n; \mathbb{Q})$	nion algebra $\mathfrak{g} = \mathfrak{sl}(n; \mathbb{Q})$ with respect to the subalgebra $\mathfrak{h} = \mathfrak{sl}(n; \mathbb{C})$.
gilmore-lie-groups_F00962	100	■ 1.000	$\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$	$\mathfrak{h} = \mathfrak{sl}(n; \mathbb{C})$	nion algebra $\mathfrak{g} = \mathfrak{sl}(n; \mathbb{Q})$ with respect to the subalgebra $\mathfrak{h} = \mathfrak{sl}(n; \mathbb{C})$.
gilmore-lie-groups_F00963	210	■ 1.000	μ	μ	The contravariant index μ can be lowered with the metric tensor, which is the
gilmore-lie-groups_F00964	126	■ 1.000	σ	σ	representation $\mathbf{J} \rightarrow \frac{1}{2}\sigma$, where σ are the Pauli spin matrices (5.14):
gilmore-lie-groups_F00969	100	■ 1.000	$\mathfrak{g}=A$	$\mathfrak{g} = A$	14. Assume that $\mathfrak{g} = A$, where A is a Lie algebra of $n \times n$ matrices on which the in-
gilmore-lie-groups_F00970	100	■ 1.000	$\operatorname{tr} A^2 \leq 0, =0 \Rightarrow A=0$	$\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$	ner product is negative-definite: $\operatorname{tr} A^2 < 0, = 0 \Rightarrow A = 0$. Then show that $\operatorname{EXP}(tA)$
gilmore-lie-groups_F00971	100	■ 1.000	$\operatorname{EXP}(tA)$	$\operatorname{EXP}(tA)$	ner product is negative-definite: $\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$. Then show that $\operatorname{EXP}(tA)$
gilmore-lie-groups_F00972	100	■ 1.000	t	t	returns to any neighborhood of the identity I_n if t becomes large enough. If the eigen-
gilmore-lie-groups_F00973	100	■ 1.000	$\lambda_i=\gamma n_i, n_i$	$\lambda_i = \gamma n_i, n_i$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or
gilmore-lie-groups_F00974	100	■ 1.000	$\gamma \neq 0$	$\gamma \neq 0$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or
gilmore-lie-groups_F00975	100	■ 1.000	t_0	t_0	irrational), $\operatorname{EXP}(tA)$ returns periodically to I_n . What is t_0 , the minimum period in t ?
gilmore-lie-groups_F00976	100	■ 0.979	$\mathfrak{so}(3)$	$\mathfrak{so}(3)$	15. Use the parameterization of $\mathfrak{so}(3)$ given in Problem 3.14. Show that the differentials
gilmore-lie-groups_F00977	100	■ 1.000	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$ in the neighborhood of the identity by
gilmore-lie-groups_F00978	100	■ 1.000	$g(x, y, z)$	$g(x, y, z)$	metric tensor $g(x, y, z)$ and the invariant measure $d\mu(x, y, z)$ on the group $SO(3)$
gilmore-lie-groups_F00979	100	■ 1.000	$d\mu(x, y, z)$	$d\mu(x, y, z)$	metric tensor $g(x, y, z)$ and the invariant measure $d\mu(x, y, z)$ on the group $SO(3)$
gilmore-lie-groups_F00980	101	■ 1.000	$\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p}$	$(\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p})$	exist in the algebra $\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p}$,
gilmore-lie-groups_F00981	102	■ 0.987	$b_i^{\dagger}$	$b_i^{\dagger}$	a set of n independent boson creation ($b_i^{\dagger}$) and annihilation (b_j) operators that obey
gilmore-lie-groups_F00982	102	■ 0.987	b_j	b_j	a set of n independent boson creation ($b_i^{\dagger}$) and annihilation (b_j) operators that obey

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gilmore-lie-groups_F00983	102	■ 0.971	$\left[b_{\{i\}}, b_{\{j\}}\right], \left[b_{\{i\}}^{\dagger}, b_{\{j\}}^{\dagger}\right], \left[b_{\{i\}}, I\right], \left[b_{\{j\}}^{\dagger}, I\right]$	$[b_i, b_j], [b_i^{\dagger}, b_j^{\dagger}], [b_i, I], [b_j^{\dagger}, I]$	with all other commutators (e.g., $[b_i, b_j], [b_i^{\dagger}, b_j^{\dagger}], [b_i, I], [b_j^{\dagger}, I]$) equal to zero.
gilmore-lie-groups_F00984	102	■ 1.000	$\mathcal{A}$	$\mathcal{A}$	matrix A the operator $\mathcal{A}$ that is a linear combination of creation and annihilation
gilmore-lie-groups_F00985	103	■ 1.000	$[A, B]=C$	$[A, B] = C$	where $[A, B] = C$. This argument is invertible. An algebra of operators bilinear in
gilmore-lie-groups_F00986	103	■ 0.999	$2n+1$	$2n+1$	Remark The $2n+1$ operators $b_i, b_j^{\dagger}, I$ ($1 \leq i, j \leq n$) span the Heisenberg
gilmore-lie-groups_F00987	103	■ 0.999	$b_{\{i\}}, b_{\{j\}}^{\dagger}, I(1 \leq i, j \leq n)$	$b_i, b_j^{\dagger}, I(1 \leq i, j \leq n)$	Remark The $2n+1$ operators $b_i, b_j^{\dagger}, I$ ($1 \leq i, j \leq n$) span the Heisenberg
gilmore-lie-groups_F00988	103	■ 1.000	$X_{\{i\} \{j\}}$	X_{ij}	Any set of operators X_{ij} that satisfies isomorphic commutation relations
gilmore-lie-groups_F00989	103	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{j\}}$	$b_i^{\dagger} b_j$	can be used in place of the bilinear combinations $b_i^{\dagger} b_j$:
gilmore-lie-groups_F00990	103	■ 0.970	$f_{\{i\}}^{\dagger}$	$f_i^{\dagger}$	creation ($f_i^{\dagger}$) and annihilation (f_j) operators for n independent modes. These oper-
gilmore-lie-groups_F00991	103	■ 0.970	$f_{\{j\}}$	f_j	creation ($f_i^{\dagger}$) and annihilation (f_j) operators for n independent modes. These oper-
gilmore-lie-groups_F00992	103	■ 1.000	$\left\{f_{\{i\}}, f_{\{j\}}\right\}, \left\{f_{\{i\}}^{\dagger}, f_{\{j\}}^{\dagger}\right\}$	$\{f_i, f_j\}, \{f_i^{\dagger}, f_j^{\dagger}\}$	with all other bilinear anticommutators (e.g., $\{f_i, f_j\}, \{f_i^{\dagger}, f_j^{\dagger}\}$) equal to zero. Bi-
gilmore-lie-groups_F00993	104	■ 1.000	$SU(n)$	$SU(n)$	subgroup $SU(n)$. The mapping to a boson operator algebra greatly simplifies the
gilmore-lie-groups_F00994	104	■ 1.000	$\mathcal{L}_1=x_2 \partial_3-x_3 \partial_2, \mathcal{L}_2=x_3 \partial_1-x_1 \partial_3, \mathcal{L}_3=x_1 \partial_2-x_2 \partial_1$	$\mathcal{L}_1=x_2 \partial_3-x_3 \partial_2, \mathcal{L}_2=x_3 \partial_1-x_1 \partial_3, \mathcal{L}_3=x_1 \partial_2-x_2 \partial_1$	where $\mathcal{L}_1=x_2 \partial_3-x_3 \partial_2, \mathcal{L}_2=x_3 \partial_1-x_1 \partial_3, \mathcal{L}_3=x_1 \partial_2-x_2 \partial_1$. The two algebras
gilmore-lie-groups_F00995	104	■ 0.705	$L_{\{i\}}$	L_i	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_F00996	104	■ 0.705	$\mathcal{L}_{\{i\}}$	$\mathcal{L}_i$	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_F00997	105	■ 1.000	$E(2)=ISO(2)$	$E(2)=ISO(2)$	As another example, we treat the Lie algebra for the group $E(2)=ISO(2)$ of
gilmore-lie-groups_F00998	105	■ 1.000	$x-y$	$x-y$	rigid motions (translations and rotations) in the $x-y$ plane, whose matrix algebra
gilmore-lie-groups_F00999	105	■ 1.000	$t_{\{1\}}$	t_1	angle θ and displacements in the x and y directions by t_1 and t_2 . The associated
gilmore-lie-groups_F01000	105	■ 1.000	$t_{\{2\}}$	t_2	angle θ and displacements in the x and y directions by t_1 and t_2 . The associated
gilmore-lie-groups_F01001	105	■ 1.000	$\mathcal{L}_z=x_1 \partial_2-x_2 \partial_1$	$\mathcal{L}_z=x_1 \partial_2-x_2 \partial_1$	where $\mathcal{L}_z=x_1 \partial_2-x_2 \partial_1$ and $\mathcal{T}_i=\partial_i$. The matrix algebra and operator algebra have
gilmore-lie-groups_F01002	105	■ 1.000	$\mathcal{T}_{\{i\}}=\partial_{\{i\}}$	$\mathcal{T}_i=\partial_i$	where $\mathcal{L}_z=x_1 \partial_2-x_2 \partial_1$ and $\mathcal{T}_i=\partial_i$. The matrix algebra and operator algebra have
gilmore-lie-groups_F01003	105	■ 0.867	(a, b)	(a, b)	the real line parameterized by points (a, b) in R^2 as follows
gilmore-lie-groups_F01004	105	■ 1.000	$x(p)$	$x(p)$	is chosen a coordinate, $x(p)$, can be introduced and the function can be written
gilmore-lie-groups_F01005	105	■ 1.000	$S^{\prime}$	S'	If a new coordinate system S' is chosen, the value of the function at p remains
gilmore-lie-groups_F01006	105	■ 1.000	$p, x^{\prime}(p)$	$p, x'(p)$	unchanged but the new coordinate of $p, x'(p)$, is different. Therefore the functions
gilmore-lie-groups_F01007	105	■ 1.000	$f_{\{S\}}$	f_S	f_S and $f_{S'}$ must be different. We ask: how is $f_{S'}$ related to f_S ?

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gilmore-lie-groups_ F01008	105	■ 1.000	$f_{S^{\prime}}$	$f_{S'}$	f_S and $f_{S'}$ must be different. We ask: how is $f_{S'}$ related to f_S ?
gilmore-lie-groups_ F01009	105	■ 1.000	$x^{\prime}(p)$	$x'(p)$	To answer this question, assume $x'(p)$ and $x(p)$ are related by an infinitesimal
gilmore-lie-groups_ F01010	106	■ 1.000	$G(x, y)$	$G(x, y)$	As a second example we consider functions $G(x, y)$ defined on the plane R^2 that
gilmore-lie-groups_ F01011	106	■ 1.000	$\left(x^{\prime}, y^{\prime}\right)$	(x', y')	where (x', y') and (x, y) are related by
gilmore-lie-groups_ F01012	107	■ 1.000	$\mathcal{A}=\sum_i \sum_j A_{ij} X_{ij}$	$\mathcal{A}=\sum_i \sum_j A_{ij} X_{ij}$	ciating to each matrix A in a matrix Lie algebra an operator $\mathcal{A}=\sum_i \sum_j A_{ij} X_{ij}$.
gilmore-lie-groups_ F01013	107	■ 1.000	$[A, B]=C \Leftrightarrow [\mathcal{A}, \mathcal{B}]=\mathcal{C}$	$[A, B]=C \Leftrightarrow [\mathcal{A}, \mathcal{B}]=\mathcal{C}$	$[A, B]=C \Leftrightarrow [\mathcal{A}, \mathcal{B}]=\mathcal{C}$. Under these conditions, complicated commutators in
gilmore-lie-groups_ F01014	107	■ 1.000	$e^A e^B=e^D \Leftrightarrow e^{\mathcal{A}} e^{\mathcal{B}}=e^{\mathcal{D}}$	$e^A e^B=e^D \Leftrightarrow e^{\mathcal{A}} e^{\mathcal{B}}=e^{\mathcal{D}}$	with a little care: $e^A e^B=e^D \Leftrightarrow e^{\mathcal{A}} e^{\mathcal{B}}=e^{\mathcal{D}}$.
gilmore-lie-groups_ F01015	107	■ 1.000	$a_i^{\dagger} a_j, 1 \leq i, j \leq 2$	$a_i^{\dagger} a_j, 1 \leq i, j \leq 2$	generate a four-dimensional Lie algebra with basis vectors $a_i^{\dagger} a_j, 1 \leq i, j \leq 2$.
gilmore-lie-groups_ F01016	107	■ 1.000	$\hat{n}=a_1^{\dagger} a_1+a_2^{\dagger} a_2$	$\hat{n}=a_1^{\dagger} a_1+a_2^{\dagger} a_2$	a. Show that $\hat{n}=a_1^{\dagger} a_1+a_2^{\dagger} a_2$ commutes with all the operators in this set.
gilmore-lie-groups_ F01017	107	■ 1.000	$\hat{n}$	$\hat{n}$	b. If $\hat{n}$ is chosen as one basis vector in this four-dimensional space, the remaining
gilmore-lie-groups_ F01018	107	■ 1.000	$a_1^{\dagger} a_1-a_2^{\dagger} a_2, a_1^{\dagger} a_2, a_2^{\dagger} a_1$	$a_1^{\dagger} a_1-a_2^{\dagger} a_2, a_1^{\dagger} a_2, a_2^{\dagger} a_1$	three operators can be chosen as $a_1^{\dagger} a_1-a_2^{\dagger} a_2, a_1^{\dagger} a_2$, and $a_2^{\dagger} a_1$. Construct their
gilmore-lie-groups_ F01019	107	■ 1.000	$a_2^{\dagger} a_1$	$a_2^{\dagger} a_1$	three operators can be chosen as $a_1^{\dagger} a_1-a_2^{\dagger} a_2, a_1^{\dagger} a_2$, and $a_2^{\dagger} a_1$. Construct their
gilmore-lie-groups_ F01020	108	■ 0.999	$\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right), a_1^{\dagger} a_2, a_2^{\dagger} a_1$	$\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right), a_1^{\dagger} a_2$	d. Show that the three operators $\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right), a_1^{\dagger} a_2$, and $a_2^{\dagger} a_1$ satisfy commutation
gilmore-lie-groups_ F01021	108	■ 1.000	$J_z, J_{\pm}$	$J_z, J_{\pm}$	$J_z, J_{\pm}$. In particular, show
gilmore-lie-groups_ F01022	108	■ 1.000	$J^2=J_x^2+J_y^2+J_z^2$	$J^2=J_x^2+J_y^2+J_z^2$	e. Evaluate $J^2=J_x^2+J_y^2+J_z^2$ in terms of the creation and annihilation operators,
gilmore-lie-groups_ F01023	108	■ 1.000	$i(i=1,2)$	$i(i=1,2)$	modes. Assume that the quantum state of mode i ($i=1,2$) is $ n_i\rangle$, where n_i is
gilmore-lie-groups_ F01024	108	■ 1.000	$ n_i\rangle$	$ n_i\rangle$	modes. Assume that the quantum state of mode i ($i=1,2$) is $ n_i\rangle$, where n_i is
gilmore-lie-groups_ F01025	108	■ 1.000	n_i	n_i	modes. Assume that the quantum state of mode i ($i=1,2$) is $ n_i\rangle$, where n_i is
gilmore-lie-groups_ F01026	108	■ 1.000	$a_i^{\dagger}$	$a_i^{\dagger}$	operators $a_i^{\dagger}$ and a_i act on state $ n_i\rangle$ in the usual way:
gilmore-lie-groups_ F01027	108	■ 1.000	a_i	a_i	operators $a_i^{\dagger}$ and a_i act on state $ n_i\rangle$ in the usual way:
gilmore-lie-groups_ F01028	108	■ 1.000	$ n_1\rangle \otimes n_2\rangle= n_1, n_2\rangle$	$ n_1\rangle \otimes n_2\rangle= n_1, n_2\rangle$	Choose as a set of basis vectors the direct product states $ n_1\rangle \otimes n_2\rangle= n_1, n_2\rangle$. Define
gilmore-lie-groups_ F01029	108	■ 1.000	$ n_1, n_2\rangle= j m\rangle$	$ n_1, n_2\rangle= j m\rangle$	a. Identify the lattice sites in Fig. 6.1 with the states $ n_1, n_2\rangle= j m\rangle$, the diagonal
gilmore-lie-groups_ F01030	108	■ 0.994	$\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right)$	$\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right)$	operator $\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right)$ with the operator J_z , and the shift operators $a_1^{\dagger} a_2, a_2^{\dagger} a_1$
gilmore-lie-groups_ F01031	108	■ 0.994	J_z	J_z	operator $\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right)$ with the operator J_z , and the shift operators $a_1^{\dagger} a_2, a_2^{\dagger} a_1$
gilmore-lie-groups_ F01032	108	■ 0.994	$a_1^{\dagger} a_2, a_2^{\dagger} a_1$	$a_1^{\dagger} a_2, a_2^{\dagger} a_1$	operator $\frac{1}{2}\left(a_1^{\dagger} a_1-a_2^{\dagger} a_2\right)$ with the operator J_z , and the shift operators $a_1^{\dagger} a_2, a_2^{\dagger} a_1$
gilmore-lie-groups_ F01033	108	■ 1.000	J_+	J_+	with J_+ and J_- .

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gilmore-lie-groups_ F01059	109	■ 1.000	$f_{\mathrm{i}}^{\dagger} f_{\mathrm{j}}$	$f_i^{\dagger} f_j$	ators $f_i^{\dagger} f_j$. What is now the dimension of this representation?
gilmore-lie-groups_ F01060	109	■ 1.000	$d, d^{\dagger}$	$d, d^{\dagger}$	5. Construct operators $d, d^{\dagger}$ defined formally from the standard creation and annihilation
gilmore-lie-groups_ F01061	109	■ 0.996	$a, a^{\dagger}$	$a, a^{\dagger}$	operators $a, a^{\dagger}$ as follows:
gilmore-lie-groups_ F01062	110	■ 0.996	$\left[d, d^{\dagger}\right]=1$	$[d, d^{\dagger}] = 1$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01063	110	■ 0.996	$[d, d]=\left[d^{\dagger}, d^{\dagger}\right]=0$	$[d, d] = [d^{\dagger}, d^{\dagger}] = 0$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01064	110	■ 0.996	$A D$	AD	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01065	110	■ 1.000	$B C=1$	$BC = 1$	$BC = 1$.
gilmore-lie-groups_ F01066	110	■ 0.998	$\operatorname{Sp}(2 ; \mathbb{R})=$	$\operatorname{Sp}(2; \mathbb{R}) =$	b. Argue that the commutation relations are invariant under the group $\operatorname{Sp}(2; \mathbb{R}) =$
gilmore-lie-groups_ F01067	110	■ 0.982	$\operatorname{Sp}(2 ; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R})$	c. Show that under $\operatorname{Sp}(2; \mathbb{R})$, linear combinations of the coordinate and differential
gilmore-lie-groups_ F01068	110	■ 1.000	x, ∂	x, ∂	operators x, ∂ preserve the commutation relations. In particular, show that
gilmore-lie-groups_ F01069	110	■ 0.946	$a_{\{1\}}, a_{\{2\}}, \ldots, a_{\{n\}}$	$a_1, a_2, \dots, a_n$	d. Replace a by $(a_1, a_2, \dots, a_n)$ and similarly for $a^{\dagger}$ and their images $d, d^{\dagger}$ under
gilmore-lie-groups_ F01070	110	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{N\}}\right\rangle$	$ n_1, n_2, \dots, n_N\rangle$	and eigenstates $ n_1, n_2, \dots, n_N\rangle$.
gilmore-lie-groups_ F01071	110	■ 1.000	$\hbar \omega(n+$	$\hbar \omega(n+$	a. Show that the degeneracy of the multiplet containing n quanta, with energy $\hbar \omega(n +$
gilmore-lie-groups_ F01072	110	■ 0.531	$\frac{N}{2}$	$\frac{N}{2}$	$\frac{N}{2}$) is $\deg(N, n) = (n + N - 1)!/n!(N - 1)!$. This solution to the Bose–Einstein
gilmore-lie-groups_ F01073	110	■ 0.531	$\operatorname{deg}(N, n)=(n+N-1)! / n !(N-1) !$	$\deg(N, n) = (n + N - 1)!/n!(N - 1)!$	$\frac{N}{2}$) is $\deg(N, n) = (n + N - 1)!/n!(N - 1)!$. This solution to the Bose–Einstein
gilmore-lie-groups_ F01074	110	■ 0.999	$f\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{N\}}\right)$	$f\left(x_1, x_2, \dots, x_N\right)$	Taylor series expansion of a function of N variables: $f(x_1, x_2, \dots, x_N)$.
gilmore-lie-groups_ F01075	110	■ 1.000	$N^{\{2\}}$	N^2	N^2 operators $a_i^{\dagger} a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^{\dagger} a_j] = 0$,
gilmore-lie-groups_ F01076	110	■ 1.000	$\mathfrak{u}(N)$	$\mathfrak{u}(N)$	N^2 operators $a_i^{\dagger} a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^{\dagger} a_j] = 0$,
gilmore-lie-groups_ F01077	110	■ 1.000	$\left[\mathcal{H}, a_{\mathrm{i}}^{\dagger} a_{\mathrm{j}}\right]=0$	$[\mathcal{H}, a_i^{\dagger} a_j] = 0$	N^2 operators $a_i^{\dagger} a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^{\dagger} a_j] = 0$,
gilmore-lie-groups_ F01078	110	■ 1.000	$\mathfrak{s u}(N)$	$\mathfrak{su}(N)$	this algebra is a direct sum of a simple Lie algebra, $\mathfrak{su}(N)$, plus the one-dimensional
gilmore-lie-groups_ F01079	110	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	algebra spanned by $\mathcal{H}$.
gilmore-lie-groups_ F01080	110	■ 1.000	$a_{\{j\}}$	a_j	single creation and annihilation operators $a_i^{\dagger}$ and a_j , as well as their commutator I ,
gilmore-lie-groups_ F01081	110	■ 1.000	$(N+1)^{\{2\}}$	$(N + 1)^2$	the resulting set of operators closes to form an $(N + 1)^2$ dimensional Lie algebra
gilmore-lie-groups_ F01082	110	■ 1.000	$n^{\prime}$	n'	state in a multiplet with n' excitations.
gilmore-lie-groups_ F01083	110	■ 1.000	$R, S, T, U, \ldots$	$R, S, T, U, \dots$	7. The set of matrices $R, S, T, U, \dots$ belong to a Lie algebra of $n \times n$ matrices, $a^{\dagger} =$
gilmore-lie-groups_ F01084	110	■ 1.000	$a^{\dagger}=$	$a^{\dagger} =$	7. The set of matrices $R, S, T, U, \dots$ belong to a Lie algebra of $n \times n$ matrices, $a^{\dagger} =$

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gilmore-lie-groups_ F01085	110	■ 0.999	$a_{1}^{\dagger}, a_{2}^{\dagger}, \ldots, a_{n}^{\dagger}$	$a_1^\dagger, a_2^\dagger, \dots, a_n^\dagger$	$(a_1^\dagger, a_2^\dagger, \dots, a_n^\dagger)$ is a row vector of creation operators for n boson modes, and a is its
gilmore-lie-groups_ F01086	111	■ 1.000	$\mathcal{R} = a^{\dagger} R$ $a = a_{i}^{\dagger} R_{ij} a_j$	$\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$	adjoint, a column vector of annihilation operators. Define $\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$, and
gilmore-lie-groups_ F01087	111	■ 1.000	$\mathcal{S}, T, U, \ldots$	$\mathcal{S}, T, U, \dots$	similarly for $\mathcal{S}, T, U, \dots$.
gilmore-lie-groups_ F01088	111	■ 1.000	$[R, S] = T$ $\mathcal{R} = a^{\dagger} R$ $a = a_{i}^{\dagger} R_{ij} a_j$	$[R, S] = T \Leftrightarrow [\mathcal{R}, S] = \mathcal{T}$	a. $[R, S] = T \Leftrightarrow [\mathcal{R}, S] = \mathcal{T}$
gilmore-lie-groups_ F01089	111	■ 1.000	$e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$ $\mathcal{R} = a^{\dagger} R$ $a = a_{i}^{\dagger} R_{ij} a_j$	$e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$	b. $e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}} \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$
gilmore-lie-groups_ F01090	111	■ 0.994	$\left[\frac{d}{dx}, e^{-x^2/2} \right] = -x e^{-x^2/2}$	$\left[\frac{d}{dx}, e^{-x^2/2} \right] = -x e^{-x^2/2}$	a. Show $\left[\frac{d}{dx}, e^{-x^2/2} \right] = -x e^{-x^2/2}$.
gilmore-lie-groups_ F01091	111	■ 1.000	$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$	$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$	d. Introduce the annihilation operator $a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right)$, define the normalized ground
gilmore-lie-groups_ F01092	111	■ 1.000	$\langle x 0 \rangle$	$\langle x 0 \rangle$	state $\langle x 0 \rangle$ by $a \langle x 0 \rangle = 0$. Solve this equation, normalize the solution, and show
gilmore-lie-groups_ F01093	111	■ 1.000	$a \langle x 0 \rangle = 0$	$a \langle x 0 \rangle = 0$	state $\langle x 0 \rangle$ by $a \langle x 0 \rangle = 0$. Solve this equation, normalize the solution, and show
gilmore-lie-groups_ F01094	111	■ 0.990	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$.
gilmore-lie-groups_ F01095	111	■ 1.000	$a^{\dagger} = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right)$	$a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right)$	e. Introduce the creation operator $a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right)$ and show
gilmore-lie-groups_ F01096	111	■ 1.000	$\psi_n(x)$	$\psi_n(x)$	where $\psi_n(x)$ is the n th normalized harmonic oscillator eigenstate $\langle x n \rangle =$
gilmore-lie-groups_ F01097	111	■ 1.000	$\langle x n \rangle =$	$\langle x n \rangle =$	where $\psi_n(x)$ is the n th normalized harmonic oscillator eigenstate $\langle x n \rangle =$
gilmore-lie-groups_ F01098	111	■ 0.999	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x 0 \rangle$	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x 0 \rangle$	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x 0 \rangle$.
gilmore-lie-groups_ F01099	111	■ 0.749	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i$	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i$	$(L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i)$ and a quadrupole interaction $(Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i)$.
gilmore-lie-groups_ F01100	111	■ 0.749	$Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i$	$Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i$	$(L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i)$ and a quadrupole interaction $(Q_{ij} = a_i^\dagger a_j + a_j^\dagger a_i)$.
gilmore-lie-groups_ F01101	111	■ 1.000	$i \leq j$	$i \leq j$	for $i \leq j$ the matrix elements are
gilmore-lie-groups_ F01102	111	■ 0.999	$\Gamma_{ji}^* = \Gamma_{ij}$	$\Gamma_{ji}^* = \Gamma_{ij}$	with $\Gamma_{ji}^* = \Gamma_{ij}$.
gilmore-lie-groups_ F01103	112	■ 0.992	$b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i' \left(b_i^\dagger b_i + \frac{1}{2} \right) +$	$b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i' \left(b_i^\dagger b_i + \frac{1}{2} \right) +$	annihilation operators $b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i' \left(b_i^\dagger b_i + \frac{1}{2} \right) +$ constant. Express
gilmore-lie-groups_ F01104	112	■ 1.000	$\Gamma(H)$	$\Gamma(H)$	the amplitudes m_{ij} in terms of the eigenvectors of $\Gamma(H)$.
gilmore-lie-groups_ F01105	114	■ 1.000	$\text{EXP}(X)$	$\text{EXP}(X)$	Lie algebra onto the entire Lie group through a single mapping of the form $\text{EXP}(X)$.
gilmore-lie-groups_ F01106	114	■ 1.000	$\text{Tr } X = 0$	$\text{Tr } X = 0$	Now choose S to diagonalize (7.2). Since $\text{Tr } X = 0$, the eigenvalues λ can only
gilmore-lie-groups_ F01107	114	■ 1.000	$\pm \theta$	$\pm \theta$	have the form $\pm \theta$ or $\pm i \theta$ (θ real)
gilmore-lie-groups_ F01108	114	■ 1.000	$\pm i \theta$	$\pm i \theta$	have the form $\pm \theta$ or $\pm i \theta$ (θ real)
gilmore-lie-groups_ F01109	116	■ 0.997	$a = b = 0$	$a = b = 0$	To see this most easily set $a = b = 0$. Points on the c -axis map onto
gilmore-lie-groups_ F01110	116	■ 1.000	$2\pi n$	$2\pi n$	and therefore points separated by $2\pi n$ along the c -axis map onto the same group

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gilmore-lie-groups_ F01111	116	0.651	$S O(2) \subset S L(2 ; R)$	$SO(2) \subset SL(2; R)$	operation in $SO(2) \subset SL(2; R)$. The complementary subspace $(a, b, 0)$ maps onto
gilmore-lie-groups_ F01112	116	0.651	$a, b, 0$	$a, b, 0$	operation in $SO(2) \subset SL(2; R)$. The complementary subspace $(a, b, 0)$ maps onto
gilmore-lie-groups_ F01113	116	1.000	$z=\cosh r \geq 1$	$z = \cosh r \geq 1$	For simplicity we have set $z = \cosh r \geq 1$ and $(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$
gilmore-lie-groups_ F01114	116	1.000	$(x, y)=(b, a) \sinh (r) / r, r^2=a^2+$	$(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$	For simplicity we have set $z = \cosh r \geq 1$ and $(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$
gilmore-lie-groups_ F01115	116	1.000	b^2	b^2	b^2 . We observe that
gilmore-lie-groups_ F01116	116	1.000	H_{2+}^2	H_{2+}^2	which is just the upper sheet of the two-sheeted hyperboloid H_{2+}^2 , shown in
gilmore-lie-groups_ F01118	117	—	$S O(1,1)$	$SO(1,1)$	
gilmore-lie-groups_ F01119	117	1.000	H_1^2	H_1^2	This equation describes the surface of the single-sheeted hyperboloid H_1^2 , shown
gilmore-lie-groups_ F01120	118	1.000	$c, 0 \leq c < 2 \pi$	$c, 0 \leq c < 2 \pi$	set parameterized by $c, 0 \leq c < 2 \pi$, is the coset. In the theory of Lie groups, cosets
gilmore-lie-groups_ F01121	118	0.999	$S L(2 ; \mathbb{R}) / S O(2)$	$SL(2; \mathbb{R})/SO(2)$	obtained from the coset decomposition $SL(2; \mathbb{R})/SO(2)$, the manifold parameteriz-
gilmore-lie-groups_ F01122	118	0.996	$R^2 \times S^1$	$R^2 \times S^1$	is the direct product $R^2 \times S^1$. A different parameterization (7.13) based on the coset
gilmore-lie-groups_ F01123	118	1.000	$[S L(2 ; \mathbb{R}) / S O(1,1)] \times S O(1,1) \left(S O(1,1) \simeq R^1\right)$	$[SL(2; \mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$	decomposition $[SL(2; \mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$ shows that the
gilmore-lie-groups_ F01124	118	1.000	$R^1 \times S^1$	$R^1 \times S^1$	boloid (equivalent to $R^1 \times S^1$) with R^1 . This product is once again $R^2 \times S^1$.
gilmore-lie-groups_ F01125	118	1.000	$R^N, N=n^2$	$R^N, N = n^2$	the underlying manifold is a subset of $R^N, N = n^2$, that is defined by algebraic
gilmore-lie-groups_ F01126	119	1.000	R^m	R^m	generators is R^m , for suitable m (m is the number of noncompact generators). The
gilmore-lie-groups_ F01127	119	0.943	$S O(2,1)$	$SO(2,1)$	the groups $SO(2,1)$ and $SU(1,1)$ with Lie algebras
gilmore-lie-groups_ F01128	119	0.993	$S O(2,1) / S O(2)$	$SO(2,1)/SO(2)$	The cosets $SO(2,1)/SO(2)$ and $SU(1,1)/U(1)$ are both isomorphic to R^2 and have
gilmore-lie-groups_ F01129	119	0.993	$S U(1,1) / U(1)$	$SU(1,1)/U(1)$	The cosets $SO(2,1)/SO(2)$ and $SU(1,1)/U(1)$ are both isomorphic to R^2 and have
gilmore-lie-groups_ F01130	119	0.969	$U(1)$	$U(1)$	a 1:1 correspondence. The subgroups $SO(2)$ and $U(1)$ have a 2:1 correspondence.
gilmore-lie-groups_ F01131	119	0.969	$2: 1$	$2:1$	a 1:1 correspondence. The subgroups $SO(2)$ and $U(1)$ have a 2:1 correspondence.
gilmore-lie-groups_ F01132	119	0.936	b_3	b_3	This can be seen by increasing b_3 by 2π and noticing that the 2×2 unitary matrix

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gilmore-lie-groups_ F01133	119	■ 1.000	$U\left(b_3+2\pi\right)=-U\left(b_3\right)$	$U\left(b_3+2\pi\right)=-U\left(b_3\right)$	in (7.19) goes to its negative: $U(b_3+2\pi)=-U(b_3)$. However, increasing a_3 by
gilmore-lie-groups_ F01134	119	■ 1.000	a_3	a_3	in (7.19) goes to its negative: $U(b_3+2\pi)=-U(b_3)$. However, increasing a_3 by
gilmore-lie-groups_ F01135	120	■ 0.984	$U(1)\subset S U(1,1)$	$U(1)\subset SU(1,1)$	$U(1)\subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2)\subset SO(2,1)$ the result is
gilmore-lie-groups_ F01136	120	■ 0.984	4π	4π	$U(1)\subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2)\subset SO(2,1)$ the result is
gilmore-lie-groups_ F01137	120	■ 0.984	$S O(2)\subset S O(2,1)$	$SO(2)\subset SO(2,1)$	$U(1)\subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2)\subset SO(2,1)$ the result is
gilmore-lie-groups_ F01138	120	■ 0.811	$\operatorname{SO}(2,1)$	$SO(2,1)$	2π . Therefore, $SU(1,1)$ is “twice as large” as $SO(2,1)$. More formally, there is a
gilmore-lie-groups_ F01139	120	■ 0.992	$2\rightarrow 1$	$2\rightarrow 1$	$2\rightarrow 1$ homomorphism of $SU(1,1)$ onto $SO(2,1)$.
gilmore-lie-groups_ F01140	120	■ 1.000	$\sqrt{a_1^2+a_2^2+a_3^2}\leq$	$\sqrt{a_1^2+a_2^2+a_3^2}\leq$	It can be checked that all points in the interior of a sphere of radius $\sqrt{a_1^2+a_2^2+a_3^2}\leq$
gilmore-lie-groups_ F01141	120	■ 0.994	$ \mathbf{a} =\pi$	$ \mathbf{a} =\pi$	π) map onto $SO(3)$ provided antipodal points at $ \mathbf{a} =\pi$ are identified
gilmore-lie-groups_ F01142	120	■ 0.999	$\operatorname{SU}(2)$	$SU(2)$	with θ the latitude, and ϕ the longitude on a sphere. For $SU(2)$ all points within a
gilmore-lie-groups_ F01143	120	■ 0.998	$2\pi\left(\sqrt{b_1^2+b_2^2+b_3^2}<2\pi\right)$	$2\pi\left(\sqrt{b_1^2+b_2^2+b_3^2}<2\pi\right)$	sphere of radius 2π ($\sqrt{b_1^2+b_2^2+b_3^2}<2\pi$) are mapped onto distinct elements of
gilmore-lie-groups_ F01144	120	■ 0.986	$-I_2$	$-I_2$	$SU(2)$ and all points at a radius of 2π are mapped onto $-I_2$. There is an easier way
gilmore-lie-groups_ F01145	121	■ 0.652	$x^2+y^2+z^2=1$	$x^2+y^2+z^2=1$	(x,y,z) subject to $x^2+y^2+z^2=1$, and $SU(2)/U(1)$, parameterized by the real
gilmore-lie-groups_ F01146	121	■ 0.652	$S U(2) / U(1)$	$SU(2)/U(1)$	(x,y,z) subject to $x^2+y^2+z^2=1$, and $SU(2)/U(1)$, parameterized by the real
gilmore-lie-groups_ F01147	121	■ 0.989	$x^{\prime}, y^{\prime}, z^{\prime}$	x', y', z'	numbers (x', y', z') subject to $x'^2+y'^2+z'^2=1$, are in 1:1 correspondence with
gilmore-lie-groups_ F01148	121	■ 0.989	$x^{\prime 2}+y^{\prime 2}+z^{\prime 2}=1$	$x'^2+y'^2+z'^2=1$	numbers (x', y', z') subject to $x'^2+y'^2+z'^2=1$, are in 1:1 correspondence with
gilmore-lie-groups_ F01149	121	■ 0.604	$S U(2) \rightarrow S O(3)$	$SU(2)\rightarrow SO(3)$	of the mapping $SU(2)\rightarrow SO(3)$ can be seen from the 2:1 nature of the rotations
gilmore-lie-groups_ F01150	121	■ 1.000	$\bar{G}$	$\bar{G}$	group, $\bar{G}$, called the universal covering group . This group is simply connected:
gilmore-lie-groups_ F01151	121	■ 1.000	$\bar{G} / D$	$\bar{G}/D$	this simply connected Lie group, or else has the form of a quotient $\bar{G}/D$, where D
gilmore-lie-groups_ F01152	121	■ 1.000	$\bar{G}: g d_i=d_i g$	$\bar{G}: gd_i=d_i g$	is a discrete invariant subgroup of $\bar{G}$ whose elements commute with $\bar{G}$: $gd_i=d_i g$
gilmore-lie-groups_ F01153	121	■ 1.000	$d_i \in D$	$d_i\in D$	for $d_i\in D$ and $g\in G$. If $\bar{G}$ is compact it is useful to determine the largest such
gilmore-lie-groups_ F01154	121	■ 0.863	D_{MAX}	D_{MAX}	subgroup, D_{MAX} , of $\bar{G}$. Then all compact Lie groups with the same Lie algebra as
gilmore-lie-groups_ F01155	121	■ 0.979	D_{MAX}	D_{MAX}	$\bar{G}$ are obtained by “dividing” $\bar{G}$ by all possible subgroups of D_{MAX} , as shown in
gilmore-lie-groups_ F01157	121	■ 1.000	$G_1=\bar{G} / D_1$	$G_1=\bar{G}/D_1$	G_2 have the same Lie algebra, $G_1=\bar{G}/D_1$ and G_1 is locally isomorphic with $\bar{G}$.
gilmore-lie-groups_ F01158	121	■ 0.848	$\bar{G} / D_{\mathrm{MAX}}$	$\bar{G}/D_{\mathrm{MAX}}$	G_1 . If $\bar{G}$ is compact, G_1 and G_2 are also locally isomorphic with $\bar{G}/D_{\mathrm{MAX}}$, which

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gilmore-lie-groups_ F01159	122	—	$S\ G=\bar{\mathsf{G}}$	$SG = \bar{G}$	
gilmore-lie-groups_ F01160	122	—	$D_{\{i\}}$	D_i	
gilmore-lie-groups_ F01161	122	■ 1.000	$\backslash\lambda I_{\{2\}}$	λI_2	λI_2 that obey $\lambda^*\lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01162	122	■ 1.000	$\backslash\lambda^*\backslash\lambda=1$	$\lambda^*\lambda = 1$	λI_2 that obey $\lambda^*\lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01163	122	■ 1.000	$\backslash\operatorname{det}\left(\backslash\lambda I_{\{2\}}\right)=+1$	$\det(\lambda I_2) = +1$	λI_2 that obey $\lambda^*\lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01164	122	■ 1.000	$\backslash\lambda=\pm 1\cdot D$	$\lambda = \pm 1.D$	λI_2 that obey $\lambda^*\lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01165	122	■ 1.000	$D=\left\{I_{\{2\}},-I_{\{2\}}\right\}$	$D = \{I_2, -I_2\}$	subgroup $D = \{I_2, -I_2\}$. For the locally isomorphic Lie group $SO(3)$, $D = \lambda I_3$
gilmore-lie-groups_ F01166	122	■ 1.000	$S\ O(3),\ D=\backslash\lambda I_{\{3\}}$	$SO(3), D = \lambda I_3$	subgroup $D = \{I_2, -I_2\}$. For the locally isomorphic Lie group $SO(3)$, $D = \lambda I_3$
gilmore-lie-groups_ F01167	122	■ 0.874	$\backslash\lambda=+1$	$\lambda = +1$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group
gilmore-lie-groups_ F01168	122	■ 0.874	$S\ U(2)\ /\left\{I_{\{2\}},-I_{\{2\}}\right\}=S\ O(3)\ /\ I_{\{3\}}=S\ O(3)$	$SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group
gilmore-lie-groups_ F01169	122	■ 1.000	$\overline{S\ O(2,1)}=$	$\overline{SO(2,1)} =$	is R^1 , the group of translations of the line. The covering group $\overline{SO(2,1)} =$
gilmore-lie-groups_ F01170	122	■ 1.000	$\overline{S\ U(1,1)}$	$\overline{SU(1,1)}$	$\overline{SU(1,1)}$ has no compact subgroup at all. Its underlying group manifold
gilmore-lie-groups_ F01171	122	■ 1.000	$\overline{S\ O(2,1)\ /\ S\ O(2)}\ \times\ \overline{S\ O(2)}=\overline{S\ U(1,1)\ /\ U(1)}\ \times\ \overline{U(1)}=[S\ O(2,1)/SO(2)]\times$	$\overline{SO(2,1)/SO(2)} \times \overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = [SO(2,1)/SO(2)] \times$	is $\overline{SO(2,1)/SO(2)} \times \overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = [SO(2,1)/SO(2)] \times$
gilmore-lie-groups_ F01172	122	■ 1.000	$\overline{S\ O(2)}=S\ U(1,1)\ /\ U(1)\ \times\ \overline{U(1)}=R^2\ \times\ R^1$	$\overline{SO(2)} = SU(1,1)/U(1) \times \overline{U(1)} = R^2 \times R^1$	$\overline{SO(2)} = SU(1,1)/U(1) \times \overline{U(1)} = R^2 \times R$. It is the only group we will encounter

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gilmore-lie-groups_F01173	122	■ 1.000	$\overline{\mathrm{SO}(2,1)}=\overline{\mathrm{SU}(1,1)}$	$\overline{SO(2,1)}=\overline{SU(1,1)}$	in this book that is not a matrix group. The covering group $\overline{SO(2,1)}=\overline{SU(1,1)}$ has
gilmore-lie-groups_F01174	123	■ 1.000	$R_{+}^{\{+\}^2}$	R_{+}^2	right half-plane R_{+}^2 into the group operator
gilmore-lie-groups_F01175	123	■ 0.974	(w, z)	(w, z)	The second maps a point (w, z) in R^2 into the group under the EXPonential map
gilmore-lie-groups_F01176	123	■ 1.000	$R_{+}^{\{+\}^2}(x>0, y)$	$R_{+}^2(x>0, y)$	We ask: is there some mapping of the half-plane $R_{+}^2(x>0, y)$ into $R^2(w, z)$
gilmore-lie-groups_F01177	123	■ 1.000	$R^2(w, z)$	$R^2(w, z)$	We ask: is there some mapping of the half-plane $R_{+}^2(x>0, y)$ into $R^2(w, z)$
gilmore-lie-groups_F01178	123	■ 0.995	$x>0$	$x>0$	These transformations are analytic for $x>0$.
gilmore-lie-groups_F01179	124	■ 0.981	$a, a^{\dagger}, I$	$a, a^{\dagger}, I$	The single-mode photon operators $a, a^{\dagger}, I$ obey isomorphic commutation relations
gilmore-lie-groups_F01180	124	■ 1.000	$l=l^{\prime}=l^{\prime\prime}, r=r^{\prime}=r^{\prime\prime}, \delta^{\prime}=\delta+\frac{1}{2}lr=\delta^{\prime\prime}+l^{\prime\prime}r^{\prime\prime}$	$l=l^{\prime}=l^{\prime\prime}, r=r^{\prime}=r^{\prime\prime}, \delta^{\prime}=\delta+\frac{1}{2}lr=\delta^{\prime\prime}+l^{\prime\prime}r^{\prime\prime}$	We see immediately that $[=l^{\prime}=l^{\prime\prime}, r=r^{\prime}=r^{\prime\prime}, \delta^{\prime}=\delta+\frac{1}{2}lr=\delta^{\prime\prime}+l^{\prime\prime}r^{\prime\prime}]$, and
gilmore-lie-groups_F01181	124	■ 1.000	$\delta=0$	$\delta=0$	obtain the Heisenberg identity (for $\delta=0$)
gilmore-lie-groups_F01182	125	■ 1.000	$\hat{n}=a^{\dagger}a, a, a^{\dagger}, I$	$\hat{n}=a^{\dagger}a, a, a^{\dagger}, I$	photon operators $[\hat{n}=a^{\dagger}a, a, a^{\dagger}, I]$ under the identification
gilmore-lie-groups_F01183	125	■ 1.000	$\operatorname{EXP}\left(\eta a^{\dagger} a+r a^{\dagger}+l a\right)$	$\operatorname{EXP}\left(\eta a^{\dagger} a+r a^{\dagger}+l a\right)$	If for some reason $\operatorname{EXP}\left(\eta a^{\dagger} a+r a^{\dagger}+l a\right)$ needed to be rewritten in the more con-
gilmore-lie-groups_F01184	125	■ 1.000	$\operatorname{EXP}\left(r^{\prime} a^{\dagger}\right) \operatorname{EXP}\left(\eta^{\prime} a^{\dagger} a+\delta^{\prime} I\right) \operatorname{EXP}\left(l^{\prime} a\right)$	$\operatorname{EXP}\left(r^{\prime} a^{\dagger}\right) \operatorname{EXP}\left(\eta^{\prime} a^{\dagger} a+\delta^{\prime} I\right) \operatorname{EXP}\left(l^{\prime} a\right)$	veniently ordered form $\operatorname{EXP}\left(r^{\prime} a^{\dagger}\right) \operatorname{EXP}\left(\eta^{\prime} a^{\dagger} a+\delta^{\prime} I\right) \operatorname{EXP}\left(l^{\prime} a\right)$, then the reparam-
gilmore-lie-groups_F01185	125	■ 0.948	$e^{\left\{l^{\prime} a\right\}} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $	$e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $	Since $e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $ and $e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle$, the expectation value is
gilmore-lie-groups_F01186	125	■ 0.948	$e^{\left\{\eta^{\prime} a^{\dagger} a\right\}} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $	$e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $	Since $e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $ and $e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle$, the expectation value is
gilmore-lie-groups_F01187	125	■ 1.000	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	As a final example we treat the Lie algebra $\mathfrak{su}(2)$. First, we show how to compute
gilmore-lie-groups_F01188	125	■ 1.000	$(j,-j\rangle)$	$(j,-j\rangle)$	$(j,-j\rangle)$
gilmore-lie-groups_F01189	126	■ 1.000	$\theta_z^{\prime}$	$\theta_z^{\prime}$	The only problem that remains is to compute $\theta_z^{\prime}$ as a function of θ . To do this
gilmore-lie-groups_F01190	126	■ 1.000	$\mathbf{J} \rightarrow \frac{1}{2} \sigma$	$\mathbf{J} \rightarrow \frac{1}{2} \sigma$	representation $\mathbf{J} \rightarrow \frac{1}{2} \sigma$, where σ are the Pauli spin matrices (5.14):
gilmore-lie-groups_F01191	126	■ 1.000	$\theta_{\pm}=\theta_1 \pm i \theta_2$	$\theta_{\pm}=\theta_1 \pm i \theta_2$	where $[\theta_{\pm}=\theta_1 \pm i \theta_2]$. Comparison of the two matrices gives immediately
gilmore-lie-groups_F01192	127	■ 1.000	$e^{\alpha J_{+}} e^{\beta J_{-}}$	$e^{\alpha J_{+}} e^{\beta J_{-}}$	The operator product $e^{\alpha J_{+}} e^{\beta J_{-}}$ is written in normally ordered form $\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$
gilmore-lie-groups_F01193	127	■ 1.000	$\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$	$\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$	The operator product $e^{\alpha J_{+}} e^{\beta J_{-}}$ is written in normally ordered form $\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$
gilmore-lie-groups_F01194	127	■ 0.999	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$ and the parameters $\alpha^{\prime}, \beta^{\prime}, n^{\prime}$ computed. We find
gilmore-lie-groups_F01195	127	■ 0.999	$\alpha^{\prime}, \beta^{\prime}, n^{\prime}$	$\alpha^{\prime}, \beta^{\prime}, n^{\prime}$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$ and the parameters $\alpha^{\prime}, \beta^{\prime}, n^{\prime}$ computed. We find
gilmore-lie-groups_F01196	127	■ 1.000	$(1+\alpha \beta)^{2 j}$	$(1+\alpha \beta)^{2 j}$	By expanding $(1+\alpha \beta)^{2 j}$ and invoking analyticity, we find

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gilmore-lie-groups_ F01197	127	■ 1.000	$\mathcal{A}, \mathcal{B}$	$\mathcal{A}, \mathcal{B}$	For example, if operators $\mathcal{A}, \mathcal{B}$ belong to two complementary subspaces in some
gilmore-lie-groups_ F01198	127	■ 0.995	$e^{\mathcal{A}} e^{\mathcal{B}}$	$e^{\mathcal{A}} e^{\mathcal{B}}$	operator Lie algebra $\mathfrak{g}$ then the operator product $e^{\mathcal{A}} e^{\mathcal{B}}$ can be reparameterized as
gilmore-lie-groups_ F01199	127	■ 0.999	$e^{\mathcal{B}'} e^{\mathcal{A}'} \left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime} \right)$	$e^{\mathcal{B}'} e^{\mathcal{A}'} (\mathcal{A}', \mathcal{B}')$	$e^{\mathcal{B}'} e^{\mathcal{A}'} (\mathcal{A}', \mathcal{B}')$ different operators in the same subspaces as $\mathcal{A}, \mathcal{B}$ by
gilmore-lie-groups_ F01200	127	■ 0.999	$\left(\mathcal{A}, \mathcal{B} \right)$	$(\mathcal{A}, \mathcal{B})$	$e^{\mathcal{B}'} e^{\mathcal{A}'} (\mathcal{A}', \mathcal{B}')$ different operators in the same subspaces as $\mathcal{A}, \mathcal{B}$ by
gilmore-lie-groups_ F01201	127	■ 1.000	$e^{\mathcal{A}} e^{\mathcal{B}}$	$e^{\mathcal{A}} e^{\mathcal{B}}$	(iii) Carrying out the matrix calculations $e^{\mathcal{A}} e^{\mathcal{B}}$ and $e^{\mathcal{B}'} e^{\mathcal{A}'}$,
gilmore-lie-groups_ F01202	127	■ 1.000	$e^{\mathcal{B}'} e^{\mathcal{A}'}$	$e^{\mathcal{B}'} e^{\mathcal{A}'}$	(iii) Carrying out the matrix calculations $e^{\mathcal{A}} e^{\mathcal{B}}$ and $e^{\mathcal{B}'} e^{\mathcal{A}'}$,
gilmore-lie-groups_ F01203	127	■ 1.000	$\mathcal{A}', \mathcal{B}'$	$\mathcal{A}', \mathcal{B}'$	(iv) determining the matrices $\mathcal{A}', \mathcal{B}'$ by comparing matrix elements; and
gilmore-lie-groups_ F01204	127	■ 1.000	$\mathcal{A}' \leftrightarrow \mathcal{A}' \mathcal{B}' \leftrightarrow \mathcal{B}'$	$\mathcal{A}' \leftrightarrow \mathcal{A}' \mathcal{B}' \leftrightarrow \mathcal{B}'$	(v) using the isomorphism $\mathcal{A}' \leftrightarrow \mathcal{A}' \mathcal{B}' \leftrightarrow \mathcal{B}'$.
gilmore-lie-groups_ F01205	127	■ 1.000	$(\mathcal{A}, \mathcal{B}) \leftrightarrow$	$(\mathcal{A}, \mathcal{B}) \leftrightarrow$	This procedure will produce a local analytic reparameterization $(\mathcal{A}, \mathcal{B}) \leftrightarrow$
gilmore-lie-groups_ F01206	127	■ 0.998	$(\mathcal{A}', \mathcal{B}')$	$(\mathcal{A}', \mathcal{B}')$	$(\mathcal{A}', \mathcal{B}')$. If the matrix group used to construct this reparameterization is simply
gilmore-lie-groups_ F01207	127	■ 1.000	$\mathcal{A}, \mathcal{B}, \dots$	$\mathcal{A}, \mathcal{B}, \dots$	groups of the operator group and the matrix group. When the operators $\mathcal{A}, \mathcal{B}, \dots$ are
gilmore-lie-groups_ F01208	128	■ 1.000	$A, B, \dots$	$A, B, \dots$	related to matrices $A, B, \dots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A} \leftrightarrow A$,
gilmore-lie-groups_ F01209	128	■ 1.000	$\mathcal{A} \leftrightarrow A$	$\mathcal{A} \leftrightarrow A$	related to matrices $A, B, \dots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A} \leftrightarrow A$,
gilmore-lie-groups_ F01210	128	■ 1.000	$t + \delta t$	$t + \delta t$	The state of the system at time $t + \delta t$ is related to the state at time t by
gilmore-lie-groups_ F01211	128	■ 1.000	$ \psi(t_f)\rangle$	$ \psi(t_f)\rangle$	$ \psi(t_f)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi(t_f)\rangle =$
gilmore-lie-groups_ F01212	128	■ 1.000	t_f	t_f	$ \psi(t_f)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi(t_f)\rangle =$
gilmore-lie-groups_ F01213	128	■ 1.000	$ \psi(t_f)\rangle =$	$ \psi(t_f)\rangle =$	$ \psi(t_f)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi(t_f)\rangle =$
gilmore-lie-groups_ F01214	128	■ 0.996	$U(t_f, t_i) \psi(t_i)\rangle$	$U(t_f, t_i) \psi(t_i)\rangle$	$U(t_f, t_i) \psi(t_i)\rangle$. The finite time unitary operator is built up from small dis-
gilmore-lie-groups_ F01215	128	■ 1.000	$H(t')$	$H(t')$	Care must be taken with the formal integration in this equation, as in general $H(t')$
gilmore-lie-groups_ F01216	128	■ 1.000	$H(t), t' \neq t$	$H(t), t' \neq t$	does not commute with $H(t)$, $t' \neq t$. It is for this reason that the symbol “ T ”
gilmore-lie-groups_ F01217	129	■ 1.000	$2j + 1$	$2j + 1$	Example 1. A Hamiltonian acts in a $2j + 1$ dimensional space through a set of
gilmore-lie-groups_ F01218	129	■ 1.000	$ j, m_j\rangle$	$ j, m_j\rangle$	wish to determine the evolution of some particular state $ j, m_j\rangle$. The Hamiltonian
gilmore-lie-groups_ F01219	129	■ 1.000	$g(t) \in SU(2)$	$g(t) \in SU(2)$	$g(t) \in SU(2)$ evolves, and then construct its unitary representation, than it is to de-
gilmore-lie-groups_ F01220	129	■ 1.000	$(2j + 1) \times (2j + 1)$	$(2j + 1) \times (2j + 1)$	termine the time evolution of the $(2j + 1) \times (2j + 1)$ unitary matrix. Specifically,
gilmore-lie-groups_ F01221	129	■ 1.000	$a(t)$	$a(t)$	the complex coefficients $a(t)$ and $b(t)$ or three equations for the real coefficients of
gilmore-lie-groups_ F01222	129	■ 1.000	$b(t)$	$b(t)$	the complex coefficients $a(t)$ and $b(t)$ or three equations for the real coefficients of

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gilmore-lie-groups_F01223	129	■ 1.000	$\sigma_1, \sigma_2, \sigma_3$	$\sigma_1, \sigma_2, \sigma_3$	the Pauli spin matrices $\sigma_1, \sigma_2, \sigma_3$. These are first order equations and can be solved
gilmore-lie-groups_F01224	129	■ 1.000	$a\left(t_i\right)=$	$a\left(t_i\right)=$	by standard integration methods (e.g., RK4). The initial conditions are $a(t_i)=$
gilmore-lie-groups_F01225	129	■ 0.999	$1, b\left(t_i\right)=0$	$1, b\left(t_i\right)=0$	$1, b(t_i)=0$. The final 2×2 unitary matrix is determined by $a(t_f), b(t_f)$. This is
gilmore-lie-groups_F01226	129	■ 0.999	$a\left(t_f\right), b\left(t_f\right)$	$a\left(t_f\right), b\left(t_f\right)$	$1, b(t_i)=0$. The final 2×2 unitary matrix is determined by $a(t_f), b(t_f)$. This is
gilmore-lie-groups_F01227	129	■ 0.971	$(2 j+1) \times$	$(2 j+1) \times$	a group operation in $SU(2)$ that can subsequently be mapped into the $(2 j+1) \times$
gilmore-lie-groups_F01228	129	■ 1.000	$(2 j+1)$	$(2 j+1)$	$(2 j+1)$ unitary irreducible representation of this group. At this point the problem
gilmore-lie-groups_F01229	129	■ 1.000	$\left \psi\left(t_i\right)\right\rangle$	$\left \psi\left(t_i\right)\right\rangle$	is solved, independent of the initial state $\left \psi\left(t_i\right)\right\rangle$.
gilmore-lie-groups_F01230	130	■ 1.000	$\left n\right\rangle, n=0,1,2, \ldots$	$\left n\right\rangle, n=0,1,2, \ldots$	space with basis vectors $\left n\right\rangle, n=0,1,2, \ldots$. The matrix on the right is a faith-
gilmore-lie-groups_F01231	130	■ 1.000	$U=$	$U=$	general unitary operator that can be constructed from these operators is $U=$
gilmore-lie-groups_F01232	130	■ 1.000	$\operatorname{EXP}\left(i\left[n(t) a^{\dagger} a+r(t) a^{\dagger}+r^{*}(t) a+d(t) I\right]\right)$	$\operatorname{EXP}\left(i\left[n(t) a^{\dagger} a+r(t) a^{\dagger}+r^{*}(t) a+d(t) I\right]\right)$	$\operatorname{EXP}(i[n(t) a^{\dagger} a+r(t) a^{\dagger}+r^{*}(t) a+d(t) I])$. This exponential is easy to compute
gilmore-lie-groups_F01233	130	■ 0.982	$\left(n, r, r^{*}, d\right)$	$\left(n, r, r^{*}, d\right)$	tions in four real variables $\left(n, r, r^{*}, d\right)$ that can be integrated by standard methods.
gilmore-lie-groups_F01234	130	■ 1.000	$d \omega(t) / d t=0$	$d \omega(t) / d t=0$	In the case that $d \omega(t) / d t=0$ the equations simplify considerably, and can almost
gilmore-lie-groups_F01235	130	■ 1.000	ρ	ρ	density operator ρ
gilmore-lie-groups_F01236	130	■ 1.000	$\rho=e^{-\beta H} / Z$	$\rho=e^{-\beta H} / Z$	hamiltonian describing the system as $\rho=e^{-\beta H} / Z$, where the normalization con-
gilmore-lie-groups_F01237	130	■ 1.000	$Z=\operatorname{tr} e^{-\beta H}$	$Z=\operatorname{tr} e^{-\beta H}$	stant, or partition function, is $Z=\operatorname{tr} e^{-\beta H}$ and $\beta=1 / k_B T$, k_B is the Boltzmann
gilmore-lie-groups_F01238	130	■ 1.000	$\beta=1 / k_B T, k_B$	$\beta=1 / k_B T, k_B$	stant, or partition function, is $Z=\operatorname{tr} e^{-\beta H}$ and $\beta=1 / k_B T$, k_B is the Boltzmann
gilmore-lie-groups_F01239	131	■ 1.000	$\left\langle e^{\Lambda}\right\rangle$	$\left\langle e^{\Lambda}\right\rangle$	generating functions for these expectation values. To do this we compute $\left\langle e^{\Lambda}\right\rangle$,
gilmore-lie-groups_F01240	131	■ 1.000	$\Lambda=\lambda \cdot \mathbf{J}$	$\Lambda=\lambda \cdot \mathbf{J}$	where $\Lambda=\lambda \cdot \mathbf{J}$. All symmetric moments can be constructed by taking derivatives
gilmore-lie-groups_F01241	131	■ 1.000	$H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$	$H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$	In these expressions $H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$, and similarly for $ H =\sqrt{(H, H)}$
gilmore-lie-groups_F01242	131	■ 1.000	$ H =\sqrt{(H, H)}$	$ H =\sqrt{(H, H)}$	In these expressions $H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$, and similarly for $ H =\sqrt{(H, H)}$
gilmore-lie-groups_F01243	131	■ 1.000	$ \Lambda =\sqrt{(\Lambda, \Lambda)}$	$ \Lambda =\sqrt{(\Lambda, \Lambda)}$	and $ \Lambda =\sqrt{(\Lambda, \Lambda)}$.
gilmore-lie-groups_F01244	131	■ 1.000	2^N	2^N	If N two-level atoms are acting incoherently, the trace over the 2^N states of all
gilmore-lie-groups_F01245	131	■ 1.000	$2 J+1$	$2 J+1$	N atoms are acting coherently, there are $2 J+1$ states, where $N=2 J$. The trace
gilmore-lie-groups_F01246	131	■ 1.000	$N=2 J$	$N=2 J$	N atoms are acting coherently, there are $2 J+1$ states, where $N=2 J$. The trace
gilmore-lie-groups_F01247	131	■ 1.000	$\mu(H, \Lambda, T)$	$\mu(H, \Lambda, T)$	where $\mu(H, \Lambda, T)$ is determined from Eq. (7.65). The thermodynamic generating
gilmore-lie-groups_F01248	131	■ 1.000	$\left\langle J_{-}\right\rangle$	$\left\langle J_{-}\right\rangle$	To construct explicit expectation values (e.g., $\left\langle J_{-}\right\rangle$) it is sufficient to differenti-

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gilmore-lie-groups_ F01249	131	■ 1.000	$\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$	$\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$	ate the generating function (e.g., $\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$) and evaluate the result at $\Lambda = 0$.
gilmore-lie-groups_ F01250	131	■ 1.000	$\Lambda = 0$	$\Lambda = 0$	ate the generating function (e.g., $\frac{\partial}{\partial \lambda^*} \langle e^\Lambda \rangle / \langle e^0 \rangle$) and evaluate the result at $\Lambda = 0$.
gilmore-lie-groups_ F01251	131	■ 1.000	$\left. \frac{\partial}{\partial \lambda^*} \log(\langle e^\Lambda \rangle) \right _{\Lambda=0}$	$\left. \frac{\partial}{\partial \lambda^*} \log(\langle e^\Lambda \rangle) \right _{\Lambda=0}$	$\left. \frac{\partial}{\partial \lambda^*} \log(\langle e^\Lambda \rangle) \right _{\Lambda=0}$.
gilmore-lie-groups_ F01252	132	■ 0.999	$\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	$\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	The density operator is $\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$. The generating function for
gilmore-lie-groups_ F01253	132	■ 1.000	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger a + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger a + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$	operator expectation values is $\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger a + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$.
gilmore-lie-groups_ F01254	132	■ 0.967	$ 0\rangle, 1\rangle, 2\rangle, \dots$	$ 0\rangle, 1\rangle, 2\rangle, \dots$	$ 0\rangle, 1\rangle, 2\rangle, \dots$. It would be insane to attempt to compute this expectation value
gilmore-lie-groups_ F01255	132	■ 0.989	H, Λ	H, Λ	(i) Write each of the operators H, Λ in the 3×3 matrix representation M (cf., Eq.
gilmore-lie-groups_ F01256	132	■ 0.999	Z_l	Z_l	(iv) Find a similarity transformation, S , that zeroes out Z_l and Z_r :
gilmore-lie-groups_ F01257	132	■ 0.999	Z_r	Z_r	(iv) Find a similarity transformation, S , that zeroes out Z_l and Z_r :
gilmore-lie-groups_ F01258	133	■ 1.000	$A < 0$	$A < 0$	(vi) Take the trace. Assuming $A < 0$ the sum converges to
gilmore-lie-groups_ F01259	133	■ 1.000	$\lambda_n =$	$\lambda_n =$	The generating function for only the creation and annihilation operators ($\lambda_n =$
gilmore-lie-groups_ F01260	133	■ 1.000	$d = 0$	$d = 0$	$d = 0$) is considerably simpler.
gilmore-lie-groups_ F01261	133	■ 1.000	$D = \text{Id}, G$	$D = \text{Id}, G$	D is a discrete invariant subgroup of $\overline{G}$. If $\overline{D} = \text{Id}, \overline{G}$ is isomorphic to $\overline{G}$, otherwise
gilmore-lie-groups_ F01262	133	■ 1.000	λI_n	λI_n	of the identity matrix, λI_n , and is simple to compute. If G_1 and G_2 have isomorphic
gilmore-lie-groups_ F01263	134	■ 0.940	$it/\hbar \leftrightarrow 1/k_B T$	$it/\hbar \leftrightarrow 1/k_B T$	(dynamics and statics) are related by a “Wick rotation”: $it/\hbar \leftrightarrow 1/k_B T$. We have
gilmore-lie-groups_ F01264	134	■ 1.000	$\phi((x_1, y_1), (x_2, y_2))$	$\phi((x_1, y_1), (x_2, y_2))$	1. Construct the analytic group mapping $\phi((x_1, y_1), (x_2, y_2))$ for the parameterization
gilmore-lie-groups_ F01265	134	■ 1.000	$\phi((w_1, z_1), (w_2, z_2))$	$\phi((w_1, z_1), (w_2, z_2))$	(7.24) of the affine group. Construct the mapping $\phi((w_1, z_1), (w_2, z_2))$ for the param-
gilmore-lie-groups_ F01266	134	■ 1.000	$a^2 + b^2 - c^2 < 0$	$a^2 + b^2 - c^2 < 0$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_ F01267	134	■ 1.000	$SO(2) \subset$	$SO(2) \subset$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_ F01268	134	■ 0.999	c_T	c_T	c_T , in the subgroup is 2π but if $a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$), the basic
gilmore-lie-groups_ F01269	134	■ 0.999	$a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$)	$a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$)	c_T , in the subgroup is 2π but if $a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$), the basic
gilmore-lie-groups_ F01270	134	■ 0.999	$2\pi\gamma$	$2\pi\gamma$	repetition period in the c -direction is increased to $2\pi\gamma$, where $\gamma = 1/\sqrt{1 - \beta^2}$ and
gilmore-lie-groups_ F01271	134	■ 0.999	$\gamma = 1/\sqrt{1 - \beta^2}$	$\gamma = 1/\sqrt{1 - \beta^2}$	repetition period in the c -direction is increased to $2\pi\gamma$, where $\gamma = 1/\sqrt{1 - \beta^2}$ and
gilmore-lie-groups_ F01272	134	■ 1.000	$\beta^2 = (a^2 + b^2)/c^2$	$\beta^2 = (a^2 + b^2)/c^2$	$\beta^2 = (a^2 + b^2)/c^2$. Compare this renormalization of periodicity with “time dilation.”
gilmore-lie-groups_ F01273	134	■ 0.947	$SU(3)$	$SU(3)$	3. Compute the maximal discrete invariant subgroup D_{MAX} of $SU(3)$ and show that it

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gilmore-lie-groups_ F01274	134	■ 0.954	$\left\{I_3, \lambda I_3, \lambda^2 I_3\right\}$	$\{I_3, \lambda I_3, \lambda^2 I_3\}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_ F01275	134	■ 0.954	$\lambda=e^{2 \pi i / 3}$	$\lambda = e^{2\pi i/3}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_ F01276	134	■ 0.954	$\operatorname{SU}(3) / D_{\text {MAX }}$	$SU(3)/D_{\text{MAX}}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_ F01277	134	■ 0.229	$\operatorname{Re} g\left(\frac{\mathfrak{s} u(3)}{3}\right)$	$[\Re g(\mathfrak{su}(3))]$	the group of real 8×8 matrices $\operatorname{EXP}[\Re g(\mathfrak{su}(3))]$ (“eight-fold way”).
gilmore-lie-groups_ F01278	134	■ 1.000	$\epsilon I_n, \epsilon=e^{2 \pi i / n}$	$\epsilon I_n, \epsilon = e^{2\pi i/n}$	and show that it is the cyclic group of order n generated by $\epsilon I_n, \epsilon = e^{2\pi i/n}$. What real
gilmore-lie-groups_ F01279	134	■ 0.988	$S U(n) / D_{\text {MAX }}$	$SU(n)/D_{\text{MAX}}$	matrix group is $SU(n)/D_{\text{MAX}}$ equivalent to?
gilmore-lie-groups_ F01280	135	■ 1.000	$a^{\dagger}, a$	$a^\dagger, a$	6. It is convenient to introduce the creation and annihilation operators $a^\dagger, a$ to study the
gilmore-lie-groups_ F01281	135	■ 1.000	$\left\langle n^{\prime} \mid x^k \mid n\right\rangle$	$\langle n' x^k n\rangle$	tor basis, $\langle n' x^k n\rangle$, can be simplified using disentangling theorems. This problem
gilmore-lie-groups_ F01282	135	■ 1.000	$e^{\lambda x}$	$e^{\lambda x}$	a. The function $e^{\lambda x}$ is a generating function for matrix elements of x^k . Show that
gilmore-lie-groups_ F01283	135	■ 1.000	x^k	x^k	a. The function $e^{\lambda x}$ is a generating function for matrix elements of x^k . Show that
gilmore-lie-groups_ F01284	135	■ 1.000	$\left[a, a^{\dagger}\right]=I$	$[a, a^\dagger] = I$	ators and their commutator $[a, a^\dagger] = I$ to show
gilmore-lie-groups_ F01285	135	■ 0.955	$e^{r a^{\dagger}} e^{\delta I} e^{l a}$	$e^{ra^\dagger} e^{\delta I} e^{la}$	$e^{ra^\dagger} e^{\delta I} e^{la}$ by constructing the matrix product of these three operators:
gilmore-lie-groups_ F01286	135	■ 0.999	$\left\langle n^{\prime} \mid x^4 \mid n\right\rangle$	$\langle n' x^4 n\rangle$	e. Use the disentangling theorem in d to compute $\langle n' x^4 n\rangle$. In particular, show
gilmore-lie-groups_ F01287	135	■ 1.000	$\left\langle e^{i k x}\right\rangle$	$\langle e^{ikx}\rangle$	potential it is necessary to compute a structure factor $\langle e^{ikx}\rangle$. The expectation value is
gilmore-lie-groups_ F01288	136	■ 1.000	$P_n \simeq e^{-n \beta \hbar \omega}$	$P_n \simeq e^{-n\beta\hbar\omega}$	thermal: $P_n \simeq e^{-n\beta\hbar\omega}$. This expectation value can be written in algebraic form as
gilmore-lie-groups_ F01289	136	■ 1.000	$k \rightarrow 0$	$k \rightarrow 0$	We concentrate on the numerator, as the denominator is obtained in the limit $k \rightarrow 0$.
gilmore-lie-groups_ F01290	136	■ 1.000	$\left\langle e^{i k x}\right\rangle=e^{\delta}$	$\langle e^{ikx}\rangle = e^\delta$	As a result $\boxed{\langle e^{ikx}\rangle = e^\delta}$.
gilmore-lie-groups_ F01291	136	■ 0.976	δ	δ	c. Compute δ using 3×3 nonunitary matrix multiplications to carry out multi-
gilmore-lie-groups_ F01292	136	■ 1.000	$\infty \times \infty$	$\infty \times \infty$	plications <i>in the group</i> rather than in an $\infty \times \infty$ unitary representation of the
gilmore-lie-groups_ F01293	136	■ 0.999	$\omega^{\prime}=\omega$	$\omega' = \omega$	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_ F01294	136	■ 0.999	α, β	α, β	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_ F01295	136	■ 0.999	γ	γ	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_ F01296	136	■ 1.000	$\left[X_{\{i\}}, X_{\{j\}}\right]=\sum_{k=1}^N C_{i j}^k X_{\{k\}}$	$[X_i, X_j] = \sum_{k=1}^N C_{ij}^k X_k$	8. A finite set of operators X_i closes under commutation: $[X_i, X_j] = \sum_{k=1}^N C_{ij}^k X_k$.
gilmore-lie-groups_ F01297	136	■ 1.000	R	R	this set of operators has two representations R and S with the following properties:
gilmore-lie-groups_ F01298	136	■ 1.000	$\left(R\left(a^{\{i\}} X_{\{i\}}\right)\right)^{\dagger}=\left(a^{\{i\}} R\left(X_{\{i\}}\right)\right)^{\dagger}=\left(a^{\{i\}}\right)^{*} R^{\dagger}\left(X_{\{i\}}\right)$	$(R(a^{\{i\}} X_i))^{\dagger} = (a^{\{i\}} R(X_i))^{\dagger} = (a^{\{i\}})^* R^\dagger(X_i)$	• R is hermitian: $\boxed{(R(a^{\{i\}} X_i))^{\dagger} = (a^{\{i\}} R(X_i))^{\dagger} = (a^{\{i\}})^* R^\dagger(X_i)}$
gilmore-lie-groups_ F01299	136	■ 1.000	$S\left(a^{\{i\}} X_{\{i\}}\right)=0 \Rightarrow a^{\{i\}}=0$	$S(a^{\{i\}} X_i) = 0 \Rightarrow a^{\{i\}} = 0$	• S is faithful: $\boxed{S(a^{\{i\}} X_i) = 0 \Rightarrow a^{\{i\}} = 0}$.

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gilmore-lie-groups_F01300	136	■ 1.000	$\mathcal{H} = R(\mathcal{H}) = R(a^i X_i)$	$\mathcal{H} = R(a^i X_i)$	a. It happens frequently that $\mathcal{H} = R(a^i X_i)$ describes the physics of some quantum
gilmore-lie-groups_F01301	136	■ 0.999	$H_1, H_2, \dots, H_r \in \mathfrak{g}$	$H_1, H_2, \dots, H_r \in \mathfrak{g}$	mechanical system. Show that if $H_1, H_2, \dots, H_r \in \mathfrak{g}$ span a maximal commutative
gilmore-lie-groups_F01302	136	■ 1.000	$[H_i, H_j] = 0, 1 \leq i, j \leq r$	$[H_i, H_j] = 0, 1 \leq i, j \leq r$	subspace, so that $[H_i, H_j] = 0, 1 \leq i, j \leq r$, then the hermitian operators $R(H_i)$
gilmore-lie-groups_F01303	136	■ 1.000	$R(H_i)$	$R(H_i)$	subspace, so that $[H_i, H_j] = 0, 1 \leq i, j \leq r$, then the hermitian operators $R(H_i)$
gilmore-lie-groups_F01304	136	■ 1.000	$[R(H_i)]_{\alpha\beta} = r_\alpha(i) \delta_{\alpha\beta}$	$[R(H_i)]_{\alpha\beta} = r_\alpha(i) \delta_{\alpha\beta}$	representation: $[R(H_i)]_{\alpha\beta} = r_\alpha(i) \delta_{\alpha\beta}$.
gilmore-lie-groups_F01305	137	■ 1.000	$[S(H_i), S(H_j)] = 0$	$[S(H_i), S(H_j)] = 0$	b. Show that $[S(H_i), S(H_j)] = 0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_F01306	137	■ 1.000	r	r	b. Show that $[S(H_i), S(H_j)] = 0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_F01307	137	■ 1.000	$S(H_i)$	$S(H_i)$	b. Show that $[S(H_i), S(H_j)] = 0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_F01308	137	■ 1.000	$U(t) = R(e^{-\frac{i}{\hbar} \mathcal{H} t}) = e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$	$U(t) = R(e^{-\frac{i}{\hbar} \mathcal{H} t}) = e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$	$U(t) = R(e^{-\frac{i}{\hbar} \mathcal{H} t}) = e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$.
gilmore-lie-groups_F01309	137	■ 1.000	$\rho(T) = e^{-\beta \mathcal{H}} / Z$	$\rho(T) = e^{-\beta \mathcal{H}} / Z$	d. Show that the density operator for thermal expectation values is $\rho(T) = e^{-\beta \mathcal{H}} / Z$
gilmore-lie-groups_F01310	137	■ 0.975	$R(e^{-\beta \mathcal{H}}) / Z = e^{-\beta R(\mathcal{H})} / Z$	$R(e^{-\beta \mathcal{H}}) / Z = e^{-\beta R(\mathcal{H})} / Z$	$R(e^{-\beta \mathcal{H}}) / Z = e^{-\beta R(\mathcal{H})} / Z$. What is Z ?
gilmore-lie-groups_F01311	137	■ 1.000	$U(t)$	$U(t)$	e. Show that the unitary time evolution operator $U(t)$ and the hermitian density
gilmore-lie-groups_F01312	137	■ 1.000	$\rho(T)$	$\rho(T)$	operator $\rho(T)$ are related by a Wick rotation $it/\hbar \leftrightarrow \beta = 1/k_B T$.
gilmore-lie-groups_F01313	137	■ 1.000	$it/\hbar \leftrightarrow \beta = 1/k_B T$	$it/\hbar \leftrightarrow \beta = 1/k_B T$	operator $\rho(T)$ are related by a Wick rotation $it/\hbar \leftrightarrow \beta = 1/k_B T$.
gilmore-lie-groups_F01314	137	■ 1.000	$G = e^{\mathfrak{g}}$	$G = e^{\mathfrak{g}}$	g. The operator product in the numerator is in the group $G = e^{\mathfrak{g}}$ or its complex exten-
gilmore-lie-groups_F01315	137	■ 1.000	H_i	H_i	in terms of the operators H_i the trace can easily be constructed. Show that for
gilmore-lie-groups_F01316	137	■ 1.000	x^i	x^i	x^i sufficiently small it is always possible to construct a similarity transformation
gilmore-lie-groups_F01317	137	■ 1.000	$S = e^{y^k X_k}$	$S = e^{y^k X_k}$	$S = e^{y^k X_k}$ with the property
gilmore-lie-groups_F01318	137	■ 1.000	$e^{x^i X_i}$	$e^{x^i X_i}$	function $e^{x^i X_i}$, and the Wick rotation.
gilmore-lie-groups_F01319	137	■ 1.000	$U(\alpha) = e^{(\alpha a^\dagger - \alpha^* a)}$	$U(\alpha) = e^{(\alpha a^\dagger - \alpha^* a)}$	The unitary transformation has the form $U(\alpha) = e^{(\alpha a^\dagger - \alpha^* a)}$, where $a^\dagger$ and a are the
gilmore-lie-groups_F01320	138	■ 0.896	$a^\dagger n\rangle =$	$a^\dagger n\rangle =$	Hilbert space and out of the nonunitary 3×3 matrix representation), $a^\dagger n\rangle =$
gilmore-lie-groups_F01321	138	■ 1.000	$ n+1\rangle \sqrt{n+1}$	$ n+1\rangle \sqrt{n+1}$	$ n+1\rangle \sqrt{n+1}$ to conclude
gilmore-lie-groups_F01322	138	■ 1.000	$\langle \beta \alpha \rangle$	$\langle \beta \alpha \rangle$	e. Compute the inner product $\langle \beta \alpha \rangle$ and show $\langle \alpha \alpha \rangle = 1$.
gilmore-lie-groups_F01323	138	■ 1.000	$\langle \alpha \alpha \rangle = 1$	$\langle \alpha \alpha \rangle = 1$	e. Compute the inner product $\langle \beta \alpha \rangle$ and show $\langle \alpha \alpha \rangle = 1$.
gilmore-lie-groups_F01324	138	■ 1.000	$a \alpha\rangle = \alpha \alpha\rangle$	$a \alpha\rangle = \alpha \alpha\rangle$	f. Show $a \alpha\rangle = \alpha \alpha\rangle$.
gilmore-lie-groups_F01325	138	■ 1.000	$\langle \alpha x \alpha \rangle = (\alpha^* + \alpha) / \sqrt{2}$	$\langle \alpha x \alpha \rangle = (\alpha^* + \alpha) / \sqrt{2}$	g. Show $\langle \alpha x \alpha \rangle = (\alpha^* + \alpha) / \sqrt{2}$
gilmore-lie-groups_F01326	138	■ 0.966	$e^{i\phi J_z} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle e^{-ij\phi}$	$e^{i\phi J_z} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle e^{-ij\phi}$	by a phase angle: $e^{i\phi J_z} \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle = \left \begin{smallmatrix} j \\ -j \end{smallmatrix} \right\rangle e^{-ij\phi}$. This simply “renormalizes” the fiducial

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gilmore-lie-groups_F01327	138	■ 1.000	$e^{i\left(\theta_x J_x + \theta_y J_y\right)}$	$e^{i(\theta_x J_x + \theta_y J_y)}$	c. Rewrite $e^{i(\theta_x J_x + \theta_y J_y)}$ in the form $e^{i\alpha_+ J_+} e^{i\alpha_z J_z} e^{i\alpha_- J_-}$ and compute the analytic relation
gilmore-lie-groups_F01328	138	■ 1.000	$e^{i\alpha_+ J_+} e^{i\alpha_z J_z} e^{i\alpha_- J_-}$	$e^{i\alpha_+ J_+} e^{i\alpha_z J_z} e^{i\alpha_- J_-}$	c. Rewrite $e^{i(\theta_x J_x + \theta_y J_y)}$ in the form $e^{i\alpha_+ J_+} e^{i\alpha_z J_z} e^{i\alpha_- J_-}$ and compute the analytic relation
gilmore-lie-groups_F01329	138	■ 0.859	$e^{i\alpha_- J_-} \left j, -j \right\rangle \left j, -j \right\rangle$	$e^{i\alpha_- J_-} \left j, -j \right\rangle = \left j, -j \right\rangle$	d. Show $e^{i\alpha_- J_-} \left j, -j \right\rangle = \left j, -j \right\rangle$.
gilmore-lie-groups_F01330	138	■ 0.952	$e^{i\alpha_z J_z} \left j, -j \right\rangle \left j, -j \right\rangle e^{-i\alpha_z J_z}$	$e^{i\alpha_z J_z} \left j, -j \right\rangle = \left j, -j \right\rangle e^{-ij\alpha_z}$	e. Show $e^{i\alpha_z J_z} \left j, -j \right\rangle = \left j, -j \right\rangle e^{-ij\alpha_z}$.
gilmore-lie-groups_F01331	139	■ 0.815	$J_- \left j, \theta_x, \theta_y \right\rangle$	$J_- \left j, \theta_x, \theta_y \right\rangle$	g. Show that $J_- \left j, \theta_x, \theta_y \right\rangle$ cannot be proportional to $\left j, \theta_x, \theta_y \right\rangle$ because the state $\left j, \theta_x, \theta_y \right\rangle$ is not
gilmore-lie-groups_F01332	139	■ 0.815	$\left j, \theta_x, \theta_y \right\rangle$	$\left j, \theta_x, \theta_y \right\rangle$	g. Show that $J_- \left j, \theta_x, \theta_y \right\rangle$ cannot be proportional to $\left j, \theta_x, \theta_y \right\rangle$ because the state $\left j, \theta_x, \theta_y \right\rangle$ is not
gilmore-lie-groups_F01333	139	■ 0.815	$\left j, \theta_x, \theta_y \right\rangle$	$\left j, \theta_x, \theta_y \right\rangle$	g. Show that $J_- \left j, \theta_x, \theta_y \right\rangle$ cannot be proportional to $\left j, \theta_x, \theta_y \right\rangle$ because the state $\left j, \theta_x, \theta_y \right\rangle$ is not
gilmore-lie-groups_F01334	139	■ 1.000	H_4	H_4	irreducible representations and the harmonic oscillator group H_4 is not compact
gilmore-lie-groups_F01335	139	■ 1.000	$e^{-i\phi} = (\theta_x - i\theta_y) / \theta$	$e^{-i\phi} = (\theta_x - i\theta_y) / \theta$	where $e^{-i\phi} = (\theta_x - i\theta_y) / \theta$, and similarly for θ' (cf., Eq. (7.46)).
gilmore-lie-groups_F01336	139	■ 1.000	θ'	θ'	where $e^{-i\phi} = (\theta_x - i\theta_y) / \theta$, and similarly for θ' (cf., Eq. (7.46)).
gilmore-lie-groups_F01337	139	■ 1.000	$ u\rangle$	$ u\rangle$	sometimes does) represent a charge, and $ u\rangle$ is an eigenfunction. In this toy example,
gilmore-lie-groups_F01338	139	■ 1.000	J_3	J_3	the operators J_3 and J_1 are assumed to belong to the Lie algebra $\mathfrak{su}(2)$ and the equation
gilmore-lie-groups_F01339	139	■ 1.000	$(E' J_3 - Z) v\rangle = 0$	$(E' J_3 - Z) v\rangle = 0$	$(E' J_3 - Z) v\rangle = 0$, where $E' = \sqrt{E^2 + p^2}$ and $ v\rangle = U u\rangle$.
gilmore-lie-groups_F01340	139	■ 1.000	$E' = \sqrt{E^2 + p^2}$	$E' = \sqrt{E^2 + p^2}$	$(E' J_3 - Z) v\rangle = 0$, where $E' = \sqrt{E^2 + p^2}$ and $ v\rangle = U u\rangle$.
gilmore-lie-groups_F01341	139	■ 1.000	$ v\rangle = U u\rangle$	$ v\rangle = U u\rangle$	$(E' J_3 - Z) v\rangle = 0$, where $E' = \sqrt{E^2 + p^2}$ and $ v\rangle = U u\rangle$.
gilmore-lie-groups_F01342	139	■ 1.000	$E = \pm \sqrt{(Z/m)^2 - p^2}$	$E = \pm \sqrt{(Z/m)^2 - p^2}$	b. Show that $E = \pm \sqrt{(Z/m)^2 - p^2}$.
gilmore-lie-groups_F01343	139	■ 0.871	$(p \rightarrow 0)$	$(p \rightarrow 0)$	c. Compare this spectrum with the unperturbed spectrum ($p \rightarrow 0$).
gilmore-lie-groups_F01344	139	■ 1.000	j, p, Z	j, p, Z	d. Under what conditions on j, p, Z are these solutions valid?
gilmore-lie-groups_F01345	139	■ 0.986	$ u\rangle = e^{i\theta J_2} \left j, m \right\rangle$	$ u\rangle = e^{i\theta J_2} \left j, m \right\rangle$	and show that $ u\rangle = e^{i\theta J_2} \left j, m \right\rangle$. Compute θ for each E .
gilmore-lie-groups_F01346	139	■ 1.000	$z = \cos(\beta)$	$z = \cos(\beta)$	where $z = \cos(\beta)$. The Wigner matrix elements D_{mn}^j are related to the Jacobi poly-
gilmore-lie-groups_F01347	139	■ 1.000	D_{mn}^j	D_{mn}^j	where $z = \cos(\beta)$. The Wigner matrix elements D_{mn}^j are related to the Jacobi poly-
gilmore-lie-groups_F01348	139	■ 1.000	$j = l$	$j = l$	nomials when $j = l$, where l is an integer.
gilmore-lie-groups_F01349	140	■ 1.000	$I_2 \in SU(2)$	$I_2 \in SU(2)$	a. Geodesics through $I_2 \in SU(2)$ focus at $-I_2$ and geodesics through $I_3 \in SO(3)$
gilmore-lie-groups_F01350	140	■ 1.000	$I_3 \in SO(3)$	$I_3 \in SO(3)$	a. Geodesics through $I_2 \in SU(2)$ focus at $-I_2$ and geodesics through $I_3 \in SO(3)$
gilmore-lie-groups_F01351	140	■ 0.835	I_3	I_3	focus at I_3 . Conclude that $SU(2)$ is a two-fold covering group of $SO(3)$.
gilmore-lie-groups_F01352	140	■ 0.999	$SU(2)/U(1) (z=1, x=y=0)$	$SU(2)/U(1) (z=1, x=y=0)$	b. Geodesics through the “north pole” of $SU(2)/U(1) (z=1, x=y=0)$ focus at

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gilmore-lie-groups_ F01353	140	■ 0.793	$z=-1, x=y=0$	$z = -1, x = y = 0$	its “south pole” ($\underline{z = -1, x = y = 0}$) and geodesics through the north pole of
gilmore-lie-groups_ F01354	140	■ 0.701	$S^0(3) / S^0(2) (z=1, x=y=0)$	$SO(3)/SO(2)(z = 1, x = y = 0)$	$SO(3)/SO(2) (z = 1, x = y = 0)$ focus at its south pole ($z = -1, x = y = 0$).
gilmore-lie-groups_ F01355	140	■ 0.701	$(z=-1, x=y=0)$	$(z = -1, x = y = 0)$	$SO(3)/SO(2) (z = 1, x = y = 0)$ focus at its south pole ($\underline{z = -1, x = y = 0}$).
gilmore-lie-groups_ F01356	140	■ 1.000	$S^U(2) / U(1)=S^2=S^0(3) / S^0(2)$	$SU(2)/U(1) = S^2 = SO(3)/SO(2)$	c. Conclude that $\underline{SU(2)/U(1) = S^2 = SO(3)/SO(2)}$ and the $2 \rightarrow 1$ nature of the
gilmore-lie-groups_ F01357	140	■ 0.986	$S^U(2) \searrow S^0(3)$	$SU(2) \downarrow SO(3)$	covering $SU(2) \downarrow SO(3)$ is contained in the subgroup of rotations about the z -
gilmore-lie-groups_ F01358	140	■ 0.918	$U(1) \searrow S^0(2)$	$U(1) \downarrow SO(2)$	axis $\underline{U(1) \downarrow SO(2)}$:
gilmore-lie-groups_ F01359	140	■ 1.000	$e^{2\pi i k / r} I_n$	$e^{2\pi i k / r} I_n$	order r , with group elements $e^{2\pi i k / r} I_n$, with n/r integer. Compute the foci in $SU(n)$.
gilmore-lie-groups_ F01360	140	■ 1.000	n / r	n/r	order r , with group elements $e^{2\pi i k / r} I_n$, with n/r integer. Compute the foci in $SU(n)$.
gilmore-lie-groups_ F01361	140	■ 1.000	$e^{2\pi i k / n} I_n$	$e^{2\pi i k / n} I_n$	How are the foci related to the group operations of the form $e^{2\pi i k / n} I_n$?
gilmore-lie-groups_ F01362	140	■ 0.999	$\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1/\lambda \end{array}\right]$	$\begin{bmatrix} -\lambda & 0 \\ 0 & -1/\lambda \end{bmatrix}$	14. Show that the matrix $\begin{bmatrix} -\lambda & 0 \\ 0 & -1/\lambda \end{bmatrix}$ in $SL(2; \mathbb{R})$ cannot be reached by exponentiating
gilmore-lie-groups_ F01363	140	■ 1.000	$\lambda > 1$	$\lambda > 1$	any element in the Lie algebra if $\lambda > 1$. Show that it can be reached by following a
gilmore-lie-groups_ F01364	140	■ 1.000	$\hbar \omega$	$\hbar \omega$	photon has energy $\hbar \omega$. The hamiltonian is chosen as
gilmore-lie-groups_ F01365	140	■ 1.000	$\sigma_z^{(i)}$	$\sigma_z^{(i)}$	The operator $\sigma_z^{(i)}$ describes the two states of atom i and the operator $a^\dagger a$ describes
gilmore-lie-groups_ F01366	140	■ 1.000	$a^\dagger a$	$a^\dagger a$	The operator $\sigma_z^{(i)}$ describes the two states of atom i and the operator $a^\dagger a$ describes
gilmore-lie-groups_ F01367	140	■ 0.999	$\sigma_+^{(j)} \left(\sigma_\pm^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	$\sigma_+^{(j)} \left(\sigma_\pm^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	the number of photons in the field mode. The operator $\sigma_\pm^{(j)} (\sigma_\pm^{(j)}) = \frac{1}{2} (\sigma_x^{(j)} \pm i \sigma_y^{(j)})$
gilmore-lie-groups_ F01368	140	■ 1.000	$\sigma_-^{(j)} a^\dagger$	$\sigma_-^{(j)} a^\dagger$	operator $\sigma_-^{(j)} a^\dagger$, describes deexcitation of an atom with emission (creation, $a^\dagger$) of a
gilmore-lie-groups_ F01369	140	■ 1.000	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	b. If all the atoms behave cooperatively it is possible to replace $\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$,
gilmore-lie-groups_ F01370	140	■ 0.999	$\sum_{i=1}^N \sigma_\pm^{(i)} \rightarrow J_\pm$	$\sum_{i=1}^N \sigma_\pm^{(i)} \rightarrow J_\pm$	$\sum_{i=1}^N \sigma_\pm^{(i)} \rightarrow J_\pm$. Show that the operators $J_z, J_\pm$ satisfy the usual $\mathfrak{su}(2)$ commuta-
gilmore-lie-groups_ F01371	140	■ 0.999	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	$\sum_{i=1}^N \sigma_\pm^{(i)} \rightarrow J_\pm$. Show that the operators $J_z, J_\pm$ satisfy the usual $\mathfrak{su}(2)$ commuta-
gilmore-lie-groups_ F01372	141	■ 1.000	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$\underline{J_- \rightarrow \langle J_-(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*}$. Show that this semiclas-
gilmore-lie-groups_ F01373	141	■ 1.000	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	of the electromagnetic field: $\underline{ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle}$. Use the disentan-
gilmore-lie-groups_ F01374	141	■ 1.000	$\alpha(t)$	$\alpha(t)$	gling theorems to compute the relation between the coherent state parameter $\alpha(t)$
gilmore-lie-groups_ F01375	141	■ 1.000	$\langle J_z(t) \rangle$	$\langle J_z(t) \rangle$	and the classical driving fields $\langle J_z(t) \rangle$ and $\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$.
gilmore-lie-groups_ F01376	141	■ 1.000	$\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$	$\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$	and the classical driving fields $\langle J_z(t) \rangle$ and $\underline{\langle J_+(t) \rangle = \langle J_-(t) \rangle^*}$.
gilmore-lie-groups_ F01377	141	■ 1.000	$ \beta\rangle$	$ \beta\rangle$	state $ \beta\rangle$, the state obtained by the action of the classical current is still a coherent

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gilmore-lie-groups_F01378	141	■ 1.000	\beta	β	state. How are the parameters β , describing the initial condition, and α , describing
gilmore-lie-groups_F01379	141	■ 1.000	$a^{\dagger} \rightarrow \langle a(t) \rangle^*$	$a^{\dagger} \rightarrow \langle a(t) \rangle^*$	considered classically. Show that this amounts to the substitutions $a^{\dagger} \rightarrow \langle a(t) \rangle^*$,
gilmore-lie-groups_F01380	141	■ 1.000	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$, and $a^{\dagger} a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$.
gilmore-lie-groups_F01381	141	■ 1.000	$a^{\dagger} a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a^{\dagger} a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$, and $a^{\dagger} a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$.
gilmore-lie-groups_F01382	141	■ 1.000	$m = -\frac{1}{2}$	$m = -\frac{1}{2}$	ground state ($m = -\frac{1}{2}$ for each atom, or $M = -J$, $J = N/2$ for the ensemble of
gilmore-lie-groups_F01383	141	■ 1.000	$M = -J$, $J = N/2$	$M = -J$, $J = N/2$	ground state ($m = -\frac{1}{2}$ for each atom, or $M = -J$, $J = N/2$ for the ensemble of
gilmore-lie-groups_F01384	141	■ 0.997	$ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$	$ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$	evolve into a coherent state: $ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$, where $J = N/2$. How are the
gilmore-lie-groups_F01385	141	■ 0.997	$J = N/2$	$J = N/2$	evolve into a coherent state: $ \theta(t)\rangle = e^{i\theta(t)\mathbf{J}} J, -J\rangle$, where $J = N/2$. How are the
gilmore-lie-groups_F01386	141	■ 1.000	$\theta(t)$	$\theta(t)$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_F01387	141	■ 1.000	$\langle a(t) \rangle$	$\langle a(t) \rangle$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_F01388	141	■ 1.000	$\langle a(t) \rangle^*$	$\langle a(t) \rangle^*$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_F01389	141	■ 1.000	$\langle a \rangle$	$\langle a \rangle$	field expectation values $\langle a \rangle$ and $\langle a \rangle^*$. Conversely, the semiclassical hamiltonian
gilmore-lie-groups_F01390	141	■ 1.000	$\langle a \rangle^*$	$\langle a \rangle^*$	field expectation values $\langle a \rangle$ and $\langle a \rangle^*$. Conversely, the semiclassical hamiltonian
gilmore-lie-groups_F01391	141	■ 1.000	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$. Construct a self-consistent model by requiring that both sets of
gilmore-lie-groups_F01392	141	■ 1.000	$T(\beta = 1/k_B T)$	$T(\beta = 1/k_B T)$	equilibrium at temperature T ($\beta = 1/k_B T$).
gilmore-lie-groups_F01393	141	■ 1.000	$\langle \sigma_+^{(i)} \rangle_T$	$\langle \sigma_+^{(i)} \rangle_T$	a. Assume $\langle \sigma_+^{(i)} \rangle_T$ has some fixed unknown value, and similarly for the other atomic
gilmore-lie-groups_F01394	141	■ 1.000	$a^{\dagger}, a, a^{\dagger} a$	$a^{\dagger}, a, a^{\dagger} a$	expectation values for the operators $a^{\dagger}, a, a^{\dagger} a$.
gilmore-lie-groups_F01395	142	■ 1.000	$\sigma_z, \sigma_+, \sigma_-$	$\sigma_z, \sigma_+, \sigma_-$	operators $\sigma_z, \sigma_+, \sigma_-$.
gilmore-lie-groups_F01396	142	■ 1.000	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$ is there a nontrivial solution (e.g., $\langle J_+ \rangle_T \neq 0$)?
gilmore-lie-groups_F01397	142	■ 1.000	$\langle J_+ \rangle_T \neq 0$	$\langle J_+ \rangle_T \neq 0$	$\epsilon, \hbar\omega, \lambda, N$ is there a nontrivial solution (e.g., $\langle J_+ \rangle_T \neq 0$)?
gilmore-lie-groups_F01398	142	■ 1.000	$\lambda^2 / \epsilon \hbar \omega$	$\lambda^2 / \epsilon \hbar \omega$	e. Show that a thermodynamic phase transition occurs as $\lambda^2 / \epsilon \hbar \omega$ increases through
gilmore-lie-groups_F01399	142	■ 1.000	$a(t), b(t)$	$a(t), b(t)$	17. The two complex parameters $a(t), b(t)$ in the evolution equation (7.58) can be ex-
gilmore-lie-groups_F01400	142	■ 1.000	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	pressed in terms of their real and imaginary parts. These obey $a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$
gilmore-lie-groups_F01401	142	■ 0.542	(a_r, a_i, b_r, b_i)	(a_r, a_i, b_r, b_i)	compute the length of the vector (a_r, a_i, b_r, b_i) and scale this length back to +1.
gilmore-lie-groups_F01402	142	■ 1.000	$\lambda_3 \rightarrow 0$	$\lambda_3 \rightarrow 0$	Simplify (7.67) by taking the limit $\lambda_3 \rightarrow 0$. Simplify (7.70) by taking the limit $\lambda_n \rightarrow 0$
gilmore-lie-groups_F01403	142	■ 1.000	$\lambda_n \rightarrow 0$	$\lambda_n \rightarrow 0$	Simplify (7.67) by taking the limit $\lambda_3 \rightarrow 0$. Simplify (7.70) by taking the limit $\lambda_n \rightarrow 0$
gilmore-lie-groups_F01404	142	■ 1.000	$d = \delta = 0$	$d = \delta = 0$	and setting $d = \delta = 0$.

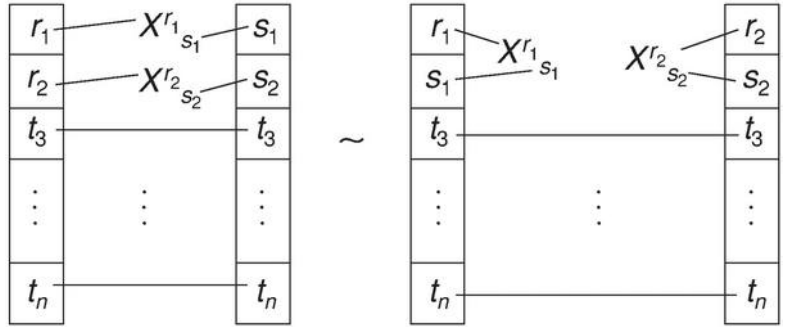
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gilmore-lie-groups_ F01485	142	■ 1.000	$\left\langle J_+ J_- + J_- J_+ \right\rangle$	$\langle J_+ J_- + J_- J_+ \rangle$	<i>metric</i> operator products such as $\langle J_+ J_- + J_- J_+ \rangle$ or $\langle a a^\dagger + a^\dagger a \rangle$ than it is to construct
gilmore-lie-groups_ F01486	142	■ 1.000	$\left\langle a a^{\dagger} + a^{\dagger} a \right\rangle$	$\langle a a^\dagger + a^\dagger a \rangle$	<i>metric</i> operator products such as $\langle J_+ J_- + J_- J_+ \rangle$ or $\langle a a^\dagger + a^\dagger a \rangle$ than it is to construct
gilmore-lie-groups_ F01487	142	■ 1.000	$\left\langle J_+ J_- \right\rangle$	$\langle J_+ J_- \rangle$	generating functions for <i>ordered</i> products of operators such as $\langle J_+ J_- \rangle$ or $\langle a^\dagger a \rangle$. Show
gilmore-lie-groups_ F01488	142	■ 1.000	$\left\langle a^{\dagger} a \right\rangle$	$\langle a^\dagger a \rangle$	generating functions for <i>ordered</i> products of operators such as $\langle J_+ J_- \rangle$ or $\langle a^\dagger a \rangle$. Show
gilmore-lie-groups_ F01489	143	■ 0.749	$X_{\{j\}} = A_{\{j\}}^s Y_{\{s\}}$	$X_j = A_j^s Y_s$	Under a change of basis $X_j = A_j^s Y_s$ this $n \times n$ matrix undergoes a similarity trans-
gilmore-lie-groups_ F01418	144	■ 1.000	$N = p \ q$	$N = pq$	This is an $n \times n$ ($n = p + q$) matrix algebra which is $N = pq$ dimensional. The
gilmore-lie-groups_ F01411	144	■ 1.000	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	3. $\mathfrak{so}(n)$ Upper triangular In this case nonzero elements occur on and above the
gilmore-lie-groups_ F01412	145	■ 1.000	$\hat{n} =$	$\hat{n} =$	Example We consider the algebra consisting of the six photon operators $\hat{n} =$
gilmore-lie-groups_ F01413	145	■ 0.993	$\frac{1}{2} \left\{ a, a^{\dagger} \right\} = a^{\dagger} a + \frac{1}{2}, a^{\dagger 2}, a^2, a^{\dagger}, a, I = [a, a^{\dagger}]$	$\frac{1}{2} \{a, a^\dagger\} = a^\dagger a + \frac{1}{2}, a^{\dagger 2}, a^2, a^\dagger, a, I = [a, a^\dagger]$	$\frac{1}{5} \{a, a^\dagger\} = a^\dagger a + \frac{1}{5}, a^{\dagger 2}, a^2, a^\dagger, a, I = [a, a]$. This is isomorphic to the algebra
gilmore-lie-groups_ F01414	145	■ 1.000	$X_{\{i \ j\}} = -X_{\{j \ i\}}, \ 1 \leq i, \ j \leq 4$	$X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$	Example The Lie algebra $\mathfrak{so}(4)$ has six generators $X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$ and
gilmore-lie-groups_ F01415	146	■ 1.000	$\mathfrak{so}(n), \ n > 4$	$\mathfrak{so}(n), n > 4$	other orthogonal Lie algebras, $\mathfrak{so}(n), n > 4$. We will have to wait until Chapter 10
gilmore-lie-groups_ F01416	146	■ 0.999	$\mathfrak{su}(n) (n \geq 2), \ \mathfrak{so}(n) (n > 4)$	$\mathfrak{su}(n) (n \geq 2), \mathfrak{so}(n) (n > 4)$	Example The Lie algebras $\mathfrak{su}(n) (n > 2), \mathfrak{so}(n) (n > 4)$, and $\mathfrak{sp}(n) (n \geq 1)$ are
gilmore-lie-groups_ F01417	146	■ 0.999	$\mathfrak{sp}(n) (n \geq 1)$	$\mathfrak{sp}(n) (n \geq 1)$	Example The Lie algebras $\mathfrak{su}(n) (n \geq 2), \mathfrak{so}(n) (n > 4)$, and $\mathfrak{sp}(n) (n \geq 1)$ are
gilmore-lie-groups_ F01418	147	■ 0.999	$\left[Z, X_i \right]$	$[Z, X_i]$	This means that $[Z, X_i]$ can be expressed as a linear combination of basis vectors
gilmore-lie-groups_ F01419	147	■ 1.000	$X_{\{j\}}$	X_j	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01420	147	■ 1.000	$j \geq i$	$j \geq i$	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01421	147	■ 1.000	$X_{\{i\}}, X_{\{i+1\}}, \ldots, X_{\{n\}}$	$X_i, X_{i+1}, \dots, X_n$	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01422	147	■ 0.593	$i=1,2, \ldots, n$	$i = 1, 2, \dots, n$	subalgebra for each value of $i = 1, 2, \dots, n$. Since this subalgebra is mapped onto
gilmore-lie-groups_ F01423	147	■ 1.000	$V_{\{i\}}$	V_i	Each V_i is an invariant subalgebra in $V_j, i > j$. The original algebra is V_1 .
gilmore-lie-groups_ F01424	147	■ 1.000	$V_{\{j\}}, \ i > j$	$V_j, i > j$	Each V_i is an invariant subalgebra in $V_j, i > j$. The original algebra is V_1 .
gilmore-lie-groups_ F01425	148	■ 1.000	$\left[V_{\{2\}}, V_{\{2\}} \right] \subseteq V_{\{2\}}, V_{\{2\}}$	$[V_2, V_2] \subseteq V_2, V_2$	Since in particular $[V_2, V_2] \subseteq V_2, V_2$ is a subalgebra in the original algebra. Further,
gilmore-lie-groups_ F01426	148	■ 1.000	$V_{\{2\}}, V_{\{2\}}$	V_2, V_2	since the commutator of anything in the original algebra with V_2 is in V_2 , V_2 is an
gilmore-lie-groups_ F01427	149	■ 1.000	$\mathfrak{g}^{\prime} = \mathfrak{g} - V_{\{0\}}$	$\mathfrak{g}' = \mathfrak{g} - V_0$	(iv) Form the difference $\mathfrak{g}' = \mathfrak{g} - V_0$. This is a Lie algebra (under the “mod” operation: set
gilmore-lie-groups_ F01428	149	■ 1.000	$\mathfrak{g}^{\prime}$	$\mathfrak{g}'$	Cartan–Killing inner product for $\mathfrak{g}'$.
gilmore-lie-groups_ F01429	150	■ 1.000	$V_{\{0\}}^{\prime} = 0$	$V'_0 = 0$	In this case $V'_0 = 0$ and the two photon algebra has the decomposition
gilmore-lie-groups_ F01438	150	■ 1.000	$a^{\dagger 2}$	$a^{\dagger 2}$	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since

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gilmore-lie-groups_ F01431	150	■ 1.000	$a^{\{2\}}$	a^2	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01432	150	■ 1.000	$a^{\{\dagger 2\}} + a^{\{2\}}$	$a^{\dagger 2} + a^2$	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01433	150	■ 1.000	$a^{\{\dagger\}} a + \frac{1}{2}$	$a^{\dagger} a + \frac{1}{2}$	the Cartan–Killing form is negative-definite on it. The other two operators, $a^{\dagger} a + \frac{1}{2}$
gilmore-lie-groups_ F01434	150	■ 1.000	$a^{\{\dagger 2\}} - a^{\{2\}}$	$a^{\dagger 2} - a^2$	and $a^{\dagger 2} - a^2$, are noncompact.
gilmore-lie-groups_ F01435	151	■ 1.000	$\hat{n}, a^{\{\dagger\}}, a, I$	$\hat{n}, a^{\dagger}, a, I$	a. The photon algebra $\hat{n}, a^{\dagger}, a, I$ (Eq. (8.5)).
gilmore-lie-groups_ F01436	151	■ 0.786	$\left[b_i, b_j^{\{\dagger\}}\right] = I \delta_{ij}, 1 \leq i, j \leq n$	$[b_i, b_j^{\dagger}] = I \delta_{ij}, 1 \leq i, j \leq n$	$[b_i, b_j^{\dagger}] = I \delta_{ij}, 1 \leq i, j \leq n$. Commutators involving bilinear (trilinear, ...) products
gilmore-lie-groups_ F01437	151	■ 1.000	$b_{\{i\}}, b_{\{j\}}^{\{\dagger\}}, I$	$b_i, b_j^{\dagger}, I$	a. $b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01438	151	■ 1.000	$b_{\{i\}}^{\{\dagger\}} b_{\{j\}}, b_{\{i\}}, b_{\{j\}}^{\{\dagger\}}, I$	$b_i^{\dagger} b_j, b_i, b_j^{\dagger}, I$	c. $b_i^{\dagger} b_j, b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01439	151	■ 1.000	$b_{\{i\}}^{\{\dagger\}} b_{\{j\}}^{\{\dagger\}}, b_{\{i\}}^{\{\dagger\}} b_{\{j\}} + \frac{1}{2} \delta_{ij}, b_i b_j$	$b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{ij}, b_i b_j$	d. $b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{ij}, b_i b_j$.
gilmore-lie-groups_ F01440	151	■ 0.998	$b_{\{i\}}^{\{\dagger\}} b_{\{j\}}^{\{\dagger\}}, b_{\{i\}}^{\{\dagger\}} b_{\{j\}} + \frac{1}{2} \delta_{ij}, b_i b_j, b_i, b_j^{\dagger}, I$	$b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{ij}, b_i b_j, b_i, b_j^{\dagger}, I$	e. $b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j + \frac{1}{2} \delta_{ij}, b_i b_j, b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01441	151	■ 1.000	$b, b^{\{\dagger\}}, b, b^{\{\dagger\}} b^{\{\dagger\}} b$	$b, b^{\dagger} b, b^{\dagger} b^{\dagger} b$	f. $b, b^{\dagger} b, b^{\dagger} b^{\dagger} b$.
gilmore-lie-groups_ F01442	151	■ 0.788	$b^{\{\dagger\}}$	$b^{\dagger}$	g. $b^{\dagger}, b^{\dagger} b, b^{\dagger} b b$.
gilmore-lie-groups_ F01443	151	■ 0.788	$b^{\{\dagger\}} b, b^{\{\dagger\}} b b$	$b^{\dagger} b, b^{\dagger} b b$	g. $b^{\dagger}, b^{\dagger} b, b^{\dagger} b b$.
gilmore-lie-groups_ F01444	151	■ 1.000	$\left\{f_i, f_j^{\{\dagger\}}\right\} = \delta_{ij}$	$\{f_i, f_j^{\dagger}\} = \delta_{ij}$	$\{f_i, f_j^{\dagger}\} = \delta_{ij}$, but their bilinear combinations close under the same commutation re-
gilmore-lie-groups_ F01445	151	■ 0.985	$f_{\{i\}}^{\{\dagger\}} f_{\{j\}}^{\{\dagger\}}$	$f_i^{\dagger} f_j^{\dagger}$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j + \frac{1}{2} \delta_{ij}, f_i f_j$.
gilmore-lie-groups_ F01446	151	■ 0.985	$f_{\{i\}}^{\{\dagger\}} f_{\{j\}} + \frac{1}{2} \delta_{ij}$	$f_i^{\dagger} f_j + \frac{1}{2} \delta_{ij}$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j + \frac{1}{2} \delta_{ij}, f_i f_j$.
gilmore-lie-groups_ F01447	151	■ 0.985	$f_{\{i\}} f_{\{j\}}$	$f_i f_j$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j + \frac{1}{2} \delta_{ij}, f_i f_j$.
gilmore-lie-groups_ F01448	151	■ 1.000	$\left(x_i\right)$	(x_i)	of the position (x_i) and momentum (∂_j) operators for n independent directions:
gilmore-lie-groups_ F01449	151	■ 1.000	$\left(\partial_j\right)$	(∂_j)	of the position (x_i) and momentum (∂_j) operators for n independent directions:
gilmore-lie-groups_ F01450	151	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}, I$	x_i, ∂_j, I	a. x_i, ∂_j, I .
gilmore-lie-groups_ F01451	151	■ 1.000	$x_{\{i\}} \partial_{\{j\}}$	$x_i \partial_j$	b. $x_i \partial_j$.
gilmore-lie-groups_ F01452	151	■ 1.000	$x_{\{i\}} \partial_{\{j\}}, x_{\{i\}}, \partial_{\{j\}}, I$	$x_i \partial_j, x_i, \partial_j, I$	c. $x_i \partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01453	151	■ 0.991	$x_{\{i\}} x_{\{j\}}, x_{\{i\}} \partial_{\{j\}} + \frac{1}{2} I \delta_{ij}, \partial_i \partial_j$	$x_i x_j, x_i \partial_j + \frac{1}{2} I \delta_{ij}, \partial_i \partial_j$	d. $x_i x_j, x_i \partial_j + \frac{1}{2} I \delta_{ij}, \partial_i \partial_j$.
gilmore-lie-groups_ F01454	151	■ 1.000	$x_{\{i\}} x_{\{j\}}, x_{\{i\}} \partial_{\{j\}}, \partial_{\{i\}} \partial_{\{j\}}, x_{\{i\}}, \partial_{\{j\}}, I$	$x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$	e. $x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01455	151	■ 1.000	$\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$	$\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$	f. $\frac{d}{dx}, x \frac{d}{dx}, x^2 \frac{d}{dx}$.

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gilmore-lie-groups_ F01456	151	■ 1.000	$x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$	$x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$	g. $x, x \frac{d}{dx}, x \frac{d^2}{dx^2}$.
gilmore-lie-groups_ F01457	151	■ 0.799	$2+1$	$2 + 1$	6. The Lorentz, Poincaré, and Galilei groups in $2 + 1$ dimensions (x, y and t) have Lie
gilmore-lie-groups_ F01458	151	■ 0.799	$(x, y$	$(x, y$	6. The Lorentz, Poincaré, and Galilei groups in $2 + 1$ dimensions (x, y and t) have Lie
gilmore-lie-groups_ F01459	151	■ 0.799	$t)$	$t)$	6. The Lorentz, Poincaré, and Galilei groups in $2 + 1$ dimensions (x, y and t) have Lie
gilmore-lie-groups_ F01463	152	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}$	x_i, ∂_j	d. For each Lie algebra, express the generators in terms of the operators x_i, ∂_j .
gilmore-lie-groups_ F01464	152	■ 0.781	$b_{\{i\}}^{\dagger}, b_{\{j\}}$	$b_i^{\dagger}, b_j$	e. For each Lie algebra, express the generators in terms of the boson operators $b_i^{\dagger}, b_j$,
gilmore-lie-groups_ F01465	152	■ 1.000	$g_{\{i\ j\}}=C_{\{i\ r\}}^{\{s\}} C_{\{j\ s\}}^{\{r\}}$	$g_{ij} = C_{ir}^s C_{js}^r$	7. In a semisimple Lie algebra the Cartan–Killing metric $g_{ij} = C_{ir}^s C_{js}^r$ is nonsingular and
gilmore-lie-groups_ F01466	152	■ 1.000	$g^{\{i\ j\}}$	g^{ij}	therefore the contravariant metric g^{ij} is well defined. Show that the bilinear operator
gilmore-lie-groups_ F01467	152	■ 0.940	$\mathcal{C}^2=g^{\{i\ j\}} X_{\{i\}} X_{\{j\}}$	$\mathcal{C}^2 = g^{ij} X_i X_j$	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $[\mathcal{C}^2, X_k] = 0$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups_ F01468	152	■ 0.940	$\left[\mathcal{C}^2, X_k\right]=0$	$[\mathcal{C}^2, X_k] = 0$	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $\boxed{[\mathcal{C}^2, X_k] = 0}$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups_ F01469	152	■ 1.000	$\mathcal{C}^2$	$\mathcal{C}^2$	therefore be proportional to $\mathcal{C}^2$.
gilmore-lie-groups_ F01470	152	■ 1.000	$C_{\{i\ j\ k\}}=C_{\{i\ j\}}^{\{r\}} g_{\{r\ k\}}$	$C_{ijk} = C_{ij}^r g_{rk}$	8. Show that $C_{ijk} = C_{ij}^r g_{rk}$ is a third order antisymmetric tensor: $C_{ijk} = C_{jki} = C_{kij} =$
gilmore-lie-groups_ F01471	152	■ 1.000	$C_{\{i\ j\ k\}}=C_{\{j\ k\ i\}}=C_{\{k\ i\ j\}}=$	$C_{ijk} = C_{jki} = C_{kij} =$	8. Show that $C_{ijk} = C_{ij}^r g_{rk}$ is a third order antisymmetric tensor: $C_{ijk} = C_{jki} = C_{kij} =$
gilmore-lie-groups_ F01472	152	■ 1.000	$-C_{\{k\ j\ i\}}=-C_{\{j\ i\ k\}}=-C_{\{i\ k\ j\}}$	$-C_{kji} = -C_{jik} = -C_{ikj}$	$-C_{kji} = -C_{jik} = -C_{ikj}$. (Hint: use the Jacobi identity.)
gilmore-lie-groups_ F01473	154	■ 0.989	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$) have the same complex extension and have their different commutation
gilmore-lie-groups_ F01474	154	■ 1.000	Z, X	Z, X	It would be very useful to find vectors Z, X in the Lie algebra that obeyed the
gilmore-lie-groups_ F01475	154	■ 1.000	$X=\sum_{i=1}^N a^{\{i\}} X_{\{i\}}$	$X = \sum_{i=1}^N a^i X_i$	we write $X = \sum_{i=1}^{\tilde{N}} a^i X_i$, where X_i form a basis set. Then
gilmore-lie-groups_ F01476	154	■ 1.000	$a^{\{i\}}$	a^i	This equation has a nonzero solution for the coefficients a^i provided the secular
gilmore-lie-groups_ F01477	154	■ 1.000	$\phi_{\{j\}}(Z)$	$\phi_j(Z)$	coefficients $\phi_j(Z)$ are homogeneous polynomials of degree j in the coefficients
gilmore-lie-groups_ F01478	154	■ 0.973	$z^{\{i\}}\left(Z=\sum z^{\{i\}} X_{\{i\}}\right)$	$z^i \left(Z = \sum z^i X_i \right)$	$\boxed{z^i (Z = \sum z^i X_i)}$ that describe Z :
gilmore-lie-groups_ F01479	155	■ 1.000	$a^{\dagger}, a, I=\left[a, a^{\dagger}\right]$	$a^{\dagger}, a, I = [a, a^{\dagger}]$	$\boxed{a^{\dagger}, a, I = [a, a^{\dagger}]}$ is
gilmore-lie-groups_ F01480	155	■ 1.000	$(-\lambda)^N=0$	$(-\lambda)^N = 0$	The converse is true. If the secular equation of an $N \times N$ matrix is $(-\lambda)^N = 0$, then
gilmore-lie-groups_ F01481	155	■ 0.852	$X=\sum a_{\{i\}} X_{\{i\}} \in \mathfrak{su}(2)$	$X = \sum a_i X_i \in \mathfrak{su}(2)$	Example For $X = \sum a_i X_i \in \mathfrak{su}(2)$ the defining 2×2 matrix representation
gilmore-lie-groups_ F01482	155	■ 0.953	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(X)$ are
gilmore-lie-groups_ F01483	155	■ 0.953	$\mathfrak{Reg}(X)$	$\mathfrak{Reg}(X)$	$\mathfrak{def}(X)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(X)$ are
gilmore-lie-groups_ F01484	155	■ 1.000	$\phi_2(\mathbf{a}) \geq 0$	$\phi_2(\mathbf{a}) \geq 0$	Since $\phi_2(\mathbf{a}) \geq 0$, this secular equation cannot be solved over the field of real num-

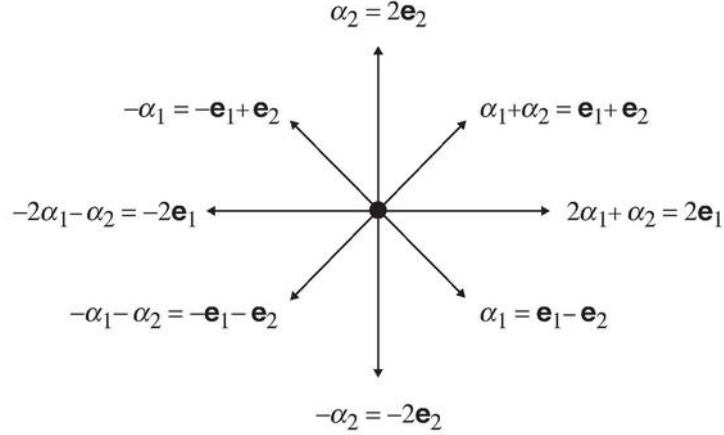
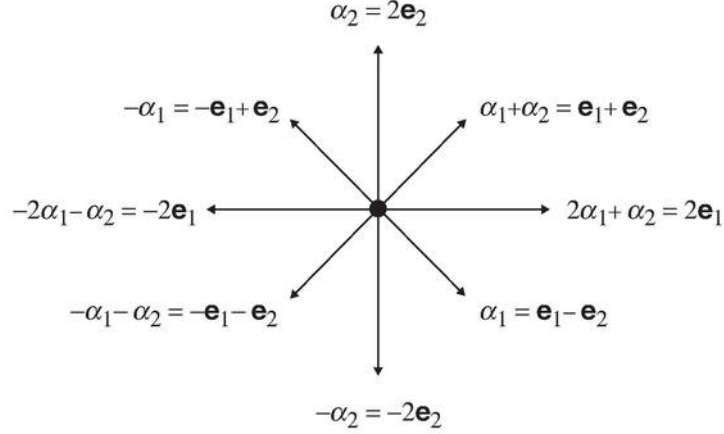
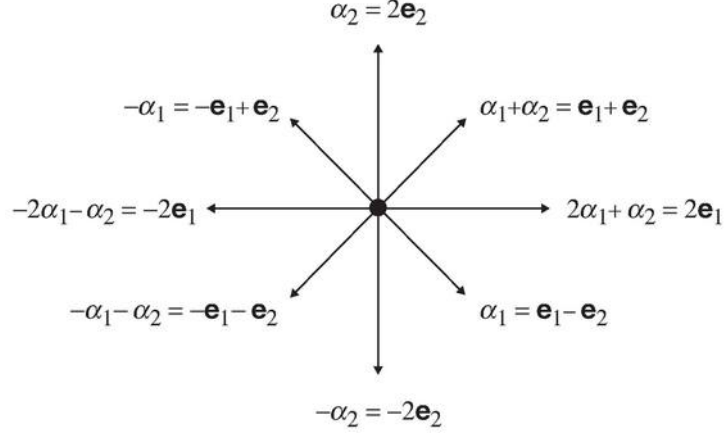
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gilmore-lie-groups_ F01485	155	0.760	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	ization to find the three (three is the dimension of $\mathfrak{su}(2)$) roots: $\lambda = 0, \lambda = \pm ia$,
gilmore-lie-groups_ F01486	155	0.760	$\lambda=0, \lambda= \pm i a$	$\lambda = 0, \lambda = \pm ia$	ization to find the three (three is the dimension of $\mathfrak{su}(2)$) roots: $\lambda = 0, \lambda = \pm ia$,
gilmore-lie-groups_ F01487	155	0.999	$a^2=+a_1^2+a_2^2+a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$.
gilmore-lie-groups_ F01488	156	0.861	$Y=\sum b_i Y_i \in \mathfrak{su}(1,1)$	$Y = \sum b_i Y_i \in \mathfrak{su}(1,1)$	Example For $Y = \sum b_i Y_i \in \mathfrak{su}(1,1)$ the defining 2×2 matrix representation
gilmore-lie-groups_ F01489	156	0.976	$\mathfrak{def}(Y)$	$\mathfrak{def}(Y)$	$\mathfrak{def}(Y)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(Y)$ are
gilmore-lie-groups_ F01490	156	0.976	$\mathfrak{Reg}(Y)$	$\mathfrak{Reg}(Y)$	$\mathfrak{def}(Y)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(Y)$ are
gilmore-lie-groups_ F01491	156	0.999	$\left(a_1, a_2, a_3\right) \rightarrow\left(i b_1, i b_2, b_3\right)$	$(a_1, a_2, a_3) \rightarrow (ib_1, ib_2, b_3)$	$(a_1, a_2, a_3) \rightarrow (ib_1, ib_2, b_3)$, the coefficient $\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$ of the secular
gilmore-lie-groups_ F01492	156	0.999	$\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$	$\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$	$(a_1, a_2, a_3) \rightarrow (ib_1, ib_2, b_3)$, the coefficient $\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$ of the secular
gilmore-lie-groups_ F01493	156	0.953	$\phi_2(\mathbf{b})=-b_1^2-b_2^2+b_3^2$	$\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2$	equation for $\mathfrak{su}(2)$ maps to $\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2$ for $\mathfrak{su}(1,1)$. This same rotation
gilmore-lie-groups_ F01494	156	1.000	ϕ_2	ϕ_2	For each algebra the functional forms of the nonzero coefficient ϕ_2 in the secular
gilmore-lie-groups_ F01495	157	0.515	$\mathfrak{Reg}$	$\mathfrak{Reg}$	equation of its regular representation, $\mathfrak{Reg}$. Since the number of independent roots
gilmore-lie-groups_ F01496	157	1.000	$\phi_j\left(z^i\right)$	$\phi_j\left(z^i\right)$	of the secular equation is equal to the number of independent coefficients $\phi_j(z^i)$,
gilmore-lie-groups_ F01497	157	1.000	$X=Z$	$X = Z$	least one zero root (put $X = Z$ in (9.1)). For simple Lie algebras of dimension N ,
gilmore-lie-groups_ F01498	157	1.000	$l^2 \sim N$	$l^2 \sim N$	$l^2 \sim N$, so describing commutation relations in terms of rank rather than dimension
gilmore-lie-groups_ F01499	157	1.000	$\mathfrak{m}$	$\mathfrak{m}$	If $\mathfrak{m}$ is some matrix Lie algebra of $n \times n$ matrices, then any operator in $\mathfrak{m}$ can be
gilmore-lie-groups_ F01500	158	—	$\phi_j(X)$	$\phi_j(X)$	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$

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gilmore-lie-groups_ F01501	158	—	$a^{\{r\}\{ \}_s}$	a^r_s	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ F01502	158	—	$X^{\{r\}\{ \}_s}$	X^r_s	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ F01503	158	—	$\phi_{\{j\}}$	ϕ_j	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ F01504	158	—	$X=a_{\{s\}}^{\{r\}} X^{\{r\}\{ \}_s}$	$X = a_s^r X^r_s$	$\phi_j(X^r_s) = \frac{1}{j!(n-j)!}$
gilmore-lie-groups_ F01505	158	■ 0.751	$R^{\{3\},=\epsilon_{ijk}=+1}$	$R^3, = \epsilon_{ijk} = +1$	(e.g., in R^3 , $= \epsilon_{ijk} = +1$ for (ijk) a cyclic permutation of (123) , -1 for a cyclic

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gilmore-lie-groups_F01506	158	0.751	(i j k)	(ijk)	(e.g., in R^3 , $= \epsilon_{ijk} = +1$ for (ijk) a cyclic permutation of (123), -1 for a cyclic
gilmore-lie-groups_F01507	159	1.000	$\phi_j(X^r_s)$	$\phi_j(X^r_s)$	these invariant operators $\phi_j(X^r_s)$ remain invariant when the matrices are replaced
gilmore-lie-groups_F01508	159	1.000	$O(n)$	$O(n)$	Example The orthogonal groups $O(n)$ and their subgroups $SO(n)$ have Lie alge-
gilmore-lie-groups_F01509	159	0.651	$d \times d$	$d \times d$	the $d \times d$ (the dimension of $\mathfrak{so}(n)$ is $d = n(n-1)/2$) regular matrix representation
gilmore-lie-groups_F01510	159	0.651	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	the $d \times d$ (the dimension of $\mathfrak{so}(n)$ is $d = n(n-1)/2$) regular matrix representation
gilmore-lie-groups_F01511	159	0.651	$d=n(n-1) / 2)$	$d = n(n-1)/2)$	the $d \times d$ (the dimension of $\mathfrak{so}(n)$ is $d = n(n-1)/2$) regular matrix representation
gilmore-lie-groups_F01512	159	1.000	$\mathfrak{def}(X)^t = -\mathfrak{def}(X)$	$\mathfrak{def}(X)^t = -\mathfrak{def}(X)$	Lie algebra consists of antisymmetric matrices, $\mathfrak{def}(X)^t = -\mathfrak{def}(X)$, and we find
gilmore-lie-groups_F01513	159	0.986	$[n / 2]$	$[n/2]$	even coefficients. Therefore the algebra $\mathfrak{so}(n)$ has rank $[n/2]$.
gilmore-lie-groups_F01514	159	0.996	$j=2$	$j = 2$	setting $j = 2$ in Fig. 9.2 for the generators X_{ij} of $SO(n)$. Since $X_{ij} = -X_{ji}$, it is
gilmore-lie-groups_F01515	159	0.996	$X_{ij} = -X_{ji}$	$X_{ij} = -X_{ji}$	setting $j = 2$ in Fig. 9.2 for the generators X_{ij} of $SO(n)$. Since $X_{ij} = -X_{ji}$, it is
gilmore-lie-groups_F01516	159	1.000	$\mathfrak{so}(5)$	$\mathfrak{so}(5)$	higher order Casimir invariant operators. For example, for $\mathfrak{so}(5)$ the fourth order
gilmore-lie-groups_F01517	159	—	$X^r_s = -X^s_r$	$X^r_s = -X^s_r$	
gilmore-lie-groups_F01518	160	0.940	$v^m = \epsilon^{ijklm} X_{ij} X_{kl}$	$v^m = \epsilon^{ijklm} X_{ij} X_{kl}$	where the components of the five-vector $\mathbf{v}$ are $v^m = \epsilon^{ijklm} X_{ij} X_{kl}$, for example
gilmore-lie-groups_F01519	160	1.000	$n / 2$	$n/2$	square. Its square root, of order $n/2$, should be taken as an appropriate functionally
gilmore-lie-groups_F01520	160	1.000	$Z=0$	$Z = 0$	the opposite extreme, the choice $Z = 0$ is not clever since all X obey the same
gilmore-lie-groups_F01521	160	1.000	$[Z, X]=0 X$	$[Z, X] = 0X$	eigenvalue equation $[Z, X] = 0X$.)
gilmore-lie-groups_F01522	160	1.000	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$	dimensional algebra of two photon operators spanned by $\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$,
gilmore-lie-groups_F01523	160	1.000	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$. A useful choice for Z is
gilmore-lie-groups_F01524	161	1.000	$\pm 2z_1, \pm z_1, 0, 0$	$\pm 2z_1, \pm z_1, 0, 0$	representation are $\pm 2z_1, \pm z_1, 0, 0$ and the commutation relations can be summa-
gilmore-lie-groups_F01525	161	0.999	$X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$	$X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$	where $X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$. We
gilmore-lie-groups_F01526	161	1.000	$k, l \in \{-2, -1, 0, +1, +2\}$	$k, l \in \{-2, -1, 0, +1, +2\}$	also see that if $k, l \in \{-2, -1, 0, +1, +2\}$
gilmore-lie-groups_F01527	161	1.000	$k+l$	$k + l$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the

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gilmore-lie-groups_F01528	161	■ 1.000	$\{-2, -1, 0, +1, +2\}$	$\{-2, -1, 0, +1, +2\}$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01529	161	■ 1.000	$k+l=0$	$k + l = 0$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01530	161	■ 0.968	$(0,0): \hat{n}+\frac{1}{2}$	$(0,0): \hat{n} + \frac{1}{2}$	$(0,0): \hat{n} + \frac{1}{2}$ and I .
gilmore-lie-groups_F01531	162	■ 1.000	$\alpha_1, \alpha_2, \dots, \alpha_l$	$\alpha_1, \alpha_2, \dots, \alpha_l$	(ii) the number of independent roots $\alpha_1, \alpha_2, \dots, \alpha_l$ of the secular equation; these l inde-
gilmore-lie-groups_F01532	162	■ 0.570	$\left(\alpha_1, \alpha_2, \dots, \alpha_l\right)$	$(\alpha_1, \alpha_2, \dots, \alpha_l)$	$(\alpha_1, \alpha_2, \dots, \alpha_l)$ in a root space;
gilmore-lie-groups_F01533	162	■ 1.000	V_{β}	V_{β}	(i) If α and β are roots with subspaces V_{α} and V_{β} in the Lie algebra, then
gilmore-lie-groups_F01534	162	■ 1.000	$V_{\alpha+\beta}$	$V_{\alpha+\beta}$	That is, the commutator of any vector in V_{α} with any vector in V_{β} is a vector in $V_{\alpha+\beta}$.
gilmore-lie-groups_F01535	162	■ 1.000	$\alpha+\beta$	$\alpha + \beta$	If $\alpha + \beta$ is not a root, the commutator vanishes.
gilmore-lie-groups_F01536	162	■ 1.000	$H_1, H_2, \dots, H_l$	$H_1, H_2, \dots, H_l$	(ii) The l basis vectors $H_1, H_2, \dots, H_l$ in the l -dimensional subspace V_0 commute:
gilmore-lie-groups_F01537	162	■ 1.000	$V_{\alpha}(\alpha \neq 0)$	$V_{\alpha}(\alpha \neq 0)$	(iii) Each subspace V_{α} ($\alpha \neq 0$) is one dimensional. Therefore each subspace V_{α} is spanned
gilmore-lie-groups_F01538	162	■ 0.995	E_{α}	E_{α}	by an operator E_{α} that can be labeled by the root α . As a result (i.e., $[V_0, V_{\alpha}] \subset V_{\alpha}$),
gilmore-lie-groups_F01539	162	■ 0.995	$\left[V_0, V_{\alpha}\right] \subset V_{\alpha}$	$[V_0, V_{\alpha}] \subset V_{\alpha}$	by an operator E_{α} that can be labeled by the root α . As a result (i.e., $[V_0, V_{\alpha}] \subset V_{\alpha}$),
gilmore-lie-groups_F01540	162	■ 1.000	$-\alpha$	$-\alpha$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01541	162	■ 1.000	$c \alpha$	$c\alpha$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01542	162	■ 1.000	$ c =1$	$ c = 1$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01543	162	■ 1.000	$E_{-\alpha}$	$E_{-\alpha}$	(v) The commutator of E_{α} and $E_{-\alpha}$ is in V_0 , so can be expanded as a linear superposition
gilmore-lie-groups_F01544	162	■ 1.000	α^i	α^i	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij}\alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01545	162	■ 1.000	α_j	α_j	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij}\alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01546	162	■ 1.000	$\alpha^i = h^{ij}\alpha_j$	$\alpha^i = h^{ij}\alpha_j$	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij}\alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01547	163	■ 1.000	$h^{ij}=\delta^{ij}$	$h^{ij} = \delta^{ij}$	then $h^{ij} = \delta^{ij}$ and we can identify α^i with α_i : $\alpha^i = \alpha_i$.
gilmore-lie-groups_F01548	163	■ 1.000	$\alpha_i: \alpha^i=\alpha_i$	$\alpha_i: \alpha^i = \alpha_i$	then $h^{ij} = \delta^{ij}$ and we can identify α^i with α_i : $\alpha^i = \alpha_i$.
gilmore-lie-groups_F01549	163	■ 1.000	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$ when $\alpha + \beta$ is a nonzero root.
gilmore-lie-groups_F01550	163	■ 1.000	$E_{\alpha}, E_{\beta}, E_{\gamma}$	$E_{\alpha}, E_{\beta}, E_{\gamma}$	$E_{\alpha}, E_{\beta}, E_{\gamma}$ of three nonzero roots that sum to zero
gilmore-lie-groups_F01551	163	■ 1.000	$\alpha, \beta+k \alpha$	$\alpha, \beta + k\alpha$	are not roots. By applying the Jacobi identity to roots α , $\beta + k\alpha$, and $-\alpha$ we obtain
gilmore-lie-groups_F01552	164	—	$\left[E_{\alpha}, E_{\beta}\right]=N_{\alpha, \beta} E_{\alpha+\beta}$	$[E_{\alpha}, E_{\beta}] = N_{\alpha, \beta} E_{\alpha+\beta}$	

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gilmore-lie-groups_ F01553	164	■ 1.000	$N_{\{\alpha, \beta+n\alpha\}}=0$	$N_{\alpha, \beta+n\alpha} = 0$	The initial condition $N_{\alpha, \beta+n\alpha} = 0$ leads to
gilmore-lie-groups_ F01554	164	■ 1.000	$N_{\{-\alpha, \beta-m\alpha\}}^2=N_{\{\alpha, \beta-(m+1)\alpha\}}^2=0$	$N_{-\alpha, \beta-m\alpha}^2 = N_{\alpha, \beta-(m+1)\alpha}^2 = 0$	The other boundary condition $N_{-\alpha, \beta-m\alpha}^2 = N_{\alpha, \beta-(m+1)\alpha}^2 = 0$ leads to
gilmore-lie-groups_ F01555	164	■ 1.000	$N_{\{\alpha, \beta+k\alpha\}}^2=(n-k)(m+k+1)(\alpha \cdot \alpha) / 2 \geq 0$	$N_{\alpha, \beta+k\alpha}^2 = (n-k)(m+k+1)(\alpha \cdot \alpha)/2 \geq 0$	(i) $N_{\alpha, \beta+k\alpha}^2 = (n-k)(m+k+1)(\alpha \cdot \alpha)/2 \geq 0$. We use this expression because it shows
gilmore-lie-groups_ F01556	164	■ 1.000	$\alpha \cdot \beta > 0$	$\alpha \cdot \beta > 0$	clearly how the boundary conditions are imposed. We note that $\alpha \cdot \beta > 0$ when $m -$
gilmore-lie-groups_ F01557	164	■ 1.000	$m -$	$m -$	clearly how the boundary conditions are imposed. We note that $\alpha \cdot \beta > 0$ when $m -$
gilmore-lie-groups_ F01558	164	■ 1.000	$n > 0$	$n > 0$	$n > 0$ and $\alpha \cdot \beta < 0$ when $m - n < 0$.
gilmore-lie-groups_ F01559	164	■ 1.000	$\alpha \cdot \beta < 0$	$\alpha \cdot \beta < 0$	$n > 0$ and $\alpha \cdot \beta < 0$ when $m - n < 0$.
gilmore-lie-groups_ F01560	164	■ 1.000	$m - n < 0$	$m - n < 0$	$n > 0$ and $\alpha \cdot \beta < 0$ when $m - n < 0$.
gilmore-lie-groups_ F01561	165	—	$A_{\{2\}}, B_{\{2\}}, C_{\{2\}}$	A_2, B_2, C_2	
gilmore-lie-groups_ F01562	165	—	$D_{\{2\}}, G_{\{2\}}$	D_2, G_2	

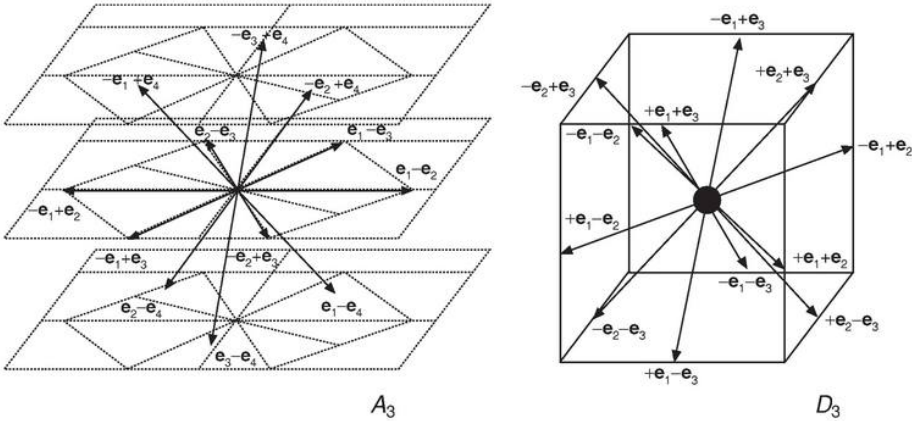
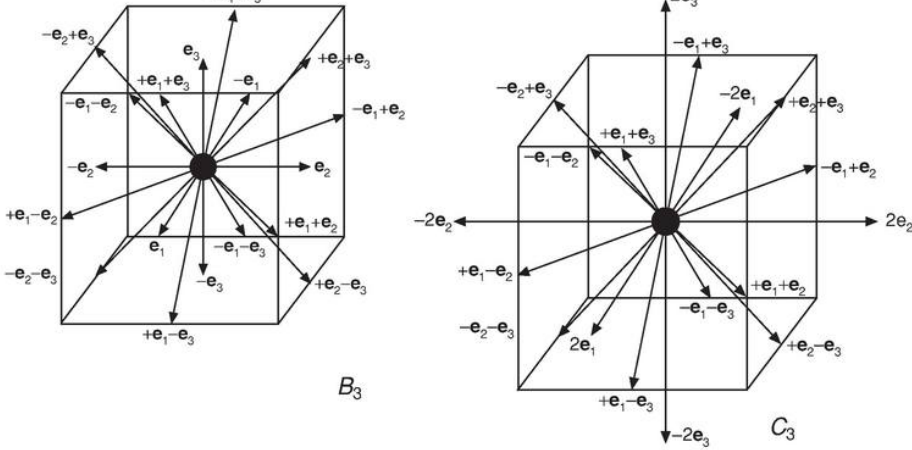
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gilmore-lie-groups_ F01563	165	■ 0.456	$l(l$	$l(l$	constants of a semisimple Lie algebra. The basis vectors are the l (l is the rank)
gilmore-lie-groups_ F01564	165	■ 0.977	$\mathbf{H}=\left(H_{\{1\}}, H_{\{2\}}, \ldots, H_{\{l\}}\right)$	$\mathbf{H} = (H_1, H_2, \dots, H_l)$	operators $\mathbf{H} = (H_1, H_2, \dots, H_l)$ and the “shift” operators E_α , one corresponding
gilmore-lie-groups_ F01565	165	■ 0.994	$\alpha=\left(\alpha_{\{1\}}, \alpha_{\{2\}}, \ldots, \alpha_{\{l\}}\right)$	$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$	to each nonzero root. The root vector $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$ has l components. The
gilmore-lie-groups_ F01566	166	—	$C_{\{2\}}$	C_2	
gilmore-lie-groups_ F01567	166	—	$\mathbf{e}_{\{1\}}, \mathbf{e}_{\{2\}}$	$\mathbf{e}_1, \mathbf{e}_2$	
gilmore-lie-groups_ F01568	166	—	$\alpha_{\{1\}}, \alpha_{\{2\}}$	α_1, α_2	
gilmore-lie-groups_ F01569	166	■ 1.000	$\mathbf{e}_{\{1\}}$	$\mathbf{e}_1$	If we choose orthogonal vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ in a two-dimensional Euclidean space,
gilmore-lie-groups_ F01570	166	■ 1.000	$\mathbf{e}_{\{2\}}$	$\mathbf{e}_2$	If we choose orthogonal vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ in a two-dimensional Euclidean space,
gilmore-lie-groups_ F01571	166	■ 1.000	$\pm 2 \mathbf{e}_{\{1\}}, \pm 2 \mathbf{e}_{\{2\}}, \pm \mathbf{e}_{\{1\}} \pm \mathbf{e}_{\{2\}}$	$\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm \mathbf{e}_1 \pm \mathbf{e}_2$	the nonzero roots for C_2 are $\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm \mathbf{e}_1 \pm \mathbf{e}_2$. The 10 basis vectors in the Lie
gilmore-lie-groups_ F01572	166	■ 0.986	$H_{\{i\}}, i=1,2$	$H_i, i = 1, 2$	algebra are $H_i, i = 1, 2$, and E_α , with α the eight nonzero roots. We normalize

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gilmore-lie-groups_ F01573	166	■ 1.000	$\sum \alpha \cdot \alpha = 2$	$\sum \alpha \cdot \alpha = 2$	these roots by $\sum \alpha \cdot \alpha = 2$ so that
gilmore-lie-groups_ F01609	168	■ 1.000	$\mathfrak{sl}(2; \mathbb{C}))$	$\mathfrak{sl}(2; \mathbb{C}))$	extension $\mathfrak{sl}(2; \mathbb{C}))$. Root spaces classify commutation relations of these complex
gilmore-lie-groups_ F01610	168	■ 1.000	α_i	α_i	real subalgebras of these complex algebras – some roots α_i and structure constants
gilmore-lie-groups_ F01611	168	■ 1.000	$\frac{1}{2} \{a^\dagger, a\}$	$\frac{1}{2} \{a^\dagger, a\}$	1. Construct the regular representation for the two-photon operator algebra: $\frac{1}{2} \{a^\dagger, a\}$,
gilmore-lie-groups_ F01612	168	■ 1.000	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$. Determine the secular equation for this matrix. Determine the rank
gilmore-lie-groups_ F01613	168	■ 1.000	$X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$	$X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$	ical basis $X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$ these operators are $\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$ and
gilmore-lie-groups_ F01614	168	■ 1.000	$\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$	$\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$	ical basis $X_{ij} = x^i \partial_j - x^j \partial_i (1 \leq i < j \leq 4)$ these operators are $\mathcal{C}_2 = \sum_{i < j} X_{ij}^2$ and
gilmore-lie-groups_ F01615	168	■ 0.980	$\mathcal{C}'_2 = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	$\mathcal{C}'_2 = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	$\mathcal{C}'_2 = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$.
gilmore-lie-groups_ F01616	168	■ 0.994	$\mathfrak{su}(4)$	$\mathfrak{su}(4)$	3. The Lie algebra $\mathfrak{su}(4)$ has a 4×4 defining matrix representation and a 15×15 regular
gilmore-lie-groups_ F01617	168	■ 1.000	$\mathfrak{so}(2n+1)$	$\mathfrak{so}(2n+1)$	4. For $\mathfrak{so}(2n+1)$ the invariant operators (Casimir operators) are of degree 2, 4, $\dots$, $2n$.
gilmore-lie-groups_ F01618	168	■ 1.000	$2, 4, \dots, 2n$	$2, 4, \dots, 2n$	4. For $\mathfrak{so}(2n+1)$ the invariant operators (Casimir operators) are of degree $2, 4, \dots, 2n$.
gilmore-lie-groups_ F01619	168	■ 1.000	$\mathfrak{so}(2n)$	$\mathfrak{so}(2n)$	This is true also for $\mathfrak{so}(2n)$, with one difference: the invariant operator of degree $2n$
gilmore-lie-groups_ F01620	168	■ 0.600	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}'_2$ for $\mathfrak{so}(4)$ and $\mathcal{C}'_3$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01621	168	■ 0.600	$\mathcal{C}'_2$	$\mathcal{C}'_2$	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}'_2$ for $\mathfrak{so}(4)$ and $\mathcal{C}'_3$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01622	168	■ 0.600	$\mathcal{C}'_3$	$\mathcal{C}'_3$	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}'_2$ for $\mathfrak{so}(4)$ and $\mathcal{C}'_3$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01623	168	■ 0.600	$\mathfrak{so}(6)$	$\mathfrak{so}(6)$	$\mathcal{C}'_n = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out $\mathcal{C}'_2$ for $\mathfrak{so}(4)$ and $\mathcal{C}'_3$ for $\mathfrak{so}(6)$.
gilmore-lie-groups_ F01624	168	■ 1.000	$\mathfrak{su}(4) = \mathfrak{so}(6)$	$\mathfrak{su}(4) = \mathfrak{so}(6)$	5. In Chapter 11 we will show that $\mathfrak{su}(4) = \mathfrak{so}(6)$. Both Lie algebras have invariant
gilmore-lie-groups_ F01625	168	■ 1.000	$a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$	$a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$	two modes. This algebra is ten dimensional. It contains the four operators $a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$
gilmore-lie-groups_ F01626	169	■ 0.999	$(1 \leq i, j \leq 2)$	$(1 \leq i, j \leq 2)$	$1 \leq i, j \leq 2$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_ F01627	169	■ 0.999	$a_i^\dagger a_j^\dagger$	$a_i^\dagger a_j^\dagger$	$(1 \leq i, j \leq 2)$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_ F01628	169	■ 0.999	$a_i a_j (a_i a_j = a_j a_i)$	$a_i a_j (a_i a_j = a_j a_i)$	$(1 \leq i, j \leq 2)$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_ F01629	169	■ 1.000	$a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+\mathbf{e}_i + \mathbf{e}_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+\mathbf{e}_i - \mathbf{e}_j} (i \neq j)$	$a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+\mathbf{e}_i + \mathbf{e}_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+\mathbf{e}_i - \mathbf{e}_j} (i \neq j)$	identification is: $a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+\mathbf{e}_i + \mathbf{e}_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+\mathbf{e}_i - \mathbf{e}_j} (i \neq j)$,
gilmore-lie-groups_ F01630	169	■ 1.000	$a_i a_j \leftrightarrow E_{-\mathbf{e}_i - \mathbf{e}_j} (i \neq j), a_i^\dagger a_i^\dagger \leftrightarrow E_{+2\mathbf{e}_i}, a_i a_i \leftrightarrow E_{-2\mathbf{e}_i}$	$a_i a_j \leftrightarrow E_{-\mathbf{e}_i - \mathbf{e}_j} (i \neq j), a_i^\dagger a_i^\dagger \leftrightarrow E_{+2\mathbf{e}_i}, a_i a_i \leftrightarrow E_{-2\mathbf{e}_i}$	$a_i a_j \leftrightarrow E_{-\mathbf{e}_i - \mathbf{e}_j} (i \neq j), a_i^\dagger a_i^\dagger \leftrightarrow E_{+2\mathbf{e}_i}, a_i a_i \leftrightarrow E_{-2\mathbf{e}_i}$.

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gilmore-lie-groups_F01631	169	■ 1.000	$D_{\{2\}}$	D_2	is six dimensional. Show that the resulting root space diagram is D_2 (Fig. 9.6). Why
gilmore-lie-groups_F01632	169	■ 0.995	$f_{\{i\}}^{\dagger} f_{\{i\}}^{\dagger} = 0$	$f_i^\dagger f_i^\dagger = 0$	the difference? (Hint: $f_i^\dagger f_i^\dagger = 0$.)
gilmore-lie-groups_F01633	169	■ 1.000	$X, Y \in \mathfrak{su}(2)$	$X, Y \in \mathfrak{su}(2)$	ular representation for the former. Choose $X, Y \in \mathfrak{su}(2)$ and compute $(X, Y) =$
gilmore-lie-groups_F01634	169	■ 0.577	$\operatorname{tr}[\mathfrak{def}(X) \mathfrak{def}(Y)]$	$\operatorname{tr}[\mathfrak{def}(X) \mathfrak{def}(Y)]$	$\operatorname{tr}[\mathfrak{def}(X) \mathfrak{def}(Y)]$ by taking the trace of the 2×2 matrices in $\mathfrak{su}(2)$ that represent
gilmore-lie-groups_F01635	169	■ 0.569	$\mathfrak{su}(2): (X, Y) = \operatorname{tr}[\mathfrak{Reg}(X) \mathfrak{Reg}(Y)]$	$\mathfrak{su}(2): (X, Y) = \operatorname{tr}[\mathfrak{Reg}(X) \mathfrak{Reg}(Y)]$	the regular matrix representation of $\mathfrak{su}(2)$: $(X, Y) = \operatorname{tr}[\mathfrak{Reg}(X) \mathfrak{Reg}(Y)]$. Show that
gilmore-lie-groups_F01636	169	■ 0.995	$\left(n^2 - 1\right) \times \left(n^2 - 1\right)$	$(n^2 - 1) \times (n^2 - 1)$	the $n \times n$ defining matrix representation and in the $(n^2 - 1) \times (n^2 - 1)$ regular matrix
gilmore-lie-groups_F01637	169	■ 1.000	$Y=X$	$Y = X$	constant? (Hint: set $Y = X$ and choose a special X , for example $X = H_1$.)
gilmore-lie-groups_F01638	169	■ 1.000	$X=H_{\{1\}}$	$X = H_1$	constant? (Hint: set $Y = X$ and choose a special X , for example $X = H_1$.)
gilmore-lie-groups_F01639	169	■ 0.999	$1 \leq i, j \leq N$	$1 \leq i, j \leq N$	form (b boson operators; f fermion operators; $1 \leq i, j \leq N$):
gilmore-lie-groups_F01640	169	■ 1.000	$\mathbf{R} = \frac{1}{2} \sum_{\alpha > 0} \alpha$	$\mathbf{R} = \frac{1}{2} \sum_{\alpha > 0} \alpha$	11. Compute $\mathbf{R} = \frac{1}{2} \sum_{\alpha > 0} \alpha$, half the sum over all positive roots, in each of the simple Lie
gilmore-lie-groups_F01641	169	■ 1.000	$B_{\{n\}}, R_{\{i\}} = \frac{1}{2}(2n+1) - i$	$B_n, R_i = \frac{1}{2}(2n+1) - i$	algebras. For example, for $B_n, R_i = \frac{1}{2}(2n+1) - i$ and the spectrum is
gilmore-lie-groups_F01642	169	■ 0.996	$\mathbf{M}$	$\mathbf{M}$	where $\mathbf{M}$ is the highest weight in the representation. For the $(2j+1)$ dimensional
gilmore-lie-groups_F01643	169	■ 1.000	$\mathfrak{so}(3), \mathbf{M} = j, \mathbf{R} = R_1 = \frac{1}{2}$	$\mathfrak{so}(3), \mathbf{M} = j, \mathbf{R} = R_1 = \frac{1}{2}$	representation of $\mathfrak{so}(3)$, $\mathbf{M} = j$, $\mathbf{R} = R_1 = \frac{1}{2}$ and $\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$
gilmore-lie-groups_F01644	169	■ 1.000	$\mathcal{C}^2(j) = \left(j + \frac{1}{2}\right)^2 - \left(0 + \frac{1}{2}\right)^2 =$	$\mathcal{C}^2(j) = \left(j + \frac{1}{2}\right)^2 - \left(0 + \frac{1}{2}\right)^2 =$	representation of $\mathfrak{so}(3)$, $\mathbf{M} = j$, $\mathbf{R} = R_1 = \frac{1}{2}$ and $\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$
gilmore-lie-groups_F01645	169	■ 1.000	$j(j+1)$	$j(j+1)$	$j(j+1)$.
gilmore-lie-groups_F01646	169	■ 1.000	$A_{\{n-1\}}$	A_{n-1}	a. Show that the Weyl group for A_{n-1} is of order $n!$, the Weyl group for D_n is of
gilmore-lie-groups_F01647	169	■ 1.000	$D_{\{n\}}$	D_n	a. Show that the Weyl group for A_{n-1} is of order $n!$, the Weyl group for D_n is of
gilmore-lie-groups_F01648	169	■ 0.919	$2^{n-1} n!$	$2^{n-1} n!$	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01649	169	■ 0.919	$B_{\{n\}}$	B_n	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01650	169	■ 0.919	$C_{\{n\}}$	C_n	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01651	169	■ 0.919	$2^n n!$	$2^n n!$	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01652	170	■ 1.000	$f_{\{r\}}(\Gamma)$	$f_r(\Gamma)$	the index $f_r(\Gamma)$.
gilmore-lie-groups_F01653	170	■ 1.000	$\mathcal{C}^r$	$\mathcal{C}^r$	In this expression $\mathcal{C}^r$ is the value of the r th Casimir invariant in the representation
gilmore-lie-groups_F01654	170	■ 1.000	$\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$	$\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$	15. The matrix Lie algebras $\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$ have the form $\sum_{ij} a^{ij} M_{ij}$, where
gilmore-lie-groups_F01655	170	■ 1.000	$\sum_{ij} a^{ij} M_{ij}$	$\sum_{ij} a^{ij} M_{ij}$	15. The matrix Lie algebras $\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$ have the form $\sum_{ij} a^{ij} M_{ij}$, where
gilmore-lie-groups_F01656	170	■ 1.000	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	M is $2n \times 2n$ for $\mathfrak{so}(2n), \mathfrak{sp}(2n)$ and $(2n+1) \times (2n+1)$ for $\mathfrak{so}(2n+1)$, and suitable
gilmore-lie-groups_F01657	170	■ 1.000	$(2n+1) \times (2n+1)$	$(2n+1) \times (2n+1)$	M is $2n \times 2n$ for $\mathfrak{so}(2n), \mathfrak{sp}(2n)$ and $(2n+1) \times (2n+1)$ for $\mathfrak{so}(2n+1)$, and suitable

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gilmore-lie-groups_ F01658	170	■ 1.000	$a^{\{i\ j\}}$	a^{ij}	reality restrictions are imposed on the coefficients a^{ij} .
gilmore-lie-groups_ F01659	170	■ 1.000	$\phi_{\{r\}}(a^{\{i\ j\}})$	$\phi_r(a^{ij})$	b. Write down the coefficients $\phi_r(a^{ij})$ that occur in the secular equation for each of
gilmore-lie-groups_ F01660	170	■ 0.999	$\epsilon_{i_1 i_2 \dots i_l}$	$\epsilon_{i_1 i_2 \dots i_l}$	d. Express the even coefficients in terms of the Levi–Civita skew tensors $\epsilon_{i_1 i_2 \dots i_l}$
gilmore-lie-groups_ F01661	170	■ 1.000	$(l=2\ n, \ 2\ n+1, 2\ n)$	$(l = 2n, 2n + 1, 2n)$	$l = 2n, 2n + 1, 2n$.
gilmore-lie-groups_ F01662	170	■ 1.000	$\phi_{2n}(a^{\{i\ j\}})$	$\phi_{2n}(a^{ij})$	g. Show that $\phi_{2n}(a^{ij})$ is a perfect square for $\mathfrak{so}(2n)$; write down its square root; show
gilmore-lie-groups_ F01663	171	■ 0.999	$\operatorname{tr} M^2 = -2\ \theta\ \cdot\ \theta$	$\operatorname{tr} M^2 = -2\theta \cdot \theta$	a. Show $\operatorname{tr} M^2 = -2\theta \cdot \theta$.
gilmore-lie-groups_ F01664	171	■ 0.987	$\operatorname{tr} \mathcal{M}^2 = -2\ \mathbf{J}\ \cdot\ \mathbf{J}$	$\operatorname{tr} \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$	b. Show $\operatorname{tr} \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$.
gilmore-lie-groups_ F01665	171	■ 1.000	$\left[\mathbf{J}, \operatorname{tr} \mathcal{M}^2\right] = 0$	$[\mathbf{J}, \operatorname{tr} \mathcal{M}^2] = 0$	c. Show $[\mathbf{J}, \operatorname{tr} \mathcal{M}^2] = 0$.
gilmore-lie-groups_ F01666	171	■ 0.999	$\operatorname{tr} \mathcal{M}^{2n+1} = 0$	$\operatorname{tr} \mathcal{M}^{2n+1} = 0$	d. Show $\operatorname{tr} \mathcal{M}^{2n+1} = 0$ and $\operatorname{tr} \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$.
gilmore-lie-groups_ F01667	171	■ 0.999	$\operatorname{tr} \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$	$\operatorname{tr} \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$	d. Show $\operatorname{tr} \mathcal{M}^{2n+1} = 0$ and $\operatorname{tr} \mathcal{M}^{2n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$.
gilmore-lie-groups_ F01668	171	■ 1.000	$\left[X_{\{i\}}, X_{\{j\}}\right] = C_{ij}^{\ \ k} X_{\{k\}}$	$[X_i, X_j] = C_{ij}^{\ k} X_k$	commutation relations $[X_i, X_j] = C_{ij}^{\ k} X_k$. There are two linear vector spaces, $V^{(1)}$
gilmore-lie-groups_ F01669	171	■ 1.000	$V^{\{1\}}$	$V^{(1)}$	commutation relations $[X_i, X_j] = C_{ij}^{\ k} X_k$. There are two linear vector spaces, $V^{(1)}$
gilmore-lie-groups_ F01670	171	■ 1.000	$V^{\{2\}}$	$V^{(2)}$	and $V^{(2)}$, that carry irreducible representations of this Lie algebra: $X_i \rightarrow \Gamma^{(1)}(X_i) =$
gilmore-lie-groups_ F01671	171	■ 1.000	$X_{\{i\}} \rightarrow \Gamma^{\{1\}}(X_{\{i\}}) =$	$X_i \rightarrow \Gamma^{(1)}(X_i) =$	and $V^{(2)}$, that carry irreducible representations of this Lie algebra: $X_i \rightarrow \Gamma^{(1)}(X_i) =$
gilmore-lie-groups_ F01672	171	■ 1.000	$Y_{\{i\}}$	Y_i	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_ F01673	171	■ 1.000	$X_{\{i\}} \rightarrow \Gamma^{\{2\}}(X_{\{i\}}) = Z_{\{i\}}$	$X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_ F01674	171	■ 1.000	$g^{\{i\ j\}} Y_{\{i\}} Z_{\{j\}}$	$g^{ij} Y_i Z_j$	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with
gilmore-lie-groups_ F01675	171	■ 0.857	$(Y+Z)_{\{k\}}$	$(Y + Z)_k$	$(Y + Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$).
gilmore-lie-groups_ F01676	171	■ 0.857	$\Gamma^{\{1\}}(X_{\{i\}}) \otimes I_{\dim V^{\{2\}}} + I_{\dim V^{\{1\}}} \otimes \Gamma^{\{2\}}(X_{\{i\}})$	$\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$	$(Y + Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$).
gilmore-lie-groups_ F01677	171	■ 1.000	$X^{\{0\}} = I_{\{n\}}$	$X^0 = I_n$	of X , starting at $X^0 = I_n$:
gilmore-lie-groups_ F01678	171	■ 1.000	$\lambda_{\{i\}}(X)$	$\lambda_i(X)$	eigenvalues $\lambda_i(X)$ or the independent coefficients $\phi_i(X)$ of the secular equation
gilmore-lie-groups_ F01679	171	■ 1.000	$\phi_{\{i\}}(X)$	$\phi_i(X)$	eigenvalues $\lambda_i(X)$ or the independent coefficients $\phi_i(X)$ of the secular equation
gilmore-lie-groups_ F01680	171	■ 0.967	$e^{i\ \phi_{\{J\}} z}$	$e^{i\phi J_z}$	d. Compute $e^{i\phi J_z}$ for the $(2j + 1)$ dimensional matrix representations of $SU(2)$ by
gilmore-lie-groups_ F01681	171	■ 0.999	$j = \frac{1}{2}, 1, \frac{3}{2}, 2$	$j = \frac{1}{2}, 1, \frac{3}{2}, 2$	computing the vanderMonde matrices. Show that for $j = \frac{1}{2}, 1, \frac{3}{2}, 2$ the resulting
gilmore-lie-groups_ F01682	172	■ 1.000	$f_{\{j\}}(\phi)$	$f_j(\phi)$	e. Invert each of these van der Monde matrices and determine the functions $f_j(\phi)$ in
gilmore-lie-groups_ F01683	172	■ 1.000	$e^{\{X\}}$	e^X	the expansions of e^X for $X \in \mathfrak{su}(2)$. In particular, show

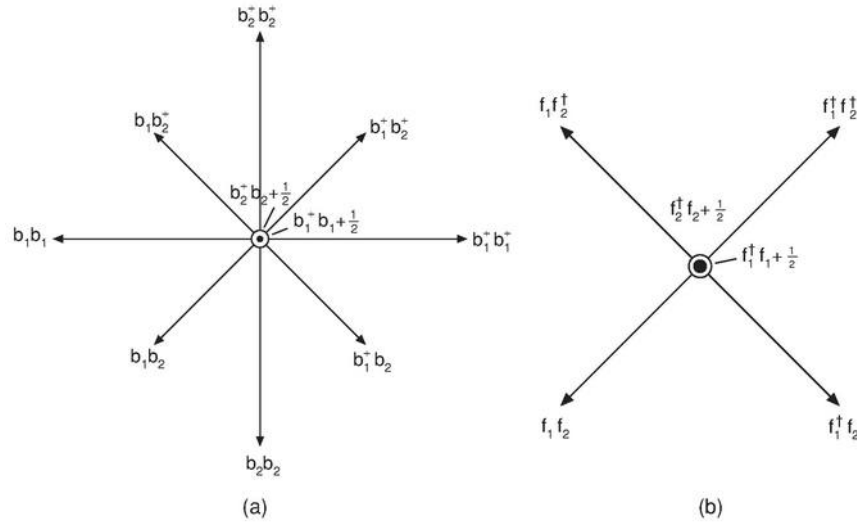
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gilmore-lie-groups_F01684	172	■ 1.000	$X \in \mathfrak{su}(2)$	$X \in \mathfrak{su}(2)$	the expansions of e^X for $X \in \mathfrak{su}(2)$. In particular, show
gilmore-lie-groups_F01686	257	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	the charge on the proton) and spin $\frac{1}{2}$. The Dirac equation $H_D \psi = E \psi$ is a matrix
gilmore-lie-groups_F01705	172	■ 1.000	$j = \frac{1}{2}$	$j = \frac{1}{2}$	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:
gilmore-lie-groups_F01706	172	■ 1.000	$l=1$	$l = 1$	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:
gilmore-lie-groups_F01707	173	■ 1.000	$l+1$	$l + 1$	grams to construct rank $l + 1$ root space diagrams. However, it is not easy
gilmore-lie-groups_F01708	174	■ 1.000	m, n	m, n	The coefficients $N_{\alpha, \beta}$ are defined in terms of the nonnegative integers m, n by
gilmore-lie-groups_F01709	174	■ 1.000	$\beta + k \alpha$	$\beta + k \alpha$	where $\beta + k \alpha$ is a root only for $k = -m, \dots, +n$. The roots are normalized by
gilmore-lie-groups_F01710	174	■ 1.000	$k = -m, \dots, +n$	$k = -m, \dots, +n$	where $\beta + k \alpha$ is a root only for $k = -m, \dots, +n$. The roots are normalized by
gilmore-lie-groups_F01711	174	■ 0.999	$\beta^{\prime}$	β'	The root β' is obtained by reflecting β in the hyperplane orthogonal to α . These two
gilmore-lie-groups_F01712	174	■ 1.000	$2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$	$2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$	If we write $2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$ and $2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$, where n and n' are
gilmore-lie-groups_F01713	174	■ 1.000	$2(\alpha \cdot \beta) / (\beta \cdot \beta) = n^{\prime}$	$2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$	If we write $2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$ and $2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$, where n and n' are
gilmore-lie-groups_F01714	174	■ 0.992	$\pm \mathbf{e}_1$	$\pm \mathbf{e}_1$	vectors $\pm \mathbf{e}_1$. To construct rank-two root spaces, add a noncollinear vector to this
gilmore-lie-groups_F01715	174	■ 1.000	$A_2, B_2 = C_2, D_2$	$A_2, B_2 = C_2, D_2$	are $A_2, B_2 \equiv C_2, D_2$ and G_2 , as shown in Fig. 9.6.
gilmore-lie-groups_F01734	175	■ 0.998	$(l-1)$	$(l - 1)$	this way from rank- $(l - 1)$ root space diagrams. We find by this building-up process
gilmore-lie-groups_F01735	175	■ 0.998	D_l	D_l	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_F01736	175	■ 0.998	$\pm(\mathbf{e}_i + \mathbf{e}_j)$	$\pm(\mathbf{e}_i + \mathbf{e}_j)$	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_F01737	175	■ 0.998	A_{l-1}	A_{l-1}	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_F01738	175	■ 0.998	B_l, C_l	B_l, C_l	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by
gilmore-lie-groups_F01739	175	■ 1.000	$\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$	$\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$	adding roots $\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$ to D_l . The root spaces A_{l-1}, D_l, B_l, C_l are all inequivalent
gilmore-lie-groups_F01740	175	■ 1.000	A_{l-1}, D_l, B_l, C_l	A_{l-1}, D_l, B_l, C_l	adding roots $\pm \mathbf{e}_i, \pm 2 \mathbf{e}_i$ to D_l . The root spaces A_{l-1}, D_l, B_l, C_l are all inequivalent

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gilmore-lie-groups_ F01741	176	—	$A_{\{3\}}, D_{\{3\}}$	A_3, D_3	
gilmore-lie-groups_ F01742	176	—	$B_{\{3\}}, C_{\{3\}}$	B_3, C_3	

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gilmore-lie-groups-F01743	178	—	$A_{\{1\}}$	A_1	
gilmore-lie-groups-F01744	178	■ 1.000	$SL(l; \mathbb{C})$	$SL(l; \mathbb{C})$	space has the same complex extension, for example, $SL(l; \mathbb{C})$ for A_{l-1} .
gilmore-lie-groups-F01745	178	■ 0.997	$\pm(\mathbf{e}_1 - \mathbf{e}_2)$	$\pm(\mathbf{e}_1 - \mathbf{e}_2)$	Remark 2. The root space D_2 consists of two orthogonal sets of roots $\pm(\mathbf{e}_1 - \mathbf{e}_2)$
gilmore-lie-groups-F01746	178	■ 1.000	$\pm(\mathbf{e}_1 + \mathbf{e}_2)$	$\pm(\mathbf{e}_1 + \mathbf{e}_2)$	and $\pm(\mathbf{e}_1 + \mathbf{e}_2)$. The decomposition is shown in Fig. 10.3. Orthogonal root spaces
gilmore-lie-groups-F01747	178	■ 0.964	B_2	B_2	Remark 3. The root spaces B_2 and C_2 are equivalent, as is easily seen by ro-
gilmore-lie-groups-F01748	178	■ 0.513	$SO(5)$	$SO(5)$	tation. The root space B_2 describes $SO(5)$ while C_2 describes $Sp(2) = U(2; \mathbb{Q})$,
gilmore-lie-groups-F01749	178	■ 0.513	$Sp(2) = U(2; \mathbb{Q})$	$Sp(2) = U(2; \mathbb{Q})$	tation. The root space B_2 describes $SO(5)$ while C_2 describes $Sp(2) = U(2; \mathbb{Q})$,
gilmore-lie-groups-F01750	178	■ 1.000	$U(2; \mathbb{Q})$	$U(2; \mathbb{Q})$	have a four-dimensional “spinor” representation based on $U(2; \mathbb{Q})$ in the same way
gilmore-lie-groups-F01751	178	■ 0.975	$SO(3)(B_1)$	$SO(3)(B_1)$	that $SO(3)(B_1)$ has a two-dimensional spinor representation based on $U(1; \mathbb{Q})$ or
gilmore-lie-groups-F01752	178	■ 0.975	$U(1; \mathbb{Q})$	$U(1; \mathbb{Q})$	that $SO(3)(B_1)$ has a two-dimensional spinor representation based on $U(1; \mathbb{Q})$ or
gilmore-lie-groups-F01753	178	■ 0.995	$SU(2)(A_1)$	$SU(2)(A_1)$	$SU(2)(A_1)$.
gilmore-lie-groups-F01754	179	—	$\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$	$\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$	

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gilmore-lie-groups_ F01755	179	—	$\alpha_2 =$	$\alpha_2 =$	
gilmore-lie-groups_ F01756	179	—	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$	

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gilmore-lie-groups_ F01757	179	—	$\mathbf{R}=\mathbf{e}_1+\mathbf{e}_2+\mathbf{e}_3$	$\mathbf{R} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$	
gilmore-lie-groups_ F01758	180	—	$G_2 \oplus B_3$	$G_2 \oplus B_3$	
gilmore-lie-groups_ F01771	184	■ 1.000	C_l	C_l	root spaces C_l and D_l :
gilmore-lie-groups_ F01773	180	■ 1.000	n_{ij}	n_{ij}	where n_{ij} is 0, 1, 2, or 3. Each fundamental root is represented by a dot. Dots i
gilmore-lie-groups_ F01774	180	■ 1.000	$0, 1, 2$	$0, 1, 2$	where n_{ij} is 0, 1, 2, or 3. Each fundamental root is represented by a dot. Dots i
gilmore-lie-groups_ F01775	180	■ 1.000	G_2+B_3	$G_2 + B_3$	represented by orthogonal root spaces $G_2 + B_3$ is shown in Fig. 10.5.
gilmore-lie-groups_ F01776	180	■ 0.665	$\mathbf{v}_i$	$\mathbf{v}_i$	Observation 2 If $\mathbf{v}_i$ is an orthonormal system of vectors in the root space and $\mathbf{u}$ is a unit
gilmore-lie-groups_ F01777	180	■ 0.665	$\mathbf{u}$	$\mathbf{u}$	Observation 2 If $\mathbf{v}_i$ is an orthonormal system of vectors in the root space and $\mathbf{u}$ is a unit
gilmore-lie-groups_ F01778	180	■ 1.000	$\mathbf{u} \cdot \mathbf{v}_i$	$\mathbf{u} \cdot \mathbf{v}_i$	vector, then the direction cosines $\mathbf{u} \cdot \mathbf{v}_i$ obey
gilmore-lie-groups_ F01779	181	■ 1.000	$\mathbf{u}_i = \alpha_i / \alpha_i $	$\mathbf{u}_i = \alpha_i / \alpha_i $	vertices. With $\mathbf{u}_i = \alpha_i / \alpha_i $ the inner product
gilmore-lie-groups_ F01780	181	■ 1.000	$2 \mathbf{u}_i \cdot \mathbf{u}_j \leq -1$	$2\mathbf{u}_i \cdot \mathbf{u}_j \leq -1$	cannot be positive since $2\mathbf{u}_i \cdot \mathbf{u}_j \leq -1$ if α_i and α_j are connected.
gilmore-lie-groups_ F01781	181	■ 0.979	(B, C, F)	(B, C, F)	For the diagrams (B, C, F) with a single double link, the Schwarz inequality
gilmore-lie-groups_ F01782	182	■ 1.000	$\mathbf{u}_i, \mathbf{v}_j$	$\mathbf{u}_i, \mathbf{v}_j$	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_j = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups_ F01783	182	■ 1.000	$\mathbf{v}_i = \alpha_j / \alpha_j $	$\mathbf{v}_i = \alpha_j / \alpha_j $	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_j = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups_ F01784	182	■ 0.985	$p \geq q$	$p \geq q$	This has the following solutions with $p \geq q$
gilmore-lie-groups_ F01785	182	■ 0.657	(D, E)	(D, E)	For the diagrams (D, E) Observation 2 applied to the vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ defined

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gilmore-lie-groups_ F01786	182	■ 0.997	$p \geq q \geq r \geq 2$	$p \geq q \geq r \geq 2$	This has the following solutions with $p \geq q \geq r \geq 2$
gilmore-lie-groups_ F01788	327	■ 0.991	$E_{\{6\}}$	E_6	E_6 , 162, 168
gilmore-lie-groups_ F01789	327	■ 0.546	$E_{\{7\}}$	E_7	E_7 , 162, 168
gilmore-lie-groups_ F01790	327	■ 0.731	$E_{\{8\}}$	E_8	E_8 , 162, 168
gilmore-lie-groups_ F01791	182	■ 1.000	$l-1$	$l-1$	inductively from each root space diagram of rank $l-1$. The completeness of this
gilmore-lie-groups_ F01792	184	—	$f_{\{1\}}^{\dagger} f_{\{1\}}^{\dagger} =$	$f_1^{\dagger} f_1^{\dagger} =$	
gilmore-lie-groups_ F01793	184	■ 1.000	$U(l)$	$U(l)$	group $U(l)$:
gilmore-lie-groups_ F01814	185	■ 0.852	D, E	D, E	Dynkin diagrams of type (D, E) to obtain the inequality of Eq. (10.19).
gilmore-lie-groups_ F01815	185	■ 1.000	$1 \leq i, j \leq n$	$1 \leq i, j \leq n$	8. A Lie algebra is spanned by n^2 operators of the form $a_i^{\dagger} a_j$, with $1 \leq i, j \leq n$. Show
gilmore-lie-groups_ F01816	185	■ 1.000	$n=3$	$n=3$	For $n=3$ the subspaces transform like an angular momentum vector and a quadrupole
gilmore-lie-groups_ F01817	185	■ 1.000	$\hat{\mathbf{Q}}$	$\hat{\mathbf{Q}}$	traceless tensor $\hat{\mathbf{Q}}$ and a scalar N :
gilmore-lie-groups_ F01818	185	■ 1.000	$\hat{N}$	$\hat{N}$	The operator $\hat{N}$ commutes with all operators $a_i^{\dagger} a_j$. Interpret these commutation re-
gilmore-lie-groups_ F01819	185	■ 0.998	$\mathbf{L} + \hat{\mathbf{Q}}$	$\mathbf{L} + \hat{\mathbf{Q}}$	simple Lie algebra $\mathbf{L} + \hat{\mathbf{Q}}$.
gilmore-lie-groups_ F01820	185	■ 0.623	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	the discussion of Problem 8 are the dimensions of the spaces $\mathbf{L}(3 \rightarrow n(n-1)/2)$, $\mathbf{Q}$
gilmore-lie-groups_ F01821	185	■ 0.623	$\mathbf{Q}$	$\mathbf{Q}$	the discussion of Problem 8 are the dimensions of the spaces $\mathbf{L}(3 \rightarrow n(n-1)/2)$, $\mathbf{Q}$
gilmore-lie-groups_ F01822	185	■ 1.000	$(6 \rightarrow n(n+1)/2)$	$(6 \rightarrow n(n+1)/2)$	$(6 \rightarrow n(n+1)/2)$, and the definition of $\hat{N}(3 \rightarrow n)$.
gilmore-lie-groups_ F01823	185	■ 1.000	$\hat{N}(3 \rightarrow n)$	$\hat{N}(3 \rightarrow n)$	$(6 \rightarrow n(n+1)/2)$, and the definition of $\hat{N}(3 \rightarrow n)$.
gilmore-lie-groups_ F01824	186	■ 1.000	$Z = r^i X_i$	$Z = r^i X_i$	Lie algebra with elements $Z = r^i X_i$ (r^i are real numbers, X_i the generators of the
gilmore-lie-groups_ F01825	186	■ 1.000	r^i	r^i	Lie algebra with elements $Z = r^i X_i$ (r^i are real numbers, X_i the generators of the
gilmore-lie-groups_ F01826	186	■ 1.000	$\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$	$\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$	of the form $\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$. This equation cannot be solved in

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gilmore-lie-groups_ F01827	186	■ 1.000	$h^{\mathfrak{i}}$	h^i	where h^i and e^α are complex numbers and the “diagonal” and “shift” operators
gilmore-lie-groups_ F01828	186	■ 1.000	$e^{\backslash\alpha}$	e^α	where h^i and e^α are complex numbers and the “diagonal” and “shift” operators
gilmore-lie-groups_ F01829	187	■ 1.000	$A_{\{n-1\}}, D_{\{n\}}, B_{\{n\}}, C_{\{n\}}$	A_{n-1}, D_n, B_n, C_n	classical simple Lie algebras A_{n-1}, D_n, B_n, C_n . These decompositions are: block
gilmore-lie-groups_ F01830	187	■ 1.000	$a_{\{1\}}, a_{\{2\}}, a_{\{3\}}$	a_1, a_2, a_3	for certain choices of the real parameters a_1, a_2, a_3 while in the compact case this
gilmore-lie-groups_ F01831	187	■ 1.000	$\backslash\left(a_{\{1\}}, a_{\{2\}}, a_{\{3\}}\right)$	(a_1, a_2, a_3)	is not possible. If the real parameters (a_1, a_2, a_3) and (b_1, b_2, b_3) are allowed to
gilmore-lie-groups_ F01832	187	■ 1.000	$\backslash\left(b_{\{1\}}, b_{\{2\}}, b_{\{3\}}\right)$	(b_1, b_2, b_3)	is not possible. If the real parameters (a_1, a_2, a_3) and (b_1, b_2, b_3) are allowed to
gilmore-lie-groups_ F01833	188	■ 0.998	$H_{\{i\}}, E_{\backslash\alpha}$	H_i, E_α	The Cartan–Killing inner product for the basis vectors H_i, E_α is
gilmore-lie-groups_ F01834	188	■ 1.000	$\backslash\frac{1}{\sqrt{2}}\backslash\left(E_{\backslash\alpha}\pm E_{-\backslash\alpha}\right)$	$\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$	of basis vectors $\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$:
gilmore-lie-groups_ F01835	189	■ 0.826	$H_{\{i\}}, \backslash\frac{1}{\sqrt{2}}\backslash\left(E_{\backslash\alpha}+E_{-\backslash\alpha}\right)$	$H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01836	189	■ 0.826	$\backslash\frac{1}{\sqrt{2}}\backslash\left(E_{\backslash\alpha}-E_{-\backslash\alpha}\right)$	$\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01837	189	■ 0.826	$\alpha \neq 0$	$\alpha \neq 0$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01838	189	■ 1.000	$\backslash\frac{1}{\sqrt{2}}\backslash\left(E_{\backslash\alpha}+E_{-\backslash\alpha}\right)$	$\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	generators H_i and $\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$ span a noncompact subspace.
gilmore-lie-groups_ F01839	189	■ 0.999	$A_{\{1\}}, \backslash\mathfrak{sl}(2;\mathbb{R})$	$A_1, \mathfrak{sl}(2; \mathbb{R})$	Two real forms of $A_1, \mathfrak{sl}(2; \mathbb{R})$ and $\mathfrak{su}(1, 1)$, are obtained as follows
gilmore-lie-groups_ F01840	189	■ 1.000	$h_{\{r\}}, h_{\{i\}}; a_{\{r\}}, a_{\{i\}}; b_{\{r\}}, b_{\{i\}}$	$h_r, h_i; a_r, a_i; b_r, b_i$	Here the six parameters $h_r, h_i; a_r, a_i; b_r, b_i$ are real.
gilmore-lie-groups_ F01841	190	■ 0.850	$n_{\{+\}}, n_{\{-\}}$	n_+, n_-	It is useful to specify how compact a real form is by specifying its index (n_+, n_-) ,
gilmore-lie-groups_ F01842	190	■ 1.000	$n_{\{+\}}$	n_+	where n_+ is the dimension of the subspace on which the nonsingular Cartan–
gilmore-lie-groups_ F01843	190	■ 1.000	$n_{\{-\}}$	n_-	Killing inner product is positive-definite and n_- is the dimension of the subspace
gilmore-lie-groups_ F01844	190	■ 1.000	$\chi=n_{\{+\}}-n_{\{-\}}$	$\chi = n_+ - n_-$	be abbreviated to a single integer, the character $\chi = n_+ - n_-$, to describe a real
gilmore-lie-groups_ F01845	190	■ 1.000	$i H_{\{i\}}, i \backslash\frac{1}{\sqrt{2}}\backslash\left(E_{\backslash\alpha}+E_{-\backslash\alpha}\right)$	$iH_i, i\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	the compact real form spanned by real linear combinations of $iH_i, i\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$,
gilmore-lie-groups_ F01846	190	■ 0.917	$g_{\backslash\mu}, \backslash\nu=\operatorname{diag}\{1,1,1,-1\}$	$g_{\mu,\nu} = \operatorname{diag}(1, 1, 1, -1)$	mation of space-time (x, y, z, ct) with indefinite metric $g_{\mu,\nu} = \operatorname{diag}(1, 1, 1, -1)$ to
gilmore-lie-groups_ F01847	190	■ 0.870	$x, y, z, i c t$	$x, y, z, i c t$	space-time with imaginary time $(x, y, z, i c t)$ and positive-definite metric $g_{\mu,\nu} =$
gilmore-lie-groups_ F01848	190	■ 0.870	$g_{\backslash\mu}, \backslash\nu=$	$g_{\mu,\nu} =$	space-time with imaginary time $(x, y, z, i c t)$ and positive-definite metric $g_{\mu,\nu} =$
gilmore-lie-groups_ F01849	190	■ 0.672	$\operatorname{diag}\{1,1,1,1\}$	$\operatorname{diag}(1, 1, 1, 1)$	$\operatorname{diag}(1, 1, 1, 1)$.
gilmore-lie-groups_ F01850	190	■ 1.000	$\mathfrak{p} \rightarrow \mathfrak{p}^{\backslash\prime}=i \mathfrak{p}$	$\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$	“Minkowski’s trick”: $\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$. The mapping, commutation relations, and
gilmore-lie-groups_ F01851	191	■ 1.000	$\mathfrak{g}^{\backslash\prime}, \mathfrak{h}$	$\mathfrak{g}', \mathfrak{h}$	In $\mathfrak{g}', \mathfrak{h}$ is the maximal compact subalgebra and $\mathfrak{p}'$ consists of all the noncompact
gilmore-lie-groups_ F01852	191	■ 1.000	$\mathfrak{p}^{\backslash\prime}$	$\mathfrak{p}'$	In $\mathfrak{g}', \mathfrak{h}$ is the maximal compact subalgebra and $\mathfrak{p}'$ consists of all the noncompact

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gilmore-lie-groups_ F01853	191	■ 1.000	$T^2=I$	$T^2 = I$	compact real form of that Lie algebra to itself that obey $T^2 = I$.
gilmore-lie-groups_ F01854	192	■ 1.000	$\mathfrak{u}(n, \mathbb{F})$	$\mathfrak{u}(n, \mathbb{F})$	In a block matrix decomposition the compact Lie algebras $\mathfrak{u}(n, \mathbb{F})$ have the form
gilmore-lie-groups_ F01855	192	■ 0.949	$A_{\{p\}}=-A_{\{p\}}^{\dagger}, A_{\{q\}}=-A_{\{q\}}^{\dagger}$	$A_p = -A_p^\dagger, A_q = -A_q^\dagger$	where $A_p = -A_p^\dagger$, $A_q = -A_q^\dagger$, and B is an arbitrary $p \times q$ matrix. Under the
gilmore-lie-groups_ F01856	192	■ 0.999	$I_{\{p+q\}}$	I_{p+q}	i . This is equivalent to changing the metric I_{p+q} that is preserved by $\mathfrak{u}(n; \mathbb{F})$ to the
gilmore-lie-groups_ F01857	192	■ 0.999	$\mathfrak{u}(n; \mathbb{F})$	$\mathfrak{u}(n; \mathbb{F})$	i . This is equivalent to changing the metric I_{p+q} that is preserved by $\mathfrak{u}(n; \mathbb{F})$ to the
gilmore-lie-groups_ F01858	192	■ 1.000	$I_{\{p, q\}}$	$I_{p,q}$	metric $I_{p,q}$ that is preserved by $\mathfrak{u}(p, q; \mathbb{F})$, where $p + q = n$. The factor i can be
gilmore-lie-groups_ F01859	192	■ 1.000	$\mathfrak{u}(p, q; \mathbb{F})$	$\mathfrak{u}(p, q; \mathbb{F})$	metric $I_{p,q}$ that is preserved by $\mathfrak{u}(p, q; \mathbb{F})$, where $p + q = n$. The factor i can be
gilmore-lie-groups_ F01860	192	■ 0.995	$\mathbb{F}=\mathbb{R}, \mathbb{C}, \mathbb{Q}$	$\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	For the fields $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$ related to the root spaces (D, B) , A, C the real forms
gilmore-lie-groups_ F01861	192	■ 0.995	$(D, B), A, C$	$(D, B), A, C$	For the fields $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$ related to the root spaces (D, B) , A, C the real forms
gilmore-lie-groups_ F01862	192	■ 0.998	$\mathfrak{sp}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$.
gilmore-lie-groups_ F01863	193	■ 1.000	$\mathfrak{sp}(n)=\mathfrak{u}(n; \mathbb{Q})$	$\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$	The transformation from $\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$ to $\mathfrak{sp}(2n; \mathbb{R})$ is somewhat less familiar.
gilmore-lie-groups_ F01864	193	■ 1.000	α, β, γ	α, β, γ	where α, β, γ and δ are real. With these replacements the Lie algebra of $n \times n$
gilmore-lie-groups_ F01865	193	■ 0.999	$\mathfrak{o}(2n)$	$\mathfrak{ou}(2n)$	these matrices $\mathfrak{ou}(2n)$, since they form an <u>o</u> rthogonal representation of the <u>u</u> nitary
gilmore-lie-groups_ F01866	193	■ 1.000	$\underline{2n} \times 2n$	$\underline{2n} \times 2n$	algebra in terms of $\underline{2n} \times 2n$ matrices. Similarly, the Lie algebra of $n \times n$ quaternion
gilmore-lie-groups_ F01867	194	■ 1.000	R^{2n}	R^{2n}	metric metric in R^{2n} . It is possible to choose coordinates $p_1, q_1, p_2, q_2, \dots, p_n, q_n$
gilmore-lie-groups_ F01868	194	■ 1.000	$p_{\{1\}}, q_{\{1\}}, p_{\{2\}}, q_{\{2\}}, \ldots, p_{\{n\}}, q_{\{n\}}$	$p_1, q_1, p_2, q_2, \dots, p_n, q_n$	metric metric in R^{2n} . It is possible to choose coordinates $p_1, q_1, p_2, q_2, \dots, p_n, q_n$
gilmore-lie-groups_ F01869	194	■ 1.000	$v_{\{i\}}^{\prime} G_{\{i j\}} v_{\{j\}}$	$v_i' G_{ij} v_j$	in this space so that the inner product between two vectors $v_i' G_{ij} v_j$ is
gilmore-lie-groups_ F01870	194	■ 0.967	$M \in \operatorname{Sp}(2n; \mathbb{R})$	$M \in \operatorname{Sp}(2n; \mathbb{R})$	Then symplectic transformations $M \in \operatorname{Sp}(2n; \mathbb{R})$ leave this metric matrix G invari-
gilmore-lie-groups_ F01871	194	■ 1.000	$M^t G M=G$	$M^t G M = G$	ant: $M^t G M = G$. Symplectic transformations leave invariant the canonical form
gilmore-lie-groups_ F01872	194	■ 1.000	$\mathfrak{p} \rightarrow i \mathfrak{p}$	$\mathfrak{p} \rightarrow i \mathfrak{p}$	Application of the map $\mathfrak{p} \rightarrow i \mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.
gilmore-lie-groups_ F01873	194	■ 1.000	$\mathfrak{so}^*(2n)$	$\mathfrak{so}^*(2n)$	Application of the map $\mathfrak{p} \rightarrow i \mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.
gilmore-lie-groups_ F01874	194	■ 1.000	$\mathfrak{su}^*(2n)$	$\mathfrak{su}^*(2n)$	Application of the map $\mathfrak{p} \rightarrow i \mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.
gilmore-lie-groups_ F01875	194	■ 0.991	A_{2n-1}	A_{2n-1}	Remark The real forms $\mathfrak{so}^*(2n)$ of D_n and $\mathfrak{su}^*(2n)$ of A_{2n-1} do not occur explicitly
gilmore-lie-groups_ F01876	195	■ 1.000	$\mathfrak{su}(2), B_1(\mathfrak{so}(3))$	$\mathfrak{su}(2)), B_1(\mathfrak{so}(3))$	the compact real form is $\mathfrak{su}(2))$, $B_1(\mathfrak{so}(3))$, and $C_1(\mathfrak{sp}(1))$ are equivalent, as are B_2
gilmore-lie-groups_ F01877	195	■ 1.000	$C_1(\mathfrak{sp}(1))$	$C_1(\mathfrak{sp}(1))$	the compact real form is $\mathfrak{su}(2))$, $B_1(\mathfrak{so}(3))$, and $C_1(\mathfrak{sp}(1))$ are equivalent, as are B_2
gilmore-lie-groups_ F01878	195	■ 1.000	$(\mathfrak{so}(5))$	$(\mathfrak{so}(5))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_ F01879	195	■ 1.000	$C_2(\mathfrak{sp}(2))$	$C_2(\mathfrak{sp}(2))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are

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gilmore-lie-groups_F01886	195	■ 1.000	$A_3(\frac{s}{u}(4))$	$A_3(\mathfrak{su}(4))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_F01881	195	■ 1.000	$D_3(\frac{s}{o}(6))$	$D_3(\mathfrak{so}(6))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are
gilmore-lie-groups_F01903	234	■ 0.679	D_3	D_3	7. One real form of D_3 is the conformal group $SO(4, 2)$.
gilmore-lie-groups_F01918	327	■ 0.999	F_4	F_4	F_4 , 162, 168
gilmore-lie-groups_F01944	196	■ 1.000	$T(\frac{p}{})=-\frac{p}{}$	$T(\mathfrak{p}) = -\mathfrak{p}$	that obeys $T(\mathfrak{p}) = -\mathfrak{p}$ is analytically continued by $\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$; the subspace $\mathfrak{h}$
gilmore-lie-groups_F01945	196	■ 1.000	$T(\frac{h}{})=+\frac{h}{}$	$T(\mathfrak{h}) = +\mathfrak{h}$	of $\mathfrak{g}$ that obeys $T(\mathfrak{h}) = +\mathfrak{h}$ is the maximal compact subalgebra of the noncompact
gilmore-lie-groups_F01946	196	■ 1.000	$\frac{g}{}^{\prime}:\frac{g}{}=\frac{h}{}+\frac{p}{}\rightarrow\frac{g}{}^{\prime}=\frac{h}{}+i\frac{p}{}$	$\mathfrak{g}' : \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}'$	real form $\mathfrak{g}': \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}$. For the simple classical (matrix) Lie alge-
gilmore-lie-groups_F01947	197	■ 1.000	$1 \leq i$	$1 \leq i$	1. Four operators $a_i^\dagger a_j$ can be constructed from boson operators for two modes ≤ 1 ,
gilmore-lie-groups_F01948	197	■ 1.000	$j \leq 2$	$j \leq 2$	$j \leq 2$. These operators close under commutation.
gilmore-lie-groups_F01949	197	■ 1.000	$a_1^\dagger a_1 + a_2^\dagger a_2$	$a_1^\dagger a_1 + a_2^\dagger a_2$	Conclude that $a_1^\dagger a_1 + a_2^\dagger a_2$ spans the maximum commutative subalgebra, $a_1^\dagger a_2 -$
gilmore-lie-groups_F01950	197	■ 1.000	$a_1^\dagger a_2 -$	$a_1^\dagger a_2 -$	Conclude that $a_1^\dagger a_1 + a_2^\dagger a_2$ spans the maximum commutative subalgebra, $a_1^\dagger a_2 -$
gilmore-lie-groups_F01951	197	■ 1.000	$a_1^\dagger a_1 - a_2^\dagger a_2$	$a_1^\dagger a_1 - a_2^\dagger a_2$	$a_2^\dagger a_1$ spans the maximal compact subalgebra, and the two generators $a_1^\dagger a_1 - a_2^\dagger a_2$
gilmore-lie-groups_F01952	197	■ 1.000	$a_1^\dagger a_2 + a_2^\dagger a_1$	$a_1^\dagger a_2 + a_2^\dagger a_1$	and $a_1^\dagger a_2 + a_2^\dagger a_1$ are noncompact.
gilmore-lie-groups_F01953	197	■ 1.000	$i \sigma_j$	$i\sigma_j$	Show that the three matrices that result are exactly $i\sigma_j$, where σ_j are the Pauli spin
gilmore-lie-groups_F01954	197	■ 1.000	σ_j	σ_j	Show that the three matrices that result are exactly $i\sigma_j$, where σ_j are the Pauli spin
gilmore-lie-groups_F01955	197	■ 0.982	$k \rightarrow 1$	$k \rightarrow 1$	2. The classical matrix groups $SO(n)$ are not simply connected, so they are $k \rightarrow 1$
gilmore-lie-groups_F01956	197	■ 0.844	$\overline{SO(n)} = Spin(n)$	$\overline{SO(n)} = Spin(n)$	images of their universal covering groups $\overline{SO(n)} = Spin(n)$, for some integer k .
gilmore-lie-groups_F01957	197	■ 1.000	$Spin(n)$	$Spin(n)$	Show that the covering groups $Spin(n)$ are classical matrix groups for $n = 3, 4, 5, 6$,
gilmore-lie-groups_F01958	197	■ 1.000	$n=3,4,5,6$	$n = 3, 4, 5, 6$	Show that the covering groups $Spin(n)$ are classical matrix groups for $n = 3, 4, 5, 6$,
gilmore-lie-groups_F01959	197	■ 1.000	$n>6$	$n > 6$	Show that for $n > 6$ the groups $Spin(n)$ are not equal to any classical matrix Lie
gilmore-lie-groups_F01960	198	■ 0.968	$H_i: n_1, n_2, \dots, n_r\rangle$	$H_i : n_1, n_2, \dots, n_r\rangle$	eigenvalues under the operators $H_i: n_1, n_2, \dots, n_r\rangle$,
gilmore-lie-groups_F01961	198	■ 1.000	$H_j \rightarrow$	$H_j \rightarrow$	c. For the orthogonal groups $SO(n)$, impose suitable reality conditions (i.e., $H_j \rightarrow$
gilmore-lie-groups_F01962	198	■ 1.000	$i H_j$	iH_j	iH_j , etc.), choose a Hilbert space containing the state $ l, 0, \dots, 0\rangle$ and show that
gilmore-lie-groups_F01963	198	■ 1.000	$ l, 0, \dots, 0\rangle$	$ l, 0, \dots, 0\rangle$	iH_j , etc.), choose a Hilbert space containing the state $ l, 0, \dots, 0\rangle$ and show that
gilmore-lie-groups_F01964	198	■ 0.989	$-m^2$	$-m^2$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_F01965	198	■ 0.989	$e^{im\phi}$	$e^{im\phi}$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).

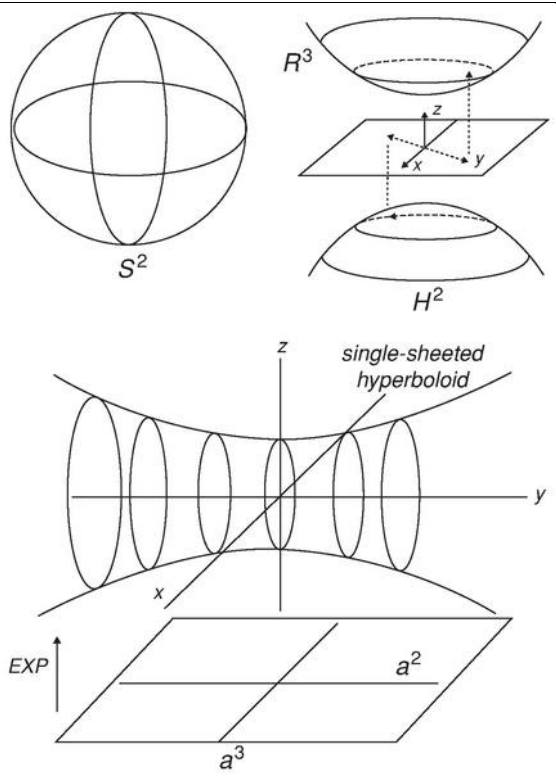
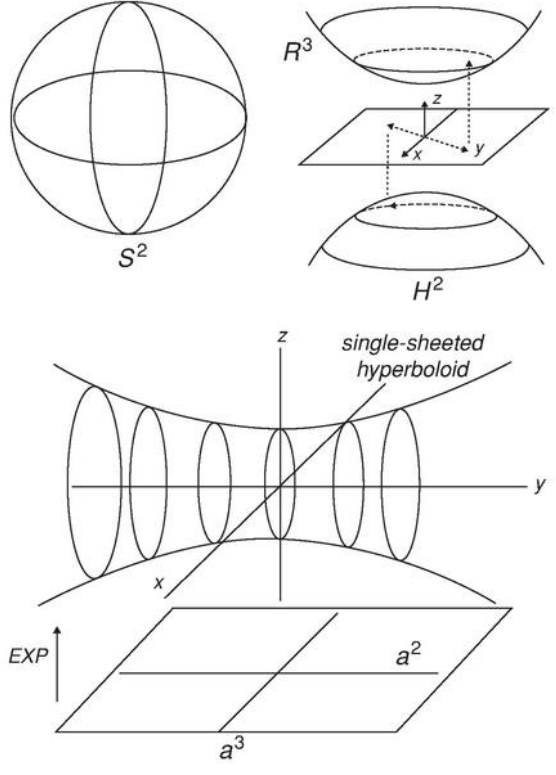
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F01966	198	■ 0.989	$-l(l+1)$	$-l(l+1)$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_F01967	198	■ 0.989	$Y_m^l(\theta, \phi)$	$Y_m^l(\theta, \phi)$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_F01968	198	■ 1.000	$J_i, i=1,2,3$	$J_i, i=1,2,3$	matrix elements of a <i>hermitian</i> representation of $U_i, i=1,2,3$. Exponentials of the
gilmore-lie-groups_F01969	198	■ 0.926	$\operatorname{EXP}\left(ir^k J_k\right)$	$\operatorname{EXP}\left(ir^k J_k\right)$	form $\operatorname{EXP}\left(ir^k J_k\right)$, with r^k real, provide <i>unitary</i> representations of the compact Lie
gilmore-lie-groups_F01970	198	■ 0.926	r^k	r^k	form $\operatorname{EXP}\left(ir^k J_k\right)$, with r^k real, provide <i>unitary</i> representations of the compact Lie
gilmore-lie-groups_F01971	198	■ 1.000	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$.
gilmore-lie-groups_F01972	199	■ 0.826	K_3	K_3	$SU(2)$ with J_3 diagonal, or in any matrix representation of $SU(1,1)$ with K_3
gilmore-lie-groups_F01973	199	■ 1.000	$e^{im\phi} \delta_{m'm}$	$e^{im\phi} \delta_{m'm}$	diagonal, the matrix is diagonal with matrix elements $e^{im\phi} \delta_{m'm}$. Show that the
gilmore-lie-groups_F01974	199	■ 1.000	$\phi \rightarrow \phi + 4\pi$	$\phi \rightarrow \phi + 4\pi$	single-valuedness condition under $\phi \rightarrow \phi + 4\pi$ requires that $m = \frac{1}{2}(n_1 - n_2)$ is
gilmore-lie-groups_F01975	199	■ 1.000	$m = \frac{1}{2}(n_1 - n_2)$	$m = \frac{1}{2}(n_1 - n_2)$	single-valuedness condition under $\phi \rightarrow \phi + 4\pi$ requires that $m = \frac{1}{2}(n_1 - n_2)$ is
gilmore-lie-groups_F01976	199	■ 1.000	$J_{\pm}$	$J_{\pm}$	integer or half-integer. Show that the shift operators $J_{\pm}$ and $K_{\pm}$ require that all m
gilmore-lie-groups_F01977	199	■ 1.000	$K_{\pm}$	$K_{\pm}$	integer or half-integer. Show that the shift operators $J_{\pm}$ and $K_{\pm}$ require that all m
gilmore-lie-groups_F01978	199	■ 1.000	$ n_1, n_2\rangle$	$ n_1, n_2\rangle$	c. Relax the assumption that all indices n_1, n_2 in the basis states $ n_1, n_2\rangle = n_1\rangle \otimes$
gilmore-lie-groups_F01979	199	■ 1.000	$ n_1, n_2\rangle = n_1\rangle \otimes$	$ n_1, n_2\rangle = n_1\rangle \otimes$	c. Relax the assumption that all indices n_1, n_2 in the basis states $ n_1, n_2\rangle = n_1\rangle \otimes$
gilmore-lie-groups_F01980	199	■ 1.000	$ n_2\rangle$	$ n_2\rangle$	$ n_2\rangle$ must be nonnegative integers. Construct the matrix elements of the operators
gilmore-lie-groups_F01981	199	■ 0.999	$J_3, J_{\pm}, K_3, K_{\pm}$	$J_3, J_{\pm}, K_3, K_{\pm}$	$J_3, J_{\pm}, K_3, K_{\pm}$ under this relaxed assumption. Show that all commutation relations
gilmore-lie-groups_F01982	199	■ 1.000	J_x	J_x	d. Show that the matrices for J_x and J_y are also hermitian provided that
gilmore-lie-groups_F01983	199	■ 1.000	J_y	J_y	d. Show that the matrices for J_x and J_y are also hermitian provided that
gilmore-lie-groups_F01984	199	■ 0.749	$(0,0)$	$(0,0)$	Show that the lattice sites in quadrants I and III of Fig. 11.2, with vertices at $(0,0)$
gilmore-lie-groups_F01985	199	■ 0.975	$ j_m\rangle = n_1, n_2\rangle, j = \frac{1}{2}(n_1 + n_2), m = \frac{1}{2}(n_1 - n_2)$	$ j_m\rangle = n_1, n_2\rangle, j = \frac{1}{2}(n_1 + n_2), m = \frac{1}{2}(n_1 - n_2)$	e. With the identification $ j_m\rangle = n_1, n_2\rangle, j = \frac{1}{2}(n_1 + n_2), m = \frac{1}{2}(n_1 - n_2)$, show
gilmore-lie-groups_F01986	199	■ 1.000	n_1	n_1	successive application of these operators to remain in QI or QIII, n_1 and n_2 must
gilmore-lie-groups_F01987	199	■ 1.000	n_2	n_2	successive application of these operators to remain in QI or QIII, n_1 and n_2 must

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gilmore-lie-groups_ F01992	200	—	$n_{\{1\}+n_{\{2\}}=$	$n_1 + n_2 =$	
gilmore-lie-groups_ F01993	200	—	$-1 < n_{\{1\}} < 0$	$-1 < n_1 < 0$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01994	200	—	$-1 < n_2 < 0$	$-1 < n_2 < 0$	
gilmore-lie-groups_ F01995	200	—	$n_1 + n_2 = -\frac{1}{2} + i\beta$	$n_1 + n_2 = -\frac{1}{2} + i\beta$	
gilmore-lie-groups_ F01996	200	■ 0.998	$\left n_1 + k, n_2 - k \right\rangle$	$ n_1 + k, n_2 - k\rangle$	i. In order to ensure that a set of states $ n_1 + k, n_2 - k\rangle$ (k integer) mapped into each
gilmore-lie-groups_ F01997	200	■ 1.000	$\left n_1, n_2 = -1 \right\rangle$	$ n_1, n_2 = -1\rangle$	(lowest m) state must be $ n_1, n_2 = -1\rangle$ for $n_1 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_ F01998	200	■ 1.000	$n_1 = 0, 1, 2, \dots$	$n_1 = 0, 1, 2, \dots$	(lowest m) state must be $ n_1, n_2 = -1\rangle$ for $n_1 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_ F01999	200	■ 0.995	$\left \{ \}_m^j \right\rangle = \left n_1, n_2 \right\rangle$	$ _m^j\rangle = n_1, n_2\rangle$	this bounded discrete series of representations are $ _m^j\rangle = n_1, n_2\rangle$ with $j + \frac{1}{2} =$
gilmore-lie-groups_ F02000	200	■ 0.995	$j + \frac{1}{2} =$	$j + \frac{1}{2} =$	this bounded discrete series of representations are $ _m^j\rangle = n_1, n_2\rangle$ with $j + \frac{1}{2} =$
gilmore-lie-groups_ F02001	200	■ 0.993	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the

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gilmore-lie-groups_ F02002	200	■ 0.993	$2\ j+1=0,1,2,3,\ \ldots$	$2j+1=0,1,2,3,\dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $\boxed{2j+1=0,1,2,3,\dots}$ and $m=j+1, j+2, \dots$. This is the
gilmore-lie-groups_ F02003	200	■ 0.993	$m=j+1,\ j+2,\ \ldots$	$m=j+1, j+2, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j+1=0,1,2,3,\dots$ and $\boxed{m=j+1, j+2,\dots}$. This is the
gilmore-lie-groups_ F02004	200	■ 1.000	$\mathcal{D}_{+}^{j}$	$\mathcal{D}_{+}^j$	discrete series of representations that is bounded below: $\mathcal{D}_{+}^j$.
gilmore-lie-groups_ F02005	200	■ 0.831	$\mathbf{j}$	j	j . In order to ensure that a set of states $ n_1+k, n_2-k\rangle$ (k integer) mapped into
gilmore-lie-groups_ F02006	200	■ 1.000	$\left n_{\{1\}}=-1,\ n_{\{2\}}\right\rangle$	$ n_1=-1, n_2\rangle$	(highest m) state must be $\boxed{n_1=-1, n_2}$ for $n_2=0,1,2,\dots$. The basis states for
gilmore-lie-groups_ F02007	200	■ 1.000	$n_{\{2\}}=0,1,2,\ \ldots$	$n_2=0,1,2,\dots$	(highest m) state must be $ n_1=-1, n_2\rangle$ for $\boxed{n_2=0,1,2,\dots}$. The basis states for
gilmore-lie-groups_ F02008	200	■ 1.000	$m=-j-1,-j-2,\ \ldots$	$m=-j-1, -j-2, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j+1=0,1,2,3,\dots$ and $\boxed{m=-j-1, -j-2,\dots}$. This is
gilmore-lie-groups_ F02009	200	■ 0.999	$\mathcal{D}_{-}^j$	$\mathcal{D}_{-}^j$	the discrete series of representations that is bounded above: $\mathcal{D}_{-}^j$.
gilmore-lie-groups_ F02010	201	■ 0.998	$\left(\mathcal{D}_{+}^j\right)$	$\left(\mathcal{D}_{+}^j\right)$	$(\mathcal{D}_{+}^j)$ or from QII $(\mathcal{D}_{-}^j)$.
gilmore-lie-groups_ F02011	201	■ 0.998	$\left(\mathcal{D}_{-}^j\right)$	$\left(\mathcal{D}_{-}^j\right)$	$(\mathcal{D}_{+}^j)$ or from QII $(\mathcal{D}_{-}^j)$.
gilmore-lie-groups_ F02012	201	■ 0.999	$\mid n_{\{1\}}+k,\ n_{\{2\}}-$	$ n_1+k, n_2-$	l . Now relax the condition that n_1 and n_2 are integers. The set of states $\boxed{n_1+k, n_2-}$
gilmore-lie-groups_ F02013	201	■ 1.000	$k\rangle(k=\ldots,-2,-1,0,+1,+2,\ \ldots)$	$k\rangle(k=\dots,-2,-1,0,+1,+2,\dots)$	$\boxed{k\rangle(k=\dots,-2,-1,0,+1,+2,\dots)}$ connected by $K_{\pm}$ carries a hermitian repre-
gilmore-lie-groups_ F02014	201	■ 1.000	$\mid p,\ q\rangle$	$ p, q\rangle$	vertices of the four quadrants. If this state is $ p, q\rangle$, with $-1 \leq p, q \leq 0$ then the
gilmore-lie-groups_ F02015	201	■ 1.000	$-1\ \leq\ p,\ q\ \leq\ 0$	$-1 \leq p, q \leq 0$	vertices of the four quadrants. If this state is $ p, q\rangle$, with $\boxed{-1 \leq p, q \leq 0}$ then the
gilmore-lie-groups_ F02016	201	■ 1.000	$\frac{1}{2}(p-q)=$	$\frac{1}{2}(p-q) =$	single-valuedness condition requires $\frac{1}{2}(p-q) =$ integer or half-integer. In the lat-
gilmore-lie-groups_ F02017	201	■ 0.891	$p-q=0$	$p-q=0$	values of the $U(1)$ index m . Therefore $p-q=0$ and $-1 \leq p=q \leq 0$. The states
gilmore-lie-groups_ F02018	201	■ 0.891	$-1\ \leq\ p=q\ \leq\ 0$	$-1 \leq p=q \leq 0$	values of the $U(1)$ index m . Therefore $p-q=0$ and $-1 \leq p=q \leq 0$. The states
gilmore-lie-groups_ F02019	201	■ 0.679	$\left.\left.\left.\right _m^j\right\rangle\right\rangle$	$ _m^j\rangle$	$ _m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j+\frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02020	201	■ 0.679	$-\frac{1}{2}\ \leq\ j+\frac{1}{2}\ \leq\ +\frac{1}{2}$	$-\frac{1}{2} \leq j+\frac{1}{2} \leq +\frac{1}{2}$	$ _m^j\rangle$ with m integer, j real and $\boxed{-\frac{1}{2} \leq j+\frac{1}{2} \leq +\frac{1}{2}}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02021	201	■ 0.679	$\mathcal{D}^p$	$\mathcal{D}^p$	$ _m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j+\frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02022	201	■ 1.000	$j+\frac{1}{2}=i\ \beta$	$j+\frac{1}{2}=i\beta$	m . By setting $j+\frac{1}{2}=i\beta$ (β real) the matrix elements in Eq. (11.30) become
gilmore-lie-groups_ F02023	201	■ 1.000	$j^{\prime}<-\frac{1}{2}$	$j'<-\frac{1}{2}$	Show that states with $\boxed{j'<-\frac{1}{2}}$ obtained by reflection through the diagonal contain-
gilmore-lie-groups_ F02024	201	■ 0.693	$\left(n_{\{1\}},\ n_{\{2\}}\right)=\left(-\frac{1}{2},-\frac{1}{2}\right)$	$(n_1, n_2) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$	ing the central point in the shaded square with coordinates $(n_1, n_2) = (-\frac{1}{2}, -\frac{1}{2})$
gilmore-lie-groups_ F02025	201	■ 1.000	$j>-\frac{1}{2}$	$j>-\frac{1}{2}$	support representations equivalent to those with index $\boxed{j>-\frac{1}{2}}$. The relation among
gilmore-lie-groups_ F02026	201	■ 0.311	$j+\frac{1}{2}=-\left(j^{\prime}+\frac{1}{2}\right)$	$j+\frac{1}{2}=-\left(j'+\frac{1}{2}\right)$	indices is $j+\frac{1}{2}=-(j'+\frac{1}{2})$ and the relation among states is $ _{m'}^{j'}\rangle \simeq _{m=m'}^j\rangle$.
gilmore-lie-groups_ F02027	201	■ 0.311	$\left k_{m^{\prime}}^{j^{\prime}}\right\rangle \simeq\left j_{m=m^{\prime}}\right\rangle$	$ k_{m'}^{j'}\rangle \simeq _{m=m'}^j\rangle$	indices is $j+\frac{1}{2}=-(j'+\frac{1}{2})$ and the relation among states is $ _{m'}^{j'}\rangle \simeq _{m=m'}^j\rangle$.

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gilmore-lie-groups_ F02028	201	■ 0.996	$X=h^{\mathrm{i}}\,H_{\mathrm{i}}+e^{\mathrm{\alpha}}\,E_{\mathrm{\alpha}}$	$X=h^iH_i+e^\alpha E_\alpha$	$X=h^iH_i+e^\alpha E_\alpha$, with h^i, e^α real. Show that the real subalgebra spanned by the
gilmore-lie-groups_ F02029	201	■ 0.996	$h^{\mathrm{i}},\,e^{\mathrm{\alpha}}$	h^i, e^α	$X=h^iH_i+e^\alpha E_\alpha$, with h^i, e^α real. Show that the real subalgebra spanned by the
gilmore-lie-groups_ F02030	201	■ 1.000	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$ linear combinations of the form $(E_\alpha-E_{-\alpha})/\sqrt{2}$ is the maximal compact
gilmore-lie-groups_ F02031	201	■ 1.000	$\left(E_{\mathrm{\alpha}}-E_{-\mathrm{\alpha}}\right)/\sqrt{2}$	$(E_\alpha-E_{-\alpha})/\sqrt{2}$	$\frac{1}{2}(n-l)$ linear combinations of the form $(E_\alpha-E_{-\alpha})/\sqrt{2}$ is the maximal compact
gilmore-lie-groups_ F02032	202	■ 1.000	$\mathfrak{sp}(p,q)$	$\mathfrak{sp}(p,q)$	6. The noncompact real form $\mathfrak{sp}(p,q)$ of the symplectic algebra was constructed from
gilmore-lie-groups_ F02033	202	■ 1.000	$\mathfrak{sp}(p+q)$	$\mathfrak{sp}(p+q)$	the compact real form $\mathfrak{sp}(p+q)$ by “Minkowski’s trick,” or analytic continuation.
gilmore-lie-groups_ F02034	203	■ 0.999	$\mathfrak{p},\,i\,\mathfrak{p}$	$\mathfrak{p}, i\mathfrak{p}$	countered subspaces $\mathfrak{p}, i\mathfrak{p}$ on which the Cartan–Killing inner product was
gilmore-lie-groups_ F02035	203	■ 1.000	$i\,\mathfrak{p}$	$i\mathfrak{p}$	negative-definite (on $\mathfrak{p}$) or positive-definite (on $i\mathfrak{p}$). In both cases these
gilmore-lie-groups_ F02036	203	■ 1.000	g_{ij}	g_{ij}	metric g_{ij} is definite, either negative or positive. Manifolds with a definite
gilmore-lie-groups_ F02037	203	■ 1.000	$\operatorname{EXP}(\mathfrak{p})$	$\operatorname{EXP}(\mathfrak{p})$	each point in the space $\operatorname{EXP}(\mathfrak{p})$ or $\operatorname{EXP}(i\mathfrak{p})$ is the image of the origin
gilmore-lie-groups_ F02038	203	■ 1.000	$\operatorname{EXP}(i\,\mathfrak{p})$	$\operatorname{EXP}(i\mathfrak{p})$	each point in the space $\operatorname{EXP}(\mathfrak{p})$ or $\operatorname{EXP}(i\mathfrak{p})$ is the image of the origin
gilmore-lie-groups_ F02039	203	■ 0.997	$S^2\sim S\,U(2)/U(1)$	$S^2\sim SU(2)/U(1)$	These were $S^2\sim SU(2)/U(1)$, which is compact, and its dual $H_{2+}^2=SL(2;\mathbb{R})/$
gilmore-lie-groups_ F02040	203	■ 0.997	$H_{2+}^2=S\,L(2\,;\,\mathbb{R})\,/\,$	$H_{2+}^2=SL(2;\mathbb{R})/$	These were $S^2\sim SU(2)/U(1)$, which is compact, and its dual $H_{2+}^2=SL(2;\mathbb{R})/$
gilmore-lie-groups_ F02041	203	■ 1.000	$S\,O(2)=S\,U(1,1)\,/\,U(1)$	$SO(2)=SU(1,1)/U(1)$	$SO(2)=SU(1,1)/U(1)$, which is the upper sheet of the two-sheeted hyperboloid.
gilmore-lie-groups_ F02042	203	■ 1.000	$H_1^2=S\,L(2\,;\,\mathbb{R})\,/\,S\,O(1,1)$	$H_1^2=SL(2;\mathbb{R})/SO(1,1)$	“Between” these two spaces occurs $H_1^2=SL(2;\mathbb{R})/SO(1,1)$, which is the single-
gilmore-lie-groups_ F02043	203	■ 0.943	$\mathfrak{su}(2)-\mathfrak{u}(1)$	$\mathfrak{su}(2)-\mathfrak{u}(1)$	The Cartan–Killing inner product in the linear vector subspace $\mathfrak{su}(2)-\mathfrak{u}(1)$ is
gilmore-lie-groups_ F02044	203	■ 0.999	$S\,U(2)\,/\,U(1)\,\sim S^2$	$SU(2)/U(1)\sim S^2$	Killing metric on the space $SU(2)/U(1)\sim S^2$, the sphere. On S^2 the Cartan–Killing
gilmore-lie-groups_ F02045	203	■ 0.991	$\mathfrak{su}(1,1)-\mathfrak{u}(1)\,\simeq\,\mathfrak{sl}(2\,;\,\mathbb{R})-\mathfrak{so}(2)$	$\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$	The Cartan–Killing inner product on $\mathfrak{su}(1,1)-\mathfrak{u}(1)\simeq\mathfrak{sl}(2;\mathbb{R})-\mathfrak{so}(2)$ is
gilmore-lie-groups_ F02046	203	■ 1.000	$H_{2+}^2=S\,U(1,1)\,/\,S\,O(2)$	$H_{2+}^2=SU(1,1)/SO(2)$	positive-definite. It maps to a positive-definite metric on $H_{2+}^2=SU(1,1)/SO(2)$.

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gilmore-lie-groups_ F02047	204	—	$S^2 = S O(3) / S O(2) = S U(2) / U(1), \quad H_{2+}^2 = S O(2,1) / S O(2) =$	$S^2 = SO(3)/SO(2) = SU(2)/U(1), \quad H_{2+}^2 = SO(2,1)/SO(2) =$	 The image contains three diagrams. The top left shows a sphere labeled S^2. The top right shows a 3D coordinate system with axes x, y, z and a hyperboloid of two sheets, with the upper sheet labeled H^2. The bottom diagram shows a single-sheeted hyperboloid in a 3D coordinate system with axes x, y, z. Below the hyperboloid is a 2D grid with labels a^2 and a^3, and an arrow labeled EXP pointing upwards.
gilmore-lie-groups_ F02048	204	—	$S U(1,1) / U(1), H_1^2 = S O(2,1) / S O(1,1) = L(2 ; \mathbb{R}) / S O(1,1)$	$SU(1,1)/U(1), H_1^2 = SO(2,1)/SO(1,1) = SL(2; \mathbb{R})/SO(1,1)$	 The image contains three diagrams. The top left shows a sphere labeled S^2. The top right shows a 3D coordinate system with axes x, y, z and a hyperboloid of two sheets, with the upper sheet labeled H^2. The bottom diagram shows a single-sheeted hyperboloid in a 3D coordinate system with axes x, y, z. Below the hyperboloid is a 2D grid with labels a^2 and a^3, and an arrow labeled EXP pointing upwards.
gilmore-lie-groups_ F02049	204	■ 0.998	$\mathfrak{su}(1,1) - \mathfrak{so}(1,1)$	$\mathfrak{su}(1,1) - \mathfrak{so}(1,1)$	obtained by exponentiating $\mathfrak{su}(1,1) - \mathfrak{so}(1,1)$. The Cartan–Killing inner product
gilmore-lie-groups_ F02050	204	■ 0.966	$\operatorname{EXP}[\mathfrak{su}(1,1) - \mathfrak{so}(1,1)] = SU(1,1) / SO(1,1)$	$\operatorname{EXP}[\mathfrak{su}(1,1) - \mathfrak{so}(1,1)] = SU(1,1)/SO(1,1)$	the topological space $\operatorname{EXP}[\mathfrak{su}(1,1) - \mathfrak{so}(1,1)] = SU(1,1)/SO(1,1)$ is indefinite.
gilmore-lie-groups_ F02051	204	■ 0.952	$\mathfrak{so}(n), \mathfrak{su}(n), \mathfrak{sp}(n)$	$\mathfrak{so}(n), \mathfrak{su}(n), \mathfrak{sp}(n)$	algebra $\mathfrak{g}$ (i.e., $\mathfrak{so}(n), \mathfrak{su}(n), \mathfrak{sp}(n)$) the Cartan decompositions have the form (11.10)
gilmore-lie-groups_ F02052	205	■ 1.000	$\mathfrak{p}(i \mathfrak{p})$	$\mathfrak{p}(i\mathfrak{p})$	On the linear vector space $\mathfrak{p}$ ($i\mathfrak{p}$) the Cartan–Killing inner product is negative

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gilmore-lie-groups_ F02053	205	■ 1.000	$P=G / H=\operatorname{EXP}\left(\frac{\mathfrak{p}}{\mathfrak{p}}\right)$	$P=G / H=\operatorname{EXP}(\mathfrak{p})$	The space $P=G / H=\operatorname{EXP}(\mathfrak{p})$ (e.g., S^2) is compact. The exponential of a
gilmore-lie-groups_ F02054	205	■ 1.000	P	P	identity. The space P is not topologically equivalent to any Euclidean space, in
gilmore-lie-groups_ F02055	205	■ 0.999	$P^{\{\prime\}}=G^{\{\prime\}} / H=\operatorname{EXP}\left(i \frac{\mathfrak{p}}{\mathfrak{p}}\right)$	$P^{\prime}=G^{\prime} / H=\operatorname{EXP}(i \mathfrak{p})$	The space $P^{\prime}=G^{\prime} / H=\operatorname{EXP}(i \mathfrak{p})$ (i.e., H_{2+}^2) is noncompact. The exponential of
gilmore-lie-groups_ F02056	205	■ 1.000	$P^{\{\prime\}}=\operatorname{EXP}\left(i \frac{\mathfrak{p}}{\mathfrak{p}}\right)$	$P^{\prime}=\operatorname{EXP}(i \mathfrak{p})$	simply goes away from this point without ever returning. The space $P^{\prime}=\operatorname{EXP}(i \mathfrak{p})$ is
gilmore-lie-groups_ F02057	205	■ 1.000	$n=\operatorname{dim} i \frac{\mathfrak{p}}{\mathfrak{p}}$	$n=\operatorname{dim} i \mathfrak{p}$	topologically equivalent to a Euclidean space R^n , where $n=\operatorname{dim} i \mathfrak{p}$. Geometrically
gilmore-lie-groups_ F02058	205	■ 1.000	$P=\operatorname{EXP}\left(\frac{\mathfrak{p}}{\mathfrak{p}}\right)$	$P=\operatorname{EXP}(\mathfrak{p})$	The Riemannian spaces $P=\operatorname{EXP}(\mathfrak{p})$ and $P^{\prime}=\operatorname{EXP}(i \mathfrak{p})$ are symmetric but not
gilmore-lie-groups_ F02059	205	■ 1.000	$\frac{\mathfrak{g}}{\mathfrak{g}}=\frac{\mathfrak{k}}{\mathfrak{k}}+\frac{\mathfrak{p}}{\mathfrak{p}}$	$\mathfrak{g}=\mathfrak{k}+\mathfrak{p}$	If $\mathfrak{g}$ is simple with a Cartan decomposition of the form $\mathfrak{g}=\mathfrak{k}+\mathfrak{p}$, with standard
gilmore-lie-groups_ F02060	205	■ 1.000	$\left[\frac{\mathfrak{k}}{\mathfrak{k}}, \frac{\mathfrak{k}}{\mathfrak{k}}\right] \subseteq \frac{\mathfrak{k}}{\mathfrak{k}},\left[\frac{\mathfrak{k}}{\mathfrak{k}}, \frac{\mathfrak{p}}{\mathfrak{p}}\right] \subseteq \frac{\mathfrak{p}}{\mathfrak{p}}$	$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k},[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$	commutation relations $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k},[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, the quotient coset $P=$
gilmore-lie-groups_ F02061	205	■ 1.000	$\left[\frac{\mathfrak{p}}{\mathfrak{p}}, \frac{\mathfrak{p}}{\mathfrak{p}}\right] \subseteq \frac{\mathfrak{k}}{\mathfrak{k}}$	$[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$	commutation relations $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k},[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, the quotient coset $P=$
gilmore-lie-groups_ F02062	205	■ 1.000	$P=$	$P=$	commutation relations $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k},[\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, the quotient coset $P=$
gilmore-lie-groups_ F02063	205	■ 1.000	G / K	G / K	G / K is a globally symmetric space as every point “looks like” every other point.
gilmore-lie-groups_ F02064	205	■ 1.000	$P=G / H$	$P=G / H$	of points (coset representatives $P=G / H$ or $P^{\prime}=G^{\prime} / H$) in the Lie group.
gilmore-lie-groups_ F02065	205	■ 1.000	$P^{\{\prime\}}=G^{\{\prime\}} / H$	$P^{\prime}=G^{\prime} / H$	of points (coset representatives $P=G / H$ or $P^{\prime}=G^{\prime} / H$) in the Lie group.
gilmore-lie-groups_ F02066	205	■ 0.993	$S O(p, q) / S O(p) \times S O(q)$	$S O(p, q) / S O(p) \times S O(q)$	resentation. The secular equations for the spaces $S O(p, q) / S O(p) \times S O(q)$,
gilmore-lie-groups_ F02067	206	■ 1.000	$B^{\dagger} B$	$B^{\dagger} B$	It is easy to check that the function ϕ_j depends on the $q \times q$ matrix $B^{\dagger} B$ or the
gilmore-lie-groups_ F02068	206	■ 1.000	$B B^{\dagger}$	$B B^{\dagger}$	$p \times p$ matrix $B B^{\dagger}$, whichever is smaller. The rank of these spaces is $\min (p, q)$.
gilmore-lie-groups_ F02069	206	■ 1.000	$\min (p, q)$	$\min (p, q)$	$p \times p$ matrix $B B^{\dagger}$, whichever is smaller. The rank of these spaces is $\min (p, q)$.
gilmore-lie-groups_ F02070	206	■ 1.000	$P \backslash \left(P^{\{\prime\}}\right)$	$P\left(P^{\prime}\right)$	(iv) the number of invariant (Laplace–Beltrami) operators defined over the space $P\left(P^{\prime}\right)$;
gilmore-lie-groups_ F02072	206	■ 0.999	$G^{\{\prime\}}$	$G^{\prime}$	form $G^{\prime} / H$, where $G^{\prime}$ is simple and noncompact and H is the maximal compact
gilmore-lie-groups_ F02078	313	■ 0.994	$n-1$	$n-1$	($n=$)st order equation.
gilmore-lie-groups_ F02082	207	■ 1.000	$S O(p, q) / S O(p) \otimes S O(q)$	$S O(p, q) / S O(p) \otimes S O(q)$	example, $S O(p, q) / S O(p) \otimes S O(q)$ and $S O(p+q) / S O(p) \otimes S O(q)$ are dual.
gilmore-lie-groups_ F02106	207	■ 1.000	$G^{\{\prime\}} / H \rightarrow G / H$	$G^{\prime} / H \rightarrow G / H$	subgroup in $G^{\prime}$. To each there is a compact real form under $G^{\prime} / H \rightarrow G / H$. For
gilmore-lie-groups_ F02107	207	■ 1.000	$S O(p+q) / S O(p) \otimes S O(q)$	$S O(p+q) / S O(p) \otimes S O(q)$	example, $S O(p, q) / S O(p) \otimes S O(q)$ and $S O(p+q) / S O(p) \otimes S O(q)$ are dual.
gilmore-lie-groups_ F02108	207	■ 1.000	$P, P^{\{\prime\}}$	$P, P^{\prime}$	The metric tensor on the spaces $P, P^{\prime}$ is computed by defining a metric at the
gilmore-lie-groups_ F02109	207	■ 0.987	$d x\left(\mathrm{Id}\right)$	$d x(\operatorname{Id})$	If $d x(\operatorname{Id})$ are infinitesimal displacements at the identity that are translated to
gilmore-lie-groups_ F02110	207	■ 1.000	$d x(p)$	$d x(p)$	infinitesimal displacements $d x(p)$ at point p , then these two sets of infinitesimals
gilmore-lie-groups_ F02111	208	■ 1.000	$M^{\{i\}}_{\{\mu\}}(p)$	$M^i_{\mu}(p)$	The matrix $M^i_{\mu}(p)$ is not easy to compute in general. For the rank-one spaces

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gilmore-lie-groups_ F02112	208	■ 0.998	$S O(n, 1) / S O(n), S U(n, 1) / U(n), S p(n, 1) / S p(n) \times S p(1)$	$SO(n,1)/SO(n), SU(n,1)/U(n), Sp(n,1)/Sp(n) \times Sp(1)$	$SO(n,1)/SO(n), SU(n,1)/U(n), Sp(n,1)/Sp(n) \times Sp(1)$ defined by
gilmore-lie-groups_ F02113	208	■ 1.000	$M^{\{i\}}_{\{\mu\}}(X)$	$M^i_{\mu}(X)$	the matrix $M^i_{\mu}(X)$ is determined from
gilmore-lie-groups_ F02114	208	■ 1.000	$W^{\{-1\}}$	W^{-1}	The matrix $M^i_{\mu}(X)$ is given by W^{-1} . Since the Cartan–Killing inner product is I_n
gilmore-lie-groups_ F02115	208	■ 1.000	$H_2^2=S O(2,1) / S O(2)$	$H_2^2 = SO(2,1)/SO(2)$	the noncompact hyperboloid $H_2^2 = SO(2,1)/SO(2)$ and the compact sphere
gilmore-lie-groups_ F02116	209	■ 0.999	$S^{\{2\}}=S O(3) / S O(2)$	$S^2 = SO(3)/SO(2)$	$S^2 = SO(3)/SO(2)$. The metric tensors in the two cases are the 2×2 matrices
gilmore-lie-groups_ F02117	209	■ 1.000	$S O(2+1) / S O(2)$	$SO(2+1)/SO(2)$	ric space $SO(2+1)/SO(2)$ is parameterized by the interior of the unit circle
gilmore-lie-groups_ F02118	209	■ 0.999	$Y^{\{2\}}=1-\left(x^{\{2\}}+y^{\{2\}}\right) \geq 0$	$Y^2 = 1 - (x^2 + y^2) \geq 0$	$Y^2 = 1 - (x^2 + y^2) \geq 0$.
gilmore-lie-groups_ F02119	209	■ 1.000	x, y	x, y	for the compact case, the sphere. Instead of using the pair x, y as coordinates,
gilmore-lie-groups_ F02120	209	■ 1.000	$x^{\{i\}}, i=1,2, \ldots, N$	$x^i, i = 1, 2, \dots, N$	we use indexed coordinates $x^i, i = 1, 2, \dots, N$, and set $N = 2$ at the end of this
gilmore-lie-groups_ F02121	209	■ 1.000	$N=2$	$N = 2$	we use indexed coordinates $x^i, i = 1, 2, \dots, N$, and set $N = 2$ at the end of this
gilmore-lie-groups_ F02122	210	■ 1.000	$g^{\{i j\}} \simeq \delta^{\{i j\}}-x^{\{i\}} x^{\{j\}}$	$g^{ij} \simeq \delta^{ij} - x^i x^j$	The inverse (contravariant metric) is $g^{ij} \simeq \delta^{ij} - x^i x^j$, but we will not need this
gilmore-lie-groups_ F02123	210	■ 0.900	$g^{\{i j\}} \rightarrow \delta^{\{i j\}}$	$g^{ij} \rightarrow \delta^{ij}$	result. In the neighborhood of the identity ($g^{ij} \rightarrow \delta^{ij}$)
gilmore-lie-groups_ F02124	210	■ 1.000	$R_{\{\mu \sigma, \alpha \beta\}}=$	$R_{\mu\sigma,\alpha\beta} =$	delta function at the origin, and the resulting fully covariant metric tensor $R_{\mu\sigma,\alpha\beta} =$
gilmore-lie-groups_ F02125	210	■ 1.000	$\delta_{\alpha \mu} \delta_{\beta \sigma}-\delta_{\alpha \sigma} \delta_{\beta \mu}$	$\delta_{\alpha\mu}\delta_{\beta\sigma} - \delta_{\alpha\sigma}\delta_{\beta\mu}$	$\delta_{\alpha\mu}\delta_{\beta\sigma} - \delta_{\alpha\sigma}\delta_{\beta\mu}$ exhibits the full spectrum of expected symmetries.
gilmore-lie-groups_ F02126	210	■ 0.874	$R=2$	$R = 2$	For $N = 2$ (sphere S^2), $R = 2$.
gilmore-lie-groups_ F02127	210	■ 1.000	H_2^2	H_2^2	The computation can be carried out just as easily for the noncompact space H_2^2 .
gilmore-lie-groups_ F02128	210	■ 0.848	$\Gamma_{\mu \nu}^{\sigma} \rightarrow-\delta_{\mu \nu} x^{\sigma}$	$\Gamma_{\mu\nu}^{\sigma} \rightarrow -\delta_{\mu\nu} x^{\sigma}$	for example $\Gamma_{\mu\nu}^{\sigma} \rightarrow -\delta_{\mu\nu} x^{\sigma}$. The end result for H_2^2 is that $R = -2$.
gilmore-lie-groups_ F02129	210	■ 0.848	$R=-2$	$R = -2$	for example $\Gamma_{\mu\nu}^{\sigma} \rightarrow -\delta_{\mu\nu} x^{\sigma}$. The end result for H_2^2 is that $R = -2$.
gilmore-lie-groups_ F02130	211	■ 0.998	$d s^{\{2\}}=g_{\{\mu \nu\}} d x^{\{\mu\}} d x^{\{\nu\}}>0$	$ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} > 0$	positive-definite ($ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} > 0$, equality $\Rightarrow dx = 0$) nor negative-
gilmore-lie-groups_ F02131	211	■ 0.998	$\rightarrow d x=0$	$\Rightarrow dx = 0$	positive-definite ($ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} > 0$, equality $\Rightarrow dx = 0$) nor negative-
gilmore-lie-groups_ F02132	211	■ 0.958	$\left(d s^{\{2\}}<0\right)$	$(ds^2 < 0)$	definite ($ds^2 < 0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-
gilmore-lie-groups_ F02133	211	■ 0.958	$ g \neq 0$	$\ g\ \neq 0$	definite ($ds^2 < 0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-
gilmore-lie-groups_ F02134	211	■ 1.000	$\mathfrak{g}^{\{\prime \prime\}}$	$\mathfrak{g}''$	To make these statements more explicit, assume a Lie algebra $\mathfrak{g}''$ (noncompact)
gilmore-lie-groups_ F02135	211	■ 1.000	$\mathfrak{h}^{\{\prime \prime\}}$	$\mathfrak{h}''$	Then $\mathfrak{h}''$ and $\mathfrak{p}''$ are orthogonal subspaces in $\mathfrak{g}''$ under the Cartan–Killing inner
gilmore-lie-groups_ F02136	211	■ 1.000	$\mathfrak{p}^{\{\prime \prime\}}$	$\mathfrak{p}''$	Then $\mathfrak{h}''$ and $\mathfrak{p}''$ are orthogonal subspaces in $\mathfrak{g}''$ under the Cartan–Killing inner
gilmore-lie-groups_ F02137	211	■ 1.000	$H^{\{\prime \prime\}}=$	$H'' =$	indefinite. The space is curved and typically multiply connected. The space $H'' =$

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gilmore-lie-groups_ F02138	211	■ 1.000	$\operatorname{EXP}\left(\mathfrak{h}^{\prime\prime}\right)$	$\operatorname{EXP}(\mathfrak{h}'')$	$\operatorname{EXP}(\mathfrak{h}'')$ is also an interesting pseudo-Riemannian symmetric space.
gilmore-lie-groups_ F02139	211	■ 1.000	T_1, T_2	T_1, T_2	spaces. Begin with a compact simple Lie algebra $\mathfrak{g}$ and suppose T_1, T_2 are two
gilmore-lie-groups_ F02140	211	■ 0.819	$T_1^2=I$	$T_1^2=I$	metric-preserving mappings of the Lie algebra onto itself that obey $T_1^2=I, T_2^2=I$
gilmore-lie-groups_ F02141	211	■ 0.819	$T_2^2=I$	$T_2^2=I$	metric-preserving mappings of the Lie algebra onto itself that obey $T_1^2=I, T_2^2=I$
gilmore-lie-groups_ F02142	211	■ 1.000	$T_1\neq T_2$	$T_1\neq T_2$	(cf. Section 11.3) and $T_1\neq T_2$. Define the eigenspaces of $\mathfrak{g}$ under T_1, T_2 as $\mathfrak{g}_{\pm,\pm}$:
gilmore-lie-groups_ F02143	211	■ 1.000	$\mathfrak{g}_{\pm,\pm}$	$\mathfrak{g}_{\pm,\pm}$	(cf. Section 11.3) and $T_1\neq T_2$. Define the eigenspaces of $\mathfrak{g}$ under T_1, T_2 as $\mathfrak{g}_{\pm,\pm}$:
gilmore-lie-groups_ F02144	212	■ 1.000	$\mathfrak{h}^{\prime\prime}, \mathfrak{p}^{\prime\prime}$	$\mathfrak{h}'', \mathfrak{p}''$	The subspaces $\mathfrak{h}'', \mathfrak{p}''$ obey commutation relations (12.20). The Cartan-Killing inner
gilmore-lie-groups_ F02145	212	■ 0.996	$\mathfrak{p}^{\prime\prime}$	$\mathfrak{p}''$	product is indefinite on both $\mathfrak{h}''$ and $\mathfrak{p}''$ as long as $T_1\neq T_2$.
gilmore-lie-groups_ F02146	212	■ 0.999	$T_1=$	$T_1=$	For $\mathfrak{su}(2)$ the only two mappings are $T_1=$ block diagonal decomposition and
gilmore-lie-groups_ F02147	212	■ 1.000	$T_2=$	$T_2=$	$T_2=$ complex conjugation. The eigenspace decomposition is
gilmore-lie-groups_ F02151	213	■ 1.000	$T_3=T_1T_2$	$T_3=T_1T_2$	generate a third, $T_3=T_1T_2$ and that $T_1T_2=T_2T_1$. Show that $T_3\neq I$ if $T_1\neq T_2$. Show
gilmore-lie-groups_ F02152	212	■ 1.000	$\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i\sigma_3, \mathfrak{g}_{-,+}=i\sigma_2, \mathfrak{g}_{-,-}=i\sigma_1$	$\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i\sigma_3, \mathfrak{g}_{-,+}=i\sigma_2, \mathfrak{g}_{-,-}=i\sigma_1$	This gives $\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i\sigma_3, \mathfrak{g}_{-,+}=i\sigma_2, \mathfrak{g}_{-,-}=i\sigma_1$. Note that each mapping
gilmore-lie-groups_ F02153	212	■ 1.000	T_i	T_i	T_i has one positive and two negative eigenvalues, and chooses a different generator
gilmore-lie-groups_ F02154	212	■ 0.999	$\mathfrak{h}'$	$\mathfrak{h}'$	for the maximal compact subalgebra $\mathfrak{h}'$ of the noncompact real form $\mathfrak{g}'$.
gilmore-lie-groups_ F02155	212	■ 0.984	$P=G/K$	$P=G/K$	Globally symmetric spaces have the form $P=G/K$, where $\mathfrak{g}$ is a real form of
gilmore-lie-groups_ F02156	212	■ 1.000	$\phi_j(B, B^\dagger)$	$\phi_j(B, B^\dagger)$	1. Show that the invariant polynomials $\phi_j(B, B^\dagger)$ in (12.4) actually depend on the invari-
gilmore-lie-groups_ F02157	212	■ 0.585	$(p-q)$	$(p-q)$	The remaining $(p-q)$ or $(q-p)$ (whichever is positive) eigenvalues of the larger
gilmore-lie-groups_ F02158	212	■ 0.585	$(q-p)$	$(q-p)$	The remaining $(p-q)$ or $(q-p)$ (whichever is positive) eigenvalues of the larger
gilmore-lie-groups_ F02159	213	■ 1.000	Δ^2	Δ^2	2. The second order Laplace–Beltrami operator Δ^2 is constructed from the second order
gilmore-lie-groups_ F02160	213	■ 1.000	$G/H=P$	$G/H=P$	$G/H=P$.
gilmore-lie-groups_ F02161	213	■ 1.000	$\Delta^2=g^{ij}(\partial_i\partial_j-\Gamma_{ij}^k\partial_k)$	$\Delta^2=g^{ij}(\partial_i\partial_j-\Gamma_{ij}^k\partial_k)$	tensor on P as $\Delta^2=g^{ij}(\partial_i\partial_j-\Gamma_{ij}^k\partial_k)$.
gilmore-lie-groups_ F02162	213	■ 1.000	$x^2+y^2\leq 1$	$x^2+y^2\leq 1$	unit disk $x^2+y^2\leq 1$.
gilmore-lie-groups_ F02163	213	■ 1.000	S^n	S^n	e. Extend these results to the sphere S^n and its dual, $H^n, n>2$.
gilmore-lie-groups_ F02164	213	■ 1.000	$H^n, n>2$	$H^n, n>2$	e. Extend these results to the sphere S^n and its dual, $H^n, n>2$.
gilmore-lie-groups_ F02165	213	■ 1.000	$T_1^2=T_2^2=I$	$T_1^2=T_2^2=I$	3. Show that the two metric-preserving mappings T_1 and T_2 that satisfy $T_1^2=T_2^2=I$
gilmore-lie-groups_ F02166	213	■ 1.000	$T_1T_2=T_2T_1$	$T_1T_2=T_2T_1$	generate a third, $T_3=T_1T_2$ and that $T_1T_2=T_2T_1$. Show that $T_3\neq I$ if $T_1\neq T_2$. Show

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gilmore-lie-groups-F02167	213	■ 1.000	$T_{\{3\}} \not= I$	$T_3 \neq I$	generate a third, $T_3 = T_1 T_2$ and that $T_1 T_2 = T_2 T_1$. Show that $T_3 \neq I$ if $T_1 \neq T_2$. Show
gilmore-lie-groups-F02168	213	■ 1.000	$\mathfrak{g} = \mathfrak{g}_{+,+} + \mathfrak{g}_{+,-} + \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-} + \mathfrak{g}_{\{-,+\}} + \mathfrak{g}_{\{-,-\}}$	$\mathfrak{g} = \mathfrak{g}_{+,+} + \mathfrak{g}_{+,-} + \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-}$	compact Lie algebra $\mathfrak{g} = \mathfrak{g}_{+,+} + \mathfrak{g}_{+,-} + \mathfrak{g}_{-,+} + \mathfrak{g}_{-,-}$ that is available by choosing
gilmore-lie-groups-F02169	213	■ 1.000	$3!/1!=6$	$3!/1! = 6$	first, one of these three involutions, and then a second (there are $3!/1!=6$ choices).
gilmore-lie-groups-F02170	213	■ 1.000	$b_{\{3\}}=0$	$b_3 = 0$	from (11.2) by setting $b_3 = 0$:
gilmore-lie-groups-F02171	213	■ 1.000	$\phi_{\{j\}}(\mathfrak{p})$	$\phi_j(\mathfrak{p})$	5. Show that the coefficients $\phi_j(\mathfrak{p})$ in the secular equation for a symmetric space are
gilmore-lie-groups-F02172	213	■ 0.992	$\phi_{\{j\}}(\mathfrak{h}, \mathfrak{p})$	$\phi_j(\mathfrak{h}, \mathfrak{p})$	obtained from the coefficients $\phi_j(\mathfrak{h}, \mathfrak{p})$ in the secular equation for the parent Lie
gilmore-lie-groups-F02173	213	■ 0.998	$\mathfrak{h}=0$	$\mathfrak{h} = 0$	algebra (Eq. (12.3)) by setting $\mathfrak{h} = 0$.
gilmore-lie-groups-F02174	213	■ 0.891	$x_{\{0\}}^2-x_{\{1\}}^2-x_{\{2\}}^2=1$	$x_0^2 - x_1^2 - x_2^2 = 1$	by the condition $x_0^2 - x_1^2 - x_2^2 = 1$ in the Lorentz 3-space with signature (1, 2).
gilmore-lie-groups-F02175	213	■ 1.000	$-d\,s^2=d\,x_{\{0\}}^2-d\,x_{\{1\}}^2-$	$-ds^2 = dx_0^2 - dx_1^2 -$	b. Show that the invariant metric is induced from the metric $-ds^2 = dx_0^2 - dx_1^2 -$
gilmore-lie-groups-F02176	213	■ 0.973	$d\,x_{\{2\}}^2$	dx_2^2	dx_2^2 in this Lorentz 3-space.
gilmore-lie-groups-F02177	214	■ 1.000	$x_{\{1\}},\,x_{\{2\}}$	x_1, x_2	c. Use coordinates x_1, x_2 to parameterize the points in H_2^2 , and show
gilmore-lie-groups-F02178	214	■ 1.000	$(r,\,\theta),\,x_{\{1\}}=r\cos(\theta),\,x_{\{2\}}=r\sin(\theta)$	$(r, \theta), x_1 = r \cos(\theta), x_2 = r \sin(\theta)$	e. Introduce polar coordinates $(r, \theta), x_1 = r \cos(\theta), x_2 = r \sin(\theta)$. Show that
gilmore-lie-groups-F02179	214	■ 1.000	$S_{\,0(1,2)}$	$SO(1, 2)$	f. Determine the action of a group operation in $SO(1, 2)$ on the point $(x_1, x_2) \in H_2^2$.
gilmore-lie-groups-F02180	214	■ 1.000	$\left(x_{\{1\}},\,x_{\{2\}}\right)\in H_{\{2\}}^2$	$(x_1, x_2) \in H_2^2$	f. Determine the action of a group operation in $SO(1, 2)$ on the point $(x_1, x_2) \in H_2^2$.
gilmore-lie-groups-F02181	214	■ 1.000	$g_{\{i\,j\}}(x)$	$g_{ij}(x)$	7. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.
gilmore-lie-groups-F02182	214	■ 1.000	$X_{\{r\,s\}}=g_{\{r\,t\}}\,x^{\{t\}}\,\partial_{\{s\}}-$	$X_{rs} = g_{rt} x^t \partial_s -$	a. Show that the generators of infinitesimal rotations at a point are $X_{rs} = g_{rt} x^t \partial_s -$
gilmore-lie-groups-F02183	214	■ 1.000	$g_{\{s\,t\}}\,x^{\{t\}}\,\partial_{\{r\}}$	$g_{st} x^t \partial_r$	$g_{st} x^t \partial_r$.
gilmore-lie-groups-F02184	214	■ 1.000	$\left[X_{\{a\,b\}},\,\Delta\right]=0$	$[X_{ab}, \Delta] = 0$	b. Show $[X_{ab}, \Delta] = 0$, where $\Delta = G^{ab;rs} X_{ab} X_{rs}$ is the Laplace–Beltrami operator
gilmore-lie-groups-F02185	214	■ 1.000	$\Delta=G^{\{a\,b\,;\,r\,s\}}\,X_{\{a\,b\}}\,X_{\{r\,s\}}$	$\Delta = G^{ab;rs} X_{ab} X_{rs}$	b. Show $[X_{ab}, \Delta] = 0$, where $\Delta = G^{ab;rs} X_{ab} X_{rs}$ is the Laplace–Beltrami operator
gilmore-lie-groups-F02186	214	■ 0.546	$G_{\{a\,b\,;\,r\,s\}}=\operatorname{tr}\left\{\frac{d}{d\epsilon}\left(X_{\{a\,b\}}\right)\frac{d}{d\epsilon}\left(X_{\{r\,s\}}\right)\right\}$	$G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{ab}) \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{rs}) \}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{ab}) \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups-F02187	214	■ 0.546	$G^{\{a\,b\,;\,r\,s\}}$	$G^{ab;rs}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{ab}) \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups-F02188	214	■ 0.546	$G_{\{a\,b\,;\,r\,s\}}$	$G_{ab;rs}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{ab}) \mathfrak{d} \mathfrak{e} \mathfrak{f} (X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups-F02189	214	■ 0.989	Δ	Δ	c. Show that Δ consists of terms that are both quadratic and linear in the operators
gilmore-lie-groups-F02190	214	■ 1.000	$\partial_{\{r\}}$	∂_r	∂_r , and that
gilmore-lie-groups-F02191	214	■ 0.765	$\Gamma_{\{r\}\,\{s\}}^{\{t\}}$	$\Gamma_{rs}^{\,t}$	The function $\Gamma_{rs}^{\,t}$ is <i>not</i> a tensor. The components of the Christoffel symbol are
gilmore-lie-groups-F02192	214	■ 1.000	$\left(r,\,\phi_{\{2\}},\,\phi_{\{3\}},\,\ldots,\,\phi_{\{n\}}\right)$	$(r, \phi_2, \phi_3, \dots, \phi_n)$	8. Use radial coordinates $(r, \phi_2, \phi_3, \dots, \phi_n)$ on the sphere $S^n \subset R^{n+1}$.

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gilmore-lie-groups_F02193	214	■ 1.000	$S^n \subset R^{n+1}$	$S^n \subset R^{n+1}$	8. Use radial coordinates $(r, \phi_2, \phi_3, \dots, \phi_n)$ on the sphere $S^n \subset R^{n+1}$.
gilmore-lie-groups_F02194	215	■ 0.995	$S^0(n+1)$	$SO(n+1)$	c. Compare this with the second order Casimir operator for $SO(n+1)$:
gilmore-lie-groups_F02195	215	■ 1.000	$f_1(\phi)$	$f_1(\phi)$	Compute $f_1(\phi)$ and $f_2(\phi)$.
gilmore-lie-groups_F02196	215	■ 1.000	$f_2(\phi)$	$f_2(\phi)$	Compute $f_1(\phi)$ and $f_2(\phi)$.
gilmore-lie-groups_F02197	215	■ 0.994	$a_i^\dagger a_j \left(\mathcal{H} = h_{ij}(t) a_i^\dagger a_j, 1 \leq i, j \leq \right.$ $\left. \right)$	$a_i^\dagger a_j \left(\mathcal{H} = h_{ij}(t) a_i^\dagger a_j, 1 \leq i, j \leq \right.$ $\left. \right)$	is a linear superposition of the bilinear products $a_i^\dagger a_j$ ($\mathcal{H} = h_{ij}(t) a_i^\dagger a_j$, $1 \leq i, j \leq$
gilmore-lie-groups_F02198	215	■ 0.985	$i \mathcal{H}$	$i \mathcal{H}$	n), so that $i \mathcal{H}$ is a time-dependent element in the Lie algebra $\mathfrak{u}(n)$. Assume the
gilmore-lie-groups_F02199	215	■ 1.000	$SU(n)/U(n-1)$	$SU(n)/U(n-1)$	trajectory exists in the rank one symmetric space $SU(n)/U(n-1)$. Write down the
gilmore-lie-groups_F02200	215	■ 1.000	$V^{(n)}$	$V^{(n)}$	10. Conformal group The inner product on an n -dimensional linear vector space $V^{(n)}$
gilmore-lie-groups_F02201	215	■ 0.999	$(x, x)_m = m_{ij} x^i x^j$	$(x, x)_m = m_{ij} x^i x^j$	is defined by $(x, x)_m = m_{ij} x^i x^j$. Define coordinates y in an $n+2$ dimensional linear
gilmore-lie-groups_F02202	215	■ 0.999	$n+2$	$n+2$	is defined by $(x, x)_m = m_{ij} x^i x^j$. Define coordinates y in an $n+2$ dimensional linear
gilmore-lie-groups_F02203	215	■ 1.000	$W^{(n+2)}$	$W^{(n+2)}$	vector space $W^{(n+2)}$ as follows
gilmore-lie-groups_F02204	215	■ 1.000	$(y, y)_M = M_{\mu\nu} y^\mu y^\nu = (sx, sx)_m - \frac{2}{2} s[s(x, x)_m] = 0$	$(y, y)_M = M_{\mu\nu} y^\mu y^\nu = (sx, sx)_m - \frac{2}{2} s[s(x, x)_m] = 0$	a. Show $(y, y)_M = M_{\mu\nu} y^\mu y^\nu = (sx, sx)_m - \frac{2}{2} s[s(x, x)_m] = 0$.
gilmore-lie-groups_F02205	215	■ 1.000	$G=O(n)$	$G=O(n)$	$G=O(n)$.
gilmore-lie-groups_F02206	215	■ 1.000	$O(n+1, 1)$	$O(n+1, 1)$	c. Show that the Lie group H that preserves inner products in $W^{(n+2)}$ is $O(n+1, 1)$.
gilmore-lie-groups_F02207	215	■ 1.000	$n_1, n_2 \left(n_1 + n_2 = n \right)$	$n_1, n_2 \left(n_1 + n_2 = n \right)$	d. If the metric m has signature $n_1, n_2 \left(n_1 + n_2 = n \right)$, show that $G=O(n_1, n_2)$ and
gilmore-lie-groups_F02208	215	■ 1.000	$G=O(n_1, n_2)$	$G=O(n_1, n_2)$	d. If the metric m has signature $n_1, n_2 \left(n_1 + n_2 = n \right)$, show that $G=O(n_1, n_2)$ and
gilmore-lie-groups_F02209	215	■ 1.000	$H=O(n_1+1, n_2+1)$	$H=O(n_1+1, n_2+1)$	$H=O(n_1+1, n_2+1)$.
gilmore-lie-groups_F02210	215	■ 1.000	$SO(n_1+1, n_2+1)/SO(n_1, n_2)$	$SO(n_1+1, n_2+1)/SO(n_1, n_2)$	f. Construct the quotient space $SO(n_1+1, n_2+1)/SO(n_1, n_2)$.
gilmore-lie-groups_F02211	215	■ 0.996	$y \rightarrow y'$	$y \rightarrow y'$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x'^i = y'^i / y'^{n+1}$.
gilmore-lie-groups_F02212	215	■ 0.996	$x \rightarrow x'$	$x \rightarrow x'$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x'^i = y'^i / y'^{n+1}$.
gilmore-lie-groups_F02213	215	■ 0.996	$x'^i = y'^i / y'^{n+1}$	$x'^i = y'^i / y'^{n+1}$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x'^i = y'^i / y'^{n+1}$.
gilmore-lie-groups_F02214	215	■ 0.997	$(+1, -1, -1, -1)$	$(+1, -1, -1, -1)$	h. The Lorentz metric $(+1, -1, -1, -1)$ leaves the four-momentum invariant:
gilmore-lie-groups_F02215	216	■ 0.988	$L_{\mu\nu}$	$L_{\mu\nu}$	The operators $L_{\mu\nu}$ are the infinitesimal generators of the Lorentz group $SO(3, 1)$
gilmore-lie-groups_F02216	216	■ 1.000	P_μ	P_μ	and P_μ generate translations. Taken together $L_{\mu\nu}$ and P_μ generate the Poincaré
gilmore-lie-groups_F02217	216	■ 1.000	K_μ	K_μ	group. The operator S generates dilations and the four operators K_μ generate
gilmore-lie-groups_F02218	216	■ 1.000	$x_\mu = g_{\mu\nu} x^\nu$	$x_\mu = g_{\mu\nu} x^\nu$	conformal transformations. Above $x_\mu = g_{\mu\nu} x^\nu$.
gilmore-lie-groups_F02219	216	■ 0.989	$e^{c^\mu K_\mu} (x^\nu) = x'^\nu = \frac{x^\nu + c^\nu(x, x)}{1 + 2(c, x) + (c, c)(x, x)}$	$e^{c^\mu K_\mu} (x^\nu) = x'^\nu = \frac{x^\nu + c^\nu(x, x)}{1 + 2(c, x) + (c, c)(x, x)}$	k. Show that $e^{c^\mu K_\mu} (x^\nu) = x'^\nu = \frac{x^\nu + c^\nu(x, x)}{1 + 2(c, x) + (c, c)(x, x)}$.

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gilmore-lie-groups_ F02220	216	■ 0.999	$P_{\mu} \rightarrow P_{\mu}^{\prime} = x_{\mu}$	$P_{\mu} \rightarrow P_{\mu}^{\prime} = x_{\mu}$	m. Discuss the duality created by $P_{\mu} \rightarrow P_{\mu}^{\prime} = x_{\mu}$ and $K_{\mu} \rightarrow K_{\mu}^{\prime} = 2(x, \partial)\partial_{\mu} -$
gilmore-lie-groups_ F02221	216	■ 0.999	$K_{\mu} \rightarrow K_{\mu}^{\prime} = 2(x, \partial)\partial_{\mu} - \partial_{\mu}$	$K_{\mu} \rightarrow K_{\mu}^{\prime} = 2(x, \partial)\partial_{\mu} -$	m. Discuss the duality created by $P_{\mu} \rightarrow P_{\mu}^{\prime} = x_{\mu}$ and $K_{\mu} \rightarrow K_{\mu}^{\prime} = 2(x, \partial)\partial_{\mu} -$
gilmore-lie-groups_ F02222	216	■ 0.719	$(\partial, \partial) x_{\mu}$	$(\partial, \partial) x_{\mu}$	$(\partial, \partial) x_{\mu}$.
gilmore-lie-groups_ F02223	216	■ 1.000	$z = x + iy$	$z = x + iy$	11. The upper half of the complex plane has coordinates $z = x + iy$. This upper half-
gilmore-lie-groups_ F02224	216	■ 1.000	$PSL(2, \mathbb{R})$	$PSL(2, \mathbb{R})$	and denoted $PSL(2, \mathbb{R})$.
gilmore-lie-groups_ F02225	216	■ 1.000	$M, -M \in SL(2; \mathbb{R})$	$M, -M \in SL(2; \mathbb{R})$	a. Show that $M, -M \in SL(2; \mathbb{R})$ generate identical transformations. The group
gilmore-lie-groups_ F02226	216	■ 1.000	$y^{\prime} > 0$	$y^{\prime} > 0$	In particular, show that $y^{\prime} > 0$ if $y > 0$ and $y^{\prime} = 0$ if $y = 0$. The transformation
gilmore-lie-groups_ F02227	216	■ 1.000	$y > 0$	$y > 0$	In particular, show that $y^{\prime} > 0$ if $y > 0$ and $y^{\prime} = 0$ if $y = 0$. The transformation
gilmore-lie-groups_ F02228	216	■ 1.000	$y^{\prime} = 0$	$y^{\prime} = 0$	In particular, show that $y^{\prime} > 0$ if $y > 0$ and $y^{\prime} = 0$ if $y = 0$. The transformation
gilmore-lie-groups_ F02229	216	■ 0.997	$(y = 0)$	$(y = 0)$	$(y = 0)$, onto itself.
gilmore-lie-groups_ F02230	217	■ 1.000	$dz^{\prime} = dz / cz + d ^2$	$dz^{\prime} = dz / cz + d ^2$	d. Show $dz^{\prime} = dz / cz + d ^2$
gilmore-lie-groups_ F02231	217	■ 1.000	$d\mu = dx dy / y^2$	$d\mu = dx dy / y^2$	e. Show that the invariant measure is $d\mu = dx dy / y^2$
gilmore-lie-groups_ F02232	217	■ 1.000	z_2	z_2	f. Show that the distance between two points z_1 and z_2 is
gilmore-lie-groups_ F02233	217	■ 1.000	$w = x + iy$	$w = x + iy$	12. The unit disk in the complex plane $w = x + iy$ consists of those points that satisfy
gilmore-lie-groups_ F02234	217	■ 1.000	$\bar{w}w = x^2 + y^2 \leq 1$	$\bar{w}w = x^2 + y^2 \leq 1$	$\bar{w}w = x^2 + y^2 < 1$. The unit disk, with a suitable metric, provides a second represen-
gilmore-lie-groups_ F02235	217	■ 1.000	$M, -M \in SU(1, 1)$	$M, -M \in SU(1, 1)$	a. Show that $M, -M \in SU(1, 1)$ generate identical mappings of the unit disk into
gilmore-lie-groups_ F02236	217	■ 1.000	$w = e^{i\phi} \rightarrow w^{\prime} = e^{i\psi}$	$w = e^{i\phi} \rightarrow w^{\prime} = e^{i\psi}$	b. Show that $w = e^{i\phi} \rightarrow w^{\prime} = e^{i\psi}$. Compute $\psi(\phi)$.
gilmore-lie-groups_ F02237	217	■ 1.000	$\psi(\phi)$	$\psi(\phi)$	b. Show that $w = e^{i\phi} \rightarrow w^{\prime} = e^{i\psi}$. Compute $\psi(\phi)$.
gilmore-lie-groups_ F02238	217	■ 1.000	w_1	w_1	e. Show that the distance between two points w_1 and w_2 in this unit disk is
gilmore-lie-groups_ F02239	217	■ 1.000	w_2	w_2	e. Show that the distance between two points w_1 and w_2 in this unit disk is
gilmore-lie-groups_ F02240	217	■ 1.000	w	w	13. Show that the mapping from z in the upper half-plane to w in the unit disk given by
gilmore-lie-groups_ F02241	217	■ 1.000	z_0	z_0	is conformal, that is, it preserves angles. Here z_0 is any point in the upper half-plane.
gilmore-lie-groups_ F02242	218	■ 1.000	$z_0 = i$	$z_0 = i$	b. Choose $z_0 = i$ and $e^{i\phi} = i$ to give the canonical map
gilmore-lie-groups_ F02243	218	■ 1.000	$e^{i\phi} = i$	$e^{i\phi} = i$	b. Choose $z_0 = i$ and $e^{i\phi} = i$ to give the canonical map
gilmore-lie-groups_ F02244	219	■ 1.000	Y_r	Y_r	called contraction . In this process a new set of basis vectors Y_r is related to the
gilmore-lie-groups_ F02245	219	■ 1.000	$Y_r =$	$Y_r =$	initial set of basis vectors X_i through a parameter-dependent change of basis: $Y_r =$
gilmore-lie-groups_ F02246	219	■ 1.000	$M_r^i(\epsilon) X_i$	$M_r^i(\epsilon) X_i$	$M_r^i(\epsilon) X_i$. The structure constants have the transformation properties of a tensor:

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gilmore-lie-groups_ F02247	219	■ 0.986	$C_{rs}^t(\epsilon) = M_r^i(\epsilon) M_s^j(\epsilon) C_{ij}^k (M(\epsilon)^{-1})_k^t$	$C_{rs}^t(\epsilon) = M_r^i(\epsilon) M_s^j(\epsilon) C_{ij}^k (M(\epsilon)^{-1})_k^t$	$C_{rs}^t(\epsilon) = M_r^i(\epsilon) M_s^j(\epsilon) C_{ij}^k (M(\epsilon)^{-1})_k^t$ (cf. Eq. (4.22)). As long as the change of
gilmore-lie-groups_ F02248	219	■ 1.000	$C_{rs}^t(\epsilon)$	$C_{rs}^t(\epsilon)$	If the transformation becomes singular, the structure constants $C_{rs}^t(\epsilon)$ may
gilmore-lie-groups_ F02249	220	■ 0.998	$\mathfrak{g} \rightarrow \mathfrak{g}'$	$\mathfrak{g} \rightarrow \mathfrak{g}'$	then the Inönü–Wigner contraction of $\mathfrak{g} \rightarrow \mathfrak{g}'$ involves the following change of
gilmore-lie-groups_ F02250	220	■ 1.000	$\dim(\mathfrak{h})$	$\dim(\mathfrak{h})$	where $\dim(\mathfrak{h})$ is the dimension of the subalgebra $\mathfrak{h}$. The commutation relations of
gilmore-lie-groups_ F02251	220	■ 1.000	$\epsilon \rightarrow 0$	$\epsilon \rightarrow 0$	$\mathfrak{g}'$ are well defined for all values of ϵ , including the singular limit $\epsilon \rightarrow 0$:
gilmore-lie-groups_ F02252	220	■ 1.000	$[\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}'$	$[\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}'$	subalgebra $\mathfrak{h}$ and the subalgebra $\mathfrak{p}'$, where $\mathfrak{p}'$ is commutative and $[\mathfrak{h}, \mathfrak{p}'] \subseteq \mathfrak{p}'$:
gilmore-lie-groups_ F02254	220	■ 0.998	$L_1 = X_{23} =$	$L_1 = X_{23} =$	The infinitesimal generators of the Lie group $SO(3)$ may be chosen as $L_1 = X_{23} =$
gilmore-lie-groups_ F02255	220	■ 1.000	$x_2 \partial_3 - x_3 \partial_2 = \epsilon_{1jk} x_j \partial_k$	$x_2 \partial_3 - x_3 \partial_2 = \epsilon_{1jk} x_j \partial_k$	$x_2 \partial_3 - x_3 \partial_2 = \epsilon_{1jk} x_j \partial_k$, with L_2 and L_3 defined by cyclic permutation. The com-
gilmore-lie-groups_ F02256	220	■ 1.000	L_2	L_2	$x_2 \partial_3 - x_3 \partial_2 = \epsilon_{1jk} x_j \partial_k$, with L_2 and L_3 defined by cyclic permutation. The com-
gilmore-lie-groups_ F02257	220	■ 1.000	L_3	L_3	$x_2 \partial_3 - x_3 \partial_2 = \epsilon_{1jk} x_j \partial_k$, with L_2 and L_3 defined by cyclic permutation. The com-
gilmore-lie-groups_ F02258	221	■ 1.000	L_1	L_1	finitesimal generator L_3) the operators L_1 and L_2 go to
gilmore-lie-groups_ F02259	221	■ 1.000	$ISO(2) = E(2)$	$ISO(2) = E(2)$	The commutation relations of the contracted algebra, $ISO(2) = E(2)$, are
gilmore-lie-groups_ F02260	221	■ 1.000	L_3, P_1, P_2	L_3, P_1, P_2	The three operators L_3, P_1, P_2 generate the group of Euclidean motions of the plane,
gilmore-lie-groups_ F02261	221	■ 0.983	$R^2, ISO(2)$	$R^2, ISO(2)$	$E(2)$, or inhomogeneous orthogonal transformations in the plane $R^2, ISO(2)$. This
gilmore-lie-groups_ F02262	221	■ 1.000	$P_1 = \partial_1$	$P_1 = \partial_1$	the origin in the x - and y -directions, generated by $P_1 = \partial_1$ and $P_2 = \partial_2$.
gilmore-lie-groups_ F02263	221	■ 1.000	$P_2 = \partial_2$	$P_2 = \partial_2$	the origin in the x - and y -directions, generated by $P_1 = \partial_1$ and $P_2 = \partial_2$.
gilmore-lie-groups_ F02264	221	■ 0.698	$x^2 + y^2 + z^2 = R^2$	$x^2 + y^2 + z^2 = R^2$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown in
gilmore-lie-groups_ F02265	221	■ 0.698	$0, 0, R$	$0, 0, R$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown in
gilmore-lie-groups_ F02266	221	■ 1.000	$R \rightarrow \infty$	$R \rightarrow \infty$	In the limit $R \rightarrow \infty$ we find
gilmore-lie-groups_ F02267	221	■ 0.992	$-P_2, +P_1$	$-P_2, +P_1$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_ F02268	221	■ 0.992	$-y$	$-y$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_ F02269	221	■ 0.992	$+x$	$+x$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_ F02270	221	■ 1.000	θ_1, θ_2	θ_1, θ_2	directions. In addition, the parameters θ_1, θ_2 and d_1, d_2 are related by
gilmore-lie-groups_ F02271	221	■ 1.000	d_1, d_2	d_1, d_2	directions. In addition, the parameters θ_1, θ_2 and d_1, d_2 are related by
gilmore-lie-groups_ F02272	221	■ 1.000	$R \theta_i (i = 1, 2)$	$R \theta_i (i = 1, 2)$	with the product $R \theta_i (i = 1, 2)$ approaching a well-defined limit. This corresponds
gilmore-lie-groups_ F02273	221	■ 1.000	$\theta_2 = d_1 / R$	$\theta_2 = d_1 / R$	to a rotation through an angle $\theta_2 = d_1 / R$ about the y -axis producing a displacement

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gilmore-lie-groups_ F02274	221	■ 1.000	$d_{\{1\}}$	d_1	of d_1 in the x -direction, and a rotation through an angle $\theta_1 = d_2/R$ about the x -axis
gilmore-lie-groups_ F02275	221	■ 1.000	$\theta_1=d_{\{2\}} / R$	$\theta_1 = d_2/R$	of d_1 in the x -direction, and a rotation through an angle $\theta_1 = d_2/R$ about the x -axis
gilmore-lie-groups_ F02276	221	■ 1.000	$-d_{\{2\}}$	$-d_2$	producing a displacement of $-d_2$ in the y -direction.
gilmore-lie-groups_ F02278	223	■ 0.418	$P_{\{i\}}$	P_i	The operators P_i describe displacements in the x -, y -, and z -directions ($i = 1, 2, 3$).
gilmore-lie-groups_ F02279	223	■ 0.418	$i=1,2,3$	$i = 1, 2, 3$	The operators P_i describe displacements in the x -, y -, and z -directions ($i = 1, 2, 3$).
gilmore-lie-groups_ F02280	223	■ 1.000	$I \; S \; 0(3)$	$ISO(3)$	The contracted group is $ISO(3)$, the Euclidean, or inhomogeneous orthogonal
gilmore-lie-groups_ F02281	223	■ 1.000	$S \; 0(3) \rightarrow I \; S \; 0(2)$	$SO(3) \rightarrow ISO(2)$	As in the case $SO(3) \rightarrow ISO(2)$, we can contract the second order Casimir
gilmore-lie-groups_ F02282	223	■ 1.000	$\mathbf{P} \cdot \mathbf{P}=\nabla^2$	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	The two invariant operators $\mathbf{P} \cdot \mathbf{P} = \nabla^2$ and $\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$ form a complete set of
gilmore-lie-groups_ F02283	223	■ 1.000	$\mathbf{L} \cdot \mathbf{P}=-\mathbf{L} \cdot \nabla$	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	The two invariant operators $\mathbf{P} \cdot \mathbf{P} = \nabla^2$ and $\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$ form a complete set of
gilmore-lie-groups_ F02285	224	■ 0.592	$[S \; 0(3,1)]$	$[SO(3,1)]$	The group $ISO(3, 1)$ consists of proper Lorentz transformations $[SO(3, 1)]$ that
gilmore-lie-groups_ F02286	224	■ 0.997	$S \; 0(3,2)$	$SO(3, 2)$	This group can be contracted from either $SO(4, 1)$ or $SO(3, 2)$.
gilmore-lie-groups_ F02287	224	■ 0.998	$\epsilon^{\alpha\beta\gamma\mu\nu}$	$\epsilon^{\alpha\beta\gamma\mu\nu}$	where $\epsilon^{\alpha\beta\gamma\mu\nu}$ is the Levi–Civita symbol (antisymmetric tensor) on five symbols,
gilmore-lie-groups_ F02288	224	■ 1.000	$W_{\{\alpha\}}$	W_α	and W_α is similarly defined. The contracted limit of W^α is nonzero only if one of
gilmore-lie-groups_ F02289	224	■ 1.000	$W^{\{\alpha\}}$	W^α	and W_α is similarly defined. The contracted limit of W^α is nonzero only if one of
gilmore-lie-groups_ F02290	224	■ 0.615	v	v	the four remaining symbols (e.g., v) is 5:
gilmore-lie-groups_ F02291	225	■ 1.000	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$ is fairly complicated. Since $W^\alpha W_\alpha$ is invariant,
gilmore-lie-groups_ F02292	225	■ 1.000	$W^{\{\alpha\}} W_{\{\alpha\}}$	$W^\alpha W_\alpha$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma}(\partial/\partial x^\mu)$ is fairly complicated. Since $W^\alpha W_\alpha$ is invariant,
gilmore-lie-groups_ F02293	225	■ 1.000	$\mathbf{P} \cdot \mathbf{P}=\sum P_{\{\mu\}} P^{\{\mu\}}=-(mc)^2$	$\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$	with $\mathbf{P} \cdot \mathbf{P} = \sum P_\mu P^\mu = -(mc)^2$.
gilmore-lie-groups_ F02294	225	■ 0.862	$\mathfrak{u}(2)$	$\mathfrak{u}(2)$	The Lie algebra $\mathfrak{u}(2)$ is spanned by infinitesimal generators $J_3, J_\pm, J_0$ with com-
gilmore-lie-groups_ F02295	225	■ 0.862	$J_{\{3\}}, J_{\{\pm\}}, J_{\{0\}}$	$J_3, J_\pm, J_0$	The Lie algebra $\mathfrak{u}(2)$ is spanned by infinitesimal generators $J_3, J_\pm, J_0$ with com-
gilmore-lie-groups_ F02296	225	■ 0.996	$h_{\{3\}}, h_{\{\pm\}}, h_{\{0\}}$	$h_3, h_\pm, h_0$	The operators $h_3, h_\pm, h_0$ are related to $J_3, J_\pm, J_0$ by the following change of basis
gilmore-lie-groups_ F02297	226	■ 1.000	$c \rightarrow 0$	$c \rightarrow 0$	In the limit $c \rightarrow 0$ the change of basis transformation becomes singular but the
gilmore-lie-groups_ F02298	226	■ 1.000	$J_{\{0\}} \rightarrow h_{\{0\}}$	$J_0 \rightarrow h_0$	Under contraction $J_0 \rightarrow h_0$ but the second Casimir operator has a more interesting
gilmore-lie-groups_ F02299	226	■ 1.000	$(h_{\{0\}} / 2 \; c)^2$	$(h_0/2c)^2$	The operator $(h_0/2c)^2$ is proportional to the square of the first Casimir operator. It
gilmore-lie-groups_ F02300	226	■ 1.000	$c \rightarrow 0, (ch_{\{3\}})^2 \rightarrow 0$	$c \rightarrow 0, (ch_3)^2 \rightarrow 0$	in the Lie algebra. In the limit $c \rightarrow 0$, $(ch_3)^2 \rightarrow 0$ and the remaining operators go
gilmore-lie-groups_ F02301	227	■ 1.000	$\hat{n}+\frac{1}{2} I, a^\dagger, a$	$\hat{n} + \frac{1}{2} I, a^\dagger, a$	This is a quadratic operator in the generators $\hat{n} + \frac{1}{2} I, a^\dagger, a$, and I of H_4 . The value
gilmore-lie-groups_ F02302	227	■ 1.000	$\mathfrak{h}_{\{4\}}$	$\mathfrak{h}_4$	An arbitrary element in the Lie algebra $\mathfrak{u}(2)$ and its counterpart in the algebra $\mathfrak{h}_4$

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gilmore-lie-groups_ F02303	227	■ 1.000	θ_0	θ_0	exist. In addition, θ_0 should diverge so that $\theta_0 - \theta_3/2c^2$ remains well defined.
gilmore-lie-groups_ F02304	227	■ 1.000	$\theta_0 - \theta_3 / 2 c^2$	$\theta_0 - \theta_3/2c^2$	exist. In addition, θ_0 should diverge so that $\theta_0 - \theta_3/2c^2$ remains well defined.
gilmore-lie-groups_ F02305	227	■ 0.999	h_3	h_3	The action of the operators h_3 on the angular momentum state $ J, M\rangle$ is
gilmore-lie-groups_ F02306	227	■ 0.999	$ J, M\rangle$	$ J, M\rangle$	The action of the operators h_3 on the angular momentum state $ J, M\rangle$ is
gilmore-lie-groups_ F02307	227	■ 1.000	$ J, -J\rangle$	$ J, -J\rangle$	It is useful to measure states from the “lowest” state $ J, -J\rangle$ in the angular mo-
gilmore-lie-groups_ F02308	227	■ 1.000	$M = -J$	$M = -J$	$M = -J$, and the n th state when
gilmore-lie-groups_ F02309	227	■ 0.985	J	J	representations J of dimension $(2J + 1)$ as c becomes smaller and smaller. Specif-
gilmore-lie-groups_ F02310	227	■ 0.985	$(2J + 1)$	$(2J + 1)$	representations J of dimension $(2J + 1)$ as c becomes smaller and smaller. Specif-
gilmore-lie-groups_ F02311	228	■ 1.000	$n = J + M$	$n = J + M$	applying the angular momentum shift up operator $n = J + M$ times to the ground
gilmore-lie-groups_ F02312	229	■ 1.000	$\lim_{c \rightarrow 0} \zeta / c \rightarrow \alpha$	$\lim_{c \rightarrow 0} \zeta / c \rightarrow \alpha$	contracts under $\lim_{c \rightarrow 0} \zeta / c \rightarrow \alpha$ to the BCH formula for H_4
gilmore-lie-groups_ F02313	229	■ 1.000	$P_m^l(\cos \theta)$	$P_m^l(\cos \theta)$	$P_m^l(\cos \theta)$ are related by (Arecchi <i>et al.</i> , 1972; Gilmore, 1974b)
gilmore-lie-groups_ F02314	229	■ 1.000	$u \rightarrow$	$u \rightarrow$	These polynomials are contracted to harmonic oscillator wavefunctions under $u \rightarrow$
gilmore-lie-groups_ F02315	229	■ 1.000	$x / \sqrt{l}$	$x / \sqrt{l}$	$x / \sqrt{l}$ and $l + m = n$:
gilmore-lie-groups_ F02316	229	■ 1.000	$l + m = n$	$l + m = n$	$x / \sqrt{l}$ and $l + m = n$:
gilmore-lie-groups_ F02317	230	■ 1.000	$c \rightarrow 0, l \rightarrow \infty, l + m = n, 2lc^2 = 1$	$c \rightarrow 0, l \rightarrow \infty, l + m = n, 2lc^2 = 1$	The limit is taken as $c \rightarrow 0, l \rightarrow \infty, l + m = n, 2lc^2 = 1$. The limit inside the
gilmore-lie-groups_ F02318	230	■ 1.000	$1 / \sqrt{\pi}$	$1 / \sqrt{\pi}$	first square root is $1 / \sqrt{\pi}$, that within the second is $(2^n n!)^{-1}$. The result of this
gilmore-lie-groups_ F02319	230	■ 1.000	$(2^n n!)^{-1}$	$(2^n n!)^{-1}$	first square root is $1 / \sqrt{\pi}$, that within the second is $(2^n n!)^{-1}$. The result of this
gilmore-lie-groups_ F02320	230	■ 1.000	$H_n(x)$	$H_n(x)$	and $H_n(x)$ is the n th Hermite polynomial.
gilmore-lie-groups_ F02321	231	■ 1.000	$ISO(2)$	$ISO(2)$	representations of $ISO(2)$. Since $ISO(2)$ is a noncompact group it has no faith-
gilmore-lie-groups_ F02322	231	■ 1.000	$P_{\pm}$	$P_{\pm}$	a. Compute the matrix elements of the operators $P_{\pm}$ in the algebra $\mathfrak{iso}(2)$ and show
gilmore-lie-groups_ F02323	231	■ 1.000	$\mathfrak{iso}(2)$	$\mathfrak{iso}(2)$	a. Compute the matrix elements of the operators $P_{\pm}$ in the algebra $\mathfrak{iso}(2)$ and show
gilmore-lie-groups_ F02324	231	■ 1.000	$J_k(x)$	$J_k(x)$	where $J_k(x)$ is the k th Bessel function (Arecchi <i>et al.</i> , 1972; Gilmore, 1974b).
gilmore-lie-groups_ F02325	231	■ 0.992	A_k^l	A_k^l	compute the coefficients A_k^l and the arguments θ', ϕ' explicitly. Contract these
gilmore-lie-groups_ F02326	231	■ 0.992	θ', ϕ'	θ', ϕ'	compute the coefficients A_k^l and the arguments θ', ϕ' explicitly. Contract these
gilmore-lie-groups_ F02327	231	■ 0.998	$\mathbf{L} \cdot \mathbf{L}$	$\mathbf{L} \cdot \mathbf{L}$	f. Show that the operator $\mathbf{L} \cdot \mathbf{L}$ contracts to ∇^2 in the plane.
gilmore-lie-groups_ F02328	231	■ 0.998	∇^2	∇^2	f. Show that the operator $\mathbf{L} \cdot \mathbf{L}$ contracts to ∇^2 in the plane.

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gilmore-lie-groups_F02329	231	■ 1.000	$\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$	$\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$	2. Under the contraction $\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$ the representations of the unitary group $U(2)$ con-
gilmore-lie-groups_F02336	232	■ 1.000	N_n	N_n	$H_n(x)$ is the n th Hermite polynomial, and N_n is the usual normalization coefficient,
gilmore-lie-groups_F02331	232	■ 0.993	$N_n=1 / \sqrt{2^n n! \sqrt{\pi}}$	$N_n = 1/\sqrt{2^n n! \sqrt{\pi}}$	$N_n = 1/\sqrt{2^n n! \sqrt{\pi}}$.
gilmore-lie-groups_F02332	232	■ 0.984	$J_{\{3\}}, J_{\{ \pm \}} \left(\left(\left(J_{\{3\}}, J_{\{ \pm \}} \right) = \pm J_{\{ \pm \}} \right) \right) = \pm J_{\{ \pm \}}, \left(\left(J_{\{3\}}, J_{\{ \pm \}} \right) = 2 J_{\{3\}} \right)$	$J_3, J_{\pm} ([J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3)$	3. Contract the Lie algebra $\mathfrak{su}(2)$ spanned by $J_3, J_{\pm} ([J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3)$
gilmore-lie-groups_F02333	232	■ 0.993	$\left(P^{\prime}, T^{\prime}, V^{\prime} \right)$	(P', T', V')	operators on the right (P', T', V') satisfy an isomorphic set of commutation relations.
gilmore-lie-groups_F02334	232	■ 0.991	$\partial_x, \partial_t, t \partial_x$	$\partial_x, \partial_t, t \partial_x$	The operators $\partial_x, \partial_t, t \partial_x$ generate the Galilean group in one dimension. Conclude
gilmore-lie-groups_F02335	232	■ 1.000	$\mathfrak{a}_1$	$\mathfrak{a}_1$	that if the Lie algebra $\mathfrak{a}_1$ is contracted with respect to one of its shift operators the
gilmore-lie-groups_F02336	232	■ 0.985	$\mathfrak{gal}(1)$	$\mathfrak{gal}(1)$	Galilean algebra $\mathfrak{gal}(1)$ results.
gilmore-lie-groups_F02337	232	■ 1.000	$S^n = S O(n+1) / S O(n)$	$S^n = SO(n+1)/SO(n)$	metric and measure on the sphere $S^n = SO(n+1)/SO(n)$ reduce to the familiar
gilmore-lie-groups_F02338	232	■ 1.000	$R^n = I S O(n) / S O(n)$	$R^n = ISO(n)/SO(n)$	metric and measure on $R^n = ISO(n)/SO(n)$.
gilmore-lie-groups_F02339	233	■ 1.000	$\tau = (\zeta / \zeta) \tan (\zeta)$	$\tau = (\zeta/ \zeta) \tan(\zeta)$	and show $\tau = (\zeta/ \zeta) \tan(\zeta)$.
gilmore-lie-groups_F02340	233	■ 1.000	$\alpha = \lim_{c \rightarrow 0} \zeta / c$	$\alpha = \lim_{c \rightarrow 0} \zeta / c$	$\alpha = \lim_{c \rightarrow 0} \zeta / c$.
gilmore-lie-groups_F02341	233	■ 1.000	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$, and a generating function for expectation values is $\langle e^{\alpha X} \rangle =$
gilmore-lie-groups_F02342	233	■ 1.000	$\langle e^{\alpha X} \rangle =$	$\langle e^{\alpha X} \rangle =$	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$, and a generating function for expectation values is $\langle e^{\alpha X} \rangle =$
gilmore-lie-groups_F02343	233	■ 1.000	$\text{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\text{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\text{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$. When the operators X and $\mathcal{H}$ are elements in a finite dimensional
gilmore-lie-groups_F02344	233	■ 0.998	$\mathcal{H} = \epsilon J_3$	$\mathcal{H} = \epsilon J_3$	a. Assume $\mathcal{H} = \epsilon J_3$ and X is in the Lie algebra $\mathfrak{su}(2)$. Show that in the 2×2 defining
gilmore-lie-groups_F02345	233	■ 1.000	$2 j+1=2$	$2j+1=2$	c. Show that in the 2×2 matrix representation with $j = \frac{1}{2}$ and $2j+1=2$,
gilmore-lie-groups_F02346	234	■ 0.974	$\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$	$\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$	are analytic continuations of $\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4 =$
gilmore-lie-groups_F02347	234	■ 0.974	$\mathcal{C}^4 =$	$\mathcal{C}^4 =$	are analytic continuations of $\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4 =$
gilmore-lie-groups_F02348	234	■ 0.812	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $SO(6)$.
gilmore-lie-groups_F02349	234	■ 0.812	$Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$	$Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $SO(6)$.
gilmore-lie-groups_F02350	234	■ 0.812	$SO(6)$	$SO(6)$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $SO(6)$.
gilmore-lie-groups_F02351	234	■ 0.995	$SO(4) \otimes SO(2)$	$SO(4) \otimes SO(2)$	b. Contract $SO(4, 2)$ with respect to the subgroup $SO(4) \otimes SO(2)$.
gilmore-lie-groups_F02352	234	■ 0.933	$A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$	$A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$	of part a. If we define $A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$ and $B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$, then show that
gilmore-lie-groups_F02353	234	■ 0.933	$B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$	$B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$	of part a. If we define $A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$ and $B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$, then show that
gilmore-lie-groups_F02354	234	■ 1.000	$SO(4, 2) / [SO(4) \otimes SO(2)]$	$SO(4, 2) / [SO(4) \otimes SO(2)]$	$SO(4, 2) / [SO(4) \otimes SO(2)]$ and $I [SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$.
gilmore-lie-groups_F02355	234	■ 1.000	$I [SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$	$I [SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$	$SO(4, 2) / [SO(4) \otimes SO(2)]$ and $I [SO(4) \otimes SO(2)] / [SO(4) \otimes SO(2)]$.

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gilmore-lie-groups_F02356	234	■ 1.000	$X_{\{\alpha\}}$	X_α	Operators X_i span $\mathfrak{h}$ and X_α span $\mathfrak{p}$.
gilmore-lie-groups_F02357	234	■ 0.987	$Y_{\{\alpha\}}=\lim_{\epsilon\rightarrow 0}\epsilon\rightarrow 0\in X_{\{\alpha\}}$	$Y_\alpha=\lim_{\epsilon\rightarrow 0}\epsilon\in X_\alpha$	b. Show that in the contracted limit $Y_\alpha=\lim_{\epsilon\rightarrow 0}\epsilon\in X_\alpha$ a metric tensor on $\mathfrak{p}$ is well
gilmore-lie-groups_F02358	236	■ 1.000	$ \psi\rangle$	$ \psi\rangle$	state $ \psi\rangle$ in their respective coordinate systems. They describe the state by mathe-
gilmore-lie-groups_F02359	236	■ 1.000	$\langle S \psi\rangle$	$\langle S \psi\rangle$	matical functions $\langle S \psi\rangle$ and $\langle S' \psi\rangle$. The two observers know the relation between
gilmore-lie-groups_F02360	236	■ 1.000	$\langle S' \psi\rangle$	$\langle S' \psi\rangle$	matical functions $\langle S \psi\rangle$ and $\langle S' \psi\rangle$. The two observers know the relation between
gilmore-lie-groups_F02361	236	■ 0.998	$\langle S' S\rangle$	$\langle S' S\rangle$	tions from one coordinate system to the other is $\langle S' S\rangle$. The set of transformations
gilmore-lie-groups_F02362	236	■ 1.000	$\langle S S'\rangle$	$\langle S S'\rangle$	S has seen, he applies the appropriate transformation, $\langle S S'\rangle$, to his mathematical
gilmore-lie-groups_F02363	237	■ 1.000	$ \psi'\rangle$	$ \psi'\rangle$	by S' to describe a <i>new</i> physical state $ \psi'\rangle$
gilmore-lie-groups_F02364	237	■ 0.997	$SO(3)$	$SO(3)$	example, a rotation in $SO(3)$ is expressed by $\langle S' H S'\rangle=\langle S' S\rangle\langle S H S\rangle\langle S S'\rangle$,
gilmore-lie-groups_F02365	237	■ 0.997	$\langle S' H S'\rangle=\langle S' S\rangle\langle S H S\rangle\langle S S'\rangle$	$\langle S' H S'\rangle=\langle S' S\rangle\langle S H S\rangle\langle S S'\rangle$	example, a rotation in $SO(3)$ is expressed by $\langle S' H S'\rangle=\langle S' S\rangle\langle S H S\rangle\langle S S'\rangle$,
gilmore-lie-groups_F02366	237	■ 1.000	$\langle S' H S'\rangle=$	$\langle S' H S'\rangle=$	the invariance under the transformation $\langle S' S\rangle$ is represented by $\langle S' H S'\rangle=$
gilmore-lie-groups_F02367	237	■ 1.000	$\langle S H S\rangle$	$\langle S H S\rangle$	$\langle S H S\rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups_F02368	237	■ 1.000	$2p_z$	$2p_z$	$\langle S H S\rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups_F02369	237	■ 1.000	$2p_x$	$2p_x$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups_F02370	237	■ 1.000	$2p_y$	$2p_y$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups_F02371	237	■ 0.996	$p^\mu p_\mu=g_{\mu\nu}p^\mu p^\nu=(mc)^2$	$p^\mu p_\mu=g_{\mu\nu}p^\mu p^\nu=(mc)^2$	the relativistic dispersion relation for the free particle: $p^\mu p_\mu=g_{\mu\nu}p^\mu p^\nu=(mc)^2$.
gilmore-lie-groups_F02372	237	■ 1.000	$\mathbf{p}$	$\mathbf{p}$	In terms of the energy E and the three-momentum $\mathbf{p}$ this is
gilmore-lie-groups_F02373	237	■ 1.000	$p_\mu\rightarrow\pi_\mu=p_\mu-(q/c)A_\mu$	$p_\mu\rightarrow\pi_\mu=p_\mu-(q/c)A_\mu$	principle of minimal electromagnetic coupling: $p_\mu\rightarrow\pi_\mu=p_\mu-(q/c)A_\mu$, where
gilmore-lie-groups_F02374	237	■ 0.974	Φ	Φ	the four-vector potential A consists of the scalar potential Φ and the vector potential
gilmore-lie-groups_F02375	237	■ 0.997	$\mathbf{B}=\nabla\times\mathbf{A}$	$\mathbf{B}=\nabla\times\mathbf{A}$	A. These obey $\mathbf{B}=\nabla\times\mathbf{A}$ and $\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$. For an electron $q=$
gilmore-lie-groups_F02376	237	■ 0.997	$\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$	$\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$	A. These obey $\mathbf{B}=\nabla\times\mathbf{A}$ and $\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$. For an electron $q=$
gilmore-lie-groups_F02377	237	■ 0.997	$q=$	$q=$	A. These obey $\mathbf{B}=\nabla\times\mathbf{A}$ and $\mathbf{E}=-\nabla\Phi-(1/c)(\partial\mathbf{A}/\partial t)$. For an electron $q=$
gilmore-lie-groups_F02378	237	■ 0.928	$-e$	$-e$	$-e$, where e is the charge on the proton, positive by convention. In the Coulomb
gilmore-lie-groups_F02379	237	■ 1.000	$\Phi=e/r$	$\Phi=e/r$	field established by a proton, $\Phi=e/r$ and $\mathbf{A}=\mathbf{0}$, so that $E\rightarrow E+e^2/r$. Here
gilmore-lie-groups_F02380	237	■ 1.000	$\mathbf{A}=\mathbf{0}$	$\mathbf{A}=\mathbf{0}$	field established by a proton, $\Phi=e/r$ and $\mathbf{A}=\mathbf{0}$, so that $E\rightarrow E+e^2/r$. Here
gilmore-lie-groups_F02381	237	■ 1.000	$E\rightarrow E+e^2/r$	$E\rightarrow E+e^2/r$	field established by a proton, $\Phi=e/r$ and $\mathbf{A}=\mathbf{0}$, so that $E\rightarrow E+e^2/r$. Here

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gilmore-lie-groups_ F02382	237	■ 1.000	$\mathbf{p} \rightarrow (\hbar/i) \nabla$	$\mathbf{p} \rightarrow (\hbar/i) \nabla$	dispersion relation to a wave equation is to replace $\mathbf{p} \rightarrow (\hbar/i) \nabla$ and allow the
gilmore-lie-groups_ F02383	237	■ 1.000	$\psi(\mathbf{x})$	$\psi(\mathbf{x})$	resulting equation to act on a spacial function $\psi(\mathbf{x})$. This prescription results in the
gilmore-lie-groups_ F02384	238	■ 0.999	$\langle S' H S' \rangle = \langle S H S \rangle$	$\langle S' H S' \rangle = \langle S H S \rangle$	variant) in form under rotations: $\langle S' H S' \rangle = \langle S H S \rangle$, where $\langle S' S \rangle \in SO(3)$.
gilmore-lie-groups_ F02385	238	■ 0.999	$\langle S' S \rangle \in SO(3)$	$\langle S' S \rangle \in SO(3)$	variant) in form under rotations: $\langle S' H S' \rangle = \langle S H S \rangle$, where $\langle S' S \rangle \in SO(3)$.
gilmore-lie-groups_ F02386	238	■ 0.623	mc^2	mc^2	Since the binding energy is about 13.6 eV and the electron rest energy mc^2 is about
gilmore-lie-groups_ F02387	238	■ 0.906	$E = mc^2 + W$	$E = mc^2 + W$	510 000 eV, it makes sense to write $E = mc^2 + W$, where the principal part of the
gilmore-lie-groups_ F02388	238	■ 1.000	W	W	relativistic energy E is the electron rest energy and the nonrelativistic energy W is
gilmore-lie-groups_ F02389	238	■ 0.884	$(\simeq 0.0025 \%)$	$(\simeq 0.0025 \%)$	a small perturbation of either $\simeq 0.0025 \%$. Under this substitution, and neglecting
gilmore-lie-groups_ F02390	238	■ 1.000	$(W + e^2/r)^2 / mc^2$	$(W + e^2/r)^2 / mc^2$	terms of order $(W + e^2/r)^2 / mc^2$, we obtain the nonrelativistic form of Eq. (14.4),
gilmore-lie-groups_ F02391	238	■ 0.727	$[A, B] / i \hbar = \{A, B\}$	$[A, B] / i \hbar = \{A, B\}$	tor brackets $([A, B] / i \hbar = \{A, B\})$ is an attempt to fit quantum mechanics into the
gilmore-lie-groups_ F02392	238	■ 0.860	(r, θ, ϕ)	(r, θ, ϕ)	useful to introduce spherical coordinates (r, θ, ϕ) . In this coordinate system the
gilmore-lie-groups_ F02393	239	■ 1.000	$\mathcal{L}^2(S^2)$	$\mathcal{L}^2(S^2)$	The second order differential operator $\mathcal{L}^2(S^2)$ is the Laplacian on the sphere S^2 . Its
gilmore-lie-groups_ F02394	239	■ 0.980	$\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$	$\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$	$\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$ The integers (l, m) satisfy $l = 0, 1, 2, \dots$
gilmore-lie-groups_ F02395	239	■ 0.980	(l, m)	(l, m)	is $\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $l = 0, 1, 2, \dots$
gilmore-lie-groups_ F02396	239	■ 0.980	$l = 0, 1, 2, \dots$	$l = 0, 1, 2, \dots$	is $\mathcal{L}^2(S^2) Y_m^l(\theta, \phi) = -l(l+1) Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $l = 0, 1, 2, \dots$
gilmore-lie-groups_ F02397	239	■ 0.999	$-l \leq m \leq +l$	$-l \leq m \leq +l$	and $-l \leq m \leq +l$. The negative sign and discrete spectrum characteristically indi-
gilmore-lie-groups_ F02398	239	■ 1.000	A, B, C	A, B, C	The values of the coefficients A, B, C that are obtained for the Klein–Gordon
gilmore-lie-groups_ F02404	239	■ 1.000	$n =$	$n =$	relativistic problems are expressed in terms of the radial quantum number $n =$
gilmore-lie-groups_ F02405	239	■ 1.000	$0, 1, 2, \dots$	$0, 1, 2, \dots$	$0, 1, 2, \dots$ and the angular momentum quantum number $l = 0, 1, 2, \dots$, mass m
gilmore-lie-groups_ F02406	239	■ 1.000	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$	$m_{\text{red}}^{-1} = m_e^{-1} + M_p^{-1}$, and the fine structure constant (Gabielse <i>et al.</i> , 2006)
gilmore-lie-groups_ F02409	271	■ 1.000	$\mathcal{L}^2$	$\mathcal{L}^2$	The angular part of the Laplacian operator, $\mathcal{L}^2$, acts on spherical harmonics on S^{D-1} ,
gilmore-lie-groups_ F02412	240	■ 1.000	$f(r)$	$f(r)$	equation for $f(r)$
gilmore-lie-groups_ F02418	240	■ 1.000	$\hbar$	$\hbar$	three “different” areas of physics: e (electromagnetism), $\hbar$ (quantum mechanics),

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gilmore-lie-groups_ F02419	241	—	$Z=50$	$Z = 50$	
gilmore-lie-groups_ F02420	241	—	$\alpha \rightarrow Z \alpha$	$\alpha \rightarrow Z\alpha$	
gilmore-lie-groups_ F02421	241	■ 1.000	$ W_1 = \frac{1}{2} m c^2 \alpha^2$	$ W_1 = \frac{1}{2} m c^2 \alpha^2$	ground state energy $ W_1 = \frac{1}{2} m c^2 \alpha^2$. These normalized energy levels decrease to
gilmore-lie-groups_ F02422	241	■ 0.999	$1 / N^2$	$1/N^2$	zero like $1/N^2$, where $N = n + l + 1$ is the principal quantum number. The ener-
gilmore-lie-groups_ F02423	241	■ 0.999	$N=n+l+1$	$N = n + l + 1$	zero like $1/N^2$, where $N = n + l + 1$ is the principal quantum number. The ener-
gilmore-lie-groups_ F02424	241	■ 1.000	$N^{\prime}$	N'	ergy $m c^2$ as N' increases. When this limit is removed these energies (also rescaled
gilmore-lie-groups_ F02425	241	■ 1.000	$\frac{1}{2} m c^2 \alpha^2$	$\frac{1}{2} m c^2 \alpha^2$	by dividing by $\frac{1}{2} m c^2 \alpha^2$) can be plotted on the same graph. At the resolution shown,
gilmore-lie-groups_ F02426	241	■ 1.000	$\frac{1}{2} m c^2 (Z \alpha)^2$	$\frac{1}{2} m c^2 (Z\alpha)^2$	energy for the same ion: $\frac{1}{2} m c^2 (Z\alpha)^2$. The energy difference between the $1s$ ground
gilmore-lie-groups_ F02427	241	■ 1.000	$1 s$	$1s$	energy for the same ion: $\frac{1}{2} m c^2 (Z\alpha)^2$. The energy difference between the $1s$ ground
gilmore-lie-groups_ F02428	242	■ 1.000	$g_i \psi\rangle$	$g_i \psi\rangle$	eigenstate of H with eigenvalue E , then $g_i \psi\rangle$ is also an eigenstate of H with the
gilmore-lie-groups_ F02429	242	■ 1.000	$ \psi\rangle=\psi_{2 p_z}(\mathbf{x})$	$ \psi\rangle = \psi_{2p_z}(\mathbf{x})$	To illustrate this idea, assume that $ \psi\rangle = \psi_{2p_z}(\mathbf{x})$. A rotation by $\pi/2$ radians about
gilmore-lie-groups_ F02430	242	■ 1.000	$\pi / 2$	$\pi/2$	To illustrate this idea, assume that $ \psi\rangle = \psi_{2p_z}(\mathbf{x})$. A rotation by $\pi/2$ radians about
gilmore-lie-groups_ F02431	242	■ 1.000	$\psi_{2 p_x}(\mathbf{x})$	$\psi_{2p_x}(\mathbf{x})$	the y -axis maps this state to $\psi_{2p_x}(\mathbf{x})$ and a rotation by $\pi/2$ radians about the x -axis

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gilmore-lie-groups_ F02432	242	■ 1.000	$-\psi_{2p_y}(\mathbf{x})$	$-\psi_{2p_y}(\mathbf{x})$	maps this state to $-\psi_{2p_y}(\mathbf{x})$. By invariance (of the hamiltonian) under the rotation
gilmore-lie-groups_ F02433	242	■ 1.000	$H=\mathbf{p}\cdot\mathbf{p}/2m-e^2/r$	$H=\mathbf{p}\cdot\mathbf{p}/2m-e^2/r$	$H=\mathbf{p}\cdot\mathbf{p}/2m-e^2/r$. The scalar $\mathbf{p}\cdot\mathbf{p}=-\hbar^2\nabla^2$ is invariant under rotations, as
gilmore-lie-groups_ F02434	242	■ 1.000	$\mathbf{p}\cdot\mathbf{p}=-\hbar^2\nabla^2$	$\mathbf{p}\cdot\mathbf{p}=-\hbar^2\nabla^2$	$H=\mathbf{p}\cdot\mathbf{p}/2m-e^2/r$. The scalar $\mathbf{p}\cdot\mathbf{p}=-\hbar^2\nabla^2$ is invariant under rotations, as
gilmore-lie-groups_ F02435	242	■ 1.000	$-e^2/r$	$-e^2/r$	is also the potential energy term $-e^2/r$. Rotation operators can be expressed in
gilmore-lie-groups_ F02436	242	■ 1.000	$i:\epsilon_{ijk}x_j\partial_k$	$i:\epsilon_{ijk}x_j\partial_k$	terms of the infinitesimal generators of rotations about axis i : $\epsilon_{ijk}x_j\partial_k$. These ge-
gilmore-lie-groups_ F02437	242	■ 1.000	$\mathbf{L}_i=(\mathbf{r}\times\mathbf{p})_i=(\hbar/i)\epsilon_{ijk}x_j\partial_k$	$\mathbf{L}_i=(\mathbf{r}\times\mathbf{p})_i=(\hbar/i)\epsilon_{ijk}x_j\partial_k$	$\mathbf{L}_i=(\mathbf{r}\times\mathbf{p})_i=(\hbar/i)\epsilon_{ijk}x_j\partial_k$. Finite rotations can be expressed as exponentials
gilmore-lie-groups_ F02438	242	■ 0.998	$\mathbf{L}=\mathbf{r}\times\mathbf{p}$	$\mathbf{L}=\mathbf{r}\times\mathbf{p}$	The angular momentum operators $\mathbf{L}=\mathbf{r}\times\mathbf{p}$ share the same commutation relations
gilmore-lie-groups_ F02439	242	■ 0.992	$\mathbf{r}\times\nabla$	$\mathbf{r}\times\nabla$	as the infinitesimal generators of rotations $\mathbf{r}\times\nabla$, up to the proportionality factor
gilmore-lie-groups_ F02440	242	■ 1.000	$\hbar/i$	$\hbar/i$	$\hbar/i$. The commutation relations are
gilmore-lie-groups_ F02441	242	■ 1.000	(L_+)	(L_+)	raising (L_+) and lowering (L_-) operators by $L_{\pm}=L_x\pm iL_y$. The commutation
gilmore-lie-groups_ F02442	242	■ 1.000	(L_-)	(L_-)	raising (L_+) and lowering (L_-) operators by $L_{\pm}=L_x\pm iL_y$. The commutation
gilmore-lie-groups_ F02443	242	■ 1.000	$L_{\pm}=L_x\pm iL_y$	$L_{\pm}=L_x\pm iL_y$	raising (L_+) and lowering (L_-) operators by $L_{\pm}=L_x\pm iL_y$. The commutation
gilmore-lie-groups_ F02444	242	■ 1.000	$L_z=\hbar\frac{1}{2}(a_1^\dagger a_1-a_2^\dagger a_2)$, $L_+=\hbar a_1^\dagger a_2$, $L_-=\hbar a_2^\dagger a_1$	$L_z=\hbar\frac{1}{2}(a_1^\dagger a_1-a_2^\dagger a_2)$, $L_+=\hbar a_1^\dagger a_2$, $L_-=\hbar a_2^\dagger a_1$	follows: $L_z=\hbar\frac{1}{2}(a_1^\dagger a_1-a_2^\dagger a_2)$, $L_+=\hbar a_1^\dagger a_2$, $L_-=\hbar a_2^\dagger a_1$. As a result, the angu-
gilmore-lie-groups_ F02445	242	■ 1.000	$ n_1 n_2\rangle$	$ n_1 n_2\rangle$	lar momentum operators have matrix representations with basis vectors $ n_1 n_2\rangle$
gilmore-lie-groups_ F02446	242	■ 0.838	$ j,m\rangle$	$ j,m\rangle$	$ j,m\rangle$, with $n_1=0,1,2,\dots$, $n_2=0,1,2,\dots$, $n_1+n_2=2j$, $n_1-n_2=2m$,
gilmore-lie-groups_ F02447	242	■ 0.838	$n_1=0,1,2,\dots$, $n_2=0,1,2,\dots$, $n_1+n_2=2j$, $n_1-n_2=2m$	$n_1=0,1,2,\dots$, $n_2=0,1,2,\dots$, $n_1+n_2=2j$, $n_1-n_2=2m$	$ j,m\rangle$, with $n_1=0,1,2,\dots$, $n_2=0,1,2,\dots$, $n_1+n_2=2j$, $n_1-n_2=2m$,
gilmore-lie-groups_ F02448	243	■ 1.000	$-j\leq m\leq+j$	$-j\leq m\leq+j$	$-j\leq m\leq+j$. These basis vectors describe the finite-dimensional irreducible
gilmore-lie-groups_ F02449	243	■ 0.994	$((x,y,z)\rightarrow(r,\theta,\phi))$	$((x,y,z)\rightarrow(r,\theta,\phi))$	operators. In spherical coordinates $((x,y,z)\rightarrow(r,\theta,\phi))$ with $x=r\sin\theta\cos\phi$
gilmore-lie-groups_ F02450	243	■ 0.994	$x=r\sin\theta\cos\phi$	$x=r\sin\theta\cos\phi$	operators. In spherical coordinates $((x,y,z)\rightarrow(r,\theta,\phi))$ with $x=r\sin\theta\cos\phi$
gilmore-lie-groups_ F02451	243	■ 1.000	$Y_{-l}^l(\theta,\phi)$	$Y_{-l}^l(\theta,\phi)$	construct the extremal function $Y_{-l}^l(\theta,\phi)$ by solving the equation $L_-Y_{-l}^l(\theta,\phi)=0$
gilmore-lie-groups_ F02452	243	■ 1.000	$L_-Y_{-l}^l(\theta,\phi)=0$	$L_-Y_{-l}^l(\theta,\phi)=0$	construct the extremal function $Y_{-l}^l(\theta,\phi)$ by solving the equation $L_-Y_{-l}^l(\theta,\phi)=0$
gilmore-lie-groups_ F02453	266	■ 1.000	$l=0$	$l=0$	the relativistic solution is clearly incorrect? (Hint: set $l=0$.)
gilmore-lie-groups_ F02466	244	■ 1.000	L_+	L_+	of the shift up operator L_+
gilmore-lie-groups_ F02467	244	■ 0.815	$(l=0,1,2,3)$	$(l=0,1,2,3)$	The lowest spherical harmonics $(l=0,1,2,3)$ are collected in Table 14.2.
gilmore-lie-groups_ F02468	244	■ 0.662	θ,ϕ	θ,ϕ	generators acting on the sphere parameterized by coordinates (θ,ϕ) , is $-l(l+1)$,
gilmore-lie-groups_ F02469	244	■ 1.000	$\hbar^2l(l+1)$	$\hbar^2l(l+1)$	of the angular momentum operator, $\mathbf{L}\cdot\mathbf{L}$, is $\hbar^2l(l+1)$. This means physically that

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gilmore-lie-groups_ F02470	244	■ 1.000	$E_{\{N\}}=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	number $N = n + l + 1$ are degenerate, with energy $E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$. There are
gilmore-lie-groups_ F02471	244	■ 1.000	$\sum_{l=0}^{N-1} (2l+1) = N^2$	$\sum_{l=0}^{N-1} (2l+1) = N^2$	$\sum_{l=0}^{N-1} (2l+1) = N^2$ states with this energy. This N^2 -fold degeneracy is larger
gilmore-lie-groups_ F02472	244	■ 1.000	$1 / r^2$	$1/r^2$	related to a constant of motion that is peculiar to $1/r^2$ force laws. This constant of
gilmore-lie-groups_ F02473	244	■ 1.000	$d \mathbf{p} / d t = -K \mathbf{r} / r^3$	$d\mathbf{p}/dt = -K\mathbf{r}/r^3$	planetary motion, for which the force law has the form $d\mathbf{p}/dt = -K\mathbf{r}/r^3$, where
gilmore-lie-groups_ F02474	245	■ 1.000	$K=G M m, G$	$K = G M m, G$	$K = G M m, G$ is the universal gravitational constant, M and m are the two attracting
gilmore-lie-groups_ F02475	245	■ 1.000	$\mathbf{r}=x \hat{\mathbf{i}}+y \hat{\mathbf{j}}+z \hat{\mathbf{k}}$	$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$	masses, and $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ is the vector from one mass to the other. The time
gilmore-lie-groups_ F02476	245	■ 1.000	$\mathbf{p} \times \mathbf{L}$	$\mathbf{p} \times \mathbf{L}$	derivative of the vector $\mathbf{p} \times \mathbf{L}$ is
gilmore-lie-groups_ F02477	245	■ 0.999	$\mathbf{L}$	$\mathbf{L}$	In going from the first line in Eq. (14.22) to the second, we use the fact that $\mathbf{L}$ is
gilmore-lie-groups_ F02478	245	■ 1.000	$1 / r$	$1/r$	law for a $1/r$ potential. In going from the second line to the third, we express the
gilmore-lie-groups_ F02479	245	■ 0.967	$\mathbf{r} \times \mathbf{L}$	$\mathbf{r} \times \mathbf{L}$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\dot{\mathbf{r}}$. We also
gilmore-lie-groups_ F02480	245	■ 0.967	$\dot{\mathbf{r}}$	$\dot{\mathbf{r}}$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\dot{\mathbf{r}}$. We also
gilmore-lie-groups_ F02481	245	■ 1.000	$(d / d t)(\mathbf{r} / r)=\dot{\mathbf{r}} / r-(\dot{\mathbf{r}} \cdot \mathbf{r}) \mathbf{r} / r^3$	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$	use the identity $d/dt(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$. The result is that the Laplace–
gilmore-lie-groups_ F02482	245	■ 0.985	$d \mathbf{M} / d t=0$	$d\mathbf{M}/dt = 0$	Runge–Lenz vector $\mathbf{M}$ is a constant of motion: $d\mathbf{M}/dt = 0$, where
gilmore-lie-groups_ F02483	245	■ 1.000	$[H, \mathbf{M}]=0$	$[H, \mathbf{M}] = 0$	with the nonrelativistic hamiltonian: $[H, \mathbf{M}] = 0$. The six operators L_i, M_j obey
gilmore-lie-groups_ F02484	245	■ 1.000	L_i, M_j	L_i, M_j	with the nonrelativistic hamiltonian: $[H, \mathbf{M}] = 0$. The six operators L_i, M_j obey
gilmore-lie-groups_ F02485	245	■ 0.993	$(E<0)$	$(E < 0)$	bound states ($E < 0$) or $SO(3, 1)$ for excited states ($E > 0$). The operators $\mathbf{L}$ and
gilmore-lie-groups_ F02486	245	■ 0.993	$(E>0)$	$(E > 0)$	bound states ($E < 0$) or $SO(3, 1)$ for excited states ($E > 0$). The operators $\mathbf{L}$ and
gilmore-lie-groups_ F02487	246	■ 1.000	$\mathbf{M}'=(-m / 2 H)^{1 / 2} \mathbf{M}$	$\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$	Laplace–Runge–Lenz vector by a scale factor as follows: $\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$.
gilmore-lie-groups_ F02488	246	■ 1.000	$E>0$	$E > 0$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02489	246	■ 1.000	$-\rightarrow +$	$- \rightarrow +$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02490	246	■ 1.000	$SO(4) \rightarrow SO(3,1)$	$SO(4) \rightarrow SO(3, 1)$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_ F02491	246	■ 0.834	$\mathbf{A}$	$\mathbf{A}$	Figs. 10.3, 10.8(b)). It is useful to introduce two vector operators $\mathbf{A}$ and $\mathbf{B}$ as
gilmore-lie-groups_ F02492	246	■ 0.834	$\mathbf{B}$	$\mathbf{B}$	Figs. 10.3, 10.8(b)). It is useful to introduce two vector operators $\mathbf{A}$ and $\mathbf{B}$ as
gilmore-lie-groups_ F02493	246	■ 0.996	$A_{\{3\}}=\frac{1}{2}\left(a_{\{1\}}^{\dagger} a_{\{1\}}-\right.$	$A_3 = \frac{1}{2} \left(a_1^\dagger a_1 -$	momentum operators $\mathbf{A}$ in terms of two independent boson modes: $A_3 = \frac{1}{2}(a_1^\dagger a_1 -$
gilmore-lie-groups_ F02494	246	■ 0.910	$a_{\{2\}}^{\dagger} a_{\{2\}}$	$a_2^\dagger a_2$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2$, $A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_ F02495	246	■ 0.910	$A_{\{+\}}=a_{\{1\}}^{\dagger} a_{\{2\}}, A_{\{-\}}=a_{\{2\}}^{\dagger} a_{\{1\}}$	$A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation

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gilmore-lie-groups_F02496	246	■ 0.910	$\hbar \rightarrow 1$	$\hbar \rightarrow 1$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2$, $A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_F02497	246	■ 1.000	$b_{\{1\}}, b_{\{2\}}$	b_1, b_2	b_1, b_2 and their creation operators is also introduced.
gilmore-lie-groups_F02498	246	■ 1.000	$\left p_{\{1\}}, p_{\{2\}} \right\rangle$	$ p_1, p_2\rangle$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02499	246	■ 1.000	$p_{\{1\}} + p_{\{2\}} = 2j_a$	$p_1 + p_2 = 2j_a$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02500	246	■ 1.000	$p_{\{1\}} - p_{\{2\}} = m_a$	$p_1 - p_2 = m_a$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02501	246	■ 1.000	$2j_a + 1$	$2j_a + 1$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02502	246	■ 0.968	$p_{\{1\}} = 2j_a, p_{\{2\}} = 0$; $p_{\{1\}} = 2j_a - 1, p_{\{2\}} = 1$	$p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$	states correspond to $p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$; etc. For $\mathbf{B}$ the basis
gilmore-lie-groups_F02503	246	■ 1.000	$\left q_{\{1\}}, q_{\{2\}} \right\rangle$	$ q_1, q_2\rangle$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02504	246	■ 1.000	$q_{\{1\}} + q_{\{2\}} = 2j_b$	$q_1 + q_2 = 2j_b$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02505	246	■ 1.000	$q_{\{1\}} - q_{\{2\}} = m_b$	$q_1 - q_2 = m_b$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02506	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02507	246	■ 1.000	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02508	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02509	246	■ 1.000	$j_a = j_b$	$j_a = j_b$	Problem 14.15), $j_a = j_b$ and the set of states related by the shift operators is $(2j +$
gilmore-lie-groups_F02510	246	■ 1.000	$(2j +$	$(2j +$	Problem 14.15), $j_a = j_b$ and the set of states related by the shift operators is $(2j +$
gilmore-lie-groups_F02511	246	■ 1.000	$2j + 1 = N = n + l + 1$	$2j + 1 = N = n + l + 1$	$1)^2$ fold degenerate, where $2j + 1 = N = n + l + 1$.
gilmore-lie-groups_F02512	246	■ 0.996	$\operatorname{good} l$	good l	States with good l and m quantum numbers can be constructed from these states
gilmore-lie-groups_F02513	247	■ 1.000	$M_+^{\prime} =$	$M_+^{\prime} =$	spherical harmonics, is determined in a straightforward way. For example, $M_+^{\prime} =$
gilmore-lie-groups_F02514	247	■ 1.000	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$	$A_+ - B_+ = a_1^\dagger a_2 - b_1^\dagger b_2$, so that
gilmore-lie-groups_F02515	247	■ 0.998	$\mathbf{M}^{\prime}$	$\mathbf{M}^{\prime}$	The operators $\mathbf{L}$ and $\mathbf{M}^{\prime}$ are infinitesimal generators for the orthogonal group $SO(4)$.
gilmore-lie-groups_F02516	247	■ 0.608	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$	the vectors $\mathbf{L}$ and $\mathbf{M}$. Their cross product $\mathbf{W} = \mathbf{L} \times \mathbf{M}$ is orthogonal to both and
gilmore-lie-groups_F02517	247	■ 1.000	$\mathbf{r} \cdot \mathbf{L} = 0$	$\mathbf{r} \cdot \mathbf{L} = 0$	since $\mathbf{r} \cdot \mathbf{L} = 0$. The momentum vector moves in the same plane, since $\mathbf{p} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_F02518	247	■ 1.000	$\mathbf{p} \cdot \mathbf{L} = 0$	$\mathbf{p} \cdot \mathbf{L} = 0$	since $\mathbf{r} \cdot \mathbf{L} = 0$. The momentum vector moves in the same plane, since $\mathbf{p} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_F02519	248	■ 0.986	$\mathbf{W}$	$\mathbf{W}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02520	248	■ 0.986	$p_z = 0, p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	$p_z = 0, p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02521	248	■ 0.986	$p_y =$	$p_y =$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$

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gilmore-lie-groups_F02522	248	■ 1.000	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$. The two nonzero components of the momentum vector are not
gilmore-lie-groups_F02523	248	■ 1.000	$m K / L$	mK/L	momentum moves in the same plane on a circular orbit (radius mK/L) with the
gilmore-lie-groups_F02524	248	■ 1.000	$m M / L$	mM/L	center displaced from the origin by mM/L .
gilmore-lie-groups_F02525	248	■ 1.000	$p_{\{0\}} = \sqrt{-2 E / m}$	$p_0 = \sqrt{-2E/m}$	With $p_0 = \sqrt{-2E/m}$, define the unit vector $\hat{\mathbf{u}} \in S^3 \subset R^4$ by the projective trans-
gilmore-lie-groups_F02526	248	■ 1.000	$\hat{\mathbf{u}} \in S^3 \subset R^4$	$\hat{\mathbf{u}} \in S^3 \subset R^4$	With $p_0 = \sqrt{-2E/m}$, define the unit vector $\hat{\mathbf{u}} \in S^3 \subset R^4$ by the projective trans-
gilmore-lie-groups_F02527	248	■ 1.000	$\hat{\mathbf{w}}$	$\hat{\mathbf{w}}$	Here $\hat{\mathbf{w}}$ is a unit vector in R^4 that is orthogonal to all vectors in the physical space R^3 .
gilmore-lie-groups_F02528	248	■ 0.995	$\hat{\mathbf{u}}$	$\hat{\mathbf{u}}$	preserves angles (conformal). It is a simple matter to check that $\hat{\mathbf{u}}$ is a unit vector.
gilmore-lie-groups_F02529	248	■ 1.000	S^3	S^3	The circular trajectory in R^3 (Eq. (14.33)) lifts to a circle in S^3 . Reversibly, circles in
gilmore-lie-groups_F02530	248	■ 0.803	$SO(4) / SO(3)$	$SO(4)/SO(3)$	$SO(4)/SO(3)$ act to change the lengths of both $\mathbf{L}$ and $\mathbf{M}$. All group operations in
gilmore-lie-groups_F02531	248	■ 1.000	$p_{\{0\}}$	p_0	$SO(4)$ keep p_0 fixed. In this way the group $SO(4)$ maps states with principal quan-
gilmore-lie-groups_F02532	249	■ 1.000	$N^2 = (n+l+1)^2$	$N^2 = (n+l+1)^2$	irreducible representations of dimension $N^2 = (n+l+1)^2$.
gilmore-lie-groups_F02533	249	■ 1.000	$-N^3$	$-N^3$	plied each energy eigenvalue by $-N^3$, where N is the principal quantum number.
gilmore-lie-groups_F02534	249	■ 1.000	$\mathbf{x}' = M \mathbf{x}$	$\mathbf{x}' = M\mathbf{x}$	linear vector space: $\mathbf{x}' = M\mathbf{x}$, with M a transformation in the group and the inner
gilmore-lie-groups_F02535	249	■ 1.000	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	product defined by $(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$. As always, the metric-preserving
gilmore-lie-groups_F02536	249	■ 0.749	$\mathbf{y}$	$\mathbf{y}$	It is useful to define a new N -vector $\mathbf{y}$ as a scaled version of the original vector:
gilmore-lie-groups_F02537	249	■ 1.000	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02538	249	■ 1.000	$z_1 = \lambda$	$z_1 = \lambda$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02539	249	■ 1.000	$z_2 =$	$z_2 =$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02540	249	■ 1.000	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$. With these definitions we find the conformal condition
gilmore-lie-groups_F02541	249	■ 1.000	$N+2$	$N+2$	The conformal condition defines an inner product in the $N+2$ dimensional linear
gilmore-lie-groups_F02542	249	■ 0.998	$\mathbf{y}, z_1, z_2$	$\mathbf{y}, z_1, z_2$	vector space that is nondiagonal in the coordinates $\mathbf{y}, z_1, z_2$ but diagonal in the
gilmore-lie-groups_F02543	249	■ 0.986	$\mathbf{y}, y_{N+1}, y_{N+2}$	$\mathbf{y}, y_{N+1}, y_{N+2}$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_F02544	249	■ 0.986	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_F02545	249	■ 0.986	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_F02546	249	■ 1.000	R^N	R^N	sional space. If the group that preserves the metric G in R^N is $SO(p, q)$, the group
gilmore-lie-groups_F02547	250	■ 0.999	R^{N+2}	R^{N+2}	that preserves the metric in R^{N+2} is $SO(p+1, q+1)$. We seek to construct a

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gilmore-lie-groups_ F02548	250	■ 0.999	$S^0(p+1, q+1)$	$SO(p+1, q+1)$	that preserves the metric in R^{N+2} is $SO(p+1, q+1)$. We seek to construct a
gilmore-lie-groups_ F02549	250	■ 1.000	$S^3 \subset R^4 (E < 0)$	$S^3 \subset R^4 (E < 0)$	lifted to the coordinates on a sphere $S^3 \subset R^4 (E < 0)$ or a two-sheeted hyperboloid
gilmore-lie-groups_ F02550	250	■ 1.000	$H^3 \subset R^4 (E > 0)$	$H^3 \subset R^4 (E > 0)$	$H^3 \subset R^4 (E > 0)$ by the following projective transformations:
gilmore-lie-groups_ F02551	250	■ 1.000	$\mathbf{u}^t G \mathbf{u} = 1$	$\mathbf{u}^t G \mathbf{u} = 1$	condition $\mathbf{u}^t G \mathbf{u} = 1$.
gilmore-lie-groups_ F02552	251	■ 0.999	$\operatorname{diag}(1, \pm I_3, -1, +1)$	$\operatorname{diag}(1, \pm I_3, -1, +1)$	lows. The group that leaves invariant the conformal metric $\operatorname{diag}(1, \pm I_3, -1, +1)$ is
gilmore-lie-groups_ F02553	251	■ 0.984	$S^0(5, 1)$	$SO(5, 1)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_ F02554	251	■ 0.984	$E < 0$	$E < 0$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_ F02555	251	■ 0.984	$S^0(2, 4)$	$SO(2, 4)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_ F02556	251	■ 1.000	$z_1 = z_2, y_4 = \lambda$	$z_1 = z_2, y_4 = \lambda$	the condition $\mathbf{u}^t G \mathbf{u} = 1$ is satisfied, so that $z_1 = z_2, y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_ F02557	251	■ 1.000	$y_5 = 0$	$y_5 = 0$	the condition $\mathbf{u}^t G \mathbf{u} = 1$ is satisfied, so that $z_1 = z_2, y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_ F02558	251	■ 0.934	$\left(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2) \right)$	$\left(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2) \right)$	coordinates are labeled $(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2))$. Transform-
gilmore-lie-groups_ F02559	251	■ 0.908	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array} \right]$	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array} \right]$	$SO(5, 1)$ or $SO(2, 4)$ of the form $\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array} \right]$, with M a 5×5 matrix that preserves the
gilmore-lie-groups_ F02560	251	■ 0.997	$\operatorname{diag}(1, \pm I_3, -1)$	$\operatorname{diag}(1, \pm I_3, -1)$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_ F02561	251	■ 0.997	R^5	R^5	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_ F02562	251	■ 0.997	$S^0(1, 4)$	$SO(1, 4)$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_ F02563	251	■ 0.761	$\mathbf{u}'$	$\mathbf{u}'$	The inner product of $\mathbf{u}'$ with itself satisfies
gilmore-lie-groups_ F02564	251	■ 1.000	$A^t G A - C^t C = G$	$A^t G A - C^t C = G$	condition (e.g., $A^t G A - C^t C = G$, etc.) it is a simple matter to show that this
gilmore-lie-groups_ F02565	251	■ 1.000	u	u	follows (Bander and Itzykson, 1966a, 1966b). For $E < 0$ introduce a four-vector u
gilmore-lie-groups_ F02566	251	■ 1.000	$(u_0 \rightarrow u_4)$	$(u_0 \rightarrow u_4)$	as usual ($u_0 \rightarrow u_4$)
gilmore-lie-groups_ F02567	252	■ 0.967	$1 / \sqrt{2m E }$	$1 / \sqrt{2m E }$	vector $\mathbf{L}$ and the scaled (by $1 / \sqrt{2m E }$) Runge–Lenz vector $\mathbf{M}'$ as follows:
gilmore-lie-groups_ F02568	252	■ 1.000	L_i, M_i'	L_i, M_i'	The operators L_i, M_i' , and B_μ are the infinitesimal generators of $SO(4, 1)$ as follows,
gilmore-lie-groups_ F02569	252	■ 1.000	B_μ	B_μ	The operators L_i, M_i' , and B_μ are the infinitesimal generators of $SO(4, 1)$ as follows,
gilmore-lie-groups_ F02570	252	■ 0.993	$a_i, a_j^\dagger$	$a_i, a_j^\dagger$	That is, for the operators $\mathbf{A}$ we introduce annihilation and creation operators $a_i, a_j^\dagger$
gilmore-lie-groups_ F02571	252	■ 1.000	$b_i, b_j^\dagger$	$b_i, b_j^\dagger$	for two independent modes, and similarly we introduce operators $b_i, b_j^\dagger$ to describe
gilmore-lie-groups_ F02572	252	■ 1.000	$ m_1, m_2; n_1, n_2\rangle$	$ m_1, m_2; n_1, n_2\rangle$	B. Basis states on which these operators act have the form $ m_1, m_2; n_1, n_2\rangle$ where,

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gilmore-lie-groups_F02573	252	■ 1.000	$j_{\{a\}}=\frac{1}{2}\left(m_{\{1\}}+m_{\{2\}}\right)$	$j_a = \frac{1}{2} (m_1 + m_2)$	directly to the condition $j_a = j_b$, where $j_a = \frac{1}{2}(m_1 + m_2)$ and $j_b = \frac{1}{2}(n_1 + n_2)$.
gilmore-lie-groups_F02574	252	■ 1.000	$j_{\{b\}}=\frac{1}{2}\left(n_{\{1\}}+n_{\{2\}}\right)$	$j_b = \frac{1}{2} (n_1 + n_2)$	directly to the condition $j_a = j_b$, where $j_a = \frac{1}{2}(m_1 + m_2)$ and $j_b = \frac{1}{2}(n_1 + n_2)$.
gilmore-lie-groups_F02575	253	■ 1.000	$N=2\ j_{\{a\}}+1=2\ j_{\{b\}}+1=\left(j_{\{a\}}+j_{\{b\}}\right)+1$	$N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$	principal quantum number $N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$ must change
gilmore-lie-groups_F02576	253	■ 1.000	$j_{\{a\}}$	j_a	$j_a = j_b$. Operators that change j_a have the form $a_i^\dagger$ or $a_i^\dagger a_j^\dagger$, but they do not simul-
gilmore-lie-groups_F02577	253	■ 1.000	$j_{\{b\}}$	j_b	taneously change j_b . Only operators that simultaneously add or subtract one exci-
gilmore-lie-groups_F02578	253	■ 0.991	$\left(H_{\{1\}},\ H_{\{2\}},\ H_{\{3\}},\ H_{\{4\}}\right)=\left(a_{\{1\}}^{\dagger}\ a_{\{1\}},\ a_{\{2\}}^{\dagger}\ a_{\{2\}},\ b_{\{1\}}^{\dagger}\ b_{\{1\}}\right)$	$(H_1, H_2, H_3, H_4) = \left(a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1\right.$	number operators for the four boson modes: $(H_1, H_2, H_3, H_4) = (a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1,$
gilmore-lie-groups_F02579	253	■ 0.829	$b_{\{2\}}^{\dagger}\ b_{\{2\}}$	$b_2^\dagger b_2$	$b_2^\dagger b_2)$. The remaining twelve operators have eigenoperator commutation relations
gilmore-lie-groups_F02580	253	■ 1.000	$\pm\frac{1}{2}$	$\pm\frac{1}{2}$	All these roots have equal length, and inner products among these roots are all $\pm\frac{1}{2}$
gilmore-lie-groups_F02581	253	■ 1.000	$\pi\ /\ 4$	$\pi/4$	or make angles of $\pi/4$ or $3\pi/4$ radians with each other. The algebra is uniquely a
gilmore-lie-groups_F02582	253	■ 1.000	$3\ \pi\ /\ 4$	$3\pi/4$	or make angles of $\pi/4$ or $3\pi/4$ radians with each other. The algebra is uniquely a
gilmore-lie-groups_F02583	253	■ 1.000	$A_{\{3\}}=D_{\{3\}}$	$A_3 = D_3$	real form of $A_3 = D_3$.
gilmore-lie-groups_F02584	254	■ 1.000	$a_{\{1\}}^{\dagger}\ a_{\{1\}}+a_{\{2\}}^{\dagger}\ a_{\{2\}}+b_{\{1\}}^{\dagger}\ b_{\{1\}}+b_{\{2\}}^{\dagger}\ b_{\{2\}}$	$a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups_F02585	254	■ 1.000	$\mathfrak{so}(4)+\mathfrak{so}(2)$	$\mathfrak{so}(4) + \mathfrak{so}(2)$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups_F02586	254	■ 0.890	$\mathfrak{so}(4,2)=\mathfrak{su}(2,2)$	$\mathfrak{so}(4,2) = \mathfrak{su}(2,2)$	$\mathfrak{so}(4,2) = \mathfrak{su}(2,2)$. This is the conformal algebra.
gilmore-lie-groups_F02587	254	■ 1.000	$\left(q_{\{1\}},\ q_{\{2\}},\ q_{\{3\}},\ q_{\{4\}}\right)$	(q_1, q_2, q_3, q_4)	Coordinates q_1, q_2, q_3, q_4 in the fictitious space R^4 are related to coordinates
gilmore-lie-groups_F02588	254	■ 0.559	$\left(Q_{\{1\}},\ Q_{\{2\}},\ Q_{\{3\}}\right)$	(Q_1, Q_2, Q_3)	(Q_1, Q_2, Q_3) in the real space by the 4×4 transformation
gilmore-lie-groups_F02589	254	■ 1.000	$Q_{\{4\}}$	Q_4	The transformation is constructed so that the “fourth” real coordinate Q_4 is identi-
gilmore-lie-groups_F02590	254	■ 1.000	$q_{\{1\}}^2+q_{\{2\}}^2+q_{\{3\}}^2+q_{\{4\}}^2\ \neq\ 0$	$q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$	cally zero. This transformation is invertible provided $q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$. The
gilmore-lie-groups_F02591	254	■ 1.000	$R=\sqrt{Q_{\{1\}}^2+Q_{\{2\}}^2+Q_{\{3\}}^2}$	$R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$	distance $R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$ in R^3 and the distance $q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$
gilmore-lie-groups_F02592	254	■ 1.000	$q=\sqrt{q_{\{1\}}^2+q_{\{2\}}^2+q_{\{3\}}^2+q_{\{4\}}^2}$	$q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$	distance $R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$ in R^3 and the distance $q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$
gilmore-lie-groups_F02593	254	■ 1.000	$R=q^2$	$R = q^2$	in R^4 are related by $R = q^2$.
gilmore-lie-groups_F02594	254	■ 1.000	$P_{\{4\}}=0$	$P_4 = 0$	A constraint condition must be applied to force $P_4 = 0$. This condition is
gilmore-lie-groups_F02595	254	■ 1.000	$P^2=P_{\{1\}}^2+P_{\{2\}}^2+P_{\{3\}}^2=\left(1\ /\ 4\ R\ p^2\right)-\left(\zeta^2/4R^2\right)\rightarrow$	$P^2 = P_1^2 + P_2^2 + P_3^2 = (1/4Rp^2) - (\zeta^2/4R^2) \rightarrow$	With this constraint we find $P^2 = P_1^2 + P_2^2 + P_3^2 = (1/4Rp^2) - (\zeta^2/4R^2) \rightarrow$
gilmore-lie-groups_F02596	254	■ 0.980	$\left(1\ /\ 4\ R\right)\left(p_{\{1\}}^2+p_{\{2\}}^2+p_{\{3\}}^2+p_{\{4\}}^2\right)$	$(1/4R) (p_1^2 + p_2^2 + p_3^2 + p_4^2)$	$(1/4R)(p_1^2 + p_2^2 + p_3^2 + p_4^2)$. With these transformations the hamiltonian in the
gilmore-lie-groups_F02597	255	■ 1.000	$\left(q_{\{1\}},\ q_{\{2\}},\ q_{\{3\}},\ q_{\{4\}}\ ;\ p_{\{1\}},\ p_{\{2\}},\ p_{\{3\}},\ p_{\{4\}}\right)$	$(q_1, q_2, q_3, q_4; p_1, p_2, p_3, p_4)$	$(q_1, q_2, q_3, q_4; p_1, p_2, p_3, p_4)$ on the left and its transpose on the right (Sadovskii

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gilmore-lie-groups_F02598	255	■ 1.000	$\backslash\mathrm{zeta}=\theta$	$\zeta = 0$	(iii) preserves $\zeta = 0$.
gilmore-lie-groups_F02599	255	■ 1.000	$M^{\mathrm{t}}G_{\mathrm{1}}M=G_{\mathrm{1}}$	$M^tG_1M = G_1$	symplectic metric-preserving condition: $M^tG_1M = G_1$. Preserving the condition
gilmore-lie-groups_F02600	255	■ 1.000	$M^{\mathrm{t}}G_{\mathrm{2}}M=G_{\mathrm{2}}$	$M^tG_2M = G_2$	$M^tG_2M = G_2$.
gilmore-lie-groups_F02601	256	■ 0.988	$\backslash\mathrm{operatorname{Sp}}\{8\ ;\ \backslash\mathrm{mathbb{R}}\}$	$\mathrm{Sp}(8;\mathbb{R})$	$(+4, -4)$. The group that preserves the antisymmetric metric is $Sp(8;\mathbb{R})$ and the
gilmore-lie-groups_F02602	256	■ 1.000	$S\ 0(4,4)$	$SO(4,4)$	group that preserves the symmetric metric is $SO(4,4)$. The group that satisfies both
gilmore-lie-groups_F02603	256	■ 0.998	$(q,\ p)$	(q,p)	coordinates (q,p) to coordinates (s,r) :
gilmore-lie-groups_F02604	256	■ 0.998	$(s,\ r)$	(s,r)	coordinates (q,p) to coordinates (s,r) :
gilmore-lie-groups_F02605	256	■ 1.000	$U(2,2)$	$U(2,2)$	The noncompact group $U(2,2)$ preserves the constraint Eq. (14.49).
gilmore-lie-groups_F02606	256	■ 0.825	$\backslash\mathrm{left}(V(r)=-e^{\mathrm{2}}\ / \ r\backslash\mathrm{right})$	$(V(r) = -e^2/r)$	charged particle interacting with an external Coulomb field $(V(r) = -e^2/r)$ can be
gilmore-lie-groups_F02607	257	■ 1.000	$\backslash\mathrm{mathfrak{s\ o}}(4,1)$	$\mathfrak{so}(4,1)$	close under commutation to span a Lie algebra that is isomorphic with $\mathfrak{so}(4,1)$.
gilmore-lie-groups_F02608	257	■ 1.000	$\backslash\mathrm{mathfrak{s\ o}}(4,2)$	$\mathfrak{so}(4,2)$	Five additional operators can be introduced that extend the algebra to $\mathfrak{so}(4,2)$.
gilmore-lie-groups_F02609	257	■ 1.000	$J_{\mathrm{i}},\ M_{\mathrm{i}},\ A_{\mathrm{i}},\ \backslash\mathrm{Gamma}_{\mathrm{i}}(i=1,2,3)$	$J_i, M_i, A_i, \Gamma_i (i = 1, 2, 3)$	The four triplets $J_i, M_i, A_i, \Gamma_i (i = 1, 2, 3)$ have transformation properties of three-
gilmore-lie-groups_F02610	257	■ 0.996	$A_{\mathrm{4}},\ \backslash\mathrm{Gamma}_{\mathrm{4}},\ \backslash\mathrm{Gamma}_{\mathrm{5}}$	A_4, Γ_4, Γ_5	vectors under rotations. The three additional operators A_4, Γ_4, Γ_5 close under com-
gilmore-lie-groups_F02611	257	■ 1.000	$\backslash\mathrm{Phi}(r)=e\ /\ r$	$\Phi(r) = e/r$	in the Coulomb field $\Phi(r) = e/r$ of a proton can be expressed in terms of operators
gilmore-lie-groups_F02612	257	■ 1.000	$A_{\mathrm{4}},\ \backslash\mathrm{Gamma}_{\mathrm{4}}$	A_4, Γ_4	of type A_4, Γ_4 , and Γ_5 . These operators are displayed in Table 14.3, along with the
gilmore-lie-groups_F02613	257	■ 1.000	$\backslash\mathrm{Gamma}_{\mathrm{5}}$	Γ_5	of type A_4, Γ_4 , and Γ_5 . These operators are displayed in Table 14.3, along with the
gilmore-lie-groups_F02614	257	■ 0.860	$(\backslash\mathrm{mathbf{E}},\ \backslash\mathrm{mathbf{B}})$	$(\mathbf{E}, \mathbf{B})$	scribed by the Dirac equation. The electromagnetic field $(\mathbf{E}, \mathbf{B})$ is described by the
gilmore-lie-groups_F02615	257	■ 1.000	$A_{\mathrm{\mu}}=(\backslash\mathrm{phi},\ \backslash\mathrm{mathbf{A}})$	$A_\mu = (\phi, \mathbf{A})$	four-vector potential $A_\mu = (\phi, \mathbf{A})$. The electron has charge $q = -e$ (where e is
gilmore-lie-groups_F02616	257	■ 1.000	$q=-e$	$q = -e$	four-vector potential $A_\mu = (\phi, \mathbf{A})$. The electron has charge $q = -e$ (where e is
gilmore-lie-groups_F02617	257	■ 1.000	$H_{\mathrm{D}}\backslash\mathrm{psi}=E\ \backslash\mathrm{psi}$	$H_D\psi = E\psi$	the charge on the proton) and spin $\frac{1}{2}$. The Dirac equation $H_D\psi = E\psi$ is a matrix
gilmore-lie-groups_F02621	260	■ 1.000	$\backslash\mathrm{Gamma}_{\mathrm{4}}$	Γ_4	Γ_4 or the compact generator Γ_5 , depending on the sign of the parameter C .
gilmore-lie-groups_F02634	258	■ 1.000	$\backslash\mathrm{pi}=\backslash\mathrm{mathbf{p}}-\backslash\mathrm{frac}\{q\}\{c\}\ \backslash\mathrm{mathbf{A}}$	$\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$	replaced by $\pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$. Under this condition the operators still close under
gilmore-lie-groups_F02635	258	■ 0.971	$\backslash\mathrm{gamma}_{\mathrm{i}}$	γ_i	The 4×4 matrices β and γ_i can be chosen as
gilmore-lie-groups_F02636	258	■ 0.982	$\backslash\mathrm{sigma}_{\mathrm{i}}$	σ_i	Here σ_i are the standard Pauli 2×2 spin matrices (cf., Eq. (3.39), Problem 3.1).
gilmore-lie-groups_F02637	259	■ 1.000	$\backslash\mathrm{mathbf{p}}\ \backslash\mathrm{rightarrow}\ \backslash\mathrm{pi}=\backslash\mathrm{mathbf{p}}-\backslash\mathrm{frac}\{q\}\{c\}\ \backslash\mathrm{mathbf{A}}$	$\mathbf{p} \rightarrow \pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$	As before, the substitution $\mathbf{p} \rightarrow \pi = \mathbf{p} - \frac{q}{c}\mathbf{A}$ is in order in the event there is a
gilmore-lie-groups_F02638	259	■ 1.000	$J_{\mathrm{i}},\ M_{\mathrm{i}},\ A_{\mathrm{i}},\ \backslash\mathrm{Gamma}_{\mathrm{i}}$	J_i, M_i, A_i, Γ_i	three-vectors J_i, M_i, A_i, Γ_i defined in Table 14.3. The Dirac hamiltonian is ex-
gilmore-lie-groups_F02639	259	■ 0.990	$\backslash\mathrm{Gamma}_{\mathrm{4}},\ \backslash\mathrm{Gamma}_{\mathrm{5}}$	Γ_4, Γ_5	Gordon operator (see Table 14.3). In short, the operators Γ_4, Γ_5 are modified but

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gilmore-lie-groups_F02640	259	■ 1.000	$D=d / d r$	$D = d/dr$	with $D = d/dr$. The operators r and D behave under commutation like the boson
gilmore-lie-groups_F02641	259	■ 1.000	$r D^{\{2\}}+r$	$rD^2 + r$	The linear combinations $rD^2 + r$ and $rD^2 - r$ are compact and noncompact, re-
gilmore-lie-groups_F02642	259	■ 1.000	$r D^{\{2\}}-r$	$rD^2 - r$	The linear combinations $rD^2 + r$ and $rD^2 - r$ are compact and noncompact, re-
gilmore-lie-groups_F02643	260	■ 1.000	$C^{\{2\}}=\Gamma_5^{\{2\}}-\Gamma_4^{\{2\}}-M_4^{\{2\}}=-a$	$C^2 = \Gamma_5^2 - \Gamma_4^2 - M_4^2 = -a$	The Casimir operator for this algebra is $C^2 = \Gamma_5^2 - \Gamma_4^2 - M_4^2 = -a$. The represen-
gilmore-lie-groups_F02644	260	■ 0.978	$a \rightarrowtail A$	$a \rightarrow A$	follows ($a \rightarrowtail A$)
gilmore-lie-groups_F02645	260	■ 1.000	$C<0$	$C < 0$	If $C < 0$ we can choose $e^{-\theta} + C e^{\theta} = 0$, so that the resulting equation becomes
gilmore-lie-groups_F02646	260	■ 1.000	$e^{\{-\backslash\theta\}}+C e^{\{\backslash\theta\}}=0$	$e^{-\theta} + C e^{\theta} = 0$	If $C < 0$ we can choose $e^{-\theta} + C e^{\theta} = 0$, so that the resulting equation becomes
gilmore-lie-groups_F02647	260	■ 1.000	$u(r)=e^{\{\backslash\theta\} M_{\{4\}}} R(r)$	$u(r) = e^{\theta M_4} R(r)$	where $u(r) = e^{\theta M_4} R(r)$. If A is the Casimir invariant of this representation
gilmore-lie-groups_F02648	260	■ 0.798	$N=-\frac{1}{2}+$	$N = -\frac{1}{2} +$	of $\mathfrak{su}(1,1)$, the discrete spectrum of the compact operator Γ_5 is $N = -\frac{1}{2} +$
gilmore-lie-groups_F02649	260	■ 1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A}+1+n, n=0,1,2, \ldots$	$\sqrt{\left(\frac{1}{2}\right)^2 - A + 1 + n}, n = 0, 1, 2, \dots$	$\sqrt{(\frac{1}{2})^2 - A + 1 + n}, n = 0, 1, 2, \dots$. This result leads directly to the eigenvalue
gilmore-lie-groups_F02650	261	■ 1.000	$W=-\frac{1}{2} m c^2 \alpha^2 \left(1 / N^2\right)$	$W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$	relativistic case) $W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$, where $N = l + 1 + k, k = 0, 1, 2, \dots$
gilmore-lie-groups_F02651	261	■ 1.000	$N=l+1+k, k=0,1,2, \ldots$	$N = l + 1 + k, k = 0, 1, 2, \dots$	relativistic case) $W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$, where $N = l + 1 + k, k = 0, 1, 2, \dots$.
gilmore-lie-groups_F02652	261	■ 1.000	$m_{\{l\}}$	m_l	tum operators, the quantum number m_l (eigenvalue of L_z) is also invariant under
gilmore-lie-groups_F02653	261	■ 0.449	$ N, \backslash n \rangle \rightarrowtail \leftarrowtail N \pm 1, l m\rangle$	$ N, \ln\rangle \leftrightarrow N \pm 1, lm\rangle$	algebra are $ N, lm\rangle \leftrightarrow N \pm 1, lm\rangle$. The states on which these operators act are
gilmore-lie-groups_F02660	261	■ 1.000	$L_{\{-\}} Y_{\{m=-l\}^{\{l\}}}(\backslash\theta, \backslash\phi)=0$	$L_- Y_{m=-l}^l(\theta, \phi) = 0$	known, this relation can be used, as was the relation $L_- Y_{m=-l}^l(\theta, \phi) = 0$, to deter-
gilmore-lie-groups_F02662	261	■ 1.000	$C>0$	$C > 0$	If $C > 0$ we can choose θ so that $e^{-\theta} - C e^{\theta} = 0$. Equation (14.66) reduces to
gilmore-lie-groups_F02663	261	■ 1.000	$e^{\{-\backslash\theta\}}-C e^{\{\backslash\theta\}}=0$	$e^{-\theta} - C e^{\theta} = 0$	If $C > 0$ we can choose θ so that $e^{-\theta} - C e^{\theta} = 0$. Equation (14.66) reduces to
gilmore-lie-groups_F02664	261	■ 1.000	$k \simeq e^{\{-\backslash\theta\}}$	$k \simeq e^{-\theta}$	$k \simeq e^{-\theta}$ with $E = \hbar^2 k^2 / 2m$. The asymptotic form of the wave function is (Gilmore
gilmore-lie-groups_F02665	261	■ 1.000	$E=\hbar\bar{r}^{\{2\}} k^{\{2\}} / 2 m$	$E = \hbar^2 k^2 / 2m$	$k \simeq e^{-\theta}$ with $E = \hbar^2 k^2 / 2m$. The asymptotic form of the wave function is (Gilmore
gilmore-lie-groups_F02666	262	■ 1.000	$\delta(j)=\arg [\Gamma(j+1-i(\alpha / k))]$	$\delta(j) = \arg[\Gamma(j + 1 - i(\alpha/k))]$	where $\delta(j) = \arg[\Gamma(j + 1 - i(\alpha/k))]$ is part of the scattering phase shift, and the
gilmore-lie-groups_F02667	262	■ 1.000	$j=-\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}$	$j = -\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A}$	expression for j is given by $j = -\frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}$.
gilmore-lie-groups_F02668	262	■ 1.000	$-1 / r$	$-1/r$	attracted than a simple $-1/r$ potential suggests. It is possible to represent this
gilmore-lie-groups_F02669	262	■ 1.000	$-1 / r^{\{2\}}$	$-1/r^2$	extra attraction by adding a term of the form $-1/r^2$ to the potential to represent
gilmore-lie-groups_F02670	262	■ 1.000	$V(r)=-e^{\{2\}} / r \rightarrowtail e^{\{2\}} / r-\mu_{\{l\}} \left(\hbar\bar{r}^{\{2\}} / 2 m\right) / r^{\{2\}}$	$V(r) = -e^2/r \rightarrow -e^2/r - \mu_l (\hbar^2/2m)/r^2$	and Klein–Gordon equations is $V(r) = -e^2/r \rightarrow -e^2/r - \mu_l (\hbar^2/2m)/r^2$. This
gilmore-lie-groups_F02671	262	■ 1.000	$j \rightarrowtail j^{\{\prime\}}=j+\Delta j$	$j \rightarrow j' = j + \Delta j$	This change produces a change in the value of $j \rightarrow j' = j + \Delta j$, where $\Delta j =$

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gilmore-lie-groups_F02672	262	■ 1.000	$\Delta j =$	$\Delta j =$	This change produces a change in the value of $j \rightarrow j' = j + \Delta j$, where $\Delta j =$
gilmore-lie-groups_F02673	262	■ 1.000	$-\mu_l/(2l+1)$	$-\mu_l/(2l+1)$	$-\mu_l/(2l+1)$ in the nonrelativistic case. This change produces a change in the
gilmore-lie-groups_F02674	262	■ 1.000	Δj	Δj	The quantum defect Δj causes the Rydberg states to be bound more strongly than in
gilmore-lie-groups_F02675	263	■ 1.000	$G \supset SO(3)$	$G \supset SO(3)$	group, we are prodded to search for a larger group $G \supset SO(3)$ that explains the
gilmore-lie-groups_F02676	264	■ 0.987	3^{deg}	3^{deg}	of relativity. Show that the existence of the 3^{deg} microwave background radiation is
gilmore-lie-groups_F02677	264	■ 1.000	$\mathbf{F} = m \mathbf{g}$	$\mathbf{F} = m \mathbf{g}$	tional force $\mathbf{F} = m \mathbf{g}$ in the “down” direction. Assume that the other elevator is in
gilmore-lie-groups_F02678	264	■ 1.000	$\mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r}$	$\mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r}$	taken as $\mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r}$, so that $\mathbf{B} = \nabla \times \mathbf{A}$. Derive the Klein–Gordon equation for an
gilmore-lie-groups_F02679	264	■ 0.996	$(r, \theta, \phi) = (\theta_3, \theta_2, \theta_1)$	$(r, \theta, \phi) = (\theta_3, \theta_2, \theta_1)$	4. Introduce spherical coordinates as follows: $(r, \theta, \phi) = (\theta_3, \theta_2, \theta_1)$ and
gilmore-lie-groups_F02680	264	■ 1.000	$\mathcal{L}^2(S^1) = \partial^2 / \partial \theta_1^2$	$\mathcal{L}^2(S^1) = \partial^2 / \partial \theta_1^2$	Show that $\mathcal{L}^2(S^1) = \partial^2 / \partial \theta_1^2$. Show that
gilmore-lie-groups_F02681	264	■ 1.000	$\mathcal{L}^2(S^3)$	$\mathcal{L}^2(S^3)$	Generalize this result to $\mathcal{L}^2(S^3)$ recursively using $\mathcal{L}^2(S^2)$ and $(\partial / \partial \cos \theta_3)^2$. Do this
gilmore-lie-groups_F02682	264	■ 1.000	$(\partial / \partial \cos \theta_3)^2$	$(\partial / \partial \cos \theta_3)^2$	Generalize this result to $\mathcal{L}^2(S^3)$ recursively using $\mathcal{L}^2(S^2)$ and $(\partial / \partial \cos \theta_3)^2$. Do this
gilmore-lie-groups_F02683	264	■ 1.000	$\mathcal{L}^2(S^n)$	$\mathcal{L}^2(S^n)$	more generally for $\mathcal{L}^2(S^n)$.
gilmore-lie-groups_F02684	265	■ 1.000	$r \rightarrow \infty$	$r \rightarrow \infty$	a. Show that the singular points of Eq. (14.9) occur at $r = 0$ and $r \rightarrow \infty$.
gilmore-lie-groups_F02685	265	■ 1.000	$\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$	$\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$	Show that $\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$ and $\lambda = \pm \sqrt{-C}$.
gilmore-lie-groups_F02686	265	■ 1.000	$\lambda = \pm \sqrt{-C}$	$\lambda = \pm \sqrt{-C}$	Show that $\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$ and $\lambda = \pm \sqrt{-C}$.
gilmore-lie-groups_F02687	265	■ 1.000	$\sqrt{(\frac{1}{2})^2 - A}$	$\sqrt{(\frac{1}{2})^2 - A}$	c. Show that if $\sqrt{(\frac{1}{2})^2 - A}$ is real, the solution with the positive sign is always square
gilmore-lie-groups_F02688	265	■ 1.000	$\pm \sqrt{-C}$	$\pm \sqrt{-C}$	with the negative sign square integrable? Show that if $C < 0$ the solution $\pm \sqrt{-C}$
gilmore-lie-groups_F02689	265	■ 1.000	$R(r) = r^\gamma e^{\lambda r} f(r)$	$R(r) = r^\gamma e^{\lambda r} f(r)$	d. Show that a solution of the form $R(r) = r^\gamma e^{\lambda r} f(r)$ can be found where the function
gilmore-lie-groups_F02690	265	■ 1.000	$f(r) = \sum_{j=0}^{\infty} f_j r^j$	$f(r) = \sum_{j=0}^{\infty} f_j r^j$	f. Represent the function $f(r)$ as an ascending power series: $f(r) = \sum_{j=0}^{\infty} f_j r^j$.
gilmore-lie-groups_F02691	265	■ 1.000	$j \rightarrow \infty$	$j \rightarrow \infty$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_F02692	265	■ 1.000	$f(r) \rightarrow e^{-2\lambda r}$	$f(r) \rightarrow e^{-2\lambda r}$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_F02693	265	■ 1.000	$\lambda < 0$	$\lambda < 0$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_F02694	265	■ 0.963	r^n	r^n	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_F02695	265	■ 0.963	$f_n \neq 0$	$f_n \neq 0$	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_F02696	265	■ 0.963	$f_{n+1} = 0 (\Rightarrow f_{n+2} =$	$f_{n+1} = 0 (\Rightarrow f_{n+2} =$	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_F02697	265	■ 1.000	$f_{n+3} = \cdots = 0$	$f_{n+3} = \cdots = 0$	$f_{n+3} = \cdots = 0$), show that the quantization condition $2\lambda(n + \gamma) + B = 0$ must be

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gilmore-lie-groups_F02698	265	■ 1.000	$2 \lambda(n+\gamma)+B=0$	$2\lambda(n+\gamma)+B=0$	$f_{n+3}=\cdots=0$), show that the quantization conditon $2\lambda(n+\gamma)+B=0$ must be
gilmore-lie-groups_F02699	266	■ 1.000	$\frac{1}{r}r^{\gamma}f(r)e^{\lambda r}$	$\frac{1}{r}r^{\gamma}f(r)e^{\lambda r}$	The radial part of the wavefunction $\frac{1}{r}r^{\gamma}f(r)e^{\lambda r}$ has exactly n nodes in the open
gilmore-lie-groups_F02700	266	■ 0.907	$(0,\infty)$	$(0,\infty)$	interval $[0,\infty)$.
gilmore-lie-groups_F02701	266	■ 1.000	Ze/r	Ze/r	show that the potential is Ze/r and the solutions of the relativistic and nonrelativistic
gilmore-lie-groups_F02702	266	■ 1.000	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$ and $N=n+l+1$
gilmore-lie-groups_F02703	266	■ 1.000	$e^{\lambda r}$	$e^{\lambda r}$	8. The radial part of the wavefunction dies off like $e^{\lambda r}$ for large r , where $\lambda<0$ for bound
gilmore-lie-groups_F02704	266	■ 1.000	λ^{-1}	λ^{-1}	states. The parameter λ^{-1} has the dimensions of length, and $a\simeq 1/ \lambda $ characterizes
gilmore-lie-groups_F02705	266	■ 1.000	$a\simeq 1/ \lambda $	$a\simeq 1/ \lambda $	states. The parameter λ^{-1} has the dimensions of length, and $a\simeq 1/ \lambda $ characterizes
gilmore-lie-groups_F02706	266	■ 0.910	(n,l)	(n,l)	the size of a bound state orbit. Show that bound states with quantum numbers (n,l)
gilmore-lie-groups_F02717	267	■ 1.000	$a_B=\hbar^2/m e^2=0.529\times 10^{-8}\text{~cm}$	$a_B=\hbar^2/me^2=0.529\times 10^{-8}\text{ cm}$	In these expressions $a_B=\hbar^2/me^2=0.529\times 10^{-8}\text{ cm}$ is the Bohr radius: the char-
gilmore-lie-groups_F02718	267	■ 0.906	n,l	n,l	sizes of the atoms in the (n,l) states differ between the relativistic and nonrelativistic
gilmore-lie-groups_F02719	267	■ 1.000	$W=-\frac{1}{2}m c^2\alpha^2/N^2$	$W=-\frac{1}{2}mc^2\alpha^2/N^2$	binding energy $W=-\frac{1}{2}mc^2\alpha^2/N^2$ is the reduced mass: $1/m=1/m_1+1/m_2$ of
gilmore-lie-groups_F02720	267	■ 1.000	$1/m=1/m_1+1/m_2$	$1/m=1/m_1+1/m_2$	binding energy $W=-\frac{1}{2}mc^2\alpha^2/N^2$ is the reduced mass: $1/m=1/m_1+1/m_2$ of
gilmore-lie-groups_F02721	267	■ 1.000	$(E=0)$	$(E=0)$	($E>0$); and parabolic at the separatrix ($E=0$). If the radial force field includes a
gilmore-lie-groups_F02722	267	■ 1.000	$1/r^3$	$1/r^3$	radial $1/r^3$ perturbation
gilmore-lie-groups_F02723	267	■ 1.000	$\alpha=\sqrt{1-\eta},\eta=C/Ka$	$\alpha=\sqrt{1-\eta},\eta=C/Ka$	where $\alpha=\sqrt{1-\eta},\eta=C/Ka$. This can be treated as an ellipse that is slowly
gilmore-lie-groups_F02724	267	■ 1.000	$\alpha\simeq 1$	$\alpha\simeq 1$	rotating, $\alpha\simeq 1$. In this case the parameters a and ϵ have their usual meanings
gilmore-lie-groups_F02725	267	■ 1.000	η	η	η is a measure of the strength of the perturbation to the strength of the Coulomb
gilmore-lie-groups_F02726	268	■ 0.651	$E=\sqrt{\left(m c^2\right)^2+\left(\mathbf{p} c\right)^2}-K / r$	$E=\sqrt{\left(m c^2\right)^2+\left(\mathbf{p} c\right)^2}-K / r$	a. Expand the relativistic energy $E=\sqrt{\left(m c^2\right)^2+\left(\mathbf{p} c\right)^2}-K / r$ to fourth order in $\mathbf{p}$
gilmore-lie-groups_F02727	268	■ 1.000	$E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$	$E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$	and show $E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$.
gilmore-lie-groups_F02728	268	■ 1.000	$-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$	$-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$	b. Replace the quartic term $-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$ by $-\left(W+K / r\right)^2 /\left(2 m c^2\right)$ and ex-
gilmore-lie-groups_F02729	268	■ 1.000	$-\left(W+K / r\right)^2 /\left(2 m c^2\right)$	$-\left(W+K / r\right)^2 /\left(2 m c^2\right)$	b. Replace the quartic term $-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$ by $-\left(W+K / r\right)^2 /\left(2 m c^2\right)$ and ex-
gilmore-lie-groups_F02730	268	■ 1.000	$K^{\prime}$	$K^{\prime}$	Evaluate $K^{\prime}$ and $C^{\prime}$ and show $K^{\prime}=K\left(1+W / m c^2\right)$ and $C^{\prime}=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_F02731	268	■ 1.000	$C^{\prime}$	$C^{\prime}$	Evaluate $K^{\prime}$ and $C^{\prime}$ and show $K^{\prime}=K\left(1+W / m c^2\right)$ and $C^{\prime}=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_F02732	268	■ 1.000	$K^{\prime}=K\left(1+W / m c^2\right)$	$K^{\prime}=K\left(1+W / m c^2\right)$	Evaluate $K^{\prime}$ and $C^{\prime}$ and show $K^{\prime}=K\left(1+W / m c^2\right)$ and $C^{\prime}=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_F02733	268	■ 1.000	$C^{\prime}=-K^2 /\left(2 m c^2\right)$	$C^{\prime}=-K^2 /\left(2 m c^2\right)$	Evaluate $K^{\prime}$ and $C^{\prime}$ and show $K^{\prime}=K\left(1+W / m c^2\right)$ and $C^{\prime}=-K^2 /\left(2 m c^2\right)$.

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gilmore-lie-groups_F02734	268	■ 1.000	$K \rightarrow K^{\prime}$	$K \rightarrow K'$	c. Argue that the classical motion involves a renormalized coupling $K \rightarrow K'$ as well
gilmore-lie-groups_F02735	268	■ 1.000	$C=2\ C^{\prime}$	$C = 2C'$	as a $1/r^3$ component to the force, with $C = 2C'$.
gilmore-lie-groups_F02736	268	■ 1.000	$\delta\theta\ \simeq\ \eta\ /\ 2$	$\delta\theta \simeq \eta/2$	d. Show that the advance in the perihelion of the orbit is $\delta\theta \simeq \eta/2$ per period.
gilmore-lie-groups_F02737	268	■ 1.000	$\epsilon=0.206$	$\epsilon = 0.206$	e. Evaluate η for the planet Mercury, for which $\epsilon = 0.206$ and the period is $T = 0.24$
gilmore-lie-groups_F02738	268	■ 1.000	$T=0.24$	$T = 0.24$	e. Evaluate η for the planet Mercury, for which $\epsilon = 0.206$ and the period is $T = 0.24$
gilmore-lie-groups_F02739	268	■ 1.000	$m=$	$m =$	correct and the mass of the particle varies with its state of motion according to $m =$
gilmore-lie-groups_F02740	268	■ 1.000	$m_{\{0\}}\ /\ \sqrt{1-(v\ /\ c)^{2}}$	$m_0/\sqrt{1-(v/c)^2}$	$m_0/\sqrt{1-(v/c)^2}$. (2) The mass of a particle is a constant of nature and Newton's
gilmore-lie-groups_F02741	268	■ 1.000	$\mathbf{F}(r)=\left(-K\ /\ r^{2}+p(r)\right)\ \hat{\mathbf{r}}$	$\mathbf{F}(r) = (-K/r^2 + p(r))\ \hat{\mathbf{r}}$	force has the form $\mathbf{F}(r) = (-K/r^2 + p(r))\hat{\mathbf{r}}$, where $p(r)$ is a small perturbation. The
gilmore-lie-groups_F02742	268	■ 1.000	$p(r)$	$p(r)$	force has the form $\mathbf{F}(r) = (-K/r^2 + p(r))\hat{\mathbf{r}}$, where $p(r)$ is a small perturbation. The
gilmore-lie-groups_F02743	268	■ 1.000	$1\ /\ r=\left(m\ K\ /\ L^{2}\right)\left(1+(M\ /\ m\ K)\ \cos\ \theta\right)$	$1/r = (mK/L^2)(1 + (M/mK)\cos\theta)$	with $1/r = (mK/L^2)(1 + (M/mK)\cos\theta)$. Here L is the particle's orbital angular
gilmore-lie-groups_F02744	268	■ 1.000	L	L	with $1/r = (mK/L^2)(1 + (M/mK)\cos\theta)$. Here L is the particle's orbital angular
gilmore-lie-groups_F02745	268	■ 1.000	$C\ /\ r^{3}$	C/r^3	momentum and T is its period. If the perturbing term is of the form C/r^3 the integral
gilmore-lie-groups_F02746	268	■ 1.000	$C\ \times\ 2\pi\ \frac{m\ K}{L^2}$	$C \times 2\pi \frac{mK}{L^2}$	is $C \times 2\pi \frac{mK}{L^2}$. The perturbations due to special and General Relativity are
gilmore-lie-groups_F02747	268	■ 1.000	$K=G\ M\ m$	$K = GMm$	For planetary motion $K = GMm$. When $M \gg m$, ω is (almost) independent of m .
gilmore-lie-groups_F02748	268	■ 1.000	$M\ \gg\ m,\ \omega$	$M \gg m, \omega$	For planetary motion $K = GMm$. When $M \gg m$, ω is (almost) independent of m .
gilmore-lie-groups_F02749	269	■ 1.000	$\omega=42^{\prime\prime}$	$\omega = 42''$	planetary distance from the Sun. What is the precession for the Earth? Use $\omega = 42''$
gilmore-lie-groups_F02750	269	■ 1.000	L_{-}	L_-	12. The action of the angular momentum shift down operator L_- on the lowest m -value
gilmore-lie-groups_F02751	269	■ 1.000	$Y_{\{m\}}^{\{l\}}(\theta,\ \phi)=P_{\{-l\}}^{\{l\}}(\theta)\ e^{-i\ l\ \phi}$	$Y_m^l(\theta, \phi) = P_{-l}^l(\theta)e^{-il\phi}$	a. Write $Y_m^l(\theta, \phi) = P_{-l}^l(\theta)e^{-il\phi}$ and show
gilmore-lie-groups_F02752	269	■ 1.000	$P_{\{-l\}}^{\{l\}}=(\sin\ \theta)^{\{l\}}$	$P_{-l}^l = (\sin\theta)^l$	b. Show $P_{-l}^l = (\sin\theta)$ satisfies this equation.
gilmore-lie-groups_F02753	269	■ 1.000	$N_{\{l\}}$	N_l	a normalization coefficient N_l and enforcing the condition
gilmore-lie-groups_F02754	269	■ 1.000	$Y_{\{\pm\ m\}}^{\{l\}}(\theta,\ \phi)$	$Y_{\pm l}^l(\theta, \phi)$	$Y_{\pm l}^l(\theta, \phi)$:
gilmore-lie-groups_F02755	269	■ 0.999	$N_{\{0\}}=\sqrt{1\ /\ 4\ \pi}$	$N_0 = \sqrt{1/4\pi}$	Compare these results with Table 14.2 using initial condition $N_0 = \sqrt{1/4\pi}$. Com-
gilmore-lie-groups_F02756	269	■ 1.000	$N_{\{3\}}$	N_3	pute N_3 .
gilmore-lie-groups_F02757	269	■ 0.999	$\left \left\langle\right.\right._{\{m'\}}^{\{l\}}\left.\left.\right L_{+}\left \right.\right._{\{m\}}^{\{l\}}\left.\right\rangle=\sqrt{\left(l+m'\right)\left(l-m\right)}$	$\langle_{m'}^l L_+ _m^l\rangle = \sqrt{(l+m')(l-m)}$	f. Use the numerical value of the matrix elements $\langle_{m'}^l L_+ _m^l\rangle = \sqrt{(l+m')(l-m)}$
gilmore-lie-groups_F02758	269	■ 1.000	$\delta_{m^{\prime},\ m+1}$	$\delta_{m',m+1}$	$\delta_{m',m+1}$ and the coordinate representation of the shift up operator L_+ to construct
gilmore-lie-groups_F02759	269	■ 1.000	$l=N-1$	$l = N - 1$	numbers: $l = N - 1$, where in general the principal quantum number $N = n + l + 1$.
gilmore-lie-groups_F02760	269	■ 0.980	$n=0$	$n = 0$	These functions have no nodes in the interval $(0, \infty)$ (since $n = 0$).

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gilmore-lie-groups_F02761	269	■ 0.848	$\mathbf{r}$	$\mathbf{r}$	15. $\mathbf{r}$ is the position vector from the sun to a planet, or from the proton to the electron
gilmore-lie-groups_F02762	270	■ 1.000	$\mathbf{M} \cdot \mathbf{L} = \mathbf{0}$	$\mathbf{M} \cdot \mathbf{L} = \mathbf{0}$	a. $\mathbf{M} \cdot \mathbf{L} = \mathbf{0}$.
gilmore-lie-groups_F02763	270	■ 1.000	$\mathbf{M} \cdot \mathbf{M} = (2\mathbf{L} \cdot \mathbf{L}/m)(\mathbf{p} \cdot \mathbf{p}/2m - K/r) + K^2$	$\mathbf{M} \cdot \mathbf{M} = (2\mathbf{L} \cdot \mathbf{L}/m)(\mathbf{p} \cdot \mathbf{p}/2m - K/r) + K^2$	b. $\mathbf{M} \cdot \mathbf{M} = (2\mathbf{L} \cdot \mathbf{L}/m)(\mathbf{p} \cdot \mathbf{p}/2m - K/r) + K^2$.
gilmore-lie-groups_F02764	270	■ 1.000	$\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L}/m - Kr$	$\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L}/m - Kr$	c. $\mathbf{M} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L}/m - Kr$.
gilmore-lie-groups_F02765	270	■ 0.999	$\mathbf{M} \cdot \mathbf{r} = Mr \cos \theta$	$\mathbf{M} \cdot \mathbf{r} = Mr \cos \theta$	d. $\mathbf{M} \cdot \mathbf{r} = Mr \cos \theta$.
gilmore-lie-groups_F02766	270	■ 1.000	$r = (\mathbf{L} \cdot \mathbf{L}/mK/1 + (M/K) \cos \theta)$	$r = (\mathbf{L} \cdot \mathbf{L}/mK/1 + (M/K) \cos \theta)$	e. $r = (\mathbf{L} \cdot \mathbf{L}/mK/1 + (M/K) \cos \theta)$.
gilmore-lie-groups_F02767	270	■ 1.000	L^2/mK	L^2/mK	in a $1/r$ potential to conclude that L^2/mK is the semimajor axis of the elliptical
gilmore-lie-groups_F02768	270	■ 1.000	$\epsilon = M/K$	$\epsilon = M/K$	orbit and $\epsilon = M/K$ is the eccentricity of the orbit.
gilmore-lie-groups_F02769	270	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = (-1/4\hbar^2)(\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$	$\mathbf{A} \cdot \mathbf{A} = (-1/4\hbar^2)(\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$	16. Show that $\mathbf{A} \cdot \mathbf{A} = (-1/4\hbar^2)(\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$. Show that $\mathbf{B} \cdot \mathbf{B}$
gilmore-lie-groups_F02770	270	■ 1.000	$\mathbf{B} \cdot \mathbf{B}$	$\mathbf{B} \cdot \mathbf{B}$	16. Show that $\mathbf{A} \cdot \mathbf{A} = (-1/4\hbar^2)(\mathbf{L} \cdot \mathbf{L} + \mathbf{M}' \cdot \mathbf{M}' + \mathbf{L} \cdot \mathbf{M}' + \mathbf{M}' \cdot \mathbf{L})$. Show that $\mathbf{B} \cdot \mathbf{B}$
gilmore-lie-groups_F02771	270	■ 0.880	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = \mathbf{0}$	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = \mathbf{0}$	$\mathbf{L} \cdot \mathbf{M} = \mathbf{M} \cdot \mathbf{L} = \mathbf{0}$.
gilmore-lie-groups_F02772	270	■ 1.000	$p_x = \mathbf{p} \cdot \mathbf{M}/M$	$p_x = \mathbf{p} \cdot \mathbf{M}/M$	18. Compute $p_x = \mathbf{p} \cdot \mathbf{M}/M$ and $p_y = \mathbf{p} \cdot \mathbf{W}/W$. Show $p_x^2 + (p_y - a)^2 = r^2$. Explicitly
gilmore-lie-groups_F02773	270	■ 1.000	$p_y = \mathbf{p} \cdot \mathbf{W}/W$	$p_y = \mathbf{p} \cdot \mathbf{W}/W$	18. Compute $p_x = \mathbf{p} \cdot \mathbf{M}/M$ and $p_y = \mathbf{p} \cdot \mathbf{W}/W$. Show $p_x^2 + (p_y - a)^2 = r^2$. Explicitly
gilmore-lie-groups_F02774	270	■ 1.000	$p_x^2 + (p_y - a)^2 = r^2$	$p_x^2 + (p_y - a)^2 = r^2$	18. Compute $p_x = \mathbf{p} \cdot \mathbf{M}/M$ and $p_y = \mathbf{p} \cdot \mathbf{W}/W$. Show $p_x^2 + (p_y - a)^2 = r^2$. Explicitly
gilmore-lie-groups_F02775	270	■ 0.900	$0, a$	$0, a$	compute the displacement vector a (i.e., $(0, a)$) and the radius r of circular motion.
gilmore-lie-groups_F02776	270	■ 1.000	$x^a y^b z^c$	$x^a y^b z^c$	19. Show that the number of independent monomials of the form $x^a y^b z^c$, with a, b, c
gilmore-lie-groups_F02777	270	■ 0.996	$a+b+c=l$	$a+b+c=l$	nonnegative integers and $a+b+c=l$ is $N(l, 3) = (l+3-1)/l!(3-1)!$. In N -
gilmore-lie-groups_F02778	270	■ 0.996	$N(l, 3) = (l+3-1)/l!(3-1)!$	$N(l, 3) = (l+3-1)/l!(3-1)!$	nonnegative integers and $a+b+c=l$ is $N(l, 3) = (l+3-1)/l!(3-1)!$. In N -
gilmore-lie-groups_F02779	270	■ 0.997	$x_1, x_2, \dots, x_N$	$x_1, x_2, \dots, x_N$	l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_F02780	270	■ 0.997	$3 \rightarrow N$	$3 \rightarrow N$	l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_F02781	270	■ 1.000	$r^l Y_m^l(\theta, \phi)$	$r^l Y_m^l(\theta, \phi)$	a. Show that the functions $r^l Y_m^l(\theta, \phi)$ are homogeneous polynomials in x, y, z of
gilmore-lie-groups_F02782	270	■ 1.000	x, y, z	x, y, z	a. Show that the functions $r^l Y_m^l(\theta, \phi)$ are homogeneous polynomials in x, y, z of
gilmore-lie-groups_F02783	270	■ 1.000	$l-2$	$l-2$	ference between the number of homogeneous polynomials of degree l and $l-2$
gilmore-lie-groups_F02784	270	■ 1.000	$\dim \{Y_m^l\} = N(l, 3) - N(l-2, 3) = 2l+1$	$\dim \{Y_m^l\} = N(l, 3) - N(l-2, 3) = 2l+1$	on three variables: $\dim \{Y_m^l\} = N(l, 3) - N(l-2, 3) = 2l+1$.
gilmore-lie-groups_F02785	270	■ 0.999	$\dim \{\mathcal{Y}_{lm}^n\} = N(n, 4) - N(n-2, 4) = (n+1)^2$	$\dim \{\mathcal{Y}_{lm}^n\} = N(n, 4) - N(n-2, 4) = (n+1)^2$	degree n is $\dim \{\mathcal{Y}_{lm}^n\} = N(n, 4) - N(n-2, 4) = (n+1)^2$.
gilmore-lie-groups_F02786	270	■ 0.995	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$ for $n = 0, 1, 2; l = 0, \dots, n-1$; and $-l \leq m \leq +l$.

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gilmore-lie-groups_ F02787	270	■ 0.995	$n=0,1,2$; $l=0, \ldots, n-1$	$n = 0, 1, 2; l = 0, \dots, n - 1$	$\psi(\mathbf{x})_{nlm}$ for $n = 0, 1, 2; l = 0, \dots, n - 1$; and $-l \leq m \leq +l$.
gilmore-lie-groups_ F02788	271	■ 1.000	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$	where a and b are half odd integers: $\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$.
gilmore-lie-groups_ F02789	271	■ 1.000	$N(l, d) =$	$N(l, d) =$	f. Show homogeneous polynomials satisfy the recursion relation: $N(l, d) =$
gilmore-lie-groups_ F02790	271	■ 1.000	$N(l, d-1) + N(l-1, d)$	$N(l, d - 1) + N(l - 1, d)$	$N(l, d - 1) + N(l - 1, d)$.
gilmore-lie-groups_ F02791	271	■ 1.000	S^{n-1}	S^{n-1}	spaces of spherical harmonics on spheres S^n and S^{n-1} :
gilmore-lie-groups_ F02792	271	■ 1.000	$(l+1)^2 = l^2 + (2l+1)$	$(l + 1)^2 = l^2 + (2l + 1)$	For the case $n = 3$ this gives $(l+1)^2 = l^2 + (2l+1)$. The initialization for all n
gilmore-lie-groups_ F02793	271	■ 1.000	$\mathcal{Y}^0(S^n) = 1 = \dim \mathcal{Y}^0(S^n)$	$\mathcal{Y}^0(S^n) = 1 = \dim \mathcal{Y}^0(S^n)$	is $\mathcal{Y}^0(S^n) = 1 = \dim \mathcal{Y}^0(S^n)$.
gilmore-lie-groups_ F02794	271	■ 1.000	$\dim \mathcal{Y}^l(S^n) = \frac{(l+n-2)!}{l!(n-1)!} (2l+n-1)$	$\dim \mathcal{Y}^l(S^n) = \frac{(l+n-2)!}{l!(n-1)!} (2l+n-1)$	h. $\dim \mathcal{Y}^l(S^n) = \frac{(l+n-2)!}{l!(n-1)!} (2l+n-1)$.
gilmore-lie-groups_ F02795	271	■ 1.000	S^{D-1}	S^{D-1}	The angular part of the Laplacian operator, $\mathcal{L}^2$, acts on spherical harmonics on S^{D-1} ,
gilmore-lie-groups_ F02796	271	■ 1.000	$\mathcal{Y}^l(S^{D-1})$	$\mathcal{Y}^l(S^{D-1})$	$\mathcal{Y}^l(S^{D-1})$. These spherical harmonics are eigenfunctions of this (Laplace–Beltrami)
gilmore-lie-groups_ F02797	271	■ 1.000	$-(l+\alpha)^2 - \alpha^2$	$-(l + \alpha)^2 - \alpha^2$	operator with eigenvalue $-(l + \alpha)^2 - \alpha^2$, and α is a quantity that depends on the
gilmore-lie-groups_ F02798	271	■ 0.709	$so(D)$	$SO(D)$	Lie algebra of $SO(D)$: it is half the sum over all positive roots of the algebra. For
gilmore-lie-groups_ F02799	271	■ 1.000	$\alpha = D - 2$	$\alpha = D - 2$	$\alpha = D - 2$.
gilmore-lie-groups_ F02800	271	■ 1.000	$\psi(\mathbf{x}) = (1/r^{(D-1)/2}) \mathcal{Y}^l$	$\psi(\mathbf{x}) = \left(1/r^{(D-1)/2}\right) \mathcal{Y}^l$	a. Show that $\psi(\mathbf{x}) = (1/r^{(D-1)/2}) \mathcal{Y}^l(\text{angles})$ is a clever ansatz that reduces the
gilmore-lie-groups_ F02801	272	■ 1.000	$a - 1/r^2$	$a - 1/r^2$	a $-1/r^2$ perturbation. Show that the bound state energy and scattering phase shifts
gilmore-lie-groups_ F02802	272	■ 0.999	$l(l+1) \rightarrow l(l+1) - \mu_l$	$l(l + 1) \rightarrow l(l + 1) - \mu_l$	are given by the substitution $l(l + 1) \rightarrow l(l + 1) - \mu_l$. Argue that electrons in the s
gilmore-lie-groups_ F02803	272	■ 0.999	s	s	are given by the substitution $l(l + 1) \rightarrow l(l + 1) - \mu_l$. Argue that electrons in the s
gilmore-lie-groups_ F02804	272	■ 0.953	$\mu_0 \gg \mu_1 > \dots$	$\mu_0 \gg \mu_1 > \dots$	d -state electrons, ...) so that $\mu_0 \gg \mu_1 > \dots$.
gilmore-lie-groups_ F02805	272	■ 1.000	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i = -L_{ji}$	$L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i = -L_{ji}$	angular momentum operators $L_{ij} = a_i^\dagger a_j - a_j^\dagger a_i = -L_{ji}$.
gilmore-lie-groups_ F02806	272	■ 1.000	$Q_{ij} =$	$Q_{ij} =$	angular momentum operators together with the quadrupole tensor operators $Q_{ij} =$
gilmore-lie-groups_ F02807	272	■ 1.000	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$.
gilmore-lie-groups_ F02808	272	■ 1.000	$[a_i, a_j^\dagger] = 1$	$[a_i, a_j^\dagger] = 1$	as the single boson operators $a_i^\dagger$ and a_j , as well as their commutator $[a_i, a_j^\dagger] = 1$.
gilmore-lie-groups_ F02809	272	■ 1.000	$a_i a_j$	$a_i a_j$	$a_i a_j$. Show that this algebra is simple and describe its structure. Show that this
gilmore-lie-groups_ F02810	273	■ 0.999	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	The electromagnetic field $\mathbf{E}(\mathbf{x}, t)$, $\mathbf{B}(\mathbf{x}, t)$ is determined by Maxwell's
gilmore-lie-groups_ F02811	273	■ 0.935	$\mathbf{E}(k), \mathbf{B}(k)$	$\mathbf{E}(k), \mathbf{B}(k)$	four linear constraints on the six amplitudes $\mathbf{E}(k)$, $\mathbf{B}(k)$. Why? At a more
gilmore-lie-groups_ F02812	274	■ 1.000	$\mathbf{E}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t)$	and magnetic fields, $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, or their components after Fourier transfor-
gilmore-lie-groups_ F02813	274	■ 1.000	$\mathbf{B}(\mathbf{x}, t)$	$\mathbf{B}(\mathbf{x}, t)$	and magnetic fields, $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, or their components after Fourier transfor-

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gilmore-lie-groups_ F02814	274	■ 1.000	$\mathbf{E}(k)$	$\mathbf{E}(k)$	mation, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_ F02815	274	■ 1.000	$\mathbf{B}(k)$	$\mathbf{B}(k)$	mation, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_ F02816	274	■ 1.000	$k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	$k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	mation, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_ F02817	274	■ 0.798	k_4	k_4	Here $\mathbf{k}$ is essentially a three-momentum vector and k_4 is essentially an energy.
gilmore-lie-groups_ F02818	274	■ 1.000	$j=1$	$j = 1$	straints so derived reduce, for $j=1$, to Maxwell's equations. This derivation is
gilmore-lie-groups_ F02819	274	■ 1.000	$k \cdot k = 0$	$k \cdot k = 0$	on four-vector k no longer obeys $k \cdot k = 0$. In this case the manifestly covariant
gilmore-lie-groups_ F02820	275	■ 0.999	$(x, y, z, i c t)$	$(x, y, z, i c t)$	This requires that the coordinates $(x, y, z, i c t)$ and $(x, y, z, i c t)'$ for observers S
gilmore-lie-groups_ F02821	275	■ 0.999	$(x, y, z, i c t)^{\prime}$	$(x, y, z, i c t)'$	This requires that the coordinates $(x, y, z, i c t)$ and $(x, y, z, i c t)'$ for observers S
gilmore-lie-groups_ F02822	275	■ 0.865	Λ	Λ	The 4×4 matrix transformations Λ belong to the Lie group $O(3, 1)$. The infinites-
gilmore-lie-groups_ F02823	275	■ 1.000	$k \cdot a = \Lambda k \cdot a$	$k \cdot a = \Lambda k \cdot a$	Homogeneous Lorentz transformations leave invariant inner products: $k \cdot a = \Lambda k \cdot \Lambda a$.
gilmore-lie-groups_ F02824	275	■ 1.000	Λa	Λa	Λa , where k and a are four vectors and $\Lambda \in O(3, 1)$. The infinitesimal generators
gilmore-lie-groups_ F02825	275	■ 1.000	$\Lambda \in O(3, 1)$	$\Lambda \in O(3, 1)$	Λa , where k and a are four vectors and $\Lambda \in O(3, 1)$. The infinitesimal generators
gilmore-lie-groups_ F02826	275	■ 0.936	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$ satisfy the following commutation relations:
gilmore-lie-groups_ F02827	276	■ 0.988	$a = (x, y, z, c t)$	$a = (x, y, z, c t)$	tion can be written as a 5×5 matrix, in terms of the 4-vector $a = (x, y, z, c t)$:
gilmore-lie-groups_ F02828	276	■ 0.995	$(\partial/\partial x, \partial/\partial y, \partial/\partial z, i \partial/\partial(c t))$	$(\partial/\partial x, \partial/\partial y, \partial/\partial z, i \partial/\partial(c t))$	subgroup are $(\partial/\partial x, \partial/\partial y, \partial/\partial z, i \partial/\partial(c t))$.
gilmore-lie-groups_ F02829	276	■ 1.000	$\{\Lambda, 0\}$	$\{\Lambda, 0\}$	groups. These are the subgroup of homogeneous Lorentz transformations $\{\Lambda, 0\}$
gilmore-lie-groups_ F02830	276	■ 1.000	$\{I, a\}$	$\{I, a\}$	and the invariant subgroup of translations $\{I, a\}$. Both their representations play a
gilmore-lie-groups_ F02831	276	■ 1.000	Γ^k	Γ^k	We may define a basis state for the one-dimensional representation Γ^k of $\{I, a\}$ as
gilmore-lie-groups_ F02832	276	■ 1.000	$ k\rangle$	$ k\rangle$	$ k\rangle$:
gilmore-lie-groups_ F02833	277	■ 1.000	$D_2 = A_1 + A_1$	$D_2 = A_1 + A_1$	The Lie algebra $D_2 = A_1 + A_1$ is semisimple: it is the direct sum of two simple
gilmore-lie-groups_ F02834	277	■ 1.000	$\mathbf{J}^{(1)}$	$\mathbf{J}^{(1)}$	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$
gilmore-lie-groups_ F02835	277	■ 1.000	D^j	D^j	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$
gilmore-lie-groups_ F02836	277	■ 1.000	$\mathbf{J}^{(2)}$	$\mathbf{J}^{(2)}$	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$
gilmore-lie-groups_ F02837	277	■ 0.970	$2j' + 1$	$2j' + 1$	has $2j' + 1$ dimensional irreducible representations $D^{j'}$. Any element in $SO(3, 1)$
gilmore-lie-groups_ F02838	277	■ 0.970	$D^{j'}$	$D^{j'}$	has $2j' + 1$ dimensional irreducible representations $D^{j'}$. Any element in $SO(3, 1)$
gilmore-lie-groups_ F02839	277	■ 1.000	$(2j + 1)(2j' + 1)$	$(2j + 1)(2j' + 1)$	can be expressed in a $(2j + 1)(2j' + 1)$ dimensional representation $D^{jj'}$ as follows
gilmore-lie-groups_ F02840	277	■ 1.000	$D^{jj'}$	$D^{jj'}$	can be expressed in a $(2j + 1)(2j' + 1)$ dimensional representation $D^{jj'}$ as follows

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gilmore-lie-groups_ F02841	278	■ 1.000	$\mathbf{J}^{\{j\}}$	$\mathbf{J}^{(j)}$	where $\mathbf{J}^{(j)}$ are the three $(2j + 1) \times (2j + 1)$ angular momentum matrices. The
gilmore-lie-groups_ F02842	278	■ 0.990	$D^{\{j \ j^{\prime}\}}(\Lambda)$	$D^{jj'}(\Lambda)$	Basis states for the action of Λ through the representation $D^{jj'}(\Lambda)$ can be com-
gilmore-lie-groups_ F02843	278	■ 1.000	$S_0(3) \subset S_0(3,1)$	$SO(3) \subset SO(3,1)$	Under restriction to the subgroup $SO(3) \subset SO(3,1)$ this representation is reducible
gilmore-lie-groups_ F02844	278	■ 1.000	$j^{\prime}=0$	$j' = 0$	This representation remains irreducible only if $j' = 0$ or $j = 0$.
gilmore-lie-groups_ F02845	278	■ 1.000	$j=0$	$j = 0$	This representation remains irreducible only if $j' = 0$ or $j = 0$.
gilmore-lie-groups_ F02846	278	■ 1.000	$T_{\mu \nu}(x)$	$T_{\mu\nu}(x)$	A field $T_{\mu\nu}(x)$ is said to be manifestly covariant (obviously covariant) under
gilmore-lie-groups_ F02847	278	■ 1.000	$\Lambda \in S_0(3,1)$	$\Lambda \in SO(3,1)$	transformations of the homogeneous Lorentz group $\Lambda \in SO(3,1)$ if
gilmore-lie-groups_ F02848	278	■ 1.000	$x^{\prime}=x \Lambda^{-1}$	$x' = x\Lambda^{-1}$	nate system changes, the coordinates of the fixed point are changed by $x' = x\Lambda^{-1}$.
gilmore-lie-groups_ F02849	279	■ 0.774	$\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array} \right\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array} \right\rangle$	$ _{\mu \mu'}^{j j'}\rangle$ separately.
gilmore-lie-groups_ F02855	280	■ 1.000	$\{\Lambda, a\}$	$\{\Lambda, a\}$	Suppose we have a representation of $\{\Lambda, a\}$ that is unitary and irreducible. Un-
gilmore-lie-groups_ F02856	280	■ 1.000	$\Gamma^k(\{I, a\})$	$\Gamma^k(\{I, a\})$	$\Gamma^k(\{I, a\})$ of $\{I, a\}$. The basis states are $ k; \xi\rangle$, where k is defined by the action of
gilmore-lie-groups_ F02857	280	■ 1.000	$ k ; \xi\rangle$	$ k; \xi\rangle$	$\Gamma^k(\{I, a\})$ of $\{I, a\}$. The basis states are $ k; \xi\rangle$, where k is defined by the action of
gilmore-lie-groups_ F02858	280	■ 1.000	ξ	ξ	and ξ is a helicity index that distinguishes different states with the same four-
gilmore-lie-groups_ F02859	280	■ 0.999	$k^{\prime}=\Lambda k$	$k' = \Lambda k$	subspace of states parameterized by $k' = \Lambda k$
gilmore-lie-groups_ F02860	280	■ 1.000	$M_{\xi^{\prime} \xi}(\Lambda)$	$M_{\xi'\xi}(\Lambda)$	where $M_{\xi'\xi}(\Lambda)$ is a matrix that remains to be determined.
gilmore-lie-groups_ F02861	280	■ 1.000	$k^{\prime}$	k'	k' with
gilmore-lie-groups_ F02862	281	■ 1.000	$M(\Lambda)$	$M(\Lambda)$	are present also. To construct the matrix $M(\Lambda)$, we choose one particular four-vector
gilmore-lie-groups_ F02863	281	■ 1.000	$k^{\{0\}}$	k^0	k^0 for each of the possible cases
gilmore-lie-groups_ F02864	281	■ 1.000	$\left k^{\{0\}} ; \xi \right\rangle$	$ k^0; \xi\rangle$	The effect of a homogeneous Lorentz transformation on the state $ k^0; \xi\rangle$ is de-
gilmore-lie-groups_ F02865	281	■ 0.822	$H_{k^{\{0\}}}$	H_{k^0}	That is, H_{k^0} is the stability subgroup of the little vector k^0 and C_k is a coset repre-
gilmore-lie-groups_ F02866	281	■ 0.822	$C_{\{k\}}$	C_k	That is, H_{k^0} is the stability subgroup of the little vector k^0 and C_k is a coset repre-
gilmore-lie-groups_ F02867	282	■ 1.000	$\theta_1=\theta_2=b_3=0, \theta_3, b_1, b_2$	$\theta_1 = \theta_2 = b_3 = 0, \theta_3, b_1, b_2$	The subalgebra leaving k^0 fixed is defined by $\theta_1 = \theta_2 = b_3 = 0, \theta_3, b_1, b_2$ arbi-
gilmore-lie-groups_ F02868	282	■ 1.000	$(0,0,1,-i)=T(0,0,1,+i)$	$(0,0,1,-i) = T(0,0,1,+i)$	$(0,0,1,-i) = T(0,0,1,+i)$ the infinitesimal generators are $Y_1 = J_1 - K_2, Y_2 =$
gilmore-lie-groups_ F02869	282	■ 1.000	$Y_{\{1\}}=J_{\{1\}}-K_{\{2\}}, Y_{\{2\}}=$	$Y_1 = J_1 - K_2, Y_2 =$	$(0,0,1,-i) = T(0,0,1,+i)$ the infinitesimal generators are $Y_1 = J_1 - K_2, Y_2 =$
gilmore-lie-groups_ F02870	282	■ 1.000	$J_{\{2\}}+K_{\{1\}}, Y_{\{3\}}=J_{\{3\}}$	$J_2 + K_1, Y_3 = J_3$	$J_2 + K_1, Y_3 = J_3$.
gilmore-lie-groups_ F02871	282	■ 1.000	$\mathbf{b}=0$	$\mathbf{b} = 0$	The subalgebra defined by $\mathbf{b} = 0$ is spanned by the angular momentum operators

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gilmore-lie-groups_ F02872	283	■ 1.000	$D_{\xi^{\prime}\xi}(H_{k^0})$	$D_{\xi^{\prime}\xi}(H_{k^0})$	irreducible if and only if the representation $D_{\xi^{\prime}\xi}(H_{k^0})$ of the little group is unitary
gilmore-lie-groups_ F02873	283	■ 0.999	$k \cdot k > 0$	$k \cdot k > 0$	were described in Problem 5 of Chapter 11. Since $k \cdot k > 0$ describes negative mass
gilmore-lie-groups_ F02874	283	■ 1.000	$j=0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$	$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	They are described by an integer or half-integer: $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. The angular
gilmore-lie-groups_ F02875	283	■ 1.000	$\kappa=\left(\kappa_1, \kappa_2\right)$	$\kappa = (\kappa_1, \kappa_2)$	can be labeled by a vector $\kappa = (\kappa_1, \kappa_2)$ in a two-dimensional Euclidean space,
gilmore-lie-groups_ F02876	283	■ 1.000	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$. If a state $ \kappa\rangle$ is in one such representation, so are all states
gilmore-lie-groups_ F02877	283	■ 1.000	$ \kappa\rangle$	$ \kappa\rangle$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$. If a state $ \kappa\rangle$ is in one such representation, so are all states
gilmore-lie-groups_ F02878	283	■ 0.999	$ \kappa^{\prime}\rangle$	$ \kappa^{\prime}\rangle$	$ \kappa^{\prime}\rangle$ for which $\kappa^{\prime} \cdot \kappa^{\prime} = \kappa \cdot \kappa$. That is, $\kappa^{\prime} = (\kappa_1^{\prime}, \kappa_2^{\prime})$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_ F02879	283	■ 0.999	$\kappa^{\prime} \cdot \kappa^{\prime} = \kappa \cdot \kappa$	$\kappa^{\prime} \cdot \kappa^{\prime} = \kappa \cdot \kappa$	$ \kappa^{\prime}\rangle$ for which $\kappa^{\prime} \cdot \kappa^{\prime} = \kappa \cdot \kappa$. That is, $\kappa^{\prime} = (\kappa_1^{\prime}, \kappa_2^{\prime})$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_ F02880	283	■ 0.999	$\kappa^{\prime} = \left(\kappa_1^{\prime}, \kappa_2^{\prime}\right)$	$\kappa^{\prime} = (\kappa_1^{\prime}, \kappa_2^{\prime})$	$ \kappa^{\prime}\rangle$ for which $\kappa^{\prime} \cdot \kappa^{\prime} = \kappa \cdot \kappa$. That is, $\kappa^{\prime} = (\kappa_1^{\prime}, \kappa_2^{\prime})$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_ F02881	283	■ 1.000	$\kappa^{\prime}=R(\theta) \kappa$	$\kappa^{\prime} = R(\theta)\kappa$	rotation: $\kappa^{\prime} = R(\theta)\kappa$. The invariant length $\kappa \cdot \kappa$ parameterizes the representation.
gilmore-lie-groups_ F02882	283	■ 1.000	$\kappa \cdot \kappa$	$\kappa \cdot \kappa$	rotation: $\kappa^{\prime} = R(\theta)\kappa$. The invariant length $\kappa \cdot \kappa$ parameterizes the representation.
gilmore-lie-groups_ F02883	283	■ 0.999	$\kappa \cdot \kappa > 0 \quad \text{quad}$	$\kappa \cdot \kappa > 0$	(i) $\kappa \cdot \kappa > 0$ little group = Identity
gilmore-lie-groups_ F02884	283	■ 0.996	$\kappa \cdot \kappa = 0 \quad \text{quad}$	$\kappa \cdot \kappa = 0$	(ii) $\kappa \cdot \kappa = 0$ little group = $ISO(2)$
gilmore-lie-groups_ F02885	283	■ 0.996	$=ISO(2)$	$=ISO(2)$	(ii) $\kappa \cdot \kappa = 0$ little group = $ISO(2)$
gilmore-lie-groups_ F02886	284	■ 1.000	κ^2	κ^2	The first case presents us with two problems. First, κ^2 is a continuous quantum
gilmore-lie-groups_ F02887	284	■ 1.000	$\kappa^2 > 0$	$\kappa^2 > 0$	$\kappa^2 > 0$ there must be an infinite number of states with this same continuous index,
gilmore-lie-groups_ F02888	284	■ 1.000	$\kappa = 0$	$\kappa = 0$	for each four-momentum value. Therefore we require $\kappa = 0$. This leaves us with the
gilmore-lie-groups_ F02889	284	■ 0.961	$\left(Y_1 \rightarrow 0, Y_2 \rightarrow 0\right)$	$(Y_1 \rightarrow 0, Y_2 \rightarrow 0)$	following physically allowable representations of the little group $(Y_1 \rightarrow 0, Y_2 \rightarrow 0)$
gilmore-lie-groups_ F02890	284	■ 1.000	$\left(j, j^{\prime}\right)$	$(j, j^{\prime})$	particle with transformation indices $(j, j^{\prime})$ contains all states of the form
gilmore-lie-groups_ F02891	285	■ 1.000	$\left\{H_{k^0}, 0\right\}$	$\{H_{k^0}, 0\}$	A. $\{H_{k^0}, 0\}$ on $ k^0; \xi\rangle$
gilmore-lie-groups_ F02892	285	■ 1.000	$H_{k^0}=\operatorname{EXP}\left(\Theta J_3+\theta_1 Y_1+\theta_2 Y_2\right)$	$H_{k^0} = \text{EXP}(\Theta J_3 + \theta_1 Y_1 + \theta_2 Y_2)$	where $H_{k^0} = \text{EXP}(\Theta J_3 + \theta_1 Y_1 + \theta_2 Y_2)$.
gilmore-lie-groups_ F02893	285	■ 0.494	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j j^{\prime}}\right\rangle\right _{\mu^{\prime}}^{\mu}$	$ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$	B. $\{H_{k^0}, 0\}$ on $ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$ The little group maps k^0 to k^0 but acts in a nontrivial way
gilmore-lie-groups_ F02894	285	■ 0.119	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j j^{\prime}}\right\rangle\right _{\mu^{\prime}}^{\mu}$	$ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$	The state $ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_ F02895	285	■ 0.119	$\xi > 0$	$\xi > 0$	The state $ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_ F02896	285	■ 0.119	$j = +\xi$	$j = +\xi$	The state $ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_ F02897	285	■ 0.983	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j j^{\prime}}\right\rangle\right _{\mu^{\prime}}^{\mu}$	$ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$	The state $ k^0\rangle _{\mu}^{j j^{\prime}}\rangle$ transforms identically to $ k^0; \xi\rangle$ if $\xi < 0$ and $j^{\prime} = -\xi$.

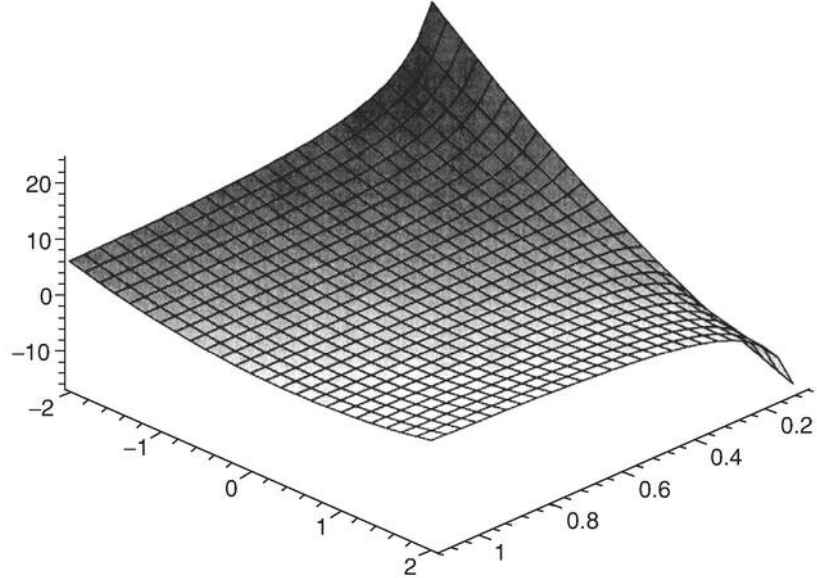
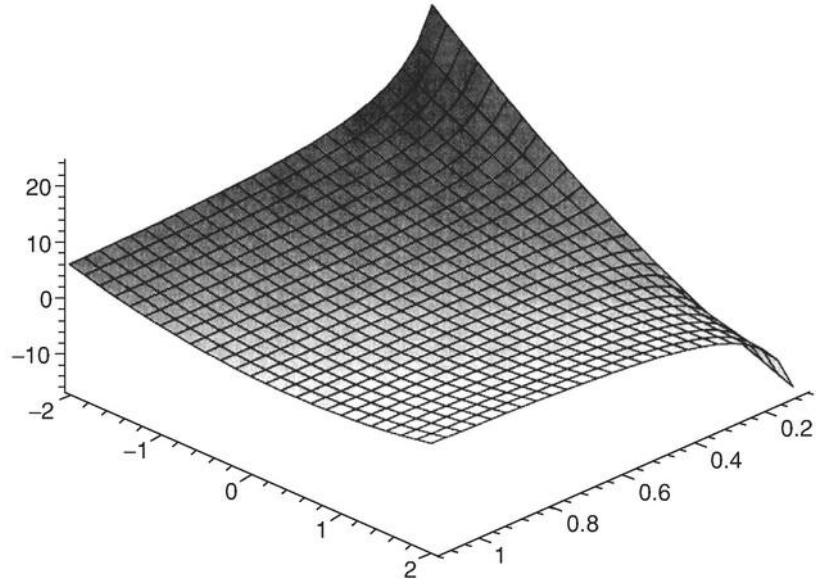
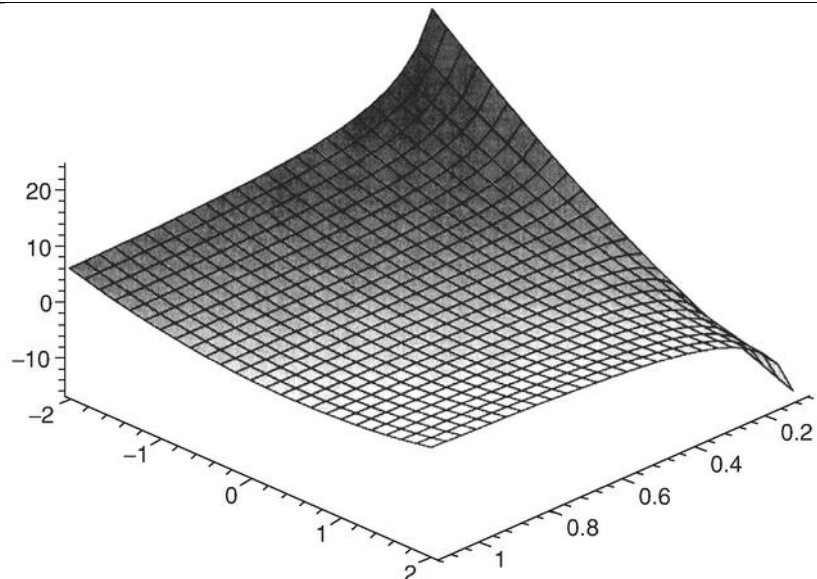
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gilmore-lie-groups_F02922	287	■ 1.000	\xi=j	$\xi = j$	For the positive helicity state $\xi = j$ the matrix $M^{jj'}(k^0) = M^{j0}(k^0)$ is given in
gilmore-lie-groups_F02923	287	■ 1.000	M^{j\ j^{\prime}}\left(k^{\{0\}}\right)=M^{j\ 0}\left(k^{\{0\}}\right)	$M^{jj'}(k^0) = M^{j0}(k^0)$	For the positive helicity state $\xi = j$ the matrix $M^{jj'}(k^0) = M^{j0}(k^0)$ is given in
gilmore-lie-groups_F02924	287	■ 0.943	k\rangle\left \begin{array}{ll}1&0\\ \mu&0\end{array}\right\rangle	$ k\rangle \left \begin{array}{cc} 1 & 0 \\ \mu & 0 \end{array} \right\rangle$	as the linear constraint that must be satisfied in the subspace $ k\rangle _{\mu\ 0}^1\ 0\rangle$. The negative
gilmore-lie-groups_F02925	287	■ 1.000	\langle x\mid k\rangle\langle k;\begin{smallmatrix}j\\m\end{smallmatrix}0\mid\psi\rangle	$\langle x k \rangle \langle k; \begin{smallmatrix} j \\ m \end{smallmatrix} 0 \psi \rangle$	If we define complex fields $\langle x k\rangle\langle k; \begin{smallmatrix} j \\ m \end{smallmatrix} 0 \psi\rangle$ by $\psi_{jm}(x)$, ($j = 1, m = +1, 0, -1$
gilmore-lie-groups_F02926	287	■ 1.000	\psi_{jm}(x),(j=1,\ m=+1,0,-1	$\psi_{jm}(x), (j = 1, m = +1, 0, -1$	If we define complex fields $\langle x k\rangle\langle k; \begin{smallmatrix} j \\ m \end{smallmatrix} 0 \psi\rangle$ by $\psi_{jm}(x), (j = 1, m = +1, 0, -1$
gilmore-lie-groups_F02927	288	■ 1.000	D^{\{01\}}	D^{01}	representation D^{01} and are
gilmore-lie-groups_F02928	289	■ 0.999	k^{\{0\}}=(0,0,1,\ i)	$k^0 = (0, 0, 1, i)$	$k^0 = (0, 0, 1, i)$ the only nonvanishing component of the field, $\langle k; \begin{smallmatrix} j=1 \\ m \end{smallmatrix} 0 \psi\rangle$, is the
gilmore-lie-groups_F02929	289	■ 0.999	\left\langle k;\begin{smallmatrix}j=1\\0\end{smallmatrix}\mid\psi\right\rangle	$\langle k; \begin{smallmatrix} j=1 \\ 0 \end{smallmatrix} \psi \rangle$	$k^0 = (0, 0, 1, i)$ the only nonvanishing component of the field, $\langle k; \begin{smallmatrix} j=1 \\ m \end{smallmatrix} 0 \psi\rangle$, is the
gilmore-lie-groups_F02930	289	■ 1.000	m=+1	$m = +1$	component with $m = +1$ (cf., Eq. (15.54)). The coordinates are $-(v_x + i v_y)$. The
gilmore-lie-groups_F02931	289	■ 1.000	-\left(v_{\{x\}}+i\ v_{\{y\}}\right)	$-(v_x + i v_y)$	component with $m = +1$ (cf., Eq. (15.54)). The coordinates are $-(v_x + i v_y)$. The
gilmore-lie-groups_F02932	289	■ 1.000	\mathbf{v}=\left(v_{\{x\}},\ v_{\{y\}},\ 0\right)	$\mathbf{v} = (v_x, v_y, 0)$	vector $\mathbf{v} = (v_x, v_y, 0)$ represented by this coordinate is orthogonal to the spacial part
gilmore-lie-groups_F02933	289	■ 1.000	\mathbf{k}^{\{0\}}=(0,0,1): \mathbf{k}^{\{0\}}\cdot\mathbf{v}=0	$\mathbf{k}^0 = (0, 0, 1) : \mathbf{k}^0 \cdot \mathbf{v} = 0$	of the little vector $\mathbf{k}^0 = (0, 0, 1)$: $\mathbf{k}^0 \cdot \mathbf{v} = 0$. Under boosts B_z and rotations, the non-
gilmore-lie-groups_F02934	289	■ 1.000	B_{\{z\}}	B_z	of the little vector $\mathbf{k}^0 = (0, 0, 1)$: $\mathbf{k}^0 \cdot \mathbf{v} = 0$. Under boosts B_z and rotations, the non-
gilmore-lie-groups_F02935	289	■ 0.976	\mathbf{k}	$\mathbf{k}$	vanishing component of the boosted field is orthogonal to the spacial part of the $\mathbf{k}$
gilmore-lie-groups_F02936	289	■ 1.000	\mathbf{k}\cdot\mathbf{v}(\mathbf{k})=0	$\mathbf{k} \cdot \mathbf{v}(\mathbf{k}) = 0$	vector: $\mathbf{k} \cdot \mathbf{v}(\mathbf{k}) = 0$. Backtransforming from the Fourier to the spacial representation,
gilmore-lie-groups_F02937	290	■ 0.999	\nabla\cdot\mathbf{E}=0	$\nabla \cdot \mathbf{E} = 0$	equations $\nabla \cdot \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$. Show this.
gilmore-lie-groups_F02938	290	■ 0.999	\nabla\cdot\mathbf{B}=0	$\nabla \cdot \mathbf{B} = 0$	equations $\nabla \cdot \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$. Show this.
gilmore-lie-groups_F02939	290	■ 1.000	\mathbf{x}(j)	$\mathbf{x}(j)$	clearly expressed in the “manifestly covariant representation.” If particle j at $\mathbf{x}(j)$ has
gilmore-lie-groups_F02940	290	■ 1.000	e_{\{j\}}	e_j	electric charge e_j and magnetic charge m_j , the electric and magnetic charge densities
gilmore-lie-groups_F02941	290	■ 1.000	m_{\{j\}}	m_j	electric charge e_j and magnetic charge m_j , the electric and magnetic charge densities
gilmore-lie-groups_F02948	290	■ 0.997	T, P, T P	T, P, TP	T, P, TP . The result is unique up to scale factor:
gilmore-lie-groups_F02949	290	■ 1.000	1 / c	$1/c$	The factor 4π is the surface area of the unit sphere in R^3 , and the factor $1/c$ on the
gilmore-lie-groups_F02950	291	■ 0.867	\phi=\pi\ / \ 2	$\phi = \pi/2$	In particular, show that for $\phi = \pi/2$ this is the dual transformation $(\mathbf{B}, \mathbf{E}) \rightarrow$
gilmore-lie-groups_F02951	291	■ 0.867	(\mathbf{B},\ \mathbf{E})\rightarrow	$(\mathbf{B}, \mathbf{E}) \rightarrow$	In particular, show that for $\phi = \pi/2$ this is the dual transformation $(\mathbf{B}, \mathbf{E}) \rightarrow$
gilmore-lie-groups_F02952	291	■ 1.000	(*)=0	$(*) = 0$	$(*) = 0$, for $*$ = anyvector. Show
gilmore-lie-groups_F02953	291	■ 1.000	*=	$*$ =	$(*) = 0$, for $*$ = anyvector. Show
gilmore-lie-groups_F02954	291	■ 0.998	D^{\{j\ j^{\prime}}+j^{\prime}\ j\}}(\Lambda)	$D^{jj'+j'j}(\Lambda)$	resentation $D^{jj'+j'j}(\Lambda)$, with $j - j' = \pm 2$. In the case with $(j, j') = (2, 0)$ a curl

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gilmore-lie-groups_ F02955	291	■ 0.998	$j-j^{\prime}=\pm 2$	$j-j^{\prime}=\pm 2$	resentation $D^{jj^{\prime}+j^{\prime}j}(\Lambda)$, with $j-j^{\prime}=\pm 2$. In the case with $(j,j^{\prime})=(2,0)$ a curl
gilmore-lie-groups_ F02956	291	■ 0.998	$\left(j,j^{\prime}\right)=(2,0)$	$(j,j^{\prime})=(2,0)$	resentation $D^{jj^{\prime}+j^{\prime}j}(\Lambda)$, with $j-j^{\prime}=\pm 2$. In the case with $\boxed{(j,j^{\prime})=(2,0)}$ a curl
gilmore-lie-groups_ F02957	291	■ 0.990	$\mathbf{G}_{\mathbf{e}}$	$\mathbf{G}_{\mathbf{e}}$	The fields $\mathbf{G}_{\mathbf{e}}$ and $\mathbf{G}_{\mathbf{m}}$ are called the gravitoelectric and gravitomagnetic fields. These
gilmore-lie-groups_ F02958	291	■ 0.990	$\mathbf{G}_{\mathbf{m}}$	$\mathbf{G}_{\mathbf{m}}$	The fields $\mathbf{G}_{\mathbf{e}}$ and $\mathbf{G}_{\mathbf{m}}$ are called the gravitoelectric and gravitomagnetic fields. These
gilmore-lie-groups_ F02959	291	■ 0.866	$\mathbf{J}\cdot\nabla$	$\mathbf{J}\cdot\nabla$	latter case the curl operator is $\mathbf{J}\cdot\nabla$, where $\mathbf{J}$ is the 5×5 angular momentum operator:
gilmore-lie-groups_ F02960	292	■ 1.000	$F_{ij}=F_{ji},\sum_iF_{ii}=0$	$F_{ij}=F_{ji},\sum_iF_{ii}=0$	The matrix components obey $\boxed{F_{ij}=F_{ji},\sum_iF_{ii}=0}$, and $\partial^iF_{ij}=0$. The gravitoelec-
gilmore-lie-groups_ F02961	292	■ 1.000	$\partial^iF_{ij}=0$	$\partial^iF_{ij}=0$	The matrix components obey $F_{ij}=F_{ji},\sum_iF_{ii}=0$, and $\partial^iF_{ij}=0$. The gravitoelec-
gilmore-lie-groups_ F02962	292	■ 0.999	$U_{ij}=\sum_km_k(\mathbf{x}_k(t)\mathbf{x}_k(t))_{ij}$	$U_{ij}=\sum_km_k(\mathbf{x}_k(t)\mathbf{x}_k(t))_{ij}$	dinates the source term for the gravitoelectric field is $U_{ij}=\sum_km_k(\mathbf{x}_k(t)\mathbf{x}_k(t))_{ij}$,
gilmore-lie-groups_ F02963	292	■ 0.999	$\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}})$	$\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}})$	$\boxed{\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}})}$ and $\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}})$. Show this.
gilmore-lie-groups_ F02964	292	■ 0.999	$\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}})$	$\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}})$	$\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{G}_{\mathbf{m}}+i\mathbf{G}_{\mathbf{e}})$ and $\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}}\rightarrow e^{i\phi}(\mathbf{J}_{\mathbf{m}}+i\mathbf{J}_{\mathbf{e}})$. Show this.
gilmore-lie-groups_ F02965	292	■ 1.000	$D^{jj}(\Lambda)$	$D^{jj}(\Lambda)$	5. Construct the source-free field equations for gravitons for the $D^{jj}(\Lambda)$ representation,
gilmore-lie-groups_ F02966	292	■ 0.871	(J,M)	(J,M)	with $j=1$. Show that there are seven constraints that correspond to (J,M) with
gilmore-lie-groups_ F02967	292	■ 1.000	$(J,M)=(0,0),(1,0),(1,\pm 1),(2,0),(2\pm 1)$	$(J,M)=(0,0),(1,0),(1,\pm 1),(2,0),(2\pm 1)$	$\boxed{(J,M)=(0,0),(1,0),(1,\pm 1),(2,0),(2\pm 1)}$. What are these equations in the stan-
gilmore-lie-groups_ F02968	293	■ 1.000	$N\lambda$	$N\lambda$	two (distance $N\lambda$), as time goes on and the distance increases the wavelength must
gilmore-lie-groups_ F02969	293	■ 1.000	$N\lambda^{\prime}$	$N\lambda^{\prime}$	also increase to $N\lambda^{\prime}$. This redshift source is sometimes confused with the Döppler
gilmore-lie-groups_ F02970	293	■ 1.000	$\rho(\mathbf{x},t)=\rho(t)$	$\rho(\mathbf{x},t)=\rho(t)$	a. Assume that the energy density in the universe has the form $\rho(\mathbf{x},t)=\rho(t)$ (time
gilmore-lie-groups_ F02971	293	■ 1.000	$m_e(t)/M_p(t)$	$m_e(t)/M_p(t)$	b. Estimate the mass dependence of the electron–proton mass ratio $m_e(t)/M_p(t)$.
gilmore-lie-groups_ F02972	293	■ 0.981	$v=\frac{1}{2}(mc^2/\hbar)\times (1/n_1^2-1/n_2^2) $	$v=\frac{1}{2}(mc^2/\hbar)\times (1/n_1^2-1/n_2^2) $	the hydrogen atom has frequency $\boxed{v=\frac{1}{5}(mc^2/\hbar)\times (1/n_1^2-1/n_2^2) }$ where n_1 and
gilmore-lie-groups_ F02973	293	■ 1.000	H_{α}	H_{α}	and m is the reduced mass of the electron–proton system. Show that H_{α} photons
gilmore-lie-groups_ F02974	294	■ 1.000	$\Phi(x)$	$\Phi(x)$	Here $\Phi(x)$ is the local Newtonian gravitational field. Find a locally linear coordinate
gilmore-lie-groups_ F02975	294	■ 1.000	$S^tg_{\text{grav}}S=g_{\text{flat}}$	$S^tg_{\text{grav}}S=g_{\text{flat}}$	transformation S that brings the curved metric to flat form: $S^tg_{\text{grav}}S=g_{\text{flat}}$. Interpret
gilmore-lie-groups_ F02976	294	■ 1.000	V	V	The integral on the left is over the surface bounding the volume V over which the
gilmore-lie-groups_ F02977	294	■ 0.867	$\mathbf{E}$	$\mathbf{E}$	integral on the right extends, $\mathbf{E}$ is the electric field and ρ is the charge density. For a
gilmore-lie-groups_ F02978	294	■ 0.523	$a,p(x)=q\delta(x)$	$a,p(x)=q\delta(x)$	charge q at the origin of a sphere of radius a , $\rho(x)=q\delta(x)$, The $\mathbf{E}$ field is spherically
gilmore-lie-groups_ F02979	294	■ 1.000	$ \mathbf{E}(a) =q/ \mathbf{a} $	$ \mathbf{E}(a) =q/ \mathbf{a} $	By completely similar arguments Gauss’ Law in the plane R^2 gives $\boxed{ \mathbf{E}(a) =q/ \mathbf{a} }$.
gilmore-lie-groups_ F02980	294	■ 0.808	$\oint\mathbf{E}\cdot d\mathbf{S}=\int2\pi\rho dA$	$\left(\oint\mathbf{E}\cdot d\mathbf{S}=\int2\pi\rho dA\right)$	Assume a Gauss law ($\oint\mathbf{E}\cdot d\mathbf{S}=\int2\pi\rho dA$) holds on the sphere S^2 . Place a charge

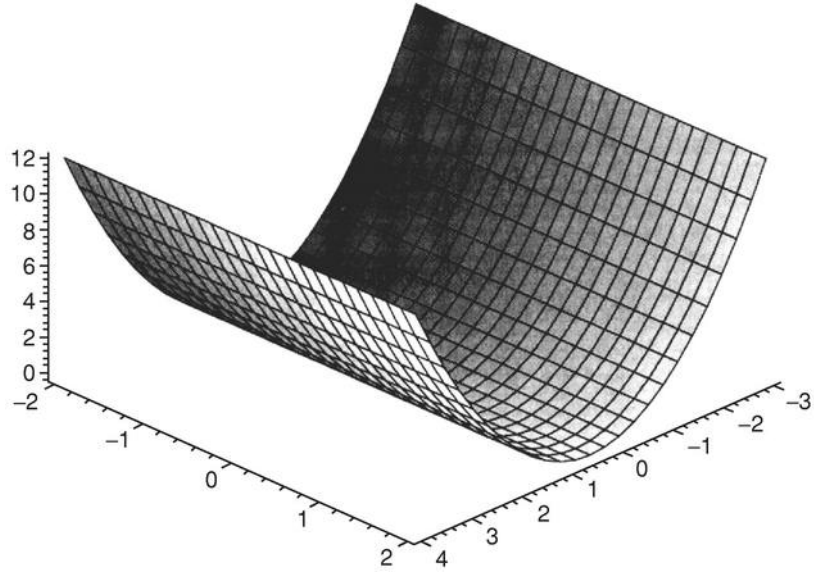
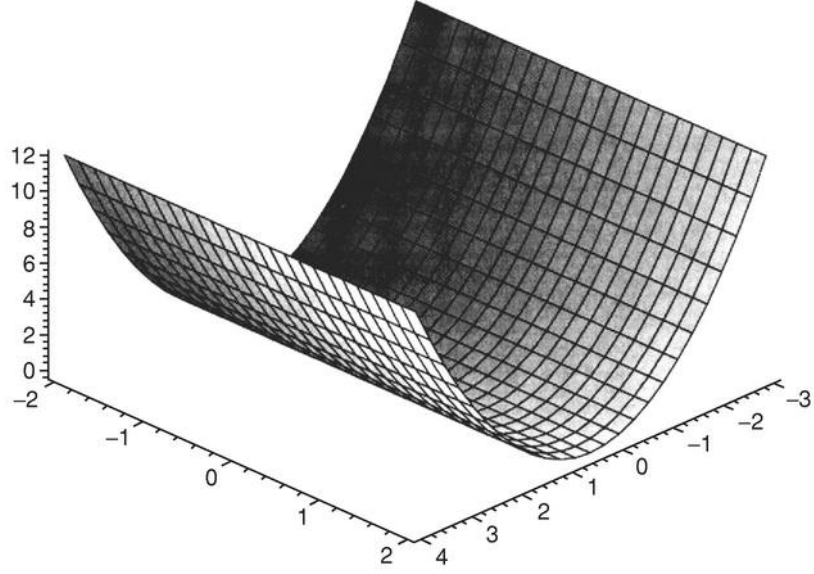
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gilmore-lie-groups_F02981	294	■ 1.000	$a=R \ \theta$	$a = R\theta$	sphere S^2 (c.f., Fig. 15.1). Show that its distance a from the north pole is $a = R\theta$
gilmore-lie-groups_F02982	295	■ 1.000	$2 \ \pi \ R \ \sin \ \theta$	$2\pi R \sin \theta$	and the circumference of a circle of latitude through this point is $2\pi R \sin \theta$. Use
gilmore-lie-groups_F02983	295	■ 1.000	$q \ / \ a$	q/a	Conclude that the field is stronger than the q/a form it would have in a plane.
gilmore-lie-groups_F02984	295	■ 1.000	$a(a=R \ \theta)$	$a(a = R\theta)$	where a ($a = R\theta$) is the distance from the charge to the observation point.
gilmore-lie-groups_F02985	295	■ 1.000	$a=c \ t$	$a = ct$	charge depends on the distance from the observation point. In particular, if $a = ct$,
gilmore-lie-groups_F02986	295	■ 1.000	$H^{\{n\}}=S \ O(n, \ 1) \ / \ S \ O(n), \ R^{\{n\}}=$	$H^n = SO(n,1)/SO(n), R^n =$	sure on the three Riemannian symmetric spaces $H^n \equiv SO(n,1)/ \ SO(n), \ R^n =$
gilmore-lie-groups_F02987	295	■ 0.999	$I \ S \ O(n) \ / \ S \ O(n)$	$ISO(n)/SO(n)$	$ISO(n)/SO(n)$, and $S^n = SO(n+1)/SO(n)$ are
gilmore-lie-groups_F02988	295	■ 1.000	$k=(-1,0,+1)$	$k = (-1, 0, +1)$	where $k \equiv (-1, 0, +1)$ for H^n, R^n, S^n and radial coordinates are used:
gilmore-lie-groups_F02989	295	■ 1.000	$H^{\{n\}}, \ R^{\{n\}}, \ S^{\{n\}}$	H^n, R^n, S^n	where $k = (-1, 0, +1)$ for H^n, R^n, S^n and radial coordinates are used:
gilmore-lie-groups_F02990	295	■ 1.000	$H^{\{n\}}, \ S^{\{n\}}$	H^n, S^n	a. Derive the metric for H^n, S^n from Eq. (12.9) and the coordinate transformation
gilmore-lie-groups_F02991	295	■ 1.000	Ω	Ω	Compute Ω , the surface area of the unit sphere $S^{n-1} \subset H^n, R^n$, or S^n . (Hint: use
gilmore-lie-groups_F02992	295	■ 1.000	$S^{\{n-1\}} \ \subset \ H^{\{n\}}, \ R^{\{n\}}$	$S^{n-1} \subset H^n, R^n$	Compute Ω , the surface area of the unit sphere $S^{n-1} \subset H^n, R^n$, or S^n . (Hint: use
gilmore-lie-groups_F02993	295	■ 1.000	$\int e^{-x^2} \ d \ x=\sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$, carry the n -fold integral out in Cartesian and radial coordinates,
gilmore-lie-groups_F02994	295	■ 1.000	$\Omega=2 \ \pi^{\{n \ / \ 2\}} \ / \ \Gamma(n \ / \ 2)$	$\Omega = 2\pi^{n/2}/\Gamma(n/2)$	and show $\Omega = 2\pi^{n/2}/\Gamma(n/2)$.)
gilmore-lie-groups_F02995	296	■ 1.000	$H^{\{n\}}$	H^n	S^n , downward in H^n , with lookback time. Give a physical interpretation involving
gilmore-lie-groups_F02996	296	■ 1.000	$R, \ c, \ t$	R, c, t	on R, c, t ?
gilmore-lie-groups_F02997	296	■ 0.839	$(l, \ b)=\left(263^{\circ}, \ 48^{\circ}\right)$	$(l, b) = (263^\circ, 48^\circ)$	the direction with galactic coordinates $(l, b) = (263^\circ, 48^\circ)$.
gilmore-lie-groups_F02998	296	■ 1.000	$T(\theta, \ \phi \ ; \ t)=$	$T(\theta, \phi; t) =$	b. Assume the temperature distribution of the microwave background is $T(\theta, \phi; t) =$
gilmore-lie-groups_F02999	296	■ 1.000	$\sum_{l, \ m} A_m^l(t) Y_m^l(\theta, \ \phi)$	$\sum_{l,m} A_m^l(t) Y_m^l(\theta, \phi)$	$\sum_{l,m} A_m^l(t) Y_m^l(\theta, \phi)$. How do you use this information to determine a frame that
gilmore-lie-groups_F03000	296	■ 0.998	$\Delta U \ \Delta(1 \ / \ T) \ \geq \ k$	$\Delta U \Delta(1/T) \geq k$	the uncertainty relations of statistical mechanics: $\Delta U \ \Delta(1/T) \geq k$ in the entropy
gilmore-lie-groups_F03001	297	■ 1.000	$j=\frac{1}{2}, \ 1, 2$	$j = \frac{1}{2}, 1, 2$	why, or why not, the special frames defined by $j = \frac{1}{2}, 1, 2$ are the same. What
gilmore-lie-groups_F03002	297	■ 1.000	$v(t)$	$v(t)$	that it moves through the microwave background radiation with a velocity $v(t)$.
gilmore-lie-groups_F03003	297	■ 1.000	$v(t) \ \simeq \ v\left(t_{\{0\}}\right) \ e^{\{-\left(t-t_{\{0\}}\right) \ / \ \tau \}}$	$v(t) \simeq v(t_0) e^{-(t-t_0)/\tau}$	Show that its velocity decays to zero according to $v(t) \simeq v(t_0)e^{-(t-t_0)/\tau}$ because
gilmore-lie-groups_F03004	297	■ 1.000	τ	τ	it is moving through a viscous medium. Estimate τ and present your answer in
gilmore-lie-groups_F03005	297	■ 0.990	$\tau \ / \ T_{\{p\}}$	τ/T_p	the form τ/T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03006	297	■ 0.990	$T_{\{p\}}$	T_p	the form τ/T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03007	297	■ 0.990	$T_{\{p\}} \ \simeq \ 13.7 \ \mathrm{BY}$	$T_p \simeq 13.7\text{BY}$	the form τ/T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To

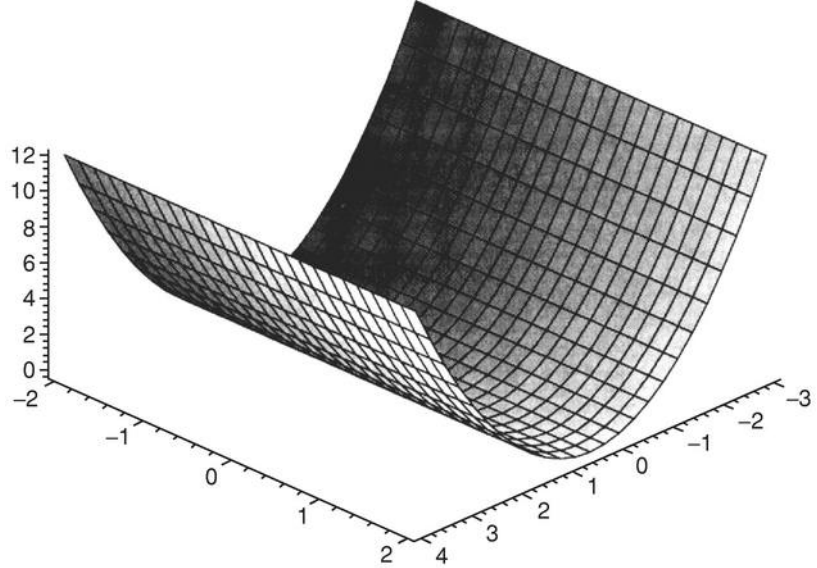
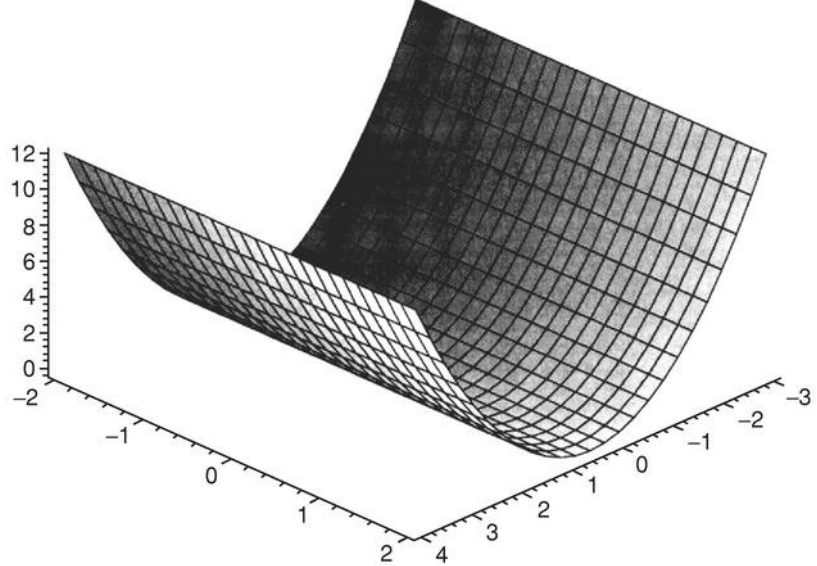
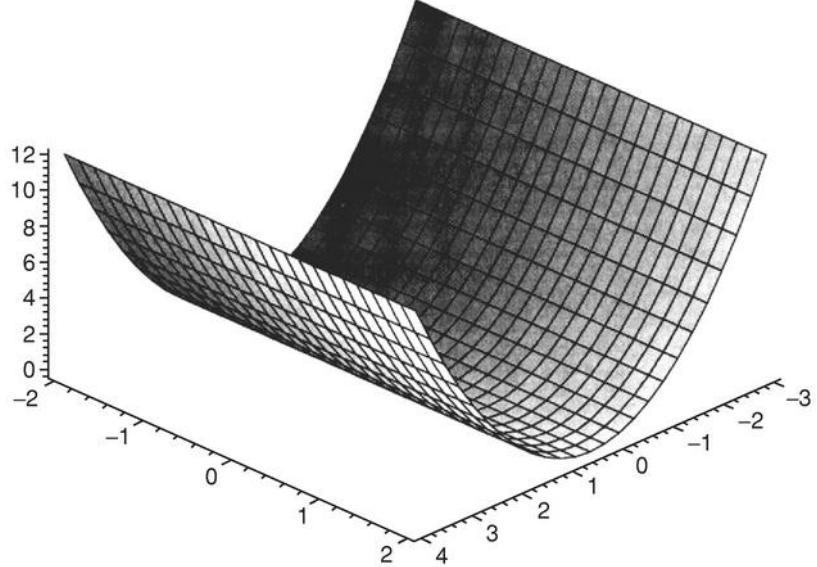
Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F03008	297	■ 1.000	T_{BH}	T_{BH}	in fact, assume it is a black hole with mass M , radius R at temperature T_{BH} .
gilmore-lie-groups_F03009	297	■ 1.000	$R=2\ G\ M\ /\ c^{\wedge\{2\}}$	$R = 2GM/c^2$	Use the standard relations for a neutral nonrotating black hole $R = 2GM/c^2$,
gilmore-lie-groups_F03010	297	■ 1.000	$T_{\text{BH}}=\hbar\bar c^{\wedge\{3\}}\ /\ 8\ \pi\ k\ G\ M$	$T_{BH} = \hbar c^3/8\pi kGM$	$T_{BH} = \hbar c^3/8\pi kGM$. You can assume that the mass M is sufficiently large that
gilmore-lie-groups_F03011	297	■ 1.000	$\gamma\pi\ R^{\wedge\{2\}}$	$\gamma\pi R^2$	(geometric) cross section for radiation on a black hole is $\gamma\pi R^2$, where $\gamma = 3^3/2^2$.
gilmore-lie-groups_F03012	297	■ 1.000	$\gamma=3^{\wedge\{3\}}\ /\ 2^{\wedge\{2\}}$	$\gamma = 3^3/2^2$	(geometric) cross section for radiation on a black hole is $\gamma\pi R^2$, where $\gamma = 3^3/2^2$.
gilmore-lie-groups_F03013	298	■ 1.000	$d\ y\ /\ d\ x=g(x)$	$dy/dx = g(x)$	constant that can be added to any indefinite integral of $dy/dx = g(x)$ is in
gilmore-lie-groups_F03014	298	■ 1.000	$y: d\ y\ /\ d\ x=g(x)$	$y: dy/dx = g(x)$	dependent variable $y: dy/dx = g(x)$. This is treated in Section 16.1. In that section
gilmore-lie-groups_F03015	298	■ 1.000	$d\ y\ /\ d\ x=p$	$dy/dx = p$	we set up the general formulation in terms of a constraint equation $dy/dx = p$ and
gilmore-lie-groups_F03016	299	■ 1.000	$F(x,\ y,\ p)=0$	$F(x,y,p) = 0$	a surface equation $F(x,y,p) = 0$. The special forms of the surface and constraint
gilmore-lie-groups_F03017	299	■ 1.000	$R,\ S,\ T$	R,S,T	metry is. Transformation to a set of canonical variables R, S, T is algorithmic. The
gilmore-lie-groups_F03018	299	■ 1.000	$R(x,\ y)$	$R(x,y)$	canonical variable $R(x,y)$ is the new independent variable (like x), $S(x,y)$ is the
gilmore-lie-groups_F03019	299	■ 1.000	$S(x,\ y)$	$S(x,y)$	canonical variable $R(x,y)$ is the new independent variable (like x), $S(x,y)$ is the
gilmore-lie-groups_F03020	299	■ 1.000	$T(x,\ y,\ p)$	$T(x,y,p)$	new dependent variable (like y), and $T(x,y,p)$ is the new constraint between S and
gilmore-lie-groups_F03021	299	■ 1.000	$d\ y\ /\ d\ x$	dy/dx	R (like dy/dx). In this new coordinate system the surface and constraint equations
gilmore-lie-groups_F03022	299	■ 0.999	$F(R,\ -,\ T)=0$	$F(R,-,T) = 0$	assume the desired forms $F(R,-,T) = 0$ and $dS/dR = f(R,-,T)$. The system
gilmore-lie-groups_F03023	299	■ 0.999	$d\ S\ /\ d\ R=f(R,\ -,\ T)$	$dS/dR = f(R,-,T)$	assume the desired forms $F(R,-,T) = 0$ and $dS/dR = f(R,-,T)$. The system
gilmore-lie-groups_F03024	299	■ 1.000	$y-G(x)=0$	$y - G(x) = 0$	If we write the solution in the form $y - G(x) = 0$, then the surface $y + c - G(x) =$
gilmore-lie-groups_F03025	299	■ 1.000	$y+c-G(x)=$	$y + c - G(x) =$	If we write the solution in the form $y - G(x) = 0$, then the surface $y + c - G(x) =$
gilmore-lie-groups_F03026	299	■ 1.000	$e^{\{c\ \partial / \ \partial y\}}$	$e^{c\partial/\partial y}$	represented by the Taylor series displacement operator $e^{c\partial/\partial y}$, for
gilmore-lie-groups_F03027	300	■ 1.000	$F(x,\ y,\ p)=p-g(x)$	$F(x,y,p) = p - g(x)$	$F(x,y,p) = p - g(x)$ for the particular case at hand. There are two relations
gilmore-lie-groups_F03028	300	■ 1.000	$x,\ y,\ p$	x,y,p	among the three variables x,y,p . They are given by the surface equation and
gilmore-lie-groups_F03029	300	■ 1.000	$\partial / \partial x,\ \partial / \partial y,\ \partial / \partial p$	$\partial/\partial x,\partial/\partial y,\partial/\partial p$	It is useful to express the action of the three partial derivatives $\partial/\partial x, \partial/\partial y, \partial/\partial p$
gilmore-lie-groups_F03030	300	■ 1.000	$F(x,\ y,\ p)$	$F(x,y,p)$	on the surface $F(x,y,p)$ defining the ordinary differential equation. It is also
gilmore-lie-groups_F03031	300	■ 0.798	$\left(d^{\wedge\{n\}}\ y\ /\ d\ x^{\wedge\{n\}},\ n=1\right)$	$(d^ny/dx^n, n = 1)$	Estabrook, 1976). We do this for equations of first order ($d^ny/dx^n, n = 1$) and
gilmore-lie-groups_F03032	300	■ 1.000	$p^{\wedge\{m\}}=(d\ y\ /\ d\ x)^{\wedge\{m\}},\ m=1$	$p^m = (dy/dx)^m, m = 1$	first degree (depends on $p^m = (dy/dx)^m, m = 1$). The results are independent of
gilmore-lie-groups_F03033	300	■ 1.000	$p-g(x)$	$p - g(x)$	$F(x,y,p) = 0$, is not of the form $p - g(x)$, so that $\frac{\partial}{\partial y}F(x,y,p) \neq 0$, then we
gilmore-lie-groups_F03034	300	■ 1.000	$\frac{\partial}{\partial y}F(x,\ y,\ p)\ \neq\ 0$	$\frac{\partial}{\partial y}F(x,y,p) \neq 0$	$F(x,y,p) = 0$, is not of the form $p - g(x)$, so that $\frac{\partial}{\partial y}F(x,y,p) \neq 0$, then we

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gilmore-lie-groups_ F03035	300	■ 0.999	(R, S, T)	(R, S, T)	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$
gilmore-lie-groups_ F03036	300	■ 0.999	$R=R(x, y)$	$R = R(x, y)$	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$
gilmore-lie-groups_ F03037	300	■ 1.000	$S=S(x, y)$	$S = S(x, y)$	is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03038	300	■ 1.000	$T=T(x, y, p)$	$T = T(x, y, p)$	is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03039	301	■ 1.000	$x \rightarrow x$	$x \rightarrow x$	In the simplest case Eq. (16.1), this one-parameter group is $x \rightarrow x$ and $y \rightarrow y + \epsilon$,
gilmore-lie-groups_ F03040	301	■ 1.000	$y \rightarrow y+\epsilon$	$y \rightarrow y + \epsilon$	In the simplest case Eq. (16.1), this one-parameter group is $x \rightarrow x$ and $y \rightarrow y + \epsilon$,
gilmore-lie-groups_ F03041	301	■ 1.000	$\xi=0, \eta=1$	$\xi = 0, \eta = 1$	so that $\xi = 0, \eta = 1$, and $\zeta = 0$.
gilmore-lie-groups_ F03042	301	■ 1.000	$\zeta(x, y, p)$	$\zeta(x, y, p)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03043	301	■ 1.000	$\xi(x, y)$	$\xi(x, y)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03044	301	■ 1.000	$\eta(x, y)$	$\eta(x, y)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03045	301	■ 1.000	$\eta_{\{x\}}=\partial \eta / \partial x$	$\eta_x = \partial \eta / \partial x$	to first order in ϵ , where $\eta_x = \partial \eta / \partial x$, etc. As a result
gilmore-lie-groups_ F03046	302	■ 1.000	$\xi(x, y), \eta(x, y)$	$\xi(x, y), \eta(x, y)$	Specifically, these equations are used to determine the functions $\xi(x, y), \eta(x, y)$,
gilmore-lie-groups_ F03047	302	■ 1.000	$y: p=p(x, y)$	$y : p = p(x, y)$	x and y : $p = p(x, y)$. This expression is substituted into the determining equation
gilmore-lie-groups_ F03048	302	■ 1.000	$X F(x, y, p(x, y))=0$	$X F(x, y, p(x, y)) = 0$	$X F(x, y, p(x, y)) = 0$, so that this equation depends only on two independent vari-
gilmore-lie-groups_ F03049	302	■ 1.000	d_{ξ}, d_{η}	d_{ξ}, d_{η}	and similarly for η . These representations are truncated at finite degrees d_{ξ}, d_{η} . The
gilmore-lie-groups_ F03050	302	■ 0.999	$X F=0$	$X F = 0$	determining equation $X F = 0$ is expanded into the form $\sum C_{ij} x^i y^j = 0$. Each
gilmore-lie-groups_ F03051	302	■ 0.999	$\sum C_{i j} x^{\{i\}} y^{\{j\}}=0$	$\sum C_{ij} x^i y^j = 0$	determining equation $X F = 0$ is expanded into the form $\sum C_{ij} x^i y^j = 0$. Each
gilmore-lie-groups_ F03052	302	■ 1.000	$C_{i j}$	C_{ij}	coefficient C_{ij} must vanish separately, by standard linear independence arguments.
gilmore-lie-groups_ F03053	302	■ 0.999	$\xi_{i j}, \eta_{i j}$	ξ_{ij}, η_{ij}	ξ_{ij}, η_{ij} . In general, there are more equations than unknowns. Since the equations
gilmore-lie-groups_ F03054	303	■ 1.000	$X(x, y, p) S(x, y)=1$	$X(x, y, p) S(x, y) = 1$	$X(x, y, p) S(x, y) = 1$. We require S to be independent of p , so the condition
gilmore-lie-groups_ F03055	303	■ 1.000	$X S=+1$	$X S = +1$	added to the solution. Further, it is not important that $X S = +1$: we could just as
gilmore-lie-groups_ F03056	303	■ 1.000	$X S=-1$	$X S = -1$	well choose a solution satisfying $X S = -1$ or, for that matter, $X S = k \neq 0$, where
gilmore-lie-groups_ F03057	303	■ 1.000	$X S=k \neq 0$	$X S = k \neq 0$	well choose a solution satisfying $X S = -1$ or, for that matter, $X S = k \neq 0$, where
gilmore-lie-groups_ F03058	303	■ 1.000	$X R=0$	$X R = 0$	transformation group. These functions obey $X R = 0$ and $X T = 0$, which are ex-
gilmore-lie-groups_ F03059	303	■ 1.000	$X T=0$	$X T = 0$	transformation group. These functions obey $X R = 0$ and $X T = 0$, which are ex-
gilmore-lie-groups_ F03060	304	■ 1.000	$R: T=T(R)$	$R : T = T(R)$	as a function of R : $T = T(R)$. This expression is used in the constraint equation,
gilmore-lie-groups_ F03061	304	■ 1.000	$x=x(R, S), y=y(R, S)$	$x = x(R, S), y = y(R, S)$	The inverse relation $x = x(R, S), y = y(R, S)$ is used to express the solution

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gilmore-lie-groups_ F03062	305	—	$x^p + y - xy^2 = 0$	$xp + y - xy^2 = 0$	
gilmore-lie-groups_ F03063	305	—	$0.1 \leq x \leq 1.1$	$0.1 \leq x \leq 1.1$	
gilmore-lie-groups_ F03064	305	—	$-2 \leq y \leq 2$	$-2 \leq y \leq +2$	
gilmore-lie-groups_ F03065	305	■ 1.000	$y \rightarrow \alpha y$	$y \rightarrow \alpha y$	formation $y \rightarrow \alpha y$ and $x \rightarrow \beta x$. Under this transformation the equation trans-

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gilmore-lie-groups_ F03066	305	■ 1.000	$x \rightarrow \beta x$	$x \rightarrow \beta x$	formation $y \rightarrow \alpha y$ and $x \rightarrow \beta x$. Under this transformation the equation trans-
gilmore-lie-groups_ F03067	305	■ 1.000	$\alpha \left(xp + y - (\alpha \beta) xy^2 \right) = 0$	$\alpha (xp + y - (\alpha \beta) xy^2) = 0$	forms to $\alpha (xp + y - (\alpha \beta) xy^2) = 0$. The equation is invariant provided $\alpha \beta = 1$.
gilmore-lie-groups_ F03068	305	■ 1.000	$\alpha \beta = 1$	$\alpha \beta = 1$	forms to $\alpha (xp + y - (\alpha \beta) xy^2) = 0$. The equation is invariant provided $\alpha \beta = 1$.
gilmore-lie-groups_ F03069	305	■ 1.000	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	ant is $x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$. Since there is a one-parameter invariance
gilmore-lie-groups_ F03070	305	■ 1.000	$X(x, y, p)$	$X(x, y, p)$	it is possible to construct the infinitesimal generator $X(x, y, p)$ from this group
gilmore-lie-groups_ F03071	305	■ 1.000	$p = y^2 - y/x$	$p = y^2 - y/x$	The surface $p = y^2 - y/x$ is shown in Fig. 16.1. The value of p clearly depends
gilmore-lie-groups_ F03072	306	■ 1.000	$p: p(x, y) =$	$p: p(x, y) =$	we use the surface equation $F(x, y, p) = 0$ to find an expression for p : $p(x, y) =$
gilmore-lie-groups_ F03073	306	■ 1.000	$-y/x + y^2$	$-y/x + y^2$	$-y/x + y^2$. This is substituted into the determining equation $XF(x, y, p) = 0$ to
gilmore-lie-groups_ F03074	306	■ 1.000	$XF(x, y, p) = 0$	$XF(x, y, p) = 0$	$-y/x + y^2$. This is substituted into the determining equation $XF(x, y, p) = 0$ to
gilmore-lie-groups_ F03075	306	■ 1.000	$\eta: \xi = \xi_{00}, \eta = \eta_{00}$	$\eta: \xi = \xi_{00}, \eta = \eta_{00}$	We first attempt zeroth degree expressions for ξ and η : $\xi = \xi_{00}, \eta = \eta_{00}$. When these
gilmore-lie-groups_ F03076	306	■ 0.999	$y/x, 1$	$y/x, 1$	coefficients of the monomials $y/x, 1$, and xy depend on the unknown parameters
gilmore-lie-groups_ F03077	306	■ 0.999	xy	xy	coefficients of the monomials $y/x, 1$, and xy depend on the unknown parameters
gilmore-lie-groups_ F03078	306	■ 1.000	ξ_{00}, η_{00}	ξ_{00}, η_{00}	ξ_{00}, η_{00} as follows:
gilmore-lie-groups_ F03079	307	■ 1.000	$\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$	$\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$	four equations we determine $\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$, and from the coefficient
gilmore-lie-groups_ F03080	307	■ 1.000	xy^2	xy^2	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_ F03081	307	■ 1.000	$-\xi_{10} - \eta_{01} = 0$	$-\xi_{10} - \eta_{01} = 0$	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_ F03082	307	■ 1.000	$\xi(x, y) = x$	$\xi(x, y) = x$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution
gilmore-lie-groups_ F03083	307	■ 1.000	$\eta(x, y) = -y$	$\eta(x, y) = -y$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution
gilmore-lie-groups_ F03084	307	■ 1.000	$\zeta = -2p$	$\zeta = -2p$	(16.8) gives $\zeta = -2p$, so that the generator of infinitesimal displacements is
gilmore-lie-groups_ F03085	307	■ 1.000	$-y dS(y)/dy = 1$	$-y dS(y)/dy = 1$	y . Such a solution can be found if the equation $-y dS(y)/dy = 1$ can be solved.
gilmore-lie-groups_ F03086	307	■ 1.000	$-\ln(y)$	$-\ln(y)$	The solution, up to an additive constant, is $-\ln(y)$. We will adopt this solution,
gilmore-lie-groups_ F03087	307	■ 1.000	$S(x, y) = \ln(y)$	$S(x, y) = \ln(y)$	neglecting the negative sign: $S(x, y) = \ln(y)$.
gilmore-lie-groups_ F03088	307	■ 1.000	$y dx = -x dy$	$y dx = -x dy$	The first equation for the new independent variable simplifies to $y dx = -x dy$ or
gilmore-lie-groups_ F03089	307	■ 1.000	$d(xy) = 0$	$d(xy) = 0$	$d(xy) = 0$, from which we conclude that $R(x, y) = xy$ is an invariant coordinate
gilmore-lie-groups_ F03090	307	■ 1.000	$R(x, y) = xy$	$R(x, y) = xy$	$d(xy) = 0$, from which we conclude that $R(x, y) = xy$ is an invariant coordinate
gilmore-lie-groups_ F03091	307	■ 1.000	$-dp/2p$	$-dp/2p$	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to
gilmore-lie-groups_ F03092	307	■ 1.000	dx/x	dx/x	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to

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gilmore-lie-groups_ F03093	307	■ 1.000	$d x / x=-d p / 2 p$	$dx/x = -dp/2p$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2 p) = 0$, so that $T(x, y, p) = x^2 p$.
gilmore-lie-groups_ F03094	307	■ 1.000	$(1 / x) d \backslash \text { left } \left(x^{\wedge}\{2\} p \backslash \text { right }\right)=0$	$(1/x)d \left(x^2 p\right) = 0$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2 p) = 0$, so that $T(x, y, p) = x^2 p$.
gilmore-lie-groups_ F03095	307	■ 1.000	$T(x, y, p)=x^{\wedge}\{2\} p$	$T(x, y, p) = x^2 p$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2 p) = 0$, so that $T(x, y, p) = x^2 p$.
gilmore-lie-groups_ F03096	308	—	$T / R+1-$	$T/R+1-$	
gilmore-lie-groups_ F03097	308	—	$R=0$	$R = 0$	

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gilmore-lie-groups_ F03098	308	—	R, S	R, S	
gilmore-lie-groups_ F03099	308	—	$-3 \leq R \leq 4$	$-3 \leq R \leq +4$	
gilmore-lie-groups_ F03100	308	—	$-2 \leq S \leq 2$	$-2 \leq S \leq +2$	
gilmore-lie-groups_ F03101	308	■ 1.000	$T=T(R, S)$	$T = T(R, S)$	pendent of S . This surface $T = T(R, S)$ is plotted in Fig. 16.2. It has the desired

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gilmore-lie-groups_ F03102	308	■ 1.000	$R: T(R)=R^2-R$	$R: T(R) = R^2 - R$	The surface equation is solved for T as a function of R : $T(R) = R^2 - R$. This
gilmore-lie-groups_ F03103	309	■ 0.999	(R, S)	(R, S)	The inverse transformation, Eq. (16.30), from (R, S) to (x, y) is finally used to
gilmore-lie-groups_ F03104	309	■ 1.000	$x d / d x$	$x d / d x$	Remarks The operator $x d / d x$ is the infinitesimal generator for scaling transfor-
gilmore-lie-groups_ F03105	309	■ 1.000	$e^{\lambda x d / d x} x = e^{\lambda} x$	$e^{\lambda x d / d x} x = e^{\lambda} x$	mations, since $e^{\lambda x d / d x} x = e^{\lambda} x$. As a result, the infinitesimal generator X has the
gilmore-lie-groups_ F03106	309	■ 0.766	(x, y, p)	(x, y, p)	following effect on the coordinates (x, y, p) :
gilmore-lie-groups_ F03107	309	■ 1.000	$\ln (y)$	$\ln (y)$	From this scaling behavior, it is easy to see that $\ln (y)$ is linear in the Lie translation
gilmore-lie-groups_ F03108	309	■ 1.000	$\ln \left(e^{-\lambda} y\right)=\ln (y)-\lambda$	$\ln \left(e^{-\lambda} y\right) = \ln (y) - \lambda$	group parameter: $\ln \left(e^{-\lambda} y\right) = \ln (y) - \lambda$. The invariant operators come right out of
gilmore-lie-groups_ F03109	309	■ 1.000	$x^2 p$	$x^2 p$	the scaling transformations: $x y$ and $x^2 p$ are unchanged by the scaling transforma-
gilmore-lie-groups_ F03110	309	■ 1.000	$\ln \left(x y^2\right)$	$\ln \left(x y^2\right)$	tion. None of these operators is unique. The operator $\ln \left(x y^2\right)$ is linear and $x^3 y p$
gilmore-lie-groups_ F03111	309	■ 1.000	$x^3 y p$	$x^3 y p$	tion. None of these operators is unique. The operator $\ln \left(x y^2\right)$ is linear and $x^3 y p$
gilmore-lie-groups_ F03112	309	■ 1.000	$F\left(x y, x^2 p\right)=0$	$F\left(x y, x^2 p\right) = 0$	Eq. (16.36) has the form $F(R,-, T)=0$ or more simply $F(x y, x^2 p)=0$. The
gilmore-lie-groups_ F03113	309	■ 1.000	$x^2 p=h(x y)$	$x^2 p = h(x y)$	$x^2 p = h(x y)$ or $d y / d x = x^{-2} h(x y)$. For the equation studied in Section 16.3,
gilmore-lie-groups_ F03114	309	■ 1.000	$d y / d x=x^{-2} h(x y)$	$d y / d x = x^{-2} h(x y)$	$x^2 p = h(x y)$ or $d y / d x = x^{-2} h(x y)$. For the equation studied in Section 16.3,
gilmore-lie-groups_ F03115	309	■ 1.000	$h(z)=-z+z^2$	$h(z) = -z + z^2$	$h(z) = -z + z^2$. For the Riccati equation $d y / d x + y^2 - 2 / x^2 = 0, h(z) = z^2 - 2$.
gilmore-lie-groups_ F03116	309	■ 1.000	$d y / d x+y^2-2 / x^2=0, h(z)=z^2-2$	$d y / d x + y^2 - 2 / x^2 = 0, h(z) = z^2 - 2$	$h(z) = -z + z^2$. For the Riccati equation $d y / d x + y^2 - 2 / x^2 = 0, h(z) = z^2 - 2$.
gilmore-lie-groups_ F03117	309	■ 1.000	$y^{\prime 2}+y^4-x^{-4}=0$	$y^{\prime 2} + y^4 - x^{-4} = 0$	For example, the first order, second degree equation $y^{\prime 2} + y^4 - x^{-4} = 0$ has
gilmore-lie-groups_ F03118	309	■ 1.000	$R^4+T^2=1$	$R^4 + T^2 = 1$	canonical form $R^4 + T^2 = 1$. The original equation has two solution branches
gilmore-lie-groups_ F03119	310	■ 1.000	$p=\pm \sqrt{x^{-4}-y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$, corresponding to the two solution branches in the canonical
gilmore-lie-groups_ F03120	310	■ 1.000	$T=\pm \sqrt{1-R^4}$	$T = \pm \sqrt{1 - R^4}$	coordinate system $T = \pm \sqrt{1 - R^4}$
gilmore-lie-groups_ F03130	310	■ 0.998	$y^{(2)}$	$y^{(2)}$	involves derivatives with respect to $y^{(2)}$ and is given by
gilmore-lie-groups_ F03234	312	■ 1.000	$F\left(x, y, y^{\prime}, y^{\prime \prime}\right)=0$	$F\left(x, y, y^{\prime}, y^{\prime \prime}\right) = 0$	Suppose a second order equation $F(x, y, y', y'') = 0$ is invariant under the scal-
gilmore-lie-groups_ F03235	312	■ 1.000	$S=\ln y$	$S = \ln y$	ing group (16.36). Then the dependent coordinate is $S = \ln y$ and the surface equa-
gilmore-lie-groups_ F03236	312	■ 1.000	$F(R,-, T, U)=0$	$F(R,-, T, U) = 0$	tion can be expressed in terms of three invariant coordinates as $F(R,-, T, U) = 0$.
gilmore-lie-groups_ F03237	312	■ 1.000	$y, T=T\left(x, y, y^{\prime}\right)$	$y, T = T\left(x, y, y^{\prime}\right)$	Here as before R depends only on x and $y, T = T(x, y, y')$, and $U = U(x, y, y', y'')$
gilmore-lie-groups_ F03238	312	■ 1.000	$U=U\left(x, y, y^{\prime}, y^{\prime \prime}\right)$	$U = U\left(x, y, y^{\prime}, y^{\prime \prime}\right)$	Here as before R depends only on x and $y, T = T(x, y, y')$, and $U = U(x, y, y', y'')$
gilmore-lie-groups_ F03239	312	■ 1.000	$d T / d R$	$d T / d R$	nate? It is simple to see that the derivative $d T / d R$ is invariant under the group. Not

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gilmore-lie-groups_ F03240	312	■ 1.000	$y^{\prime \prime}$	y''	transformations Eq. (16.36) that is of first degree in y'' is
gilmore-lie-groups_ F03241	313	■ 1.000	$F\left(x, y, \ldots, y^{(n)}\right)=0$	$F\left(x, y, \ldots, y^{(n)}\right)=0$	equation $F(x, y, \ldots, y^{(n)})=0$. As usual, we seek an infinitesimal generator
gilmore-lie-groups_ F03242	313	■ 1.000	$T\left(x, y, y^{(1)}\right)$	$T\left(x, y, y^{(1)}\right)$	In fact, only the first two invariant coordinates $R(x, y)$ and $T(x, y, y^{(1)})$ need be
gilmore-lie-groups_ F03243	313	■ 1.000	$d T^{(j)} / d R^{(j)}, j=0$	$d T^{(j)} / d R^{(j)}, j=0$	computed. The remaining invariant coordinates are $d T^{(j)} / d R^{(j)}, j=0$ (for T) and
gilmore-lie-groups_ F03244	313	■ 1.000	$j=1,2, \ldots, n-1$	$j=1,2, \ldots, n-1$	$j=1,2, \ldots, n-1$. Each of these latter is of first degree in $y^{(j+1)}$. As a result,
gilmore-lie-groups_ F03245	313	■ 1.000	$y^{(j+1)}$	$y^{(j+1)}$	$j=1,2, \ldots, n-1$. Each of these latter is of first degree in $y^{(j+1)}$. As a result,
gilmore-lie-groups_ F03246	314	■ 1.000	$x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$	$x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$	scaling transformations $x^i \rightarrow \lambda x^i, u \rightarrow \alpha u$. Under the scaling transformation
gilmore-lie-groups_ F03247	314	■ 1.000	$\delta(x) \rightarrow \delta(\lambda x)=\lambda^{-n} \delta(x)$	$\delta(x) \rightarrow \delta(\lambda x)=\lambda^{-n} \delta(x)$	$\delta(x) \rightarrow \delta(\lambda x)=\lambda^{-n} \delta(x)$ so that
gilmore-lie-groups_ F03248	314	■ 1.000	$\alpha=\lambda^{2-n}$	$\alpha=\lambda^{2-n}$	The equation is invariant provided $\alpha=\lambda^{2-n}$. The infinitesimal generators of sym-
gilmore-lie-groups_ F03249	314	■ 1.000	$R=R(x, u)$	$R=R(x, u)$	A new independent coordinate $R=R(x, u)$ satisfies $X R=0$, where X is any linear
gilmore-lie-groups_ F03250	314	■ 1.000	$R \sim u x ^{n-2}$	$R \sim u x ^{n-2}$	combination of the generators in Eq. (16.52). A solution is $R \sim u x ^{n-2}$. As a result,
gilmore-lie-groups_ F03251	314	■ 1.000	$u \sim x ^{2-n}=k x ^{2-n}$	$u \sim x ^{2-n}=k x ^{2-n}$	$u \sim x ^{2-n}=k x ^{2-n}$. The constant of proportionality can be computed using the
gilmore-lie-groups_ F03252	314	■ 1.000	$V\left(S^{n-1}\right)=2 \pi^{n / 2} / \Gamma\left(\frac{n}{2}\right)$	$V\left(S^{n-1}\right)=2 \pi^{n / 2} / \Gamma\left(\frac{n}{2}\right)$	Here $V\left(S^{n-1}\right)=2 \pi^{n / 2} / \Gamma\left(\frac{n}{2}\right)$ is the surface area of a unit sphere in R^n . As a result,
gilmore-lie-groups_ F03253	314	■ 1.000	$R^n(n \neq 2)$	$R^n(n \neq 2)$	the solution of Laplace's equation in R^n ($n \neq 2$) with unit source term at the origin
gilmore-lie-groups_ F03254	314	■ 1.000	$u(x, t)$	$u(x, t)$	The heat equation on R^n for $u(x, t)$ with source term
gilmore-lie-groups_ F03255	314	■ 1.000	$u \rightarrow \alpha u, t \rightarrow \beta t$	$u \rightarrow \alpha u, t \rightarrow \beta t$	X_{ij} are Lie symmetries. Under the scaling transformation $u \rightarrow \alpha u, t \rightarrow \beta t$, and
gilmore-lie-groups_ F03256	314	■ 1.000	$x^i \rightarrow \lambda x^i$	$x^i \rightarrow \lambda x^i$	$x^i \rightarrow \lambda x^i$ the equation transforms as follows:
gilmore-lie-groups_ F03257	314	■ 1.000	$\alpha \lambda^n=1$	$\alpha \lambda^n=1$	the three scaling variables (since there is only one equation): $\alpha \lambda^n=1$ and $\beta / \lambda^2=1$.
gilmore-lie-groups_ F03258	314	■ 1.000	$\beta / \lambda^2=1$	$\beta / \lambda^2=1$	the three scaling variables (since there is only one equation): $\alpha \lambda^n=1$ and $\beta / \lambda^2=1$.
gilmore-lie-groups_ F03259	315	■ 1.000	$n+1$	$n+1$	From these relations it is possible to construct $n+1$ additional Lie symmetries, so
gilmore-lie-groups_ F03260	315	■ 1.000	x^i, t	x^i, t	An invariant coordinate depending on the x^i, t and u is $R=u t^{n / 2} e^{ x ^2 / 4 t}$, from
gilmore-lie-groups_ F03261	315	■ 1.000	$R=u t^{n / 2} e^{ x ^2 / 4 t}$	$R=u t^{n / 2} e^{ x ^2 / 4 t}$	An invariant coordinate depending on the x^i, t and u is $R=u t^{n / 2} e^{ x ^2 / 4 t}$, from
gilmore-lie-groups_ F03262	315	■ 1.000	$F(R,-, T)=$	$F(R,-, T)=$	3. Under what conditions is it possible to solve the canonical surface equation $F(R,-, T)=$
gilmore-lie-groups_ F03263	315	■ 0.980	$\int f(R,-, T(R)) d R$	$\int f(R,-, T(R)) d R$	$\int f(R,-, T(R)) d R$
gilmore-lie-groups_ F03265	316	■ 1.000	$\zeta(x, y, p)=\eta^{(1)}\left(x, y, y^{(1)}\right)$	$\zeta(x, y, p)=\eta^{(1)}\left(x, y, y^{(1)}\right)$	(ii) Construct the first prolongation $\zeta(x, y, p)=\eta^{(1)}(x, y, y^{(1)})$ from the functions defin-
gilmore-lie-groups_ F03266	316	■ 1.000	$X=\xi \partial / \partial x+\eta \partial / \partial y+\zeta \partial / \partial p$	$X=\xi \partial / \partial x+\eta \partial / \partial y+\zeta \partial / \partial p$	(iii) Introduce the operator $X=\xi \partial / \partial x+\eta \partial / \partial y+\zeta \partial / \partial p$. This describes a Taylor series

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gilmore-lie-groups_F03267	316	■ 1.000	$F=0$	$F = 0$	(iv) Solve the determining equation $XF = 0$ when $F = 0$ for the functions $\xi(x, y)$ and
gilmore-lie-groups_F03268	316	■ 1.000	$X\ R=0, \ X\ S=1, \ X\ T=0$	$XR = 0, XS = 1, XT = 0$	(v) Solve the determining equations $XR = 0, XS = 1, XT = 0$ for the canonical coor-
gilmore-lie-groups_F03269	316	■ 1.000	$F \rightarrow F(R, -, T)=0$	$F \rightarrow F(R, -, T) = 0$	equation is independent of the new dependent variable: $F \rightarrow F(R, -, T) = 0$.
gilmore-lie-groups_F03270	317	■ 1.000	$S: S=\int f(R, -, T(R))+c$	$S: S = \int f(R, -, T(R)) + c$	(viii) Solve the constraint equation for S : $S = \int f(R, -, T(R)) + c$.
gilmore-lie-groups_F03271	317	■ 1.000	$x=x(R, S)$	$x = x(R, S)$	(ix) Backsubstitute the original coordinates for the new coordinates, $x = x(R, S)$ and
gilmore-lie-groups_F03272	317	■ 1.000	$y=y(R, S)$	$y = y(R, S)$	$y = y(R, S)$, to obtain the solution of the original equation.
gilmore-lie-groups_F03273	317	■ 1.000	$n=0,1$	$n = 0, 1$	Show that the first two terms $n = 0, 1$ are exactly the determining equations
gilmore-lie-groups_F03274	317	■ 1.000	$V(\mathbf{x})$	$V(\mathbf{x})$	mass m in the presence of a potential $V(\mathbf{x})$ is
gilmore-lie-groups_F03275	317	■ 1.000	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$. Assume also that the potential is homogeneous of degree
gilmore-lie-groups_F03276	317	■ 1.000	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$ (Landau and Lifshitz, 1960). Under this scaling transformation
gilmore-lie-groups_F03277	317	■ 1.000	$\alpha^1 \beta^{2-k}$	$\alpha^1 \beta^{2-k}$	a. Show that the scaled equation is identical to the original provided $\alpha^1 \beta^{2-k}$
gilmore-lie-groups_F03278	317	■ 1.000	$\gamma^{-2}=1$	$\gamma^{-2} = 1$	$\gamma^{-2} = 1$.
gilmore-lie-groups_F03279	317	■ 1.000	$\alpha=1$	$\alpha = 1$	b. Set $\alpha = 1$. Show that trajectories are invariant under the scaling transformation with
gilmore-lie-groups_F03280	317	■ 1.000	$\gamma^2=\beta^{2-k}$	$\gamma^2 = \beta^{2-k}$	$\gamma^2 = \beta^{2-k}$. Show that in the cases $k = -1, k = 0, k, = +1, k = +2$ the following
gilmore-lie-groups_F03281	317	■ 1.000	$k=-1, k=0, k,=+1, k=+2$	$k = -1, k = 0, k, = +1, k = +2$	$\gamma^2 = \beta^{2-k}$. Show that in the cases $k = -1, k = 0, k, = +1, k = +2$ the following
gilmore-lie-groups_F03286	318	■ 0.785	γ^2	γ^2	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'
gilmore-lie-groups_F03287	318	■ 0.785	β^3	β^3	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'
gilmore-lie-groups_F03288	318	■ 0.785	$R^{\prime}$	R'	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'
gilmore-lie-groups_F03289	318	■ 1.000	$T^{\prime}$	T'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and
gilmore-lie-groups_F03290	318	■ 1.000	$P^{\prime}$	P'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and
gilmore-lie-groups_F03291	318	■ 1.000	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$. The second line is a statement of the integral of
gilmore-lie-groups_F03292	318	■ 0.860	(β)	(β)	traveled (β) is proportional to the time elapsed (γ). The third line is a statement that
gilmore-lie-groups_F03293	318	■ 0.860	(γ)	(γ)	traveled (β) is proportional to the time elapsed (γ). The third line is a statement that
gilmore-lie-groups_F03294	318	■ 1.000	$V=m\ g\ z$	$V = mgz$	in a local gravitational potential of the form $V = mgz$, the distance fallen increases
gilmore-lie-groups_F03295	318	■ 1.000	$\gamma=1$	$\gamma = 1$	c. Fix $\gamma = 1$ and construct a table relating the mass and orbital scale under the four
gilmore-lie-groups_F03296	318	■ 1.000	$\beta=1$	$\beta = 1$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.
gilmore-lie-groups_F03297	318	■ 1.000	$\sqrt{M}$	$\sqrt{M}$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.

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gilmore-lie-groups_F03298	318	■ 0.994	$\alpha \beta^2 \gamma^{-2} = \beta^k$	$\alpha \beta^2 \gamma^{-2} = \beta^k$	f. Show that the kinetic energy scales like $\alpha \beta^2 \gamma^{-2} = \beta^k$ (use a). Since the potential
gilmore-lie-groups_F03299	318	■ 1.000	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d\mathbf{x} = 0$. Show that if a scaling transformation leaves the Lagrangian invari-
gilmore-lie-groups_F03300	318	■ 1.000	$v_{\{i\}}$	v_i	Show that the following six differential operators v_i are infinitesimal generators of
gilmore-lie-groups_F03301	318	■ 1.000	$e^{\epsilon v_i} f(x, t)$	$e^{\epsilon v_i} f(x, t)$	the invariance group of this equation. Show that $e^{\epsilon v_i} f(x, t)$ has the action shown for
gilmore-lie-groups_F03307	319	■ 1.000	$u \partial_u$	$u \partial_u$	all the rest, and is $u \partial_u$.
gilmore-lie-groups_F03317	319	■ 1.000	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$	
gilmore-lie-groups_F03318	319	■ 1.000	$\lambda^2 = 1/(1 + 4\epsilon t)$	$\lambda^2 = 1/(1 + 4\epsilon t)$	where $\lambda^2 = 1/(1 + 4\epsilon t)$
gilmore-lie-groups_F03319	319	■ 1.000	$D_2 = u \partial_u$	$D_2 = u \partial_u$	Show that $D_2 = u \partial_u$ commutes with all remaining generators. Construct the commu-
gilmore-lie-groups_F03320	319	■ 1.000	$SO(2+1, 1+1) = SO(3, 2)$	$SO(2+1, 1+1) = SO(3, 2)$	$SO(2+1, 1+1) = SO(3, 2)$.
gilmore-lie-groups_F03321	319	■ 1.000	$SO(3+1, 1+1) = SO(4, 2)$	$SO(3+1, 1+1) = SO(4, 2)$	$SO(3+1, 1+1) = SO(4, 2)$ (Bateman, 1910). The extra generator commutes with
gilmore-lie-groups_F03322	319	■ 1.000	$u_{xx} - u_t = 0$	$u_{xx} - u_t = 0$	8. The heat equation in one dimension is $u_{xx} - u_t = 0$. The infinitesimal generator of
gilmore-lie-groups_F03323	319	■ 1.000	$X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \dots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \dots$	$X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \dots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \dots$	symmetries for this equation is $X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \dots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \dots$.
gilmore-lie-groups_F03324	320	■ 1.000	$h(x, t)$	$h(x, t)$	Here $h(x, t)$ is any function that satisfies the homogeneous heat equation. Construct
gilmore-lie-groups_F03325	320	■ 0.986	$\left(t \rightarrow t' = T(t, x, \epsilon) = t + \right.$	$(t \rightarrow t' = T(t, x, \epsilon) = t +$	Under a one-parameter family of change of variables $(t \rightarrow t' = T(t, x, \epsilon) = t +$
gilmore-lie-groups_F03326	320	■ 1.000	$\left. \epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x) \right)$	$\epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x)$	$\xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x)$ the action integral transforms to
gilmore-lie-groups_F03327	320	■ 1.000	$dt'/dt = \partial T / \partial t + (\partial T / \partial x_i) dx_i / dt$	$dt'/dt = \partial T / \partial t + (\partial T / \partial x_i) dx_i / dt$	where $dt'/dt = \partial T / \partial t + (\partial T / \partial x_i) dx_i / dt$. Show that if you differentiate the action
gilmore-lie-groups_F03328	320	■ 1.000	$\epsilon = 0$	$\epsilon = 0$	integral with respect to ϵ , then set $\epsilon = 0$ the result is
gilmore-lie-groups_F03329	321	■ 1.000	$L[u] =$	$L[u] =$	the dynamics of a system is derivable from an action integral of the form $L[u] =$
gilmore-lie-groups_F03330	321	■ 1.000	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$, and suppose the infinitesimal generators that leave the
gilmore-lie-groups_F03331	321	■ 1.000	$d\rho(g)$	$d\rho(g)$	12. Representation theory G is a compact Lie group with invariant measure $d\rho(g)$
gilmore-lie-groups_F03332	321	■ 0.995	$\text{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$	$\text{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$	and volume $\text{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$ are the irreducible representations of G con-
gilmore-lie-groups_F03333	321	■ 1.000	$\phi(g), \psi(g)$	$\phi(g), \psi(g)$	structed by reduction of tensor products (Wigner–Stone theorem), and $\phi(g), \psi(g)$
gilmore-lie-groups_F03334	322	■ 1.000	$\psi(g) = \langle g \psi \rangle$	$\psi(g) = \langle g \psi \rangle$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \psi \rangle$ and $\langle \lambda_{\mu\nu} \psi \rangle =$
gilmore-lie-groups_F03335	322	■ 1.000	$\langle \lambda_{\mu\nu} \psi \rangle =$	$\langle \lambda_{\mu\nu} \psi \rangle =$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \psi \rangle$ and $\langle \lambda_{\mu\nu} \psi \rangle =$
gilmore-lie-groups_F03336	322	■ 1.000	$\int d\rho(g) \langle \lambda_{\mu\nu} g \rangle \langle g \psi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} g \rangle \langle g \psi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} g \rangle \langle g \psi \rangle$ (and similarly for $\phi(g) = \langle g \phi \rangle$) using the Dirac represen-

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gilmore-lie-groups_ F03337	322	■ 1.000	$\langle \phi(g) = \langle g \mid \phi \rangle$	$\phi(g) = \langle g \mid \phi \rangle$	$\int d\rho(g) \langle \lambda_{\mu\nu} g \rangle \langle g \psi \rangle$ (and similarly for $\phi(g) = \langle g \phi \rangle$) using the Dirac represen-
gilmore-lie-groups_ F03338	322	■ 1.000	$\int \phi^*(g) \psi(g) d\rho(g)$	$\int \phi^*(g) \psi(g) d\rho(g)$	tation. Write down the Parseval equality for the inner product $\int \phi^*(g) \psi(g) d\rho(g)$
gilmore-lie-groups_ F03339	323	■ 1.000	$\mathrm{He}^{\{+\}}$	He^+	O. BELY (1966), Quantum defect theory III. Electron scattering by He^+ , <i>Proc. Phys. Soc.</i>
gilmore-lie-groups_ F03340	324	■ 0.868	$A_{\{n\}} ; D_{\{n\}}, B_{\{n\}}, C_{\{n\}}, J$	$A_n; D_n, B_n, C_n, J$	$A_n; D_n, B_n, C_n, J$, <i>Math. Phys.</i> 11 , 513–523.
gilmore-lie-groups_ F03341	325	■ 0.866	$\mathrm{mx} / \sim$	$\mathrm{mx} / \sim$	$\mathrm{mx} / \sim$ mcintosh/comun/symm/symm.html.
gilmore-lie-groups_ F03342	327	■ 0.995	$A(p\ q),\ 39, 48$	$A(pq), 39, 48$	$A(p\ q), 39, 48$
gilmore-lie-groups_ F03343	327	■ 0.978	$A_{\{1\}},\ 161$	$A_1, 161$	$A_1, 161$
gilmore-lie-groups_ F03344	327	■ 0.991	$A_{\{2\}}$	A_2	$A_2, 151, 160$
gilmore-lie-groups_ F03345	327	■ 0.865	$A_{\{3\}},\ 46, 161, 162$	$A_3, 46, 161, 162$	$A_3, 46, 161, 162$
gilmore-lie-groups_ F03346	327	■ 0.946	$A_{\{n\}},\ 46, 49, 161, 166$	$A_n, 46, 49, 161, 166$	$A_n, 46, 49, 161, 166$
gilmore-lie-groups_ F03347	327	■ 0.992	$B_{\{1\}},\ 161$	$B_1, 161$	$B_1, 161$
gilmore-lie-groups_ F03348	327	■ 0.749	$B_{\{2\}},\ 151, 160, 161, 164$	$B_2, 151, 160, 161, 164$	$B_2, 151, 160, 161, 164$
gilmore-lie-groups_ F03349	327	■ 0.761	$B_{\{3\}},\ 162$	$B_3, 162$	$B_3, 162$
gilmore-lie-groups_ F03350	327	■ 0.729	$\boldsymbol{B}_{\{n\}},\ 161, 166, 168$	$\boldsymbol{B}_n, 161, 166, 168$	$B_n, 161, 166, 168$
gilmore-lie-groups_ F03351	327	■ 1.000	$C_{\{1\}},\ 161$	$C_1, 161$	$C_1, 161$
gilmore-lie-groups_ F03352	327	■ 0.523	$C_{\{2\}},\ 151, 160, 161, 164$	$C_2, 151, 160, 161, 164$	$C_2, 151, 160, 161, 164$
gilmore-lie-groups_ F03353	327	■ 0.712	$C_{\{3\}},\ 162$	$C_3, 162$	$C_3, 162$
gilmore-lie-groups_ F03354	327	■ 0.997	$C_{\{n\}},\ 161, 166, 168$	$C_n, 161, 166, 168$	$C_n, 161, 166, 168$
gilmore-lie-groups_ F03355	327	■ 0.840	$D_{\{2\}},\ 151, 160, 162$	$D_2, 151, 160, 162$	$D_2, 151, 160, 162$
gilmore-lie-groups_ F03356	327	■ 0.435	$G\ L(1 ; \mathbb{Q}),\ 40, 47$	$GL(1; \mathbb{Q}), 40, 47$	$GL(1; \mathbb{Q}), 40, 47$
gilmore-lie-groups_ F03357	327	■ 0.979	$G\ L(2 ; \mathbb{C}),\ 47$	$GL(2; \mathbb{C}), 47$	$GL(2; \mathbb{C}), 47$
gilmore-lie-groups_ F03358	327	■ 0.788	$G\ L(2 ; \mathbb{R})$	$GL(2; \mathbb{R})$	$GL(2; \mathbb{R}), 43$
gilmore-lie-groups_ F03359	327	■ 0.731	$G\ L(2 ; \mathbb{Z}),\ 45, 49$	$GL(2; \mathbb{Z}), 45, 49$	$GL(2; \mathbb{Z}), 45, 49$
gilmore-lie-groups_ F03360	327	■ 0.903	$G\ L(3 ; \mathbb{Z}),\ 45$	$GL(3; \mathbb{Z}), 45$	$GL(3; \mathbb{Z}), 45$
gilmore-lie-groups_ F03361	327	■ 0.471	$G\ L(n ; \mathbb{F}),\ 34, 36, 74$	$GL(n; \mathbb{F}), 34, 36, 74$	$GL(n; \mathbb{F}), 34, 36, 74$
gilmore-lie-groups_ F03362	327	■ 0.698	$G\ L(n ; \mathbb{Q})$	$GL(n; \mathbb{Q})$	$GL(n; \mathbb{Q}), 47$
gilmore-lie-groups_ F03363	327	■ 0.973	$G_{\{2\}},\ 151, 160, 162, 165$	$G_2, 151, 160, 162, 165$	$G_2, 151, 160, 162, 165$
gilmore-lie-groups_ F03364	327	■ 0.933	$\operatorname{HT}(p,\ q),\ 37, 48$	$HT(p, q), 37, 48$	$HT(p, q), 37, 48$
gilmore-lie-groups_ F03365	327	■ 0.634	$\operatorname{Nil}(n),\ 38, 48$	$Nil(n), 38, 48$	$Nil(n), 38, 48$

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gilmore-lie-groups_F03369	327	■ 0.991	$O(n)$, 40, 43, 145	$O(n)$, 40, 43, 145	$O(n)$, 40, 43, 145
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gilmore-lie-groups_F03372	327	■ 0.737	$O(p, q)$	$O(p, q)$	$O(p, q)$, 43
gilmore-lie-groups_F03373	327	■ 0.939	P_n , 45	P_n , 45	P_n , 45
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gilmore-lie-groups_F03375	327	■ 0.985	$SL(2; \mathbb{R})$, 26, 28, 29, 30, 41, 43, 56, 58, 62, 100, 102	$SL(2; \mathbb{R})$, 26, 28, 29, 30, 41, 43, 56, 58, 62, 100, 102	$SL(2; \mathbb{R})$, 26, 28, 29, 30, 41, 43, 56, 58, 62, 100, 102,
gilmore-lie-groups_F03376	327	■ 0.956	$SL(n; \mathbb{R})$, 30	$SL(n; \mathbb{R})$, 30	$SL(n; \mathbb{R})$, 30
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gilmore-lie-groups_F03378	327	■ 0.519	$SL(n; \mathbb{Q})$	$SL(n; \mathbb{Q})$	$SL(n; \mathbb{Q})$, 43
gilmore-lie-groups_F03379	327	■ 0.747	$SL(n; \mathbb{R})$	$SL(n; \mathbb{R})$	$SL(n; \mathbb{R})$, 43, 47, 164
gilmore-lie-groups_F03380	327	■ 0.880	$SL(n; \mathbb{Z})$, 45	$SL(n; \mathbb{Z})$, 45	$SL(n; \mathbb{Z})$, 45
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gilmore-lie-groups_F03383	327	■ 0.820	$SO(2)$, 48, 164	$SO(2)$, 48, 164	$SO(2)$, 48, 164
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gilmore-lie-groups_F03387	327	■ 0.979	$SO(3)$, 49, 90, 106	$SO(3)$, 49, 90, 106	$SO(3)$, 49, 90, 106
gilmore-lie-groups_F03388	327	■ 0.875	$SO(3) / SO(2)$, 107	$SO(3) / SO(2)$, 107	$SO(3) / SO(2)$, 107
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gilmore-lie-groups_F03390	327	■ 0.463	$SO(n)$	$SO(n)$	$SO(n)$, 43, 145
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gilmore-lie-groups_F03392	327	■ 0.998	$SU(1, 1)$, 38, 43, 48, 105	$SU(1, 1)$, 38, 43, 48, 105	$SU(1, 1)$, 38, 43, 48, 105
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gilmore-lie-groups_F03411	328	■ 1.000	$U(2\ ;\ \mathbb{Q})$, 164	$U(2;\mathbb{Q})$, 164	$U(2;\mathbb{Q})$, 164
gilmore-lie-groups_F03412	328	■ 0.804	$U(n)$, 40,43,90	$U(n)$, 40, 43, 90	$U(n)$, 40, 43, 90
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gilmore-lie-groups_F03456	331	■ 0.857	$\mathfrak{s} \mathfrak{u}(n)$, 80,132,178	$\mathfrak{su}(n)$, 80, 132, 178	$\mathfrak{su}(n)$, 80, 132, 178
gilmore-lie-groups_F03457	331	■ 1.000	$\mathfrak{s} \mathfrak{u}(p,q)$, 85,178	$\mathfrak{su}(p,q)$, 85, 178	$\mathfrak{su}(p,q)$, 85, 178
gilmore-lie-groups_F03458	331	■ 0.882	$\mathfrak{s} \mathfrak{u}^*(2n)$, 180	$\mathfrak{su}^*(2n)$, 180	$\mathfrak{su}^*(2n)$, 180
gilmore-lie-groups_F03459	331	■ 0.999	$\mathfrak{usp}(2n)$, 179	$\mathfrak{usp}(2n)$, 179	$\mathfrak{usp}(2n)$, 179
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gilmore-lie-groups_F03462	331	■ 1.000	$\mathfrak{ut}(p,q)$, 75,131	$\mathfrak{ut}(p,q)$, 75, 131	$\mathfrak{ut}(p,q)$, 75, 131
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